PATHOLOGY — REVISION NOTES

– MCQ-focused · Orthobullets + OB/TO question explanations + TO lecture notes · 67 chapters (1 from TO notes) –

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Every chapter, same order: 1 Basics → 2 Mechanism → 3 Presentation → 4 Diagnosis & Classification → 5 Management → 6 Complications → X vs Y → How it's asked

Always included where sources have them: criteria (diagnostic, operative) · genes & lab markers · key radiology findings

Key: TO = Target Ortho fact (must-know) · ↑ increased · ↓ decreased · → leads to / then · ✗ Wrong options = MCQ distractors · ✓ correct answer · "TO — verify" = TO figure that differs from standard teaching

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ADAMANTINOMA

– Pathology · 1 / 66 –

1 Basics

Epidemiology

Rare	<300 cases documented · **<1% of malignant primary bone tumours**
Age	young adults **20–40 yr** · 75% in 2nd–4th decades
Site	**mid-tibial diaphysis, anterior cortex** (tibia **80–90%**) · synchronous ipsilateral **fibula 10–50%** · other long bones rare
Nature	**low-grade malignant** · aetiology unknown · arises from **aberrant epithelial nests** (TO)
TO profile	tibia, **3rd decade**, **males** (TO)

Osteofibrous dysplasia association

OFD link	controversial · once thought **precursor** → now doubted · some regard as a **continuum**
Why it matters	OFD **benign → observe** · adamantinoma **malignant → wide resection**

Genes & lab markers

Cytokeratin (keratin) +	epithelial nests stain positive · marks epithelial differentiation

3 Presentation

Symptoms	**pain of months to years** · indolent course
Signs	**anterior tibial bowing** or **palpable firm mass** · no soft-tissue mass, no lymphadenopathy

4 Diagnosis & Classification

Key radiology findings

X-ray	**multiple sharply circumscribed lucencies ("soap bubble")** with interspersed sclerosis · **eccentric, intracortical, lytic** · anterior mid-tibia

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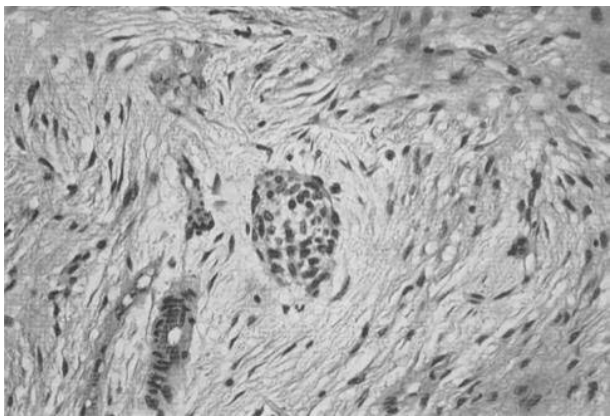


Soap-bubble lytic lesions, anterior tibial cortex

Evolution	serial films show growth and **erosion through the cortex** → favours adamantinoma over OFD
Periosteal reaction	typically **none** (unlike other primary bone tumours)
MRI	cortical erosion / destruction of anterior tibial cortex

Histology

Biphasic	**epithelial + fibrous mesenchymal** cells
Epithelial nests	**palisading or glandular** pattern · basaloid, spindle, tubular, squamous patterns (basaloid + tubular commonest) · keratin +
Stroma	background **fibrous (spindle-cell) stroma** · may have nuclear atypia, mitoses



Epithelial nests in fibrous stroma

Histology – which pattern is which?

Histology	Diagnosis
Biphasic glandular epithelial cells + spindle cells	✓ **Adamantinoma**
Woven trabeculae **with osteoblastic rimming**, fibrous stroma ("Chinese letters")	Osteofibrous dysplasia
Trabeculae **without osteoblastic rimming** in fibrous stroma	Fibrous dysplasia
Multinucleated giant cells, stromal nuclei = giant-cell nuclei	Giant cell tumour
Herringbone pattern, pleomorphic cells	Fibrosarcoma

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Differential of a tibial diaphyseal lesion

	Tibial diaphysis	Wide resection alone
Adamantinoma	Yes	Yes
Osteofibrous dysplasia	Yes	No (observe)
Chondrosarcoma (conventional)	–	Yes
Parosteal osteosarcoma	–	Yes

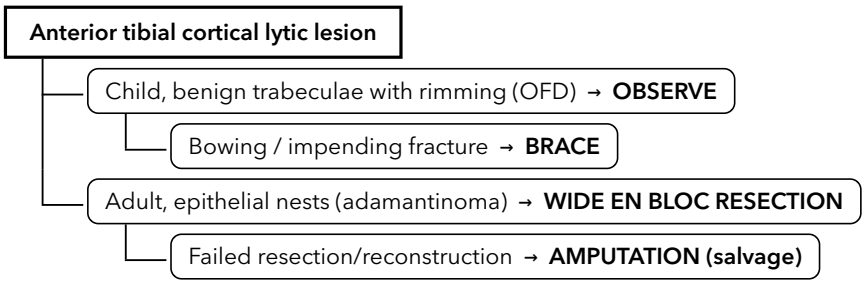
Ewing medullary, aggressive periosteal reaction, small round blue cells

Chondromyxoid fibroma anterior **proximal** tibia, **medullary**, thins cortex

Osteoid osteoma cortical, **<1 cm**, sclerotic rim

5 Management

Treatment – wide resection



Wide-margin en bloc resection standard of care · prevents recurrence + metastasis

Reconstruction **intercalary** resection → **allograft** or intercalary megaprosthesis · amputation usually unnecessary

Biological reconstruction preferred (TO) **intercalary resection + vascularised fibula** transfer

No chemo / no radiotherapy low-grade, not effective for local control

Not indicated observation · curettage + graft · external beam radiation · chemo + excision

6 Complications

Metastasis **lungs ~25%** (Orthobullets) · up to **20%** (OB explanation) · **late** → long-term follow-up

Local recurrence **high unless wide margin** · uncommon after negative-margin excision

X vs Y

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	Adamantinoma	Osteofibrous dysplasia
Age	**Adult 20–40**	**Child** (<10)
Behaviour	**Low-grade malignant**	Benign
Imaging	Cortical, may **erode through cortex**, progresses	Eccentric intracortical lytic, anterior bowing
Histology	**Epithelial nests (keratin +)** in fibrous stroma	Woven bone **with osteoblastic rimming** · no epithelial nests
Treatment	**Wide resection**	**Observe** · brace if bowing · surgery delayed to maturity (recurs)

How it's asked

"Adult, anterior mid-tibial soap-bubble cortical lesion, epithelial nests in fibrous stroma — diagnosis?"	→ **adamantinoma** ✗ Wrong options: OFD · osteomyelitis · periosteal OS (no osteoid) · fibrous dysplasia
"Histology seen only in adamantinoma, not OFD?"	→ **biphasic glandular epithelial cells surrounded by spindle cells**
"Best treatment?"	→ **wide en bloc resection + reconstruction** ✗ Wrong options: observation · bracing · chemo/radiation · AKA first
"Epithelial nests stain for?"	→ **cytokeratin**
"Site of metastasis?"	→ **lungs (~25%)** · late → long follow-up

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ANEURYSMAL BONE CYST

– Pathology · 2 / 66 –

1 Basics

Epidemiology

Age	**75% <20 yr** · first 2–3 decades · **female predominance** · **honeycomb** appearance (TO)
Site	long bones **>60%** (**femur, tibia** commonest) · **metaphysis** · spine **15%** (**posterior elements** – lamina, spinous process) · pelvis posterior elements
Foot	**metatarsal** (metaphysis) and **calcaneus** commonest (✗ Wrong options: calcaneal UBC = incidental)
Nature	benign, **locally aggressive/destructive**, may extend into soft tissue · same sites as **telangiectatic OS**

Primary vs secondary

Primary ABC	**neoplasm** · USP6 rearrangement
Secondary ABC	**not a neoplasm** (no translocation) · arises in another tumour **~30%**: **GCT**, **chondroblastoma**, fibrous dysplasia, chondromyxoid fibroma, NOF

Genes & lab markers – USP6

Gene	**USP6 (TRE2/TRE17) on 17p13** ↑ (ubiquitin-specific protease) **(TO)**
Translocation	**t(16;17)(q22;p13)** → **CDH11 promoter** (16q22) drives USP6
Frequency	"invariably" found → neoplastic origin **(TO)** · only **~70%** show USP6 ↑ (OB) – read the options
Key use	USP6 ↑ separates ABC from **telangiectatic OS** (no USP6) and GCT

Gene / translocation	Tumour (distractors)
USP6 – CDH11	✓ **ABC**
ASPL–TFE3 t(X;17)	Alveolar soft part sarcoma
EWS–ATF1 t(12;22)	Clear cell sarcoma (OB text: "clear cell chondrosarcoma")
EWS–FLI1 t(11;22)	Ewing sarcoma
EWS–WT1 t(11;22)	Desmoplastic small round cell tumour
EWS–CHN t(9;22)	Extraskeletal myxoid chondrosarcoma
PAX3–FKHR t(2;13)	Alveolar rhabdomyosarcoma

3 Presentation

Symptoms	**pain + swelling**, slow-growing · may present with **pathologic fracture**
Spine	neurologic deficit · even **cauda equina** (posterior-element ABC)

4 Diagnosis & Classification

Key radiology findings

X-ray

expansile, eccentric, lytic, metaphyseal · bony septae ("bubbly") · thin rim of periosteal new bone · no matrix



Expansile septated lytic calcaneal lesion

Width

can expand **wider than the physis** ✓ (UBC and GCT not wider than physis)

MRI / CT

multiple fluid–fluid levels (best on axial/sagittal **T2**) · may expand into soft tissue



Fluid–fluid levels on MRI

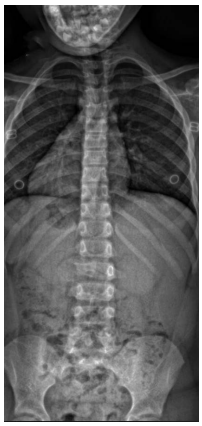
Fluid levels ≠ diagnostic

also in **GCT, chondroblastoma, telangiectatic OS**, vascular tumours **(TO)** · primary ABC = fluid levels **throughout, no solid component (TO)**

Spine X-ray

"winking owl" sign (loss of pedicle) → **order MRI**
✗ Wrong options: scoliosis curve alone · Scheuermann kyphosis alone

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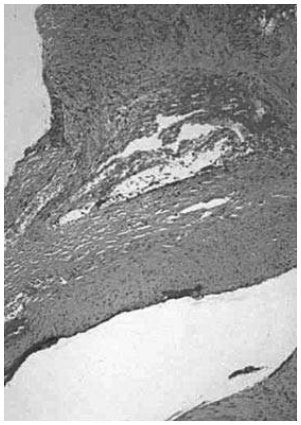


Winking owl sign – absent pedicle

Histology

Lakes of blood — **cavernous blood-filled spaces without endothelial lining**

Cyst walls — **benign giant cells** + **spindle cells** · thin strands of **woven bone** · histiocytes, hemosiderin · villous projections, clefts



Blood-filled spaces without endothelial lining

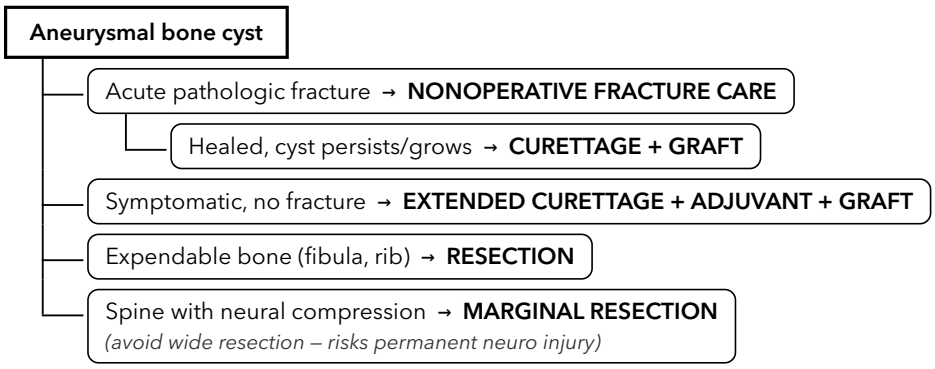
Biopsy — required to confirm (needle biopsy acceptable)

Differential

	Bubbly lytic X-ray	Lakes of blood (histology)	Curettage + grafting
ABC	✓	✓	✓
UBC	✓	–	–
NOF	✓	–	–
GCT	–	–	✓
Chondroblastoma (epiphyseal)	–	–	✓
Chondromyxoid fibroma	–	–	✓
Osteoblastoma	–	–	✓
Telangiectatic OS	–	✓	–

5 Management

Treatment algorithm



Curettage & adjuvants

Standard	**extended (aggressive) curettage + bone grafting** (allograft / substitute) **(TO – FNB 2019)** ✗ Wrong options: aspiration + methylprednisolone · curettage + cement · aspiration + autograft
Adjuvants	**phenol**, **liquid nitrogen**, **argon beam**, **high-speed burr**, hydrogen peroxide, bone cement → kill residual cells, ↓ recurrence
Sclerotherapy	**3% polidocanol** via J-needle (aspirate, break septa, scrape wall) → "current standard of care, surpassing extended curettage" **(TO)** · OB: curettage + burr + adjuvant **or** polidocanol both suitable (>90% cure) – read the options
Intralesional calcitonin + methylprednisolone	high healing rates reported **(TO)**
Embolization	**spine and pelvis** (limited access) **(TO)** · preop → ↓ blood loss · alone may resolve lesion
Radionuclide ablation	evidence for axial ABCs
Curopsy	**limited curettage** at biopsy, also effective · **path fracture → curettage + graft** (TO)
Proximal femur, painful	**curettage + grafting** (structural support) ✗ Wrong options: observation → fracture risk · denosumab (GCT only) · proximal femoral replacement

Not used

Chemotherapy	no role
Radiation	seldom used · **malignant transformation**, damage to physes, gonads, cord
Phenol alone	not used (adjuvant only) · **RFA / observation** – not for this (osteoid osteoma)
Wide resection + chemo	not for this (osteosarcoma) · wide/en bloc resection nearly eliminates recurrence but sacrifices function → reserve for expendable bones

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6 Complications

Recurrence risk factors

Local recurrence	**5–50%** · up to **25%** (Orthobullets) · spine 25–30%
Risk factors	✓ **young age**, ✓ **open physes**, stage, type of removal / margin
Not a factor	chemo, radiation sensitivity, graft material, tumour necrosis (OS prognosis), giant cells, embolization

X vs Y

	ABC	UBC	Telangiectatic OS	GCT
Age	<20	Child, open physis	Adolescent	**Skeletally mature** 20–40
Site	**Eccentric metaphysis**	**Central** metaphysis, prox humerus / femur, calcaneus	Metaphysis	**Epiphysis**–metaphysis
Width	**Wider than physis**	Not wider than physis	–	Not wider than physis
MRI	Fluid levels, no solid part	Single fluid, no levels	Fluid levels + **solid nodular** part	± secondary ABC
Histology	Lakes of blood, benign giant cells	Thin fibrous lining, giant cells, hemosiderin	Lakes of blood + **pleomorphic malignant cells, osteoid**	Giant-cell nuclei = stromal nuclei
USP6	Yes	No	No	No
Treatment	Curettage + adjuvant + graft	Steroid injection / observe	**Chemo + wide resection**	Curettage + adjuvant · denosumab

How it's asked

"Expansile eccentric metaphyseal lesion, fluid–fluid levels, lakes of blood – diagnosis?"	→ **ABC** ✗ Wrong options: GCT despite giant cells · UBC · NOF
"Gene in ABC aetiology?"	→ **USP6 (17p13)** ✗ Wrong options: ASPL · ATF1 · EWS · PAX3
"True of primary ABC?"	→ **USP6 rearrangement on chromosome 17 (TO)** ✗ Wrong options: posterior vertebral body · fluid levels confirm diagnosis
"Management of ABC (FNB)?"	→ **extended curettage + bone grafting (TO)** ✗ Wrong options: aspiration + methylprednisolone
"ABC with acute pathologic fracture – next step?"	→ **nonoperative fracture management** ✗ Wrong options: ORIF · open biopsy

"Spinal ABC with cauda equina in a child?"	→ **marginal resection** ✗ Wrong options: wide resection · aspiration + steroid
"Large ABC of the fibula?"	→ **resection alone** ✗ Wrong options: + chemo/radiation
"Pubic ramus ABC treatment?"	→ **surgery alone** (± embolization)
"Factor associated with local recurrence?"	→ **open physes / young age** ✗ Wrong options: chemo response · graft type
"Spine X-ray needing MRI?"	→ **winking owl sign** ✗ Wrong options: scoliosis curve · Scheuermann
"Curettage + grafting is NOT used for?"	→ **osteofibrous dysplasia** (✓ for GCT, ABC, chondroblastoma, CMF)

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ANGIOSARCOMA

– Pathology · 3 / 66 –

1 Basics

Epidemiology

Demographics	**elderly** · **male > female**
Origin	malignant, aggressive tumour of **vascular endothelium**
Bone involvement	**<10%** · of those, **60% long bones**

Risk factors

Classic associations	**chronic vascular stasis** (lymphoedema) · trauma · **polyvinyl chloride (PVC) exposure**

Genes & lab markers

CD31	**endothelial marker +** · sensitive for angiosarcoma

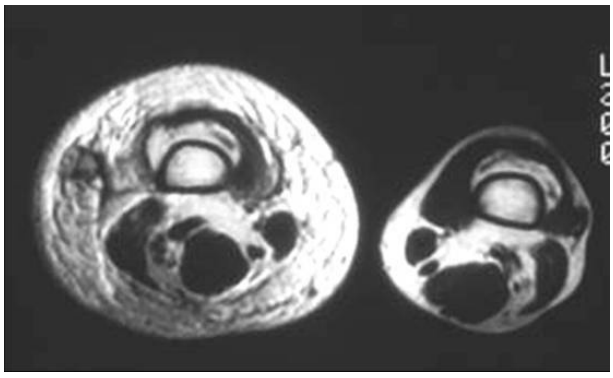
3 Presentation

Symptoms	insidious **pain**
Hallmark sign	**overlying skin changes** (central necrosis) · hand: nail-fold lesions + multiple palmar masses

4 Diagnosis & Classification

Key radiology findings

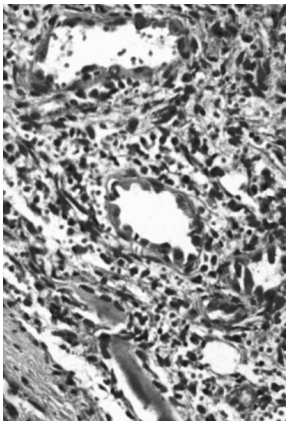
X-ray	bone invasion · **eccentric, purely lytic, metaphyseal/diaphyseal** · **no matrix**
MRI	**study of choice** for the soft-tissue mass
Angiography	diffuse multifocal involvement
CT chest	lung metastases



Angiosarcoma soft-tissue mass on MRI

Histology

Pattern	**malignant cells lining vascular channels** · variable anaplasia · abundant eosinophilic cytoplasm, large nuclei, mitoses



Malignant cells forming vascular channels

5 Management

Wide surgical resection	**treatment of choice** · relatively **insensitive to chemo + radiation** (Orthobullets summary: resection **with** radiation)
Amputation	for local control of a very aggressive tumour · forequarter amputation reported for the upper limb
Adjuncts	radiation ± chemotherapy depending on extent/metastasis

6 Complications

Prognosis	**poor** · high local failure → amputation often needed
Spread	**lymphatic** spread (unusual for sarcoma) · **lung metastases** common

How it's asked

"PVC exposure, nail-fold + palmar lesions, lung mets, CD31 + – diagnosis?"	→ **angiosarcoma** ✗ Wrong options: epithelioid sarcoma · synovial sarcoma · chondrosarcoma · osteosarcoma
"Risk factors?"	→ **chronic lymphoedema / vascular stasis, PVC, trauma**
"Endothelial immunomarker?"	→ **CD31**

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BIOPSY PRINCIPLES

– Pathology · 4 / 66 –

1 Basics

Purpose & prerequisites

Purpose	**confirm a suspected diagnosis** · not a substitute for history, exam, labs, imaging
Prerequisites	**CBC, platelets, coagulation** · **cross-sectional imaging** of local anatomy · **staging before biopsy** (new lesion + prior cancer → CT chest/abdomen/pelvis + bone scan)
Who	the **treating (definitive) surgeon / centre** capable of diagnosis + treatment · biopsy tract resected in continuity with tumour
Golden rule	all tissue exposed, and any **haematoma**, is **contaminated** → must be resected at definitive surgery
Timing (TO)	**only after complete imaging** (X-ray, MRI, CT/PET) → imaging picks the site, solid vs cystic part, soft-tissue extension · biopsy **confirms** a differential, not finds one
Biopsy = separate, planned procedure (TO)	never combined with curettage, excision, nailing or arthroscopic excision · **path fracture → splint, wait 5–7 days** for report before fixation
Mankin "Hazards of biopsy, revisited" (TO)	JBJS 1996, 597 patients · **diagnostic error 17.8%** · errors, complications, change in outcome **2–12× higher** when biopsy done **outside a treatment centre** · 18 unnecessary amputations · same as 1982 study → **refer before biopsy**
Pathologist (TO)	whole specimen to **one musculoskeletal pathologist** · report not fitting → send **same slides + blocks** for review · long descriptive report ("cannot be ruled out") = uncertain → call pathologist

"Touch-me-not" lesions — no biopsy (TO)

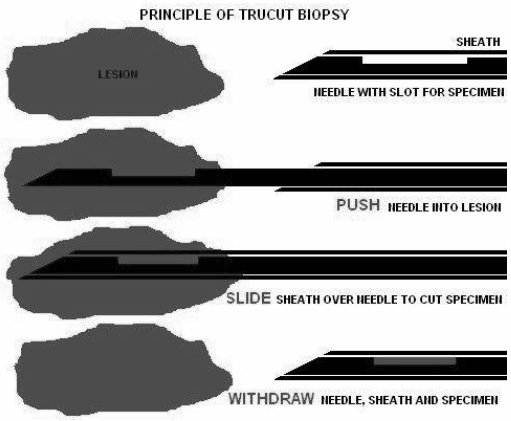
Classic radiology → no biopsy	**NOF** · **FD (ground glass)** · **osteochondroma** · **enchondroma** · **calcaneal lipoma (cockade sign)** · **bone infarct** · biopsy if atypical features (TO)

Indications for biopsy (criteria)

Biopsy indicated	Biopsy not needed
Aggressive bone or soft-tissue lesion	Asymptomatic **latent** or symptomatic active bone lesion that looks **entirely benign**
Soft tissue **>5 cm**, **deep to fascia**, or over bone / NV structures	Soft-tissue lesion **completely benign on MRI** (lipoma, haemangioma)
Unclear diagnosis in a symptomatic patient · indeterminate MRI	
Solitary bone lesion with history of carcinoma	

Types of biopsy

Type	Specimen	Use / fact
Fine needle aspiration (FNA/ FNAC)	**Cytology only** (separated cells) · no architecture or stroma	**Carcinoma** · not **sarcoma** · **no role in bone tumours** (paucicellular, no IHC, done by pathologist unaware of incisions) **(TO)** · high false-negative
Core (Tru-cut)	Cytology **+ stroma / architecture**	**Soft-tissue sarcoma** · **85–95% accurate** · CT-guided core increasingly first-line
Incisional (open)	Large sample	**Lowest non-diagnostic rate** · after non-diagnostic core
Excisional	Whole lesion	**Small (<3 cm), superficial** (to fascia) soft-tissue masses only



Principle of Tru-cut core biopsy

Open vs needle — open → more contamination → **higher local recurrence after definitive surgery (TO)**

Needle vs open (TO) — needle **as accurate as open**, fewer complications · core preferred (cost-effective, minimal contamination) · open only if more tissue needed or pathologist cannot report a core

Reamings (TO) — **not reliable** for histology · diagnosis in only **~60%** (90 cases) → never rely on reamings

Excision biopsy criteria (TO) — **all** of: **superficial (above fascia) on MRI** · **< 3 cm** · **MRI before surgery** · **en bloc** with good margins · piecemeal removal ≠ excision biopsy

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Never divide the sample	heterogeneous tumours → discordant grades · for second opinion, review slides/blocks **(TO)**
Frozen section	confirms **adequate viable tissue** · definitive diagnosis not required at that time · send sample for **culture** too

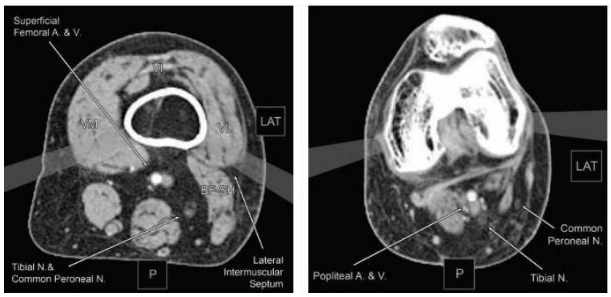
5 Management

Open incisional biopsy technique

Step	Do	Don't
Incision	**Longitudinal** in extremity, in line with future resection, no larger than needed	Not transverse
Plane	**Intramuscular** – through **one muscle belly / involved compartment** · biopsy the soft-tissue mass of a bone lesion	Avoid **intermuscular / internervous** planes (contaminate 2 compartments) · do not **raise flaps (TO)**
Structures	Stay away from NV bundle	Do not **expose neurovascular structures**, violate joint or go through quadriceps / **rectus femoris** (morbid resection)
Bone	**Oval** cortical window	
Tourniquet	**May be used** · release before closure	No **compressive exsanguination**
Haemostasis	**Meticulous** · close pseudocapsule	Haematoma = contaminated
Drain	**Exit in line with incision**, close to it	No multiple drains, no drain away from incision

Biopsy approach by site

Site	Approach
Distal femur	**Through vastus medialis or vastus lateralis** (posteromedial/posterolateral border) – medial **or** lateral acceptable
Proximal humerus	**Anterior deltoid** (not deltopectoral)
Distal humerus	Lateral triceps / brachialis
Wrist	Dorsal 2nd compartment
Iliac wing	Through superior abductors
Supra-acetabular	Smith-Petersen through TFL / rectus femoris
Femoral head / neck	**Transtrochanteric** osseous approach
Tibia	**Subcutaneous** border
Posterior thigh mass	**Direct posterior**, drain in line



Biopsy tracts through vastus medialis / lateralis

Non-diagnostic biopsy & referral

Suspected sarcoma / aggressive lesion

- Community hospital → **REFER TO ORTHOPAEDIC ONCOLOGIST before biopsy**
- Deep soft-tissue mass → **CORE NEEDLE (CT-guided) ± CULTURE**
  - Non-diagnostic (necrosis) → **OPEN BIOPSY, same intramuscular tract**
- New bone lesion + prior cancer → **RESTAGE, then BIOPSY**
  (no nail / hemiarthroplasty / radiation before tissue diagnosis)

Repeat core allowed but **not in a new direction** (contaminates another compartment)

Needle technique & lesion type (TO)

Lesion	Technique (TO)
Solid	J-needle into **solid, enhancing** part · rotate, suction, press over tract
Cystic (fluid levels)	**Heparinised** needle + syringe under C-arm → empty into heparinised bowl → **scrape cyst walls** → sieve through mop · under **spinal / GA** (scraping painful)
Bone + soft-tissue component	Sample the **soft-tissue part** · **biopsy gun** = most atraumatic · no decalcification
Intra-articular / notch lesion	**US- or CT-guided needle** · **never arthroscopic** (contaminates joint; synovial sarcoma misread as PVNS)

![Isolating tumour tissue from a cystic lesion](../../TO notes/attachments/TOn-notes-pitfalls-in-bone-biopsy-p07-f019.jpg)

Single entry point (TO) angle the needle in **different directions through the same entry** · two sites = twice the skin + muscle to excise

Fluid levels + solid areas (TO) **secondary ABC** (e.g. **GCT with secondary ABC**) → biopsy the **solid** part · cystic part gives "cyst wall, haemorrhage, fibrin" = non-diagnostic

![Fluid-fluid levels with solid areas — GCT with secondary ABC](../../TO notes/attachments/TOn-notes-pitfalls-in-bone-biopsy-p01-f002.jpg)

Knee tract (TO) distal femur **medial or lateral** · proximal tibia **medial or lateral to patellar tendon** · **never anterior (rectus femoris) or posterior (NV bundle)** · midline TKR-type incision contaminates rectus · mark the incision for the radiologist · tibia: anteromedial preferred

Closure (TO) **3-0 / 4-0, small bites** · no large hatch marks · avoid drains unless needed

Sclerotic lesion in a child (TO) **OS until proven otherwise** even without periosteal reaction · FD is ground glass, not ivory · "no malignant osteoid seen" may be sampling error

– – Active space – –

Chronic destructive monoarthritis (endemic TB region) — **open biopsy** of joint + bone for **histology (granulomas) + AFB/fungal culture**
✗ Wrong options: empirical antibiotics · steroid trial · fusion first

Expanding mass with fluid + solid parts	think **sarcoma**, not haematoma → **core biopsy** ✗ Wrong options: compression · IR drain · excise first

6 Complications

Poorly performed biopsy consequences

Bad biopsy	multiple tracts + **haematoma** spreading through compartments → **limb salvage impossible** → **chemotherapy + amputation** (osteosarcoma)
Radiation	no: osteosarcoma is radio-resistant
Unplanned surgery examples (TO)	clear cell CS curetted as "ABC" → amputation · OS debrided as Brodie abscess → wide resection · giant-cell-rich OS curetted as GCT → soft-tissue recurrence · UPS nailed, reamings sent → hindquarter amputation
Consequences (TO)	↑ morbidity · wider excision / amputation · local recurrence · flaps + grafts (e.g. hemisoleus flap after open Ewing tibia biopsy) · **delayed chemo**

How it's asked

"Diagnostic yield of reamings sent from a pathological-fracture nailing?"	→ **~60% — unreliable** (TO)
"Excision biopsy acceptable when?"	→ **superficial, < 3 cm, MRI done, en bloc** (TO) ✗ Wrong options: piecemeal excision · deep lesion
"Mankin — complications when biopsy done outside a tumour centre?"	→ **2–12× higher** (TO)
"Classic NOF / FD / cockade lipoma on X-ray — next?"	→ **no biopsy ("touch-me-not")** (TO)
"True statement about bone tumour biopsy?"	→ **open biopsy → higher local recurrence than needle (TO)** ✗ Wrong options: divide sample · raise flaps · FNAC preferred
"Important factor in a proper biopsy?"	→ **longitudinal incision** ✗ Wrong options: intermuscular planes · multiple drains · expose NV structures · avoid tourniquet
"Principle that does NOT apply?"	→ **"tourniquet should not be used"** — tourniquet allowed, no exsanguination
"Distal femur biopsy approach?"	→ **through vastus medialis or lateralis (intramuscular)** ✗ Wrong options: rectus femoris · VM–RF interval · VL–biceps interval · posterior

"Lateral distal femur lesion?"	→ **lateral longitudinal intramuscular** ✗ Wrong options: intermuscular · arthroscopic
"Proximal calf deep mass?"	→ **longitudinal incision through the involved (medial) compartment**
"Where to put the drain?"	→ **in line with the incision**
"Core biopsy shows necrosis only — next?"	→ **open biopsy through the same lateral intramuscular tract** ✗ Wrong options: FNA · core in new direction · resection
"Lung-cancer survivor, new subtrochanteric lesion?"	→ **biopsy** ✗ Wrong options: prophylactic nail · radiation
"Destructive humeral lesion at a community hospital?"	→ **refer to orthopaedic oncologist** ✗ Wrong options: biopsy there
"Painless growing thigh mass, sarcoma-like MRI — initial management?"	→ **CT-guided core needle biopsy** (+ CT chest/abdomen/pelvis)
"High-grade OS after two outside biopsies + leg haematoma?"	→ **chemotherapy + amputation**

– – Active space – –

– – Active space – –

BONE INFARCT

– Pathology · 5 / 66 –

1 Basics

Pathophysiology

Spectrum	same disease spectrum as **osteonecrosis** · medullary lesion
Mechanism	**interruption of blood supply** by intrinsic/extrinsic factors (atherosclerosis, arteritis, thrombosis)
Site	**metaphysis or diaphysis** of long bones (femur, tibia)
Healing	progressive but **incomplete resorptive replacement** → thick **calcified fibro-osseous wall**

Risk factors

Risk factor	Bone infarct?
Sickle cell disease	✓
Thalassemia	✓
Gaucher disease	✓
Steroid use (chronic prednisone)	✓
Connective tissue disorders	✓
Trauma	✓
Infection	**not a risk factor**

3 Presentation

Usually asymptomatic	**incidental** on films taken for other reasons (twisted knee, fall)
Pain	if painful → **seek another aetiology**

4 Diagnosis & Classification

Key radiology findings

X-ray	medullary **sheet-like central lucency + surrounding sclerosis with serpiginous border → "smoke up the chimney"**



Smoke up the chimney — medullary infarct

MRI key: **central signal = normal marrow** · T1 **peripheral low-signal rim** (granulation tissue, sclerosis), may enhance · T2 **double line**: inner bright granulation ring + outer dark sclerotic ring · acute infarct ill-defined ↑ T2



Bone infarct MRI — central marrow signal preserved

Bone scan **cold early** · **hot** in late resorptive / revascularisation phase

Biopsy usually unnecessary · mineralised necrotic marrow

5 Management

Observation asymptomatic, **no treatment** required

6 Complications

Malignant transformation rare · **MFH / undifferentiated pleomorphic sarcoma** at infarct site

X vs Y

	Bone infarct	Enchondroma
Matrix	Serpiginous peripheral sclerotic rim, "smoke up the chimney"	**Rings and arcs** (chondroid) throughout
MRI	Central normal marrow fat	Lobulated T2-bright cartilage
Risk factors	Sickle, Gaucher, steroids	None

– – Active space – –

How it's asked

"Incidental medullary metaphyseal lesion, serpiginous sclerotic border — diagnosis?"	→ **bone infarct** ✗ Wrong options: enchondroma · metastasis · osteosarcoma
"NOT a risk factor for this lesion?"	→ **infection** (✓ sickle cell, thalassemia, Gaucher, chronic steroids)
"Treatment?"	→ **observation**

BONE TUMOR STAGING SYSTEMS

– Pathology · 6 / 66 –

1 Basics

Prognostic variables

Prognosis depends on	overall **stage** · **metastasis** (incl. **skip / discontinuous** lesions in same bone) · **histologic grade** · **tumour size** · tumour type
Two systems	**MSTS (Enneking)** – most useful to orthopaedic surgeons · **AJCC** – adds size, depth, nodes
Enneking limitation	only for tumours of **mesenchymal origin**

Compartment & grade

Intracompartmental (T1)	bone tumour **confined within the cortex**
Extracompartmental (T2)	extends **beyond the cortex**
Grade	by **% cellular atypia** · low grade = low metastatic potential (**parosteal OS**) · high grade = **intramedullary OS, Ewing, dedifferentiated chondrosarcoma**

4 Diagnosis & Classification

MSTS (Enneking) – malignant (Roman numerals)

Stage	Grade	Site	Metastasis
IA	Low	T1 intracompartmental	M0
IB	Low	T2 extracompartmental	M0
IIA	High	T1 intracompartmental	M0
IIB	High	**T2 extracompartmental**	M0
III	Any	T1 or T2	**M1 (regional or distant)**

Commonest presenting stage **IIB** (e.g. high-grade OS with soft-tissue extension, no mets)

Stage	Grade	Site (1)	Metastasis
IA	Low Grade	T1 - intracompartmental	M0 (none)
IB	Low Grade	T2 - extracompartmental	M0 (none)
IIA	High Grade	T1 - intracompartmental	M0 (none)
IIB	High Grade	T2 - extracompartmental	M0 (none)
III	Metastatic	T1 - intracompartmental	M1 (regional or distant)
III	Metastatic	T2 - extracompartmental	M1 (regional or distant)

Enneking staging

– – Active space – –

MSTS (Enneking) – benign (Arabic numerals)

Stage	Behaviour	Examples
1	Latent	**NOF**, **enchondroma**
2	Active	**ABC, UBC**, chondromyxoid fibroma, chondroblastoma · **GCT without cortical breakthrough**
3	Aggressive	**GCT** (cortical breakthrough, soft-tissue extension)

Radiological rim (TO) **latent = thick sclerotic rim · active = thin rim · aggressive = breaching cortex**

AJCC – bone sarcoma

Stage	Grade	Size / feature	Metastasis	5-yr survival (Orthobullets)
IA	Low	**≤8 cm**	None	98%
IB	Low	**>8 cm**	None	–
IIA	High	≤8 cm	None	82%
IIB	High	>8 cm	None	82%
III	Any	**Skip (discontinuous) lesions in same bone**	None	52%
IVA	Any	Any	**Lung**	–
IVB	Any	Any	**Non-pulmonary** (other bone) or **nodes**	30%



AJCC Staging System

Stage	Grade	Size	Depth	Node	Metastasis	5 yr. survival
IA	Low Grade	< 8cm	any	none	none	98%
IB	Low Grade	> 8cm			none	
IIA	High Grade	< 8cm	any	none	none	82%
IIB	High Grade	> 8cm	superficial	none	none	82%
III	any	discontinuous (skip) lesion	deep	none	none	52%
IVA	any	any		none	lung	
IVB	any	any	any	present	other than lung	30%

AJCC staging

AJCC – soft-tissue prognosis factors

Grade low G1–G2 · high G3–G4

Size T1 <8 cm · T2 >8 cm (as listed in Orthobullets)

Nodes N0 / N1 · **distant mets → automatically stage IV**

Staging investigations (TO)

Systemic **bone scan + CT chest (2 mm cuts)** or **whole-body PET-CT (head to toe)** · **Ewing → add bone marrow biopsy** (TO)

AJCC (TO) grade, **size (8 cm cut-off)**, metastasis · **pulmonary → IVA** · **extrapulmonary → IVB**

5 Management

Surgical margins — examples (TO)

Margin	Example (TO)
Intralesional	**Curettage (GCT)** · debulking (metastasis)
Marginal	Through reactive zone · **pedunculated osteochondroma** excision
Wide	En bloc with cuff of normal tissue
Radical	Whole compartment · e.g. **BKA for foot sarcoma**

Radicality — defined by the **compartment, not the level** · **amputation is not always radical** (TO)

Limb salvage principles & contraindications

Three strikes rule — 4 components of a viable limb: **bone, nerves, vessels, soft-tissue envelope** · 1–2 resected → salvageable · **3 involved → salvage not worth considering (TO)**

Barriers (relative, not absolute) — **poorly placed biopsy** · **major vascular involvement** · **encasement of major motor nerve** · **pathological fracture (TO)**

Two of four components involved — **not** a contraindication **(TO)**

LSS requirements (TO) — tumour allows **wide excision** · major NV **not encased** · **no uncontrolled infection** · patient fit · contraindicated: unclear margins, improper-biopsy compartment contamination, **fungating / uncontrolled local disease**

Priority (TO) — save the patient, then the limb, then function · **margin decides survival, reconstruction decides quality of life**

Megaprosthesis indications (TO) — large segmental loss · joint resected · need **immediate stability + early mobilisation** · limits: infection, aseptic loosening, **extensor mechanism** (knee)

Rotationplasty indications (TO) — child needing durable reconstruction · massive resection where prosthesis will fail · **poor soft-tissue envelope**

Soft-tissue cover (TO) — local / free flaps · **gastrocnemius flap** for knee · cover ↓ infection

Complications (TO) — early: infection, wound necrosis, NV injury · late: **aseptic loosening**, implant breakage, local recurrence, biological failure

Nerves sacrificed during proximal humerus resection (axillary + radial) — **continue with endoprosthesis (TO)** · axillary usually sacrificed anyway · radial → **later tendon transfers** (or repair same sitting) · amputation indicated for **intra-op tumour spillage**, not nerve loss

How it's asked

"High-grade distal femur OS with soft-tissue extension, no mets — MSTS stage?"	→ **IIB**
"Enneking IIB means?"	→ **high grade, extracompartmental, no metastases**

– – Active space – –

"Most common presenting stage – which patient?"	→ **high-grade OS extending into soft tissue, no mets** ✗ Wrong options: confined chondrosarcoma = IIA · lung mets = III
"OS distal femur + biopsy-proven contralateral femur lesion – AJCC?"	→ **IVB** (non-pulmonary metastasis) ✗ Wrong options: III = skip lesion in same bone
"GCT without cortical breakthrough – Enneking?"	→ **2 (benign active)** ✗ Wrong options: Roman numerals = malignant
"Limb salvage contraindication – all EXCEPT?"	→ **two of four components involved** (✓ bad biopsy, ✓ path fracture, ✓ motor nerve encasement) **(TO)**
"Amputation – always a radical margin?"	→ **No – the compartment defines radicality** (TO)
"Axillary + radial nerve sacrificed during humeral OS resection – next?"	→ **proceed with planned endoprosthesis (TO)** ✗ Wrong options: amputation

– – Active space – –

CHEMOTHERAPY

– Pathology · 7 / 66 –

1 Basics

Apoptosis mechanism

Primary mechanism conventional chemo kills cancer cells by **apoptosis** (programmed cell death) · targets **rapidly dividing cells**

Side effects gut lining, **bone marrow**, hair, skin (also rapidly dividing)

Targeted agents tyrosine kinase inhibitors: **imatinib** (CML) · **gefitinib** (EGFR; lung, breast) · **erlotinib** (EGFR; NSCLC, pancreas)

Role eliminates **lung micrometastases**

Therapy	Mechanism
Doxorubicin neoadjuvant chemo	✓ **Apoptosis** (blocks topoisomerase II, DNA/RNA synthesis)
Phenol in GCT bed	**Lysis** of residual cells (not apoptosis)
TNF-α inhibitor (RA)	Immune modulation
Colchicine (gout)	Immune / crystal inhibition
Vancomycin	Cell-wall synthesis inhibition

Genes & lab markers

MDR (multi-drug resistance) gene pumps drug out of cell → **very poor prognosis** · in **25% primary**, **50% metastatic** lesions

Necrosis >98% after chemo – **good prognostic sign** (osteosarcoma)

5 Management

Indications

Chemo is integral (with surgery) **osteosarcoma** (intramedullary + periosteal) · **Ewing / PNET** · **MFH (UPS)** · **dedifferentiated chondrosarcoma**

Controversial soft-tissue sarcoma

Schedule **preop 8–12 weeks** → surgery → **maintenance 6–12 months**

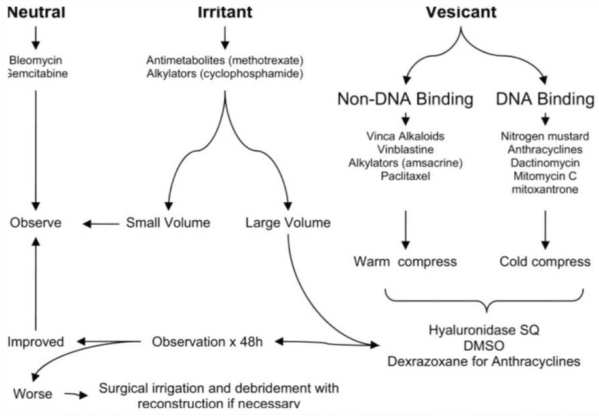
Agents, toxicities & antidotes

Agent	Dose-limiting / classic toxicity
Doxorubicin (Adriamycin)	✓ **Cardiomyopathy / CHF** (cumulative dose) · ~7%
Ifosfamide	**Encephalopathy** (neurotoxicity)
Cyclophosphamide	**Myelosuppression, urotoxicity**
Bleomycin	**Pulmonary fibrosis**

– – Active space – –

Doxorubicin — **anthracycline** antibiotic, cytostatic · delayed LV wall thinning → dilated cardiomyopathy · **dexrazoxane** protective

Extravasated agent	Antidote
Mechlorethamine / cisplatin	**Sodium thiosulfate**
Doxorubicin / epirubicin	**Dexrazoxane**
Vinca alkaloids (vincristine, vinblastine)	**Hot compress + hyaluronidase**
Other vesicants	Cold compress



Chemotherapy agents and antidotes

How it's asked

"Major (dose-limiting) side effect of doxorubicin?"	→ **cardiomyopathy / cardiac toxicity** ✗ Wrong options: pulmonary fibrosis (bleomycin) · myelosuppression/ urotoxicity (cyclophosphamide) · encephalopathy (ifosfamide) · hearing loss · neuropathy
"Pharmacologically induced apoptosis occurs with?"	→ **doxorubicin neoadjuvant chemotherapy** ✗ Wrong options: phenol in GCT · anti-TNF · colchicine · vancomycin
"True of apoptosis?"	→ **primary mechanism by which chemotherapy kills cancer cells**
"Drug protecting against doxorubicin cardiotoxicity?"	→ **dexrazoxane**

CHONDROBLASTOMA

– Pathology · 8 / 66 –

1 Basics

Epidemiology

Rare	~**1%** of osseous neoplasms · **M:F 2:1**
Age	**80% <25 yr** · 10–25 typical (2nd decade) · average diagnosis 19–23 · rare after physeal closure
Synonym	**"Codman tumour" (TO)**
Site	**epiphysis of immature long bones** · may **cross the physis into the metaphysis** · **knee** (distal femur, proximal tibia) commonest → **proximal humerus** distant third · proximal femur ✗ Wrong options: metaphysis **(TO)**
Epiphyseal equivalents	**apophysis** (greater tuberosity, greater trochanter) · **patella** · **talus, calcaneus** · flat bones
Mnemonic "ECG"	**E**piphysis: **C**hondroblastoma, **G**iant cell tumour (+ clear cell chondrosarcoma) **(TO)**

Pathophysiology

Benign, locally destructive proliferation of **immature cartilage cells (chondroblasts)** · from epiphyseal plate / secondary ossification centre (controversial)

Matrix	chondroid matrix · **hyaline cartilage rarely** produced
Natural history	progressive bone destruction · **does not heal spontaneously**
Secondary ABC	up to **1/3** (15–32% on imaging)

Genes & lab markers

Genetics	**histone H3.3** mutations · **chromosome 5 and 8** abnormalities
IHC	**S-100 +** · **SOX9 +** · vimentin, neuron-specific enolase + (GCT, ABC: no S-100)
Bloods	CBC, ESR, LDH normal (vs osteomyelitis, Ewing)

3 Presentation

Symptoms	**progressive pain** (often night pain, not relieved by NSAIDs) · **joint stiffness** · **limp**
Signs	local muscle atrophy · tenderness · ↓ ROM · **effusion**

4 Diagnosis & Classification

Key radiology findings

X-ray views	AP, lateral, oblique · **chest radiograph** (lung mets) at diagnosis

– – Active space – –

Pathology

X-ray

geographic, well-circumscribed round/oval lytic epiphyseal lesion with **thin sclerotic rim** · may cross physis · 1–4 cm, lobular · expansile cortical remodelling · **metaphyseal periosteal reaction** (no periosteum over epiphysis) · effusion



Lytic epiphyseal lesion with sclerotic rim — proximal tibia

Calcification

stippled matrix calcification, often **only on CT (25–45%)** · chondroid matrix in **~50%** (TO)

MRI

extensive bone-marrow + soft-tissue oedema around the lesion — distinguishing feature **(TO)** · T1 intermediate · T2 heterogeneous · fluid levels = secondary ABC · oedema within treated lesion → recurrence



Peritumoral oedema on MRI

Bone scan

not required · ↑ uptake, especially adjacent uninvolved bone

CT chest

if chest film suspicious

Soap bubble

classically GCT, but chondroblastoma also in DD **(TO)** · DD: ABC, GCT, chondroblastoma, enchondroma, CMF, fibrous dysplasia, ameloblastoma, plasmacytoma, chordoma

Benign lesions with marrow oedema (TO)

osteoid osteoma, osteoblastoma, chondroblastoma, LCH
✗ Wrong options: osteochondroma · enchondroma · GCT

Histology

Cells

sheets of **polygonal mononuclear chondroblasts**, well-defined borders · **"cobblestone"** arrangement

Nuclei

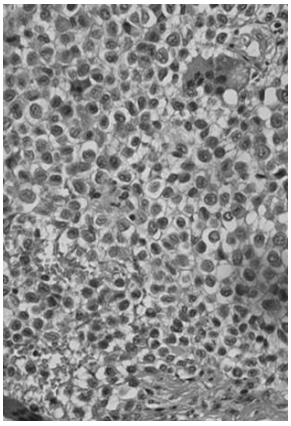
large, central, **longitudinal groove → "coffee-bean"** · scarce mitoses

Calcification

fine pericellular **"chicken-wire"** dystrophic calcification

Also scattered **multinucleated osteoclast-type giant cells** (almost always) · islands of amorphous **chondroid matrix**

– – Active space – –



Chondroblasts with chicken-wire calcification

Recurrence may show atypia → not malignant transformation

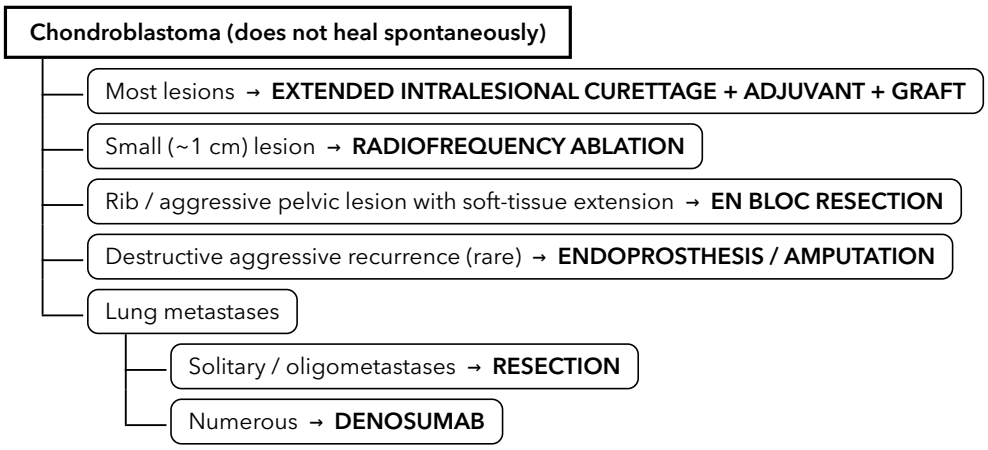
Differential of an epiphyseal lesion

Lesion	Key differentiating finding
Chondroblastoma	**Skeletally immature**, progressive pain, matrix mineralization, **extensive peritumoral oedema**
Giant cell tumour	**Skeletally mature, 20–40**, closed physes · starts in metaphysis → epiphysis · pathologic fracture common · giant-cell nuclei = stromal nuclei · S-100 negative
Clear cell chondrosarcoma	**3rd–5th decade**, **>4 cm**, **less oedema**, malignant · clear cells
Brodie abscess	Immature, often metaphyseal · **penumbra** sign on MRI · neutrophils
Intraosseous ganglion	Middle age · no periosteal reaction · cystic, less oedema
Eosinophilic granuloma	Variable appearance
Brown tumour	>50, CKD / hyperparathyroidism
ABC	<20, multicystic, fluid levels · not typically epiphyseal
Osteoblastoma	2nd–3rd decade, rarely epiphyseal, mineralization, oedema
Osteoid osteoma	Diaphyseal, NSAID-relieved night pain, nidus on CT
TB arthritis	Epiphyseal possible · **granulomas**

– – Active space – –

5 Management

Treatment algorithm



Curettage & bone grafting

Standard of care	**intralesional curettage + local adjuvant + bone graft** (or PMMA)
Approach	through **cortical window** · extend with **high-speed burr** · **spare the physis** if possible
Adjuvants	**phenol**, **hydrogen peroxide**, **argon beam**, **cryotherapy / liquid nitrogen**
Graft	fill cavity · **structural osteochondral graft** for weight-bearing surfaces
RFA	first-line option for small lesions (~**1 cm**) **(TO)** · recurrence ≈ curettage · success on MRI = necrosis beyond margin + resolution of oedema
Not indicated	observation (no medical therapy, no spontaneous healing) · **denosumab** (GCT) · chemo · radiation · wide resection (non-expendable bone) · steroid injection · TKA

6 Complications

Local recurrence	**10–15%** (OB review 14–18%) · surgery and RFA alike · **purely epiphyseal lesions recur more** · due to inadequate resection · **surveillance X-rays 5 yr**
Limb length discrepancy	physeal damage by tumour or treatment → contralateral epiphysiodesis vs lengthening
Degenerative joint disease	subchondral/cartilage damage → THA/TKA in adulthood

Lung metastasis – chest radiograph

"Benign" pulmonary metastasis	**<1%** (Orthobullets) · **2%** (one OB explanation) · like GCT · indolent → resect
Chest X-ray	the benign lesion that **needs a chest film at screening** ✗ Wrong options: enchondroma · NOF · osteochondroma · UBC
Malignant transformation	exceedingly rare · median age **52** · permeative growth, atypia, **no osteoid** · wide resection

X vs Y

	Chondroblastoma	GCT	Clear cell chondrosarcoma
Age	**10–25, open physis**	**20–40, closed physis**	20–40 / 3rd–5th decade
Site	Epiphysis (± crosses physis)	Metaphysis → epiphysis, subchondral	Epi-metaphysis (proximal femur/humerus)
MRI oedema	**Extensive**	Less	**Less**
Histology	Chondroblasts, coffee-bean nuclei, chicken-wire Ca	Giant-cell nuclei = stromal nuclei	**Clear cells**, malignant
Marker	**S-100, SOX9**	**H3F3A**, RANKL	–
Treatment	Curettage + adjuvant + graft / RFA	Curettage + adjuvant · denosumab	**Wide excision**

	Chondroblastoma	Chondromyxoid fibroma
Site	**Epiphysis**	**Metaphysis** (anterior proximal tibia)
Histology	Polygonal chondroblasts, chicken-wire Ca	**Stellate cells** in chondroid/myxoid/fibrous lobules

How it's asked

"Statement NOT true of chondroblastoma?"	→ **"commonly involves metaphysis"** (✓ Codman tumour, ✓ RFA for small lesions, ✓ chicken-wire Ca) **(TO)**
"15-yr-old, epiphyseal proximal tibial soap-bubble lesion?"	→ **chondroblastoma (TO)** ✗ Wrong options: GCT (20–40) · CMF (metaphyseal) · osteoid osteoma (diaphyseal)
"Benign tumour with extensive surrounding marrow/ soft-tissue oedema on MRI?"	→ **chondroblastoma (TO)** ✗ Wrong options: GCT · enchondroma · osteochondroma
"Adolescent, epiphyseal lytic lesion, S-100 +, normal bloods – histology?"	→ **chondroblasts with chicken-wire calcification** ✗ Wrong options: GCT · ABC · Ewing · osteomyelitis
"16-yr-old, medial heel / calcaneal epiphyseal-equivalent lesion?"	→ **chondroblastoma** ✗ Wrong options: GCT · CMF (stellate cells) · osteoblastoma
"18-yr-old, greater tuberosity lesion?"	→ **apophyseal chondroblastoma**
"Best treatment?"	→ **extended intralesional curettage + bone graft ± adjuvant** ✗ Wrong options: denosumab · chemo · observation · radiation · wide resection · TKA

– – Active space – –

"Benign lesion requiring chest radiograph at initial work-up?"	→ **chondroblastoma** ✗ Wrong options: enchondroma · NOF · osteochondroma · UBC
"Recurrence rate after curettage?"	→ **10–15%** · higher if purely epiphyseal

CHONDROMYXOID FIBROMA

– Pathology · 9 / 66 –

1 Basics

Epidemiology

Rare	**<1%** of bone tumours · benign but painful, **locally aggressive**
Demographics	**males** · **2nd–3rd decades** (10–30) · up to 75 yr
Site	**metaphysis** of long bones · **proximal tibia** commonest (around knee: proximal tibia, proximal fibula, distal femur) · **pelvis** (ilium) · **foot/ankle**, hands, wrist (distal radius)
Origin	may arise from **physeal remnants**

Genes & lab markers

Chromosome 6 (q13)	genetic rearrangement

3 Presentation

History	**long-standing pain, months–years** · may be **incidental**
Symptoms	pain + mild swelling · activity pain → rest pain

4 Diagnosis & Classification

Key radiology findings

X-ray	**lytic, eccentric, metaphyseal** · sharply demarcated, **narrow zone of transition** · **scalloped sclerotic rim** · lobulated / bubbly · cortical expansion · **calcification rare** · 2–10 cm



Eccentric lytic metaphyseal lesion with sclerotic rim

vs NOF	radiographically often indistinguishable · CMF **larger** and **active past skeletal maturity**
MRI	**T1 low**, **T2 high**, lobular growth · distinct margin

Pathology

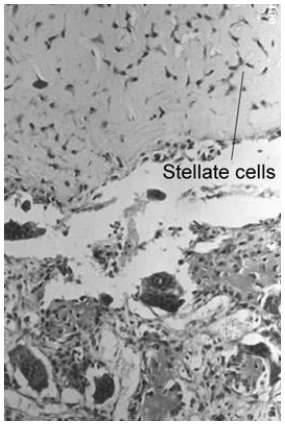
– – Active space – –

CT	low attenuation, peripheral bone rim, subtle matrix
Bone scan	↑ uptake

Histology (diagnostic criteria)

Histology mandatory for diagnosis

Low power – biphasic	**hypercellular** lobules of fibromyxoid tissue (spindle/stellate cells, hyperchromatic nuclei; giant cells + fibrovascular tissue between lobules) + **hypocellular chondroid** areas
High power	myxoid stroma with **stellate cells** · may have pleomorphic bizarre nuclei



Stellate cells in myxoid stroma

Differential

Radiographic DD	Histologic DD
ABC · chondroblastoma · **NOF**	Chondroblastoma · enchondroma · **chondrosarcoma**

Lesion	Distinguishing feature
CMF	**Stellate cells**, spindle cells, myxoid-cartilaginous matrix
NOF	**Storiform** bland fibroblasts, scattered giant cells
Enchondroma	Calcifications · bland hyaline cartilage, pyknotic nuclei
GCT	Radiolucent, not bubbly · giant-cell nuclei = stromal nuclei
Chondrosarcoma (conventional)	**Older (50+)** · uncommon in hands/feet · cortical erosion, soft-tissue mass
Atypical cartilaginous tumour	(= grade 1 extremity chondrosarcoma, WHO) · like large enchondroma + endosteal scalloping

5 Management

Intralesional curettage + bone grafting (or PMMA)	**mainstay**
Not indicated	observation (painful) · chemo + wide excision · radiotherapy · radical resection

6 Complications

– – Active space – –

Local recurrence	**~25%** (Orthobullets) · 20–30% · 20–25% (OB)
Negative prognostic factors	**children** · **more lobulated, abundant myxoid** tumour
Metastasis	**not reported**

X vs Y

	CMF	Chondroblastoma
Site	**Metaphysis** (proximal tibia)	**Epiphysis**
Histology	**Stellate cells**, lobules of fibromyxoid tissue	Polygonal chondroblasts, **chicken-wire** Ca
Calcification	Rare	Common (CT)
Lung mets	**None reported**	<1%
Treatment	Curettage + graft / PMMA	Curettage + adjuvant + graft / RFA

How it's asked

"Adolescent wrist/foot pain, eccentric lytic metaphyseal lesion with sclerotic rim, spindle/stellate cells in myxoid matrix?"	→ **chondromyxoid fibroma** ✗ Wrong options: NOF (storiform) · enchondroma · chondrosarcoma · osteosarcoma
"21-yr-old, large lobulated eccentric metaphyseal lesion with sclerotic rim?"	→ **CMF** ✗ Wrong options: NOF · ABC · atypical cartilaginous tumour
"Ilium lesion, biphasic histology with stellate cells — treatment?"	→ **curettage + bone grafting** ✗ Wrong options: chemo + wide excision · radiotherapy
"Recurrence rate?"	→ **~25%** · higher in children

– – Active space – –

CHONDROSARCOMA

– Pathology · 10 / 66 –

1 Basics

Epidemiology

Rank	**2nd commonest malignant primary bone tumour** (after OS) · commonest primary bone sarcoma in the US (OB) · commonest primary bone malignancy **of the hand** though rare there **(TO)**
Age	conventional **40–75**, most **>50** · peak primary **40–60**, secondary **25–45 (TO)** · slight **male** predominance
Site	**pelvis (25%)**, proximal femur, distal femur, proximal humerus, scapula, ribs · **proximal/axial** · metaphyseal or diaphyseal (epiphyseal only = **clear cell**)
Location rule	cartilage tumours of **hands/feet mostly benign** · **flat bones (pelvis, scapula, ribs, sternum) malignant until proven otherwise** · subperiosteal or joint-lining cartilage almost always benign **(TO)**

Primary vs secondary

Primary	sporadic · **conventional >90%** (central intramedullary 99%, periosteal ~1%) · dedifferentiated <10% · clear cell <5% · mesenchymal <1%
Secondary	from pre-existing benign cartilage lesion · **osteochondroma >80%** of secondary · younger (3rd–4th decade) · usually **low grade** · ~40% of MHE secondaries in **ilium** · histologically = conventional

Malignant transformation risk

Precursor	Orthobullets	OB explanation	TO
Solitary osteochondroma	**<1%**	1%	<1% (~1% lifetime)
Solitary enchondroma	1–9%	1%	1%
MHE	**5–10%**	~10%	1–10% · ~5% lifetime
Ollier disease	**25–30%**	30%	5–50% · ~25% by age 40
Maffucci syndrome	**>50%**	**~100%** (highest)	25–100%

Maffucci	multiple enchondromas + **soft-tissue haemangiomas** (and lymphangiomas) · also pancreatic/hepatic adenocarcinoma, ovarian, glioma/astrocytoma
Axial/pelvic osteochondroma	more likely to transform than peripheral
Other precursors (TO)	**synovial chondromatosis**, CMF, periosteal chondroma, chondroblastoma, **prior radiation**, fibrous dysplasia, **Paget** (OS + chondrosarcoma) · not **chronic osteomyelitis → squamous cell carcinoma** (Marjolin)

Genes & lab markers

IDH1 / IDH2 mutations — primary + secondary conventional chondrosarcoma **and enchondromas**

HEY1–NCOA2 fusion — **mesenchymal** chondrosarcoma

t(12;22)	listed for **clear cell** (Orthobullets) · clear cell: **S-100 +, PAS +** (glycogen → clear cytoplasm)

Telomerase activity ↑ (RT-PCR) — **directly correlates with recurrence**
✗ Wrong options: MHC II · gadolinium uptake · apoptosis · technetium uptake

3 Presentation

Pain	commonest · **pain attributable to the lesion** (night pain, not activity related) separates low-grade CS from enchondroma (vs pain from cuff tear / meniscus)
Pelvis	slowly growing mass · **bowel/bladder obstruction** from mass effect
Secondary	**new pain or growth of an osteochondroma after skeletal maturity**
Dedifferentiated	**pathologic fracture ~50%**

4 Diagnosis & Classification

Work-up & referral

Incidental chondroid lesion on MRI (meniscus work-up) — **first get orthogonal radiographs →** often simple enchondroma
✗ Wrong options: CT CAP · whole-limb MRI · core biopsy · curettage

Suspicious lesion at community hospital	**refer to orthopaedic oncologist** ✗ Wrong options: observe · curettage · chemo/RT
Staging	**CT chest** (lungs = main site) + MRI whole bone + bone scan ✗ Wrong options: CT abdomen/pelvis (bone sarcomas rarely go to viscera) · SPEP/UPEP (myeloma)
Unknown lytic lesion >40 yr	CT chest/abdomen/pelvis (carcinoma work-up)
Biopsy	core needle or open (gold standard) · target **area of ↓ T2 signal** (focus of dedifferentiation) · TO: needle biopsy not indicated for cartilage tumours (heterogeneous) · diagnosis = **clinical + radiographic + histologic correlation**

Key radiology findings

X-ray	orthogonal views mandatory · usually **>4 cm** · lytic with **"rings and arcs" / "popcorn"** calcification (punctate, comma-shaped) · seen on X-ray ~70%

– – Active space – –

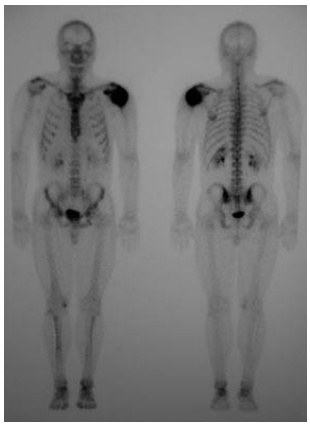


Popcorn calcification

Aggressive signs	**endosteal scalloping >2/3 cortex** · cortical destruction · periosteal reaction · bone expansion · soft-tissue mass · permeative borders · higher grade → less mineral, more lytic
CT	**best for low-grade cartilage lesions** (cortical involvement) · matrix seen more easily · **deep endosteal scalloping** → CS · **90% cortical breach**
MRI	**lobulated**, T1 low–intermediate, **T2 high** (cartilage water) with low-signal septa · **"rings and arcs" septal enhancement** · defines soft-tissue extent
MRI sequence	**plain T2** images essential (T2-hyperintense lobulated cartilage tumour) (TO)
Transformation clues (TO)	expansion · cortical erosion · **endosteal scalloping** · size **>5 cm** · **marrow oedema** · soft-tissue component

PET-CT SUV (TO)	Reading
< 2	Benign
2–3	Borderline → correlate clinically + radiologically
> 3	Malignant

PET-CT in low grade	used to **avoid under-grading** before extended curettage (TO)
Bone scan	very hot in all grades · **uptake > ASIS** characteristic (enchondroma < ASIS)



Bone scan uptake greater than ASIS

Secondary from osteochondroma

cartilage cap >2 cm (≥2 cm) on MRI/CT in a mature patient · lysis of calcification · cortical destruction · cap **14 mm, no growth, mild symptoms → serial imaging + exam**
✗ Wrong options: FNA · chest CT · bone scan · wide excision

Enchondroma vs low-grade chondrosarcoma (diagnostic criteria)

	Enchondroma	Chondrosarcoma
Age	**<50**	**>50**
Size	**<5 cm**	**>5 cm**
Site	**Hands/feet common** · axial/pelvis rare	**Axial/pelvis common** · hands/feet rare
Pain	No (or other pathology)	**Yes, attributable to lesion**
Soft-tissue mass	No	Yes
Cortical destruction	No	Yes
Periosteal reaction	No	Yes
Endosteal scalloping	**<2/3 cortex**	**>2/3 cortex**
Expansion	No	Yes
Bone scan	**< ASIS**	**> ASIS**

Histology & grading (conventional)

Graded by

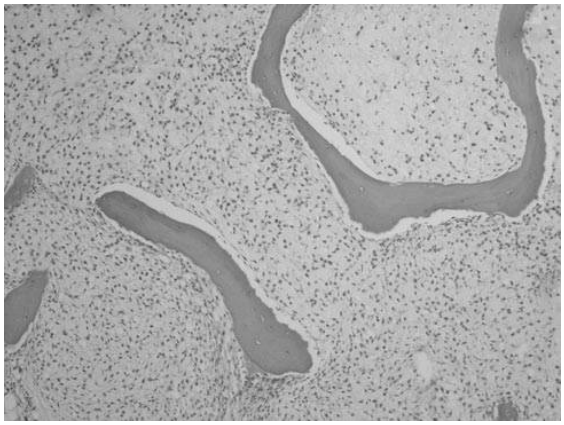
cellularity + cytological atypia (nuclear size, hyperchromasia), myxoid stroma · mitoses rare, not used **(TO)**

Diagnostic of malignancy

permeation and entrapment of pre-existing bone trabeculae · hypercellularity, plump nuclei, >occasional **binucleate** cells

Any malignant osteoid

→ **chondroblastic osteosarcoma**, not CS **(TO)**



Entrapment of host trabeculae

Pathology

– – Active space – –

Grade	Share	Histology	Metastasis	5-yr survival
1	60%	Like enchondroma · bland hyaline, low cellularity, binucleate · needs clinico-radiological correlation	Almost never	**90%**
2	25%	Plump **bi-/tri-nucleated** cells, ↑ cellularity, atypia, myxoid change	**10–15%**	**60–70%**
3	5%	Hypercellular permeative stroma, little hyaline, **myxoid**, pleomorphic, mitoses · "blue balls"	**>50%**	**30–50%**
Dedifferentiated ("4")	~10%	Bimorphic	High	**<20%** (10% at 10 yr)

WHO 2020 **grade 1 extremity CS = atypical cartilaginous tumour (ACT)** · axial cartilage tumours = separate, more aggressive category

Subtypes

Subtype	Age	Site	Imaging	Histology	Treatment / prognosis
Conventional	40–75	Pelvis, prox femur, humerus	Rings & arcs, scalloping	Hyaline, graded 1–3	**Wide resection alone**
Dedifferentiated	~60	As conventional	**Bimorphic**: aggressive **unmineralized lytic** mass next to mineralized chondroid · path fracture	Low-grade cartilage **abruptly** next to **high-grade spindle sarcoma** (OS > fibrosarcoma > MFH) · not admixed	Wide resection ± **chemo** · **<20% 5-yr**, **10% 10-yr**
Clear cell	**20–40**	**Epiphysis** · 2/3 **humeral/ femoral head**	Well-defined lytic epi-metaphyseal, thin sclerotic rim, mixed lucent/ sclerotic, no periosteal reaction · mimics **chondroblastoma**	**Large vacuolated clear cells**, giant cells · ~50% low-grade CS areas · S-100, PAS +	**Wide resection** (no chemo) · 10-yr ~80% · mets ~15%
Mesenchymal	**20–40**	**Flat bones** (ilium, ribs, skull) · **30% soft tissue**	Moth-eaten / permeative, extraosseous	**Small round blue cells** + hyaline islands · **staghorn (haemangiopericytomatous) vessels** · HEY1–NCOA2	Wide resection + **chemo ± RT** · <50% 5-yr
Periosteal (juxtacortical)	2nd–4th decade	Surface, metaphysis femur/ humerus	Juxtacortical mass, rings & arcs · **cortex eroded/ saucerized** (vs periosteal chondroma)	Low grade	Wide resection · indolent
Secondary	3rd–4th decade	Ilium (MHE)	Cap >2 cm	As conventional	As primary · usually low grade



Dedifferentiated chondrosarcoma — bimorphic histology

Differential of chondrosarcoma

	Malignant, older patient	Similar chondroid histology	Sacral lesion, older	Wide resection alone
Chondrosarcoma	✓	✓	✓	✓
Metastasis · lymphoma · myeloma · MFH	✓	–	✓	–
Enchondroma / Ollier / Maffucci · periosteal chondroma · MHE	–	✓	–	–
Parosteal OS · adamantinoma	–	–	–	✓
Chordoma	–	–	✓	✓

Imaging distractors **bone infarct** (smoke up the chimney, cold, asymptomatic) · **fibrous dysplasia** (ground glass) · **osteosarcoma** (younger, cloud-like bone) · **plasmacytoma** (uniformly T2 bright, **cold** scan) · Ewing (lytic, no mineral)

5 Management

Treatment algorithm

- **Chondrosarcoma (chemo- and radio-resistant)**
  - Grade 1 / ACT, extremity, confined to bone → **EXTENDED CURETTAGE + ADJUVANT + GRAFT ± FIXATION**
    - Asymptomatic, no aggressive features → **OBSERVE**
  - Grade 2–3, or ANY pelvic/axial lesion → **WIDE RESECTION ALONE**
  - Mesenchymal → **WIDE RESECTION + CHEMO (± RT)**
  - Dedifferentiated → **WIDE RESECTION ± CHEMO**
  - Surgically inaccessible → **RADIOTHERAPY**

Treatment — wide resection (chemo/radioresistant)

Why resistant **low mitotic rate**, **poor perfusion / low blood supply**, chondroid/**myxoid matrix** limits drug diffusion, few cells in growth phase

– – Active space – –

Pathology

Wide excision	better survival than marginal or intralesional · grade 2 distal femur → **wide excision** ✗ Wrong options: curettage · marginal · chemo first · hip disarticulation
Pelvic CS	**internal hemipelvectomy or hindquarter amputation** (✗ Wrong options: + chemo · + RT) · **wide resection alone** (✗ Wrong options: brachytherapy)
Proximal fibula CS	**wide resection + reconstruct LCL and biceps femoris** (transfer to lateral tibia, staple) → prevents varus instability + flexion weakness ✗ Wrong options: curettage + liquid N_2 + cement (GCT) · denosumab (GCT) · + chemo (OS)
Ollier patient with CS	**wide resection only**
Clear cell after "benign" curettage	**wide resection alone** ✗ Wrong options: chemo · curettage · forequarter
Lung metastases	resect if possible **(TO)**

Low-grade lesion – extended curettage

Grade 1 (TO)	**extended curettage + cementing** possible · surgery mainstay · **no RT, no chemo** except **mesenchymal / dedifferentiated** · clear cell = **low grade**, radiographs may look benign
ACT / grade 1 extremity	**extended intralesional curettage + high-speed burr + adjuvant + bone-void graft ± prophylactic fixation** · same recurrence as wide resection, better function
Adjuvants	cryotherapy / **liquid nitrogen**, argon beam, phenol, ethanol, hydrogen peroxide
Pelvis / axial grade 1	**wide excision** (higher recurrence + metastasis, poor biopsy concordance)
Not for ACT	neoadjuvant RT 50–60 Gy · proximal humeral resection + megaprosthesis (old standard) · forequarter amputation

Dedifferentiated chondrosarcoma

Treatment	**wide resection + endoprosthetic reconstruction** · chemo possible role (controversial) · RT palliative/adjuvant only
IM nail through dedifferentiated CS	seeds entire femur → **amputation** ✗ Wrong options: proximal femoral replacement · chemo · RT
Curettage + cement + THA, histology dedifferentiated	resect whole bed → **hemipelvectomy**
Pathologic fracture + high-grade CS	next test **CT chest**

6 Complications

Local recurrence	grade 1: rare after wide resection, **5–15% after curettage + adjuvant** · grade 2: margin dependent · grade 3: **25%** recurrence, **>30%** mets · high-grade recurrence → mets + ↓ survival

Prognostic factors

Most important	**histological grade (TO – FNB 2016)** ✗ Wrong options: site · matrix calcification · age

Poor prognosis	**high grade** · **axial / proximal** (pelvis) · **advanced age** · **inadequate margins** · **↑ telomerase** (recurrence)
Not prognostic	gender · negative margins, younger age, tumour-centre biopsy = **favourable**
Spine/sacrum CS	worse local control/survival with **increasing age**

X vs Y

	Chondrosarcoma (rings & arcs)	Osteosarcoma (cloud-like)	Bone infarct
Age	>50	Adolescent	Any (risk factors)
Mineral	Rings, arcs, popcorn	Fluffy/cloud-like osteoid	Serpiginous rim, smoke up chimney
Bone scan	Hot (> ASIS)	Hot	Not hot
Symptoms	Night pain	Pain, mass	Asymptomatic

	Chondrosarcoma	Mesenchymal CS	Ewing
Round blue cells	No	✓ + hyaline islands, staghorn vessels	✓ sheets, CD99
Chemo	No	Yes	Yes

How it's asked

"52-yr-old, night shoulder pain, rings-and-arcs, hot bone scan, lobular T2-bright lesion?"	→ **chondrosarcoma** ✗ Wrong options: infarct · OS · Ewing · FD · plasmacytoma
"Tumour resistant to chemo AND radiation?"	→ **classic intramedullary chondrosarcoma** ✗ Wrong options: Ewing · synovial sarcoma · chordoma (RT) · DFSP
"Pelvic CS — treatment?"	→ **wide resection alone** (internal hemipelvectomy or hindquarter amputation)
"Grade 2 distal femoral CS?"	→ **wide excision**
"Symptomatic grade 1 ACT proximal humerus?"	→ **extended intralesional curettage + graft ± prophylactic fixation**
"Proximal fibula CS — best function?"	→ **wide resection + LCL + biceps reconstruction**
"Correct about dedifferentiated CS?"	→ **~10% 10-yr survival** ✗ Wrong options: de novo · homogeneous · curettage · adolescents
"73-yr-old, pathologic fracture, high-grade spindle cells next to benign cartilage?"	→ **dedifferentiated CS** ✗ Wrong options: synovial sarcoma (keratin glands) · metastatic prostate · periosteal OS

– – Active space – –

"Dedifferentiated CS with path fracture – staging test?"	→ **CT chest** ✗ Wrong options: CT CAP · SPEP/UPEP
"IM nail placed through dedifferentiated CS?"	→ **amputation**
"38-yr-old, subchondral femoral head lesion, clear vacuolated cells?"	→ **clear cell chondrosarcoma** ✗ Wrong options: chondroblastoma · GCT
"Highest malignant risk?"	→ **Maffucci** > Ollier > MHE > solitary lesions
"MHE – malignant histology?"	→ **low-grade chondrosarcoma** (EXT1 worst)
"Secondary CS NOT seen in?"	→ **chronic osteomyelitis** (→ SCC) (✓ Paget, ✓ synovial chondromatosis, ✓ osteochondroma) **(TO)**
"FALSE about CS (FNB 2024)?"	→ source gives no answer · **"clear cell CS is predominantly metaphyseal"** is false (it is **epiphyseal**) **(TO)**
"Most important prognostic factor?"	→ **histological grade (TO)**
"Poor prognostic factor in pelvic CS?"	→ **pelvic location** ✗ Wrong options: female · negative margins · younger age
"Recurrence most directly related to?"	→ **↑ telomerase activity**
"Incidental 'possible chondrosarcoma' on knee MRI, no images – next?"	→ **orthogonal radiographs**
"Adult osteochondroma, 14 mm cap, stable?"	→ **serial imaging + exam** (cap ≥2 cm = worry)

CHORDOMA

– Pathology · 11 / 66 –

1 Basics

Epidemiology

Rank	**commonest primary malignant spinal tumour in adults** · commonest primary malignant tumour of the **sacrum (TO)** · 1–4% of primary bone tumours
Demographics	**M:F 3:1** · **>50 yr** (5th–6th decade) · middle-aged Caucasian men · TO: **40–60 yr**
Spinal rank	commonest primary malignant spinal tumour in adults (Orthobullets) · **2nd commonest primary spinal malignancy** (TO) – read the options
Midline vs ala (TO)	chordoma arises from the **midline** · **sacral GCT** arises from the **ala** (eccentric)
Site	**sacrum + coccyx 50%** (S2–S3 predilection) · **spheno-occipital (skull base) 35%** · **mobile spine 15%**

Pathoanatomy

Origin	malignant transformation of **residual notochordal cells** (same lineage as **nucleus pulposus**) → **midline** location
Behaviour	slow-growing, locally aggressive, low-grade malignant · diagnosis often delayed

Genes & lab markers

Keratin (pan-cytokeratin) +	separates from chondrosarcoma (keratin –)

EMA + (TO) · **S-100 weakly +**

3 Presentation

Pain	insidious **low back / hip pain**, **worse sitting**
Bowel/bladder	**constipation**, faecal incontinence, bladder change · UTI symptoms (1/3) · cauda equina 10% **(TO)**
Motor/sensory deficit	**rare** (most lesions **S1 or distal**) · S1–S2 lesion may lose L5–S1 function · radiculopathy (sciatic) up to 1/3
Rectal exam	**>50% palpable**

4 Diagnosis & Classification

Key radiology findings

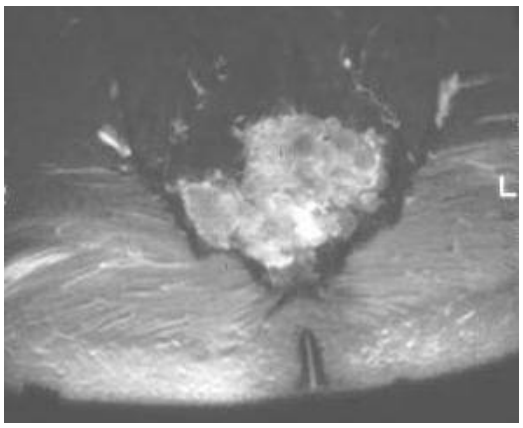
X-ray	**hard to see** (overlying **bowel gas**) · large lytic lesion **crossing the midline** (>50%) → SI joints, ilium

Pathology

– – Active space – –

CT — **midline bone destruction + soft-tissue mass** · **calcifications** in the soft-tissue component

MRI — **bright on T2** · soft-tissue extension



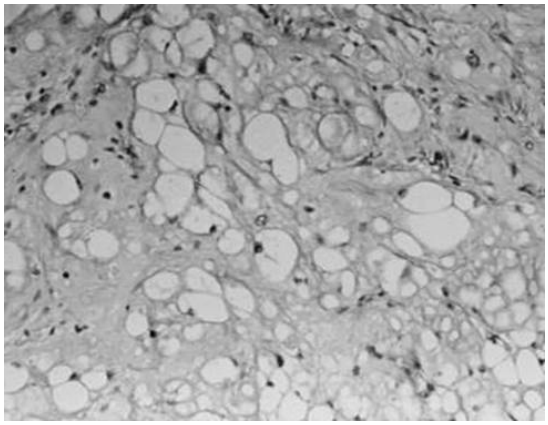
Sacral chordoma — T2-bright midline mass

Biopsy & histology

Biopsy required — MRI alone insufficient (lymphoma, chondrosarcoma mimic) · **transrectal biopsy contraindicated**

Gross — **lobular, gelatinous**

Histology — **physaliferous cells**: large, **foamy, vacuolated** (eosinophilic) cytoplasm, vesicular nucleus — **pathognomonic** · cords/lobules in **mucoid matrix** separated by fibrous septa · distinct nodules



Physaliferous cells

Differential

	Sacral lesion, older patient	Keratin +	Similar X-ray	Wide resection alone
Chordoma	Yes	Yes	Yes	Yes
Chondrosarcoma	Yes	No	Yes	Yes
Metastatic carcinoma	Yes	Yes	Yes	–
Lymphoma of bone	Yes	–	Yes	–
Myeloma · MFH	Yes	–	–	–
Adamantinoma (tibia)	–	Yes	–	Yes
Synovial sarcoma · epithelioid sarcoma	–	Yes	–	–

Other sacral tumours myeloma, **GCT**, ABC, metastasis

5 Management

Treatment & local recurrence

Standard **wide-margin en bloc resection** · **sacrifice sacral roots** to get margins · wide tumour-free margin ∝ tumour-free survival

Radiation **add if margin not achieved** (positive margins) · **inoperable** tumours

TO answer sacral chordoma → **wide resection alone (TO)** · chemo- and relatively radio-resistant; RT/SRS for recurrence
✗ Wrong options: + EBRT · + chemo · curettage + bisphosphonate

Mobile spine **en bloc corpectomy / total vertebrectomy** → best local control · needs anterior + posterior reconstruction

Not indicated chemotherapy (chemoresistant) · debulking (intralesional) · RT alone · observation

Radiosurgery (CyberKnife)

Stereotactic radiosurgery recurrent/residual **chordoma** and **chondrosarcoma** after failed surgery + fractionated RT **(TO)**

CyberKnife extracranial → **sacral chordoma** ✓ · **Gamma knife** – intracranial only **(TO)**

Proton–photon beam may help recurrence · **proton therapy** when unresectable or surgery → **severe morbidity** (TO)

Metastatic chordoma **imatinib** (palliative) (TO)

– – Active space – –

Sacral nerve root sparing & function

Roots preserved / resected	Bowel & bladder
Bilateral S3 preserved	Near-normal (Orthobullets) · **S3 most critical**
Both S2 preserved (at least)	Near-normal (OB)
Unilateral S2, S3, S4 (± S5) preserved	Near-normal ✓ (OB answer: unilateral S2–S5)
Bilateral S4–S5 resected	**Near-normal** (S2–S3 splanchnics kept)
Bilateral S2–S5 resected	**Complete loss**
Bilateral S1 / L5 only	Not enough

Pelvic splanchnic nerves — **S2–S4** (S3 predominance) → parasympathetic to pelvic viscera

6 Complications

Local recurrence — **~50%**, margin dependent → **close surveillance**
✗ Wrong options: nodal · liver · bone mets · malignant transformation

Metastasis — **30–50%**, **late** (lung, rarely bone) · OB: 30–50% already metastatic at diagnosis

Survival — **60% 5-yr**, **25% long-term** (25–50%) · local extension may be fatal

Poor prognosis (TO) — high histological grade · volume **>10 cm / 200 cm³**

Post-op — loss of bowel/bladder function (root sacrifice)

X vs Y

	Chordoma	Chondrosarcoma
Origin	Notochord, **midline**	Cartilage, often **eccentric** (pelvis)
Cells	**Physaliferous**	Atypical chondrocytes in lacunae
Keratin / EMA	**+**	**–**
S-100	Weak +	+
Treatment	Wide resection ± RT (margin +)	Wide resection

How it's asked

"Sacral lytic lesion, 58-yr-old, lobules with vacuolated physaliferous cells?" → **chordoma**
✗ Wrong options: chondrosarcoma · metastatic RCC · CMF · GCT

"All true of sacral chordoma EXCEPT?" → **"MRI is sufficient without biopsy"** (✓ notochordal like nucleus pulposus, ✓ mets 30–50%, ✓ motor deficit rare, ✓ resection ± RT)

"Treatment of sacral chordoma?"	→ **wide resection alone** (✗ Wrong options: + EBRT · + chemo) **(TO)** · OB: **wide surgical excision** (✗ Wrong options: chemo · debulking · RT alone)
"L5 en bloc corpectomy is right for which histology?"	→ **chordoma** ✗ Wrong options: osteomyelitis · myeloma · LCH · lung adenocarcinoma
"Why surveillance after resection?"	→ **high local recurrence**
"Keratin + and S-100 + sacral mass?"	→ **chordoma**
"Tumour treated by CyberKnife?"	→ **chordoma (TO)** ✗ Wrong options: vertebral osteoblastoma · sacral OS · GCT
"Near-normal bowel/bladder if which roots spared?"	→ **unilateral S2–S5** (or both S2 / bilateral S3)
"Bilateral S4–5 vs bilateral S2–5 resection?"	→ **S4–5 patient near-normal**, S2–5 patient loses function

– – Active space – –

– – Active space – –

CONVENTIONAL INTRAMEDULLARY OSTEOSARCOMA

– Pathology · 12 / 66 –

1 Basics

Epidemiology & genetics

Rank	**most common primary sarcoma of bone** · **commonest non-haematologic primary bone malignancy** · ~20% of primary bone cancers **(TO)**
Commonest primary malignant tumour of bone overall	**multiple myeloma (TO – FNB 2021)**
Age	**bimodal**: 1st peak **2nd decade (10–14)**, **75% <20** · 2nd peak **>65** (secondary: Paget, infarcts, prior osteochondroma/osteoblastoma, **radiation**)
Sex / race	**M:F 1.5:1** · ↑ Black, Hispanic
Site	**metaphysis** of long bones (10% diaphyseal, <1% epiphyseal) · **distal femur > proximal tibia > proximal humerus > axial** (spine/sacrum) · fastest-growing physes
Diaphyseal OS	**10%** (Orthobullets) · **7–9% (~8%)** (TO) – read the options
Incidence	~**1 per million per year** (TO) · **male preponderance in all types except parosteal** (female) (TO)
Hallmark	**malignant osteoid** made by tumour cells defines OS (TO)
Childhood tumour facts (TO)	commonest tumour of infancy / extracranial / abdominal: **neuroblastoma** · commonest **solid tumour** of childhood: **brain tumour** · commonest **malignancy**: **leukaemia (ALL)** · commonest benign bone lesion: **NOF** · commonest benign bone tumour: **osteochondroma** · commonest malignant bone tumour: **osteosarcoma**

Genes & lab markers

Association	Gene / inheritance	Other features
Hereditary retinoblastoma	**RB1** (13q14), germline · **500–1000×** risk **(TO)**	Somatic RB1 loss in **60–70%** of sporadic OS
Li-Fraumeni	**TP53** (17p13), AD	Breast, brain, adrenocortical cancer · somatic TP53 in 20–50%
Rothmund-Thomson	**RECQL4**, AR	**Poikiloderma** (sun-sensitive rash), alopecia, juvenile cataracts, dental anomalies, short stature · OS, fibrosarcoma, gastric adenoCa, BCC/SCC
Bloom	**BLM**, AR	UV facial rash, short stature · OS, leukaemia, lymphoma
Werner	**WRN**, AR	**"Adult progeria"** – premature ageing, cataracts, osteoporosis
Fibrous dysplasia	–	Polyostotic > monostotic

Pathogenesis (TO)	**RB1 + TP53 pathway inactivation nearly universal**, via complex structural variants / **chromothripsis** · **no pathognomonic translocation** · amplification **MDM2, CDK4, MYC** · (✗ Wrong options: MDM2/CDK4 = primary driver of conventional OS (that is **low-grade central / parosteal**) · RECQL4 in sporadic elderly OS) ✗ Wrong options: like Ewing
Not OS genes	EWS-FLI1 (Ewing) · EGFR, IGF-R · VEGF (angiogenesis; overexpression = poor prognosis)
Labs	**↑ ALP and ↑ LDH** → aggressive disease, **poor prognosis** · very high LDH may mean mets
IHC	**SATB2 +** (all OS subtypes) **(TO)**

3 Presentation

Symptoms	**pain + swelling**, night/rest pain · often blamed on sports injury ("red herring" fall) · median **4 months** to diagnosis · may have fever, weight loss
Exam	mass, tenderness, ↓ ROM, NV compression · pain with axial loading
Pathologic fracture	~10%

4 Diagnosis & Classification

Key radiology findings

X-ray (AP + lateral, whole bone)	metaphyseal **medullary + cortical destruction** · **mixed blastic/lytic** (may be purely either) · **"sunburst" / "hair-on-end"** · **Codman triangle** · **cloud-like / fluffy** osteoid mineral · soft-tissue mass · usually **doesn't cross physis**



Codman triangle and sunburst periosteal reaction

MRI (± contrast, entire bone + joint above and below)	soft-tissue + **NV involvement** · resection level · **skip metastases** (= stage III) · **gold standard** local study
CT chest	**required at presentation** (lung mets) ✗ Wrong options: chest X-ray (less sensitive) · CT abdomen/pelvis (adult unknown-lesion work-up)
Bone scan (Tc-99m)	**very hot** · bone mets, skip lesions · **superior to PET in OS** (PET better in Ewing) · PET-CT misses lung nodules <5–9 mm

– – Active space – –

FNB 2024 image (Codman triangle, distal femur) "least likely appearance?" — answer not given · **onion peel** (Ewing) and **stippled calcification** (chondroid) don't fit OS · sunburst + Codman do **(TO)**

X-ray on 7 parameters (TO)

Parameter	Conventional OS
Age	**Skeletally immature** (physis open)
Location	**Metaphyseal** (distal femur)
Type (pattern of destruction)	**Permeative**, lytic + sclerotic
Matrix (what lesion makes)	**Bone-forming / sclerotic (osteoid)** → first differential = OS
Zone of transition	**Wide** (hallmark of malignancy)
Periosteal reaction	**Codman triangle + sunburst** (both may coexist)
Soft-tissue component	Yes, often **mineralised**

![Distal femur OS read on 7 parameters](../../TO notes/attachments/TOn-notes-osteosarcoma-jan-2026-p02-f004.jpg)

Physis classically respected, but OS **can cross the physis** (esp. long-standing) (TO)

Skip vs satellite lesion (TO) **satellite** lies in the **reactive zone** · **skip** has **normal bone between** it and tumour → worse prognosis, resected → **MRI of the whole bone** (whole femur, not distal femur)

PET-CT (TO) staging + **chemo response**: scan before and after 2 cycles · **>50% fall in SUV = good response**

Staging work-up

Local **MRI of whole involved bone** (e.g. knee + tibia) · **Systemic** — **CT chest + whole-body bone scan**

Pre-chemo baseline CBC, BMP, LFTs, UA · **echocardiogram** (doxorubicin) + **hearing screen** (cisplatin)

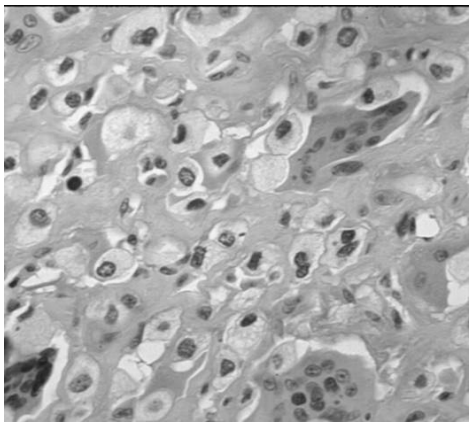
Not needed bone marrow biopsy (Ewing) · skeletal survey (myeloma / NAT) · lymph node biopsy · PFTs · contralateral MRI · angiography · reactive-zone biopsy · EXT1/2 testing (MHE)

Biopsy required · by/after discussion with **definitive surgeon** · core needle preferred · sample **peripheral soft-tissue part** · (✗ Wrong options: FNA · biopsy by IR at outside hospital (path fracture → sling, analgesia, stage, refer) · empirical antibiotics without biopsy)
✗ Wrong options: necrotic centre · Codman triangle = reactive bone

Histology — diagnostic criteria

Two criteria **tumour cells produce osteoid** + **stromal cells frankly malignant** (atypia, high N:C, abnormal mitoses)

Hallmark **"lacy" osteoid / woven bone** laid down directly by pleomorphic spindle cells



Malignant cells producing lacy osteoid

Conventional subtype	Matrix	Mimic
Osteoblastic (commonest)	Abundant **lace-like osteoid**, plasmacytoid cells **(TO)**	–
Chondroblastic	Malignant hyaline cartilage + **focal osteoid** · **"whorled" pattern on MRI** · responds **less to chemo** (TO)	Chondrosarcoma (any osteoid → OS)
Fibroblastic	Herringbone/storiform collagen, sparse osteoid	Fibrosarcoma, UPS
Giant-cell-rich	Many giant cells	**GCT** (no atypia/osteoid)

MSTS staging

Commonest	**IIB** (high grade, extracompartmental, no mets) · **90–95% high grade**, penetrate cortex early
Metastasis at presentation	**20–25%** (Enneking III) · **lung ~80%** (haematogenous) > **bone 16%** · bone mets without lung mets rare (5%)
Skip metastases	intraosseous (same bone, usually proximal) or **trans-articular** · **2–6%** (OB) · **10%** (Orthobullets) · prognosis = lung mets

– – Active space – –

Osteosarcoma subtypes

Location	Grade	Subtype	Key feature
Intramedullary	High	**Conventional** (this topic)	Mixed tissues
	High	**Telangiectatic**	Purely lytic, **ABC-like** blood lakes · **malignant cells in septa** · **3–10%** of OS, fluid levels on MRI (TO)
	High	**Small cell**	OS + Ewing look · **SATB2 +, CD99 – / focal**, no EWSR1 · treated with **Ewing-protocol chemo** (TO)
	Low	**Low-grade central** (FD-like, desmoplastic-fibroma-like)	Mimics **fibrous dysplasia** · **MDM2/CDK4** amplification
Surface	Low	**Parosteal**	**Posterior distal femur**, "stuck-on" · **middle-aged females (3rd–4th decade)** · MDM2/CDK4
	Intermediate	**Periosteal**	**Diaphysis femur/tibia** · **chondroblastic** lobules, sunburst
	High	High-grade surface	Like conventional, on surface · medullary involvement common · **least common** subtype (TO)
Intracortical	–	Rarest bony type	–
Extraskeletal	High	Soft-tissue OS	<5% · treat like STS

Chemo rule (TO) **low grade** (low-grade intramedullary, parosteal) → **surgery alone** · anything not low grade, **incl. intermediate periosteal** → chemo

Fluid levels on MRI not only ABC → think **telangiectatic OS** (TO)

![Telangiectatic OS vs ABC — both show fluid levels](../../TO notes/attachments/TOn-notes-osteosarcoma-jan-2026-p03-f007.jpg)

Secondary osteosarcoma (TO)

Who **elderly** (2nd peak) · causes: **Paget (commonest)**, post-radiation, **bone infarct**, **chronic osteomyelitis**, **fibrous dysplasia**

Prognosis **worse** than conventional, partly because elderly **tolerate chemo poorly**

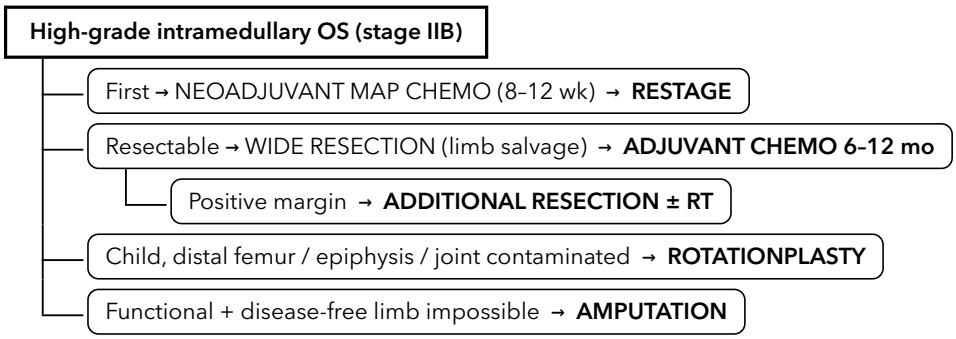
Arlen latency usually **≥5 yr** (up to 30 yr) · no strict cut-off but **not months**

Differential

Lesion	Imaging	Histology
Osteosarcoma	Mixed blastic/lytic, Codman, soft-tissue mass	**Lacy osteoid**, malignant stroma
Osteomyelitis	Lags 2 wk, lucency, sequestrum/ involucrum	Neutrophils (acute), lymphocytes (chronic)
Eosinophilic granuloma	"Great mimicker", vertebra plana	Langerhans cells, eosinophils, no atypia
Ewing	Permeative, moth-eaten, onion skin	**Small round blue cells**, t(11;22)
Dedifferentiated CS	Bimorphic	Cartilage + high-grade spindle
Fibrosarcoma	Lytic, >55 yr	**Herringbone** spindle cells
Lymphoma of bone	Diaphyseal, moth-eaten, 35–55 yr	Mixed round cells
Osteoblastoma	Posterior spine, nidus	Benign osteoblasts, giant cells
Myositis ossificans	**Mature periphery**	Mature bone, no atypia
UPS	–	Spindle cells, **no osteoid**

5 Management

Treatment – neoadjuvant chemotherapy & wide resection



Sequence — **neoadjuvant chemo → wide resection → adjuvant chemo** · surgery 3–4 wk after last preop dose · chemo restarts within ~2 wk of surgery
✗ Wrong options: surgery + RT · chemo + RT

MAP — **high-dose Methotrexate (leucovorin rescue) + Adriamycin (doxorubicin) + cisPlatin** (US standard) · ± ifosfamide · 3-drug > 2-drug (AP) by ~10% survival despite toxicity **(TO)** · **EURAMOS-1**: adding ifosfamide + etoposide (MAPIE) for poor responders → no benefit
✗ Wrong options: vincristine/carboplatin regimens (Ewing)

– – Active space – –

Cycles (TO) neoadjuvant **2–3** → limb salvage → adjuvant **3–4** · **total 6** (3 + 3 or 2 + 4) · **IAP** (ifosfamide, Adriamycin, cisplatin) alternative · 2-drug: AP or ifosfamide + Adriamycin · dose by **BSA**

Drug (TO)	Dose
Adriamycin (backbone of sarcoma chemo)	**75 mg/m²** over 3 days
Cisplatin	**100 mg/m²** over 2 days
High-dose methotrexate	**12 g/m²** over 4 h (vs 15 mg/week in RA)
Ifosfamide	**9 g/m²** over 5 days (1.8 g/m²/day)

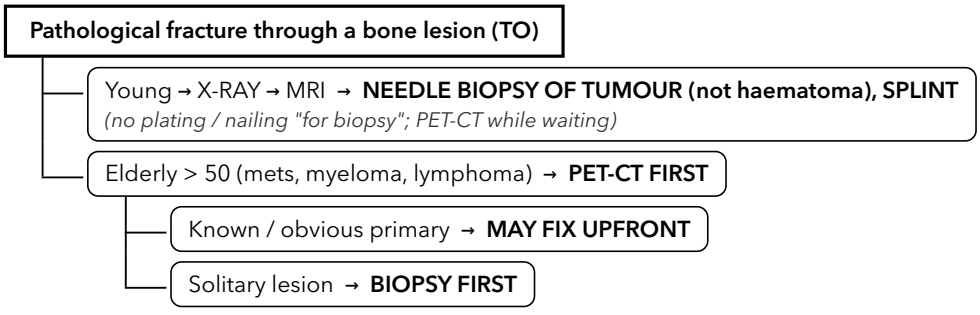
MAP cycle (TO) **5 weeks (35 days)** · day 1–3 Adriamycin · day 1–2 cisplatin · **day 21 + day 28 HD-MTX** · day 35 next cycle

Agent	Mechanism	Toxicity
Methotrexate	Dihydrofolate reductase inhibitor	Myelosuppression, mucositis
Doxorubicin	Topoisomerase II inhibitor	**Cardiotoxicity** → echo
Cisplatin	Covalent DNA binding	**Renal failure, hearing loss**, neurotoxicity → audiogram
Ifosfamide	DNA alkylating	Renal failure, **haemorrhagic cystitis**

Chemo effectiveness same preop or postop (OB)

Radiation **radio-resistant** (NF-κB inhibits JNK-mediated apoptosis) · only palliation, unresectable sites, close margins (pelvis/spine) · **ECRT** – extracorporeal 50 Gy single fraction to sterilise and re-implant bone **(TO)** · **extraskeletal OS radiosensitive** (exception)
✗ Wrong options: curative for sacral OS **(TO)**

Biopsy rules (TO) **needle (core) after imaging** · **several cores through the same entry point** · sample the **soft-tissue component** (representative, no decalcification, quicker report) · **FNAC has no role** · open only if pathologist cannot report a core · **"giant-cell-rich lesion" ≠ GCT** → curettage of giant-cell-rich OS is disastrous



Metastatic fracture fixation (TO) assume **last surgery for that bone** · must **bear weight next day**

Pathologic fracture limb salvage still possible if stabilised (ex-fix/cast) and good chemo response (healing, ↓ pain/swelling, ↑ ossification, loss of tumour blush on angiogram)

Limb salvage principles

Biopsy tract	**resected en bloc** with specimen, never exposed **(TO)** ✗ Wrong options: routine sacrifice of popliteal vessels · keep capsule/ pseudocapsule · cut at visible tumour edge
Bone cut	**2–3 cm beyond intramedullary extent on T1 MRI (TO)** · marrow sampled at cut
Limb salvage achievable	**85–90%** · survival **= amputation**
Contraindications (TO)	NV bundle encasement · path fracture with massive haematoma contaminating compartments · no clean margin · infection
Amputation indications	functional + disease-free limb impossible (only absolute) · NV encasement · growing on chemo next to NV bundle · bad biopsy · infected tumour
Physeal-sparing	not when tumour crosses physis (knee OS usually sacrifices articular surface) · transepiphyseal resection only if epiphysis free

Resection margins (Enneking)

Margin	Plane
Intralesional	Through tumour · **tumour at marrow margin** = intralesional
Marginal	Through **reactive zone / pseudocapsule**
Wide	Cuff of normal tissue 360° ("orange rind")
Radical	Whole compartment (e.g. entire femur)

Rotationplasty

Indications	distal femur / knee / proximal tibia / hip sarcomas in **young children** · **joint contamination** · epiphyseal involvement · failed salvage with intact leg · PFFD · avoids LLD
Requirement	**functioning, uninvolved sciatic nerve**
Technique	resect segment, **rotate leg 180°**, fix tibia to femur · **ankle → knee** · gastrocsoleus = quadriceps, anterior compartment = hamstrings · keep sural branches
Winkelmann	**A** (ankle = knee): **AI distal femur**, AII proximal tibia · **B** (knee = hip): BI proximal femur sparing hip, **BII involving hip**, BIII total femur
Outcome	**highest self-reported function, fewest revisions** in pre-adolescents · cosmesis is the drawback ✗ Wrong options: endoprosthesis (loosening, LLD) · allograft (fracture, infection) · AKA (energy cost)

– – Active space – –



Van Nes rotationplasty

Rotationplasty function (TO) patient functions as a **below-knee** rather than above-knee amputee · longer lever arm, better socket tolerance, improves with time · drawback **cultural stigma** (foot faces backwards)

Reconstruction options (TO)

Non-biological	Biological
Megaprosthesis (commonest) · **expandable** in children	**Allograft** (rarely available in India)
Nail–cement spacer (palliative)	Autograft: **ECRT** or **cryotherapy** · **vascularised fibula**
	Rotationplasty · **Canadell procedure**

ECRT resected tumour bone → **50 Gy single fraction** (15–20 min) → soft tissue stripped → **canal augmented with cement** (devascularised) → refixed with plates (TO)

Cryotherapy tumour bone in **liquid nitrogen at −196 °C**, re-implanted (TO)

Pedicled cryotherapy **proximal osteotomy only**, tumour debulked and dipped without distal cut → **distal blood supply preserved** → **lower graft fracture risk** · saves ~22–24 cm of native bone (TO)

Canadell procedure (TO) child, tumour **close to open physis**, epiphysis salvageable · step 1 **ex-fix physeal distraction 10–15 days** · step 2 **resection through the physis** → **ECRT** → reimplant · keeps epiphysis + growth potential

![Pedicled cryotherapy](../../TO notes/attachments/TOn-notes-osteosarcoma-jan-2026-p08-f020.jpg)

Megaprosthesis outcomes

Modular endoprosthesis gold standard reconstruction · rotating hinge (ligaments sacrificed) · cemented if poor bone or early chemo · press-fit for long survivorship

Henderson failure types (TO) 1 soft-tissue · 2 **aseptic loosening** · 3 structural · 4 **infection (5–15%)** · 5 tumour recurrence

Infection **2–10%** (Orthobullets) · 12–19% at 5–10 yr (proximal tibia) · risks **op time ≥8 h**, **drain ≥14 d** · ~50% → amputation

PARITY trial	**5-day ≠ better than 1-day** antibiotics · **flaps ↓ infection** · **NPWT → ↑ reoperation, not infection** · longer op time → ↑ complications · **primary malignant tumour → higher short-term reoperation** than benign aggressive
Aseptic loosening	~6% · commonest failure · **tibia** commonest mechanical failure
Proximal tibial EPR + medial gastrocnemius flap	good function (flexion 98–110°, lag <10°) but **infection, extensor mechanism failure (3–17%), reoperation** significant · patella alta up to 70%
Allograft	fracture at host–graft junction, nonunion (chemo, RT, ECRT risk factors)

Extraskeletal osteosarcoma

vs conventional	<5% of OS · **older (5th–6th decade)** · nearly always **high grade** · **worse survival (~50% vs ~75%)** · chemo unproven · **radiation is part of standard treatment** (wide resection + RT, like STS)

6 Complications

Prognostic factors (necrosis, MDR, margins, metastasis)

Most important **stage / metastasis at diagnosis**

Chemo response **>90% necrosis = good** (5-yr ~85%)
✗ Wrong options: 60% or 70% necrosis = poor

Huvos grade	Necrosis	Response
I	0–50%	Poor
II	51–90%	Poor (e.g. 88% → continue MAP)
III	91–99%	Good
IV	100%	Complete

Huvos pass mark (TO) **90% necrosis, not 50%** · alt. reporting: **≤10% viable cells = good**, >10% = poor

Picci grade (TO)	Finding	Response
1	**Macroscopic** viable tumour	Poor
2	**Microscopic** viable tumour	Poor
3	**No viable tumour**	Good

Bone vs lung mets (TO)	**bone mets worse** (lung = first filter, so reaching bone = more aggressive tumour)
Ranking (TO)	**stage > grade > size** · **size: volume >200 mL** poor · grade prognostic in OS but **not in Ewing** (always high grade)
Poor prognosis	mets · skip lesions · **<90% necrosis** · **age >40** · male · axial/pelvic site · **large size (≥5 cm)** · **MDR / P-glycoprotein** (25% primary, 50% mets) · **VEGF** · **↑ ALP / ↑ LDH** · vascular involvement · **positive margins** · radiation-induced or Paget OS
Not poor (TO)	✓ **>90% necrosis** is good · urinary heat-shock protein, serum proteoglycan C, unchanged DNA ploidy listed as poor-prognosis distractors

– – Active space – –

Local recurrence — most directly **surgical margins**
✗ Wrong options: site, size, grade, neoadjuvant chemo

Radiation-induced OS (TO) — **worse prognosis** than conventional · **Arlen criteria**: different prior histology, in radiation field, long latency, histological proof

Situation	5-yr survival
Localised, standard treatment	**~65%** (extremity 65%, **pelvis 55%**)
Good chemo response	**85%**
Pulmonary mets at diagnosis	**15–20%** (resectable 15–25%)
Late (>1 yr) lung mets + thoracotomy	~30%
Bone mets	Incurable · <5% long term
Paget OS	<15% long term

Pelvic OS — larger, more mets, local recurrence **up to 44%** despite negative margins

Metastasectomy — complete resection of primary + mets → **5×** long-term survival

Surveillance (NCCN) — q3 mo yr 1–2 · q4 mo yr 3 · q6 mo yr 4–5 · yearly after · exam, imaging of site + chest, ± bone scan/PET, ALP

X vs Y

	Conventional OS	Parosteal OS	Periosteal OS
Grade	**High**	**Low**	Intermediate
Age	10–20	**3rd–4th decade, F**	Slightly older
Site	Metaphysis, medullary	**Posterior distal femur**, surface	**Diaphysis** femur/tibia
X-ray	Sunburst, Codman	**Stuck-on**, heavily ossified, central > peripheral	Sunburst on surface, saucerized
Histology	Lacy osteoid, malignant stroma	Bland spindle cells, parallel trabeculae	Chondroblastic lobules
Treatment	**Chemo + wide resection + chemo**	**Wide resection alone**	Chemo (NCCN) + wide resection

	Osteosarcoma	Myositis ossificans
Ossification	Dense **centre**	**Mature periphery (zonal)**
Histology	Atypia, lacy osteoid	Mature bone, bland fibroblasts

How it's asked

"12-yr-old, distal femoral metaphysis mixed sclerosis/ lucency, aggressive periosteal reaction?"	→ **high-grade intramedullary OS (TO)** ✗ Wrong options: parosteal · periosteal · telangiectatic
"Spindle stroma producing lace-like osteoid with malignant osteoblasts – subtype?"	→ **osteoblastic OS (TO)**
"Accurate statement on OS genetics?"	→ **RB1 + TP53 inactivation nearly universal, structural variants (TO)**

"Gene mutation predisposing to OS?"	→ **RB** ✗ Wrong options: EWS-FLI1 · EGFR · IGF-R · VEGF
"Rothmund-Thomson predisposes to?"	→ **osteosarcoma**
"Commonest sites in order?"	→ **distal femur > proximal tibia > proximal humerus > axial**
"Commonest metastatic site?"	→ **lung** (then bone)
"Initial staging?"	→ **bone scan + MRI of knee AND whole tibia + CT chest** ✗ Wrong options: chest X-ray · PET alone · reactive-zone biopsy
"Next test for proximal tibial OS?"	→ **MRI of whole tibia incl. knee + ankle** ✗ Wrong options: skeletal survey · CT CAP · bone marrow biopsy · FNA
"Additional work-up?"	→ **whole-body Tc bone scan** ✗ Wrong options: CT abdomen/pelvis · angiography
"Before MAP chemo?"	→ **echocardiogram + hearing screen** ✗ Wrong options: marrow biopsy · PFTs
"Child with proximal humeral path fracture before transfer — NOT appropriate?"	→ **IR biopsy at referring hospital**
"Febrile teen, aggressive femoral lesion — next?"	→ **biopsy** ✗ Wrong options: IV antibiotics
"First step in management?"	→ **neoadjuvant chemotherapy** ✗ Wrong options: resection first · AKA · RT · repeat biopsy
"Protocol maximizing survival?"	→ **neoadjuvant chemo → excision → adjuvant chemo**
"Systemic regimen?"	→ **high-dose methotrexate + doxorubicin + cisplatin (MAP)** ✗ Wrong options: + etoposide · vincristine
"Which patient gets neoadjuvant chemo + wide resection + adjuvant chemo?"	→ **17-yr-old pelvic OS** ✗ Wrong options: chondrosarcoma · adamantinoma · synovial sarcoma t(X;18) · myxoid liposarcoma t(12;16)
"Parosteal vs intramedullary OS treatment?"	→ **parosteal: wide resection alone · intramedullary: neoadjuvant chemo then resection**
"Distal femoral OS in an 11-yr-old?"	→ **chemo → wide resection distal 1/3 femur + custom (expandable) implant → chemo** ✗ Wrong options: physeal-sparing intercalary allograft when physis crossed

– – Active space – –

"8-yr-old, knee joint contaminated — best local control + function?"	→ **rotationplasty** ✗ Wrong options: through-knee amputation (contaminated margin) · AKA
"Pre-adolescent, best self-reported outcome + fewest revisions?"	→ **Van Nes rotationplasty**
"Correct rotationplasty for distal femur?"	→ **Winkelmann AI** ✗ Wrong options: BII
"Most vital limb-salvage principle?"	→ **resect biopsy tract en bloc (TO)**
"Tumour at marrow resection margin — margin type?"	→ **intralesional**
"Greatest risk for local recurrence?"	→ **surgical margins**
"Poor prognostic factor?"	→ **60% necrosis / MDR expression / ↑ ALP** ✗ Wrong options: 95% necrosis · stage IA · 4.5 cm · normal LDH
"All poor prognostic EXCEPT?"	→ **>90% tumour necrosis (TO)**
"True statement (all others false)?"	→ **radiation-induced OS has worse prognosis than conventional (TO)** ✗ Wrong options: MAP not preferred · curative RT for sacral OS · parosteal in adolescent girls
"Extraskeletal vs conventional OS?"	→ **radiation is part of standard treatment** ✗ Wrong options: younger · low grade · better survival
"Megaprosthesis complications — true?"	→ **primary malignant tumour → higher short-term reoperation** ✗ Wrong options: 5-day antibiotics help · NPWT ↑ infection · flap ↑ infection
"Most common site of parosteal OS?"	→ **back of distal femur (TO)**
"Correct subtype correlation?"	→ keyed: **periosteal = intermediate-grade surface, chondroid** (stem adds "early metastasis" though the explanation says lower metastatic rate) ✗ Wrong options: small cell strongly CD99 + · parosteal high grade on tibia · jaw OS more mets **(TO)**
"Most common tumour in children?"	→ **brain tumour (TO)** ✗ Wrong options: neuroblastoma · OS
"Most common primary malignant bone tumour?"	→ **multiple myeloma (TO)** ✗ Wrong options: metastasis (commonest bone malignancy, but secondary) · osteosarcoma (2nd overall, 1st in children)

"Marker separating low-grade intramedullary OS from fibrous dysplasia?"	→ **MDM2 / CDK4** (TO)
"Good response to neoadjuvant chemo?"	→ **≥90% necrosis (Huvos 3–4)** (TO) ✗ Wrong options: 50% necrosis
"Child, distal femur OS near open physis, joint salvageable — biological option preserving growth?"	→ **Canadell procedure** (TO)
"45-yr-old, proximal humeral path fracture, moth-eaten?"	→ **lymphoma (TO)** ✗ Wrong options: OS · Ewing · chordoma

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DERMATOFIBROSARCOMA PROTUBERANS

– Pathology · 13 / 66 –

1 Basics

Epidemiology

Incidence	**1% of soft-tissue sarcomas** · rare, **low-grade fibrogenic cutaneous** sarcoma arising in the **dermis**
Demographics	**20–50 yr** (peak 4th decade) · **male** (3:2) · **African Americans 2:1**
Site	**trunk (47%) > proximal extremities > head/neck (14%)**

Variants

Bednar tumour	**pigmented** · **7.5× more common in African Americans** · melanin-containing dendritic cells
Fibrosarcomatous (DFSP-FS)	5–15% · **herringbone** pattern · more aggressive, poor prognosis

Genes & lab markers

t(17;22)	**COL1A1–PDGFB** fusion → PDGF-β overexpression → target for **imatinib**

3 Presentation

Mass	**slowly enlarging, painless**, firm irregular **nodules** · dark red / blue discoloration · early pink/violet plaques with telangiectatic skin
Late	ulceration, attachment to deep structures · symptoms 6 months to 30 years · often misdiagnosed and **excised before sarcoma suspected**

4 Diagnosis & Classification

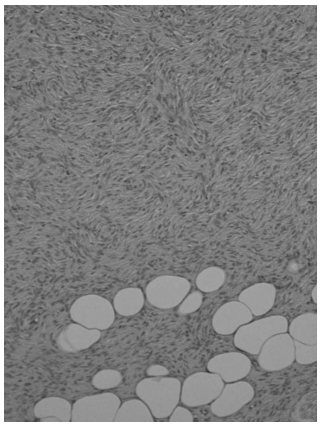
Key radiology findings

MRI	extent + depth · preop for large, atypical, recurrent lesions
CXR	pulmonary mets in high-risk cases (recurrence, fibrosarcomatous)
CT	bone involvement / mets · **US** – local + nodal monitoring · **PET** – metastatic monitoring

Histology

Biopsy required

Classic	**uniform fibroblasts in a storiform pattern** around inconspicuous vessels · diffusely infiltrates dermis/subcutis · **interdigitates with fat ("honeycomb")** at deep margin



Storiform fibroblasts

5 Management

Treatment — wide excision

Resectable DFSP	**wide excision** (with frozen sections) · **2–3 cm margins** (2.5–3 cm skin cuff) **including deep fascia**
After "lumpectomy"	**wide tumour bed re-excision** ✗ Wrong options: observation · marginal resection · RT alone · chemo
Adjuvant RT (50–70 Gy)	**positive margins**, large tumours, or when wide excision → major cosmetic/ functional deficit
Imatinib	PDGF-receptor tyrosine kinase inhibitor · **unresectable, recurrent, metastatic** adult DFSP · **65% response** · **no response without t(17;22)**
Not indicated	intralesional excision + RT · neoadjuvant chemo · Mohs referral (OB answer: wide excision)

6 Complications

Local recurrence common	body **15.7%** · **head/neck 51.8%**
Distant metastasis rare (<5%)	**lung** (haematogenous) · usually after **multiple local recurrences**
Poor prognosis	regional nodes · **fibrosarcomatous** change · ↑ mitoses, cellularity, aneuploidy, **TP53** overexpression · **age >50**

X vs Y

	Classic DFSP	Fibrosarcomatous DFSP	Bednar
Pattern	**Storiform**	**Herringbone**	Storiform + melanin dendritic cells
Behaviour	Low grade	Aggressive, can metastasize	Low grade

– – Active space – –

How it's asked

"45-yr-old, leg mass with skin ulceration, storiform spindle cells?"	→ **dermatofibrosarcoma** ✗ Wrong options: Ewing/PNET · extraskeletal myxoid CS · myxoid liposarcoma · synovial sarcoma (deep, rarely ulcerate)
"Shoulder skin mass after excisional biopsy shows DFSP — next?"	→ **wide tumour-bed resection**
"Abdominal wall nodular mass, t(17;22), herringbone areas?"	→ **wide excision** (± RT) ✗ Wrong options: Mohs · marginal · neoadjuvant chemo
"Systemic agent for unresectable DFSP?"	→ **imatinib** (only if t(17;22))
"Tumour treated with chemo + RT (in a list of chemo/RT-resistant tumours)?"	→ **DFSP** (vs classic chondrosarcoma resistant to both)

DIFFERENTIAL GROUPS

– Pathology · 14 / 66 –

1 Basics

Clinical clues (TO)

Triad — **clinical + radiology + pathology must correlate** · if one does not fit, rethink the diagnosis (TO)

Age (TO)	Likely lesions
Children / adolescents	SBC, ABC, FD, NOF, osteochondroma, chondroblastoma, osteoid osteoma, osteoblastoma, LCH · malignant: **Ewing, OS, leukaemia**
Young adult / middle age	**GCT**, FD, enchondroma, chondroblastoma (to 25), **adamantinoma**, chondrosarcoma (from middle age)
Elderly	**Metastasis (commonest)**, **myeloma**, lymphoma, chondrosarcoma, **secondary OS**

Age — strongest single diagnostic clue · most bone tumours **10–25 yr** · older: chondrosarcoma, haemangioma, metastasis (TO)

Pain pattern (TO) — **night pain → osteoid osteoma** · intermittent → FD · **painless → enchondroma, osteochondroma**

Site clues (TO) — knee: GCT, OS, Ewing · scapula/pelvis: Ewing (youth), chondrosarcoma (adult) · wrist (distal radius epiphysis): **GCT** · hand: **enchondroma until proven otherwise** · proximal humerus cyst: **SBC**

Commonest bone tumour overall — **metastasis** (TO) · breast → **mixed lytic-sclerotic** mets (TO)

4 Diagnosis & Classification

X-ray: 7-point systematic read (TO)

Checkpoints — **age (skeletal maturity) · location · type of lesion · matrix · zone of transition · periosteal reaction · soft-tissue component** (TO)

Longitudinal location (TO)	Lesions
Epiphyseal	**GCT, chondroblastoma, clear cell chondrosarcoma**
Metaphyseal (majority)	Osteochondroma, enchondroma, UBC, ABC, NOF, FD, OS, chondrosarcoma, Ewing (50%)
Diaphyseal	NOF, FD, OFD, **adamantinoma**, **Ewing**, myeloma, metastasis, osteoid osteoma, osteoblastoma, occasional OS

Transverse location (TO)	Lesions
Cortical	NOF, **adamantinoma**, OFD, **osteoid osteoma**, infection, rare metastasis
Medullary	Enchondroma, chondrosarcoma, myeloma, lymphoma, metastasis

– – Active space – –

Type of destruction (TO)	Lesions
Geographic (well defined)	UBC, ABC, **GCT**, NOF
Moth-eaten	**Myeloma** (Approach note) · **osteosarcoma** (Overview note) – read the options
Permeative	OS, **Ewing**, leukaemia, metastasis · Overview note adds chondrosarcoma

Matrix (TO)	Appearance	Lesion
Fibrous	**Ground glass**	Fibrous dysplasia
Chondroid	**Popcorn / rings & arcs** calcification	Enchondroma, chondrosarcoma
Osteoid	Ivory / cloud-like ossification	**Osteosarcoma**
None	–	GCT, ABC, Ewing

Zone of transition (TO) — **narrow = benign · wide = malignant**

Periosteal reaction (TO)	Lesion
Solid	NOF, osteoid osteoma, infection
Lamellated (onion peel)	**Ewing**
Spiculated (sunburst)	**Osteosarcoma**
Codman triangle	Any aggressive lesion

Soft-tissue component — **GCT** and malignant bone tumours (TO)

MRI (TO)

Uses — extent of disease · relation to NV bundle · **skip lesions → always full-length MRI** · ideal biopsy site (solid enhancing part)

Fluid-fluid levels — **ABC vs telangiectatic OS** – always differentiate (TO)

T2-hyperintense lobulated lesion — **chondroid** tumour (TO)

Benign lesion pearls (TO)

Fibrous cortical defect — **commonest lesion of bone** · 10–25 yr · **distal femoral metaphysis** · eccentric lytic, incidental, no treatment · **NOF = larger version** (TO)

Chondromyxoid fibroma — NOF-like X-ray but **proximal tibial metaphysis**, **no calcification** (TO)

Simple bone cyst — central, symmetrical expansion · aspirate **straw-coloured** · steroid injection first, then curettage + graft (TO)

ABC — eccentric, expansile, septated · aspirate **blood** · extended curettage, **liquid nitrogen = most potent adjuvant** (TO)

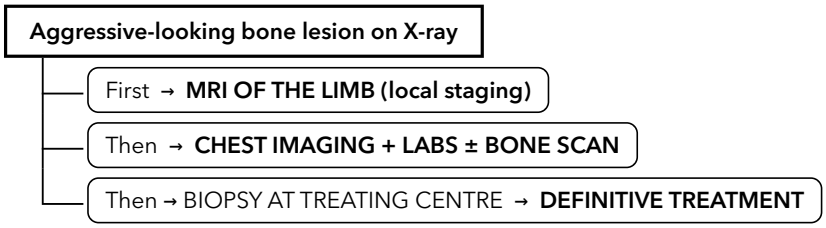
Pathological fracture through a cyst — treat the fracture; fracture **often heals the cyst** (TO) · Orthobullets: fracture healing usually does **not** resolve a UBC – read the options

Osteochondroma — symptomatic only → **extraperiosteal excision + wedge of normal bone** (TO)

Enchondroma of hand	DD **spina ventosa (TB dactylitis)** (TO)
Enchondroma vs bone infarct	enchondroma has **no central lucency** (TO)
GCT sandwich technique	lesion **<1 cm from subchondral bone** (TO) · **pulsatile** soap-bubble epiphyseal lesion, 20–40 yr (TO)

Workup of incidental lesion

Principle	a radiograph suggesting a **potentially malignant process** needs **complete staging before any treatment**, even if it looks like GCT
Next step	**cross-sectional imaging (MRI) of the limb** · then chest imaging, labs, ± bone scan → sarcoma centre referral if needed
Severe pain, insists on treatment today	**admit for parenteral analgesia + tumour work-up** ✗ Wrong options: outpatient opioids
Not appropriate first	intra-articular steroid injection · physiotherapy + NSAIDs · hip radiograph
Starting treatment before oncologic work-up	can be devastating (contamination, lost limb salvage)



How it's asked

"50-yr-old, obese, knee pain, aggressive proximal tibial lesion, wants treatment today – next?"	→ **MRI of the lower extremity** ✗ Wrong options: steroid injection · opioids · PT + NSAIDs · hip X-ray
"Fluid-fluid levels on MRI – must differentiate?"	→ **ABC vs telangiectatic osteosarcoma** (TO)
"Epiphyseal lesions?"	→ **GCT, chondroblastoma, clear cell chondrosarcoma** (TO)
"Onion peel vs sunburst?"	→ **Ewing vs osteosarcoma** (TO)

– – Active space – –

ENCHONDROMAS

– Pathology · 15 / 66 –

1 Basics

Epidemiology

Rank	**2nd commonest benign cartilage lesion** (osteochondroma 1st)
Demographics	**M = F** · **20–50 yr** (3rd–4th decade)
Site	**medullary, metaphysis or diaphysis** · **hand 60%** > feet · distal femur 20% > proximal humerus 10% > tibia · metaphyseal when first seen → appears diaphyseal as bone grows
Hand	**commonest primary bone tumour of the hand** (42% in small tubular bones) ✗ Wrong options: periosteal chondroma · subungual exostosis · osteoid osteoma

"Most common" list (TO)

Category	Answer
Tumour of the hand (overall)	**Squamous cell carcinoma (TO keyed answer)**
Bone tumour of hand bones	**Enchondroma**
Malignant bone tumour of hand bones	**Chondrosarcoma**
Tumour of bone	**Metastasis**
Primary bone tumour	**Myeloma > OS > chondrosarcoma**
Benign bone tumour	**Osteochondroma**
True benign bone tumour	Osteoid osteoma
Lesion of bone	Fibrous cortical defect
Malignant bone tumour, 1st decade	Ewing
Radiation-induced tumour	**OS > fibrosarcoma > MFH**
Most radio-/chemosensitive	**Ewing**
Bone tumour of mandible	Ameloblastoma
Malignant bone tumour of chest wall	Chondrosarcoma (chest wall)

Pathophysiology

Origin	**incomplete endochondral ossification** · physeal chondroblasts **escape into the metaphysis** and proliferate ✗ Wrong options: intramembranous

Genes & lab markers

IDH1 / IDH2	enchondroma, Ollier (and chondrosarcoma)

Ollier / Maffucci	**sporadic, no inheritance pattern (TO)** · (✗ Wrong options: GNAS (McCune-Albright) · p53 · chromosome 6q13 (CMF) · PGE2/COX (osteoid osteoma)) ✗ Wrong options: AD (that is MHE/EXT)

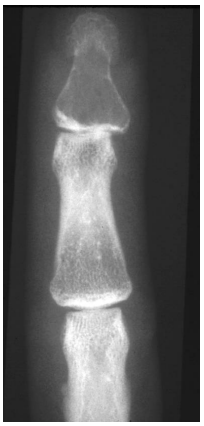
3 Presentation

Asymptomatic	**incidental** on films (long bones, foot)
Pathologic fracture	typical in the **hand** after minor trauma
Pain	uncommon · usually **another cause** (cuff tendinopathy, cervical radiculopathy, ankle sprain, vascular claudication) · **rest/night pain** → think chondrosarcoma
Syndromes	shortening, bowing, angular deformity · bluish angiomas (Maffucci)

4 Diagnosis & Classification

Key radiology findings

X-ray (usually diagnostic)	well-defined, **lucent, central medullary** lesion, **1–10 cm**, calcifies over time · **"popcorn" stippling, rings, arcs, whorls** · endosteal erosion **<50% (<2/3) of cortex** · hands/feet: expansion + thinning, may be purely lytic



Enchondroma of the distal phalanx

MRI	not needed for diagnosis · **lobular, T2 bright** (water), **no marrow oedema, no periosteal reaction** · "sled-runner tracks" · **medullary fill >90% → chondrosarcoma**
Bone scan	↑ uptake (remodelling) but **< ASIS** (only 20% > ASIS) · identifies **polyostotic** disease
CT	optional, for endosteal scalloping
Skeletal survey	if polyostotic suspected (Ollier dysplasia, Maffucci phleboliths)

Histology

Gross	blue-grey lobulated hyaline cartilage, scattered calcification
Long bones	**hypocellular, bland mature hyaline cartilage ("blue balls") separated by normal marrow** · encased by lamellar bone · abundant matrix, **no myxoid** component · **small bland chondrocytes in lacunae**, no pleomorphism/ mitoses/binucleation

– – Active space – –

Hand / small tubular bones / fibula — **hypercellular, mild atypia** → not malignant

Biopsy — rarely helpful · sampling error, can't separate from low-grade CS (SLICE study: poor reliability)

Enchondroma vs chondrosarcoma

	Enchondroma	Chondrosarcoma
Age	20–50	>50
Size	<5 cm	**>5 cm**
Pain	None / other cause	**Rest + night pain**
Endosteal scalloping	**<50% (<2/3) cortex**	**>2/3 cortex**
Cortex	Intact	Thickened, destroyed, periosteal reaction
Soft-tissue mass	No	Yes
MRI	No oedema	Oedema, >90% medullary fill
Bone scan	< ASIS	> ASIS
Histology	Hypocellular, lacunar cells, normal marrow between	Hypercellular, plump binucleate nuclei, permeation

Bone infarct — **smoke up the chimney**, **low T2** (vs bright T2 enchondroma)

Enneking staging

Enchondroma = Enneking stage 1 (latent) (with NOF)
✗ Wrong options: ABC, CMF, chondroblastoma = 2 · GCT = 3

Cartilage-origin tumours (TO)

WHO — osteochondroma, enchondroma, **chondroblastoma, CMF** all cartilage tumours → "all are cartilage tumours EXCEPT" → **none (TO)** · WHO 5th ed. moved chondroblastoma + CMF to **benign** (no intermediate cartilage tumour now)

5 Management

Latent lesion – observation vs low-grade chondrosarcoma

Enchondroma

- Asymptomatic / pain from another cause → **OBSERVE, X-rays at 6 + 12 mo (3–6 mo)**
- Growth / change on serial X-ray, true lesional pain → **EXTENDED INTRALESIONAL CURETTAGE + GRAFT**
- Hand pathologic fracture → **CURETTAGE + GRAFT (immediate = delayed)**
- Aggressive features (scalloping >2/3, mass) → **REFER ORTHO ONCOLOGY, treat as chondrosarcoma**

Observation	vast majority · serial radiographs · long-term follow-up for syndromes · no biopsy, MRI, CT, bone scan or CT chest/abdomen (even with prostate cancer history if benign-looking)
Curettage + graft (or cement) indications	**change on serial X-rays** · **symptomatic** · **pathologic fracture** · suspicious for low-grade CS · large lesion at risk of re-fracture · local recurrence unusual
Not indicated	chemo, RT (radiation-induced sarcoma), RFA, wide resection + arthroplasty (over-treatment), amputation

Pain from another source

Treat the real cause	shoulder pain with overhead activity → **cuff** (PT + observe lesion) · deltoid paraesthesia + elbow flexion weakness → **cervical spine X-rays (C5)** · ankle sprain instability → **functional rehab, proprioception** · walking pain relieved by standing still, ABI 0.8 → **vascular claudication**

Hand enchondroma & pathologic fracture

Painful/growing thumb lesion	**curettage + bone grafting** ✗ Wrong options: amputation · chemo · RT
Pathologic fracture	**curettage + grafting** · **immediate = delayed** (immobilise then curette) for consolidation, DASH, recurrence, complications – only difference is **longer recovery** with delayed · void fillers: autograft, calcium phosphate/sulfate, PMMA, coralline HA
Hand chondrosarcoma	wide resection / **ray resection**

6 Complications

Ollier disease & Maffucci syndrome

	Ollier	**Maffucci**
Lesions	**Multiple enchondromas** (metaphyses + diaphyses, hands)	Multiple enchondromas + **soft-tissue haemangiomas/angiomas**
Inheritance	**Sporadic, none**	Sporadic, none
Bones	**Dysplastic, short, bowed**, LLD	Marked expansion · **calcified phleboliths**
Chondrosarcoma risk	**25–30%** (Orthobullets) · 10–30% · "<30%" · ~25% (OB)	**25–30%** (Orthobullets) · **>50%** (OB)
Other malignancy	–	**Visceral + CNS** (astrocytoma, GI adenocarcinoma) → any malignancy **up to 100%**

Solitary enchondroma malignant transformation **<1%** (~1%)
✗ Wrong options: 5–8%

Secondary chondrosarcoma from enchondroma – usually **low grade (TO)**

– – Active space – –



Ollier disease — multiple hand enchondromas

X vs Y

	Enchondroma	Osteochondroma	Periosteal chondroma
Location	**Medullary**	**Surface**, exophytic, cortical/medullary continuity	Surface, shallow cortical defect
Origin	Escaped physeal cartilage	Zone of Ranvier (EXT1/2 in MHE)	Periosteum
Treatment	Observe	Observe / marginal excision	Marginal excision

	Enchondroma	Bone infarct
X-ray	Rings and arcs throughout	Serpiginous sclerotic rim
T2 MRI	**Bright**	Low / central fat

How it's asked

"70-yr-old, incidental proximal femoral cartilage lesion after fall?"	→ **observation**
"Prostate-cancer history, incidental benign-looking tibial chondroid lesion?"	→ **reassure + repeat X-ray 3–6 mo** ✗ Wrong options: biopsy · MRI · CT CAP + referral
"Shoulder pain with overhead activity + calcified humeral lesion?"	→ **observation** (treat cuff) ✗ Wrong options: biopsy · curettage
"Incidental lesion grew + new vague pain at 6 mo?"	→ **extended intralesional curettage + grafting** ✗ Wrong options: RFA · chemo · RT + wide resection
"Night + rest pain, failed PT, humeral enchondroma?"	→ **curettage** ✗ Wrong options: CT CAP + bone scan · resection + TSA
"Painful growing thumb enchondroma?"	→ **curettage + bone grafting**

"Pathologic fracture through 5th metacarpal enchondroma?"	→ **curettage + bone grafting** ✗ Wrong options: core biopsy · CT · wide excision
"Immediate vs delayed curettage for phalangeal fracture?"	→ **no difference in outcomes**
"True of a hand enchondroma?"	→ **curettage + grafting if pathologic fracture** ✗ Wrong options: 5–8% malignancy · MRI needed · intramembranous · Enneking 2
"Hand lesion where curettage + graft is definitive?"	→ **distal phalanx enchondroma** ✗ Wrong options: Ewing · subungual exostosis · chondrosarcoma · osteochondroma
"Commonest bone tumour in the hand?"	→ **enchondroma**
"Most common tumour of the hand (FNB)?"	→ **squamous cell carcinoma (TO)**
"Hypocellular mature hyaline cartilage, bland lacunar cells — which X-ray?"	→ **central lucent medullary lesion with rings/arcs**
"Enchondromas + haemangiomas?"	→ **Maffucci** ✗ Wrong options: Ollier · Trevor = epiphyseal osteochondroma · Hunter
">50% risk of malignancy?"	→ **Maffucci** ✗ Wrong options: Ollier · MHE · FD · McCune-Albright
"Multiple enchondromas, bowing — inheritance?"	→ **no known inheritance pattern**
"Ollier — underlying mechanism?"	→ **failure of normal endochondral ossification** ✗ Wrong options: EXT · p53 · 6q13 · PGE2
"Ollier — associated with?"	→ **10–30% chondrosarcoma risk** ✗ Wrong options: GNAS · angiomas · visceral malignancy
"FALSE about chondroid lesions?"	→ **"Ollier is autosomal dominant"** (✓ grade 1 appendicular → extended curettage, ✓ mesenchymal needs chemo, ✓ secondary CS low grade) **(TO)**
"Enneking stage 1 lesion?"	→ **enchondroma**

– – Active space – –

– – Active space – –

EOSINOPHILIC GRANULOMA

– Pathology · 16 / 66 –

1 Basics

Epidemiology

Age	**80% <20 yr** · commonly <10 · **M:F 2:1**
Site	skull, ribs, clavicle, scapula, mandible, pelvis · **spine (thoracic commonest)** · **diaphysis** of long bones
Nature	**self-limiting**, benign histiocytic lesion (Langerhans cell histiocytosis, LCH) · reticuloendothelial system

LCH spectrum

Form	Age	Features	Prognosis
Eosinophilic granuloma	Children/ adolescents	**Single**, self-limited bone lesion	Excellent · spine lesions resolve spontaneously
Hand-Schüller-Christian	**>3 yr**	Chronic disseminated · **triad: multiple lytic skull lesions + diabetes insipidus + exophthalmos** · visceral (lung, liver, spleen, skin, nodes) → abdominal pain	Depends on chemo response · worse with more extraskeletal disease
Letterer-Siwe	**<3 yr**	Fulminant · skin rash from birth, multiple lesions	**Fatal** in early childhood

Genes & lab markers

Genetics	no clear inheritance or locus

CD1a + Langerhans cells · **Birbeck granules** on EM

3 Presentation

Bone	pain + swelling · **limp** (pelvis / lower limb)
Spine	local or diffuse back/neck pain · progressive **kyphosis** · radiculopathy if aggressive
HSC	**thirst / polydipsia** (DI) · **exophthalmos** · weight loss, skin lesions, abdominal pain

4 Diagnosis & Classification

Key radiology findings

"Great mimicker"	DD osteomyelitis, leukaemia, lymphoma, fibrous dysplasia, **Ewing**
Diaphysis	**well-defined intramedullary lytic "punched-out"** lesion · cortex thinned/ expanded/destroyed · **periosteal reaction** possible
Metaphysis	up to, **not through, the physis** · less central

Spine **vertebra plana** (flattened vertebral body) · ↑ kyphosis



Vertebra plana

Skull **multiple punched-out lytic lesions**

MRI adjacent soft-tissue mass · **Bone scan** – ↑ uptake · **Whole-body PET/CT** – staging if multifocal · skeletal survey finds extra lesions

Vertebra plana – differential (TO)

Cause	Vertebra plana?
Eosinophilic granuloma / LCH	✓ **Commonest**
Ewing sarcoma	Yes
Lymphoma / leukaemia	Yes
Gaucher disease	Yes
ABC	Yes
Infection (spondylitis)	Yes
Metastasis (neuroblastoma)	✓ occasionally
Lyme disease	No

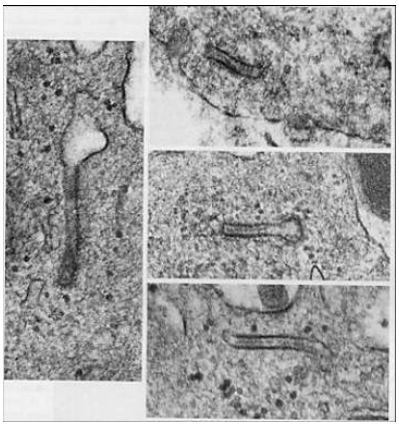
Histology

Langerhans cells mononuclear **histiocyte-like** cells, **oval nuclei** with **prominent nuclear groove ("coffee-bean")**, eosinophilic cytoplasm

Stain **CD1a +**

EM **Birbeck granules** – **tennis-racquet** shaped cytoplasmic organelles, diagnostic
✗ Wrong options: Langerhans cell itself · plasma cell · Kupffer cell

– – Active space – –



Birbeck granules on electron microscopy

Background **eosinophils** (bilobed nuclei), mixed inflammatory cells, giant cells

No nuclear atypia, no atypical mitoses separates from Ewing, lymphoma, metastatic neuroblastoma

Differential

	Destructive lesion, young	Multiple lesions, young	Observation alone
EG	✓	– (HSC ✓)	✓ (self-limited)
Osteomyelitis · OS · Ewing · desmoplastic fibroma	✓	–	–
Lymphoma · leukaemia	✓	✓	–
FD · Ollier/Maffucci · MHE · NOF/Jaffe-Campanacci	–	✓	✓
Haemangioendothelioma	–	✓	–

5 Management

Eosinophilic granuloma

- Isolated, stable → **OBSERVATION ± protected weight bearing, repeat X-ray 6 wk**
  - Symptomatic → **CORTICOSTEROID INJECTION (± after curettage)**
- Spine, no neuro deficit → **BRACE / COLLAR + OBSERVE**
  - Instability / neuro compromise → **LOW-DOSE RT (600–800 cGy)**
  - Progressive deformity despite brace (~10%) → **SURGICAL CORRECTION**
- Impending fracture / articular surface at risk → **CURETTAGE + GRAFT**
- Disseminated HSC → **CHEMOTHERAPY**

Treatment – observation

Observation self-limited · heals spontaneously or after biopsy · femoral diaphysis: **crutches + repeat X-ray in 6 wk**

✗ Wrong options: wide excision · chemo · RT alone · curettage first

Cervical spine EG, no neuro deficit	**cervical collar + observation** ✗ Wrong options: antibiotics · chemo/RT · resection
Bracing	prevents kyphosis · corrects deformity **90%** · vertebrae regain ~**50%** height
Low-dose RT 600–800 cGy	spinal instability / neurology · lesions not amenable to injection/surgery
Chemotherapy	**only diffuse/disseminated HSC**
Surgical decompression	rare cord compression

X vs Y

	EG / LCH	Ewing	Osteomyelitis
Cells	Langerhans cells, coffee-bean nuclei, eosinophils	Small round blue cells, t(11;22)	Neutrophils
Marker	**CD1a**, Birbeck	CD99	Culture
Atypia	None	Malignant	None
Treatment	Observe	Chemo + local control	Antibiotics ± debridement

How it's asked

"5-yr-old, skin lesions, limp, weight loss, lytic femoral lesion with histiocytes?"	→ **eosinophilic granuloma** ✗ Wrong options: neuroblastoma · Ewing · ALL · fifth disease
"2-yr-old, thirst, exophthalmos, lytic skull lesions?"	→ **LCH (Hand-Schüller-Christian)** ✗ Wrong options: abuse · NF · metastatic neuroblastoma · adamantinoma
"6-yr-old, limp, skull + pelvic lesions, thirst, abdominal pain?"	→ **Hand-Schüller-Christian** ✗ Wrong options: McCune-Albright · Maffucci · Ollier
"10-month-old, multiple skeletal lesions + rash since birth, CD1a +?"	→ **LCH** ✗ Wrong options: osteomyelitis · Ewing · lymphoma · neuroblastoma
"Vertebra plana — EM structure (tennis racquet)?"	→ **Birbeck granule**
"7-yr-old, femoral diaphyseal EG?"	→ **crutches + repeat X-ray in 6 weeks**
"7-yr-old, cervical EG, no neuro deficit?"	→ **cervical collar + observation**
"Causes of vertebra plana — all EXCEPT?"	→ **Lyme disease** (✓ EG, Ewing, lymphoma) **(TO)**

– – Active space – –

EPITHELIOID SARCOMA

– Pathology · 17 / 66 –

1 Basics

Epidemiology

Rank	**commonest soft-tissue sarcoma of the hand and wrist** (others in hand: MFH, synovial sarcoma, clear cell sarcoma) ✗ Wrong options: chondrosarcoma · rhabdomyosarcoma · myxoid liposarcoma · haemangiopericytoma
Demographics	adolescents + young adults **10–35** (10–40) · **M:F 2:1**
Site	hand/wrist, fingers · forearm, buttock/thigh, knee, foot
Depth	superficial → **ulcerates**, mimics **skin carcinoma, rheumatoid nodule, granuloma** · deep → attached to **tendons, tendon sheaths, fascia**

Spread

Regional lymph nodes	**common** · unique, more frequent than other sarcomas (lungs still the commonest overall site)

Spreads along lymphatics and **fascial planes** · **30% metastasized at presentation**

Lymphatic sarcomas – "RACES"	**R**habdomyosarcoma · **A**ngiosarcoma · **C**lear cell sarcoma · **E**pithelioid sarcoma · **S**ynovial sarcoma ✗ Wrong options: liposarcoma · fibrosarcoma · leiomyosarcoma · UPS

Genes & lab markers

Genetics	unknown
IHC	**keratin +**, **EMA +**, vimentin +, **CD34 +**

3 Presentation

Mass	**small (3–6 cm), firm, painless, slow-growing** nodule · upper extremity · may **ulcerate**
History	often a **previous "benign" excision → recurrence** (misdiagnosed)

4 Diagnosis & Classification

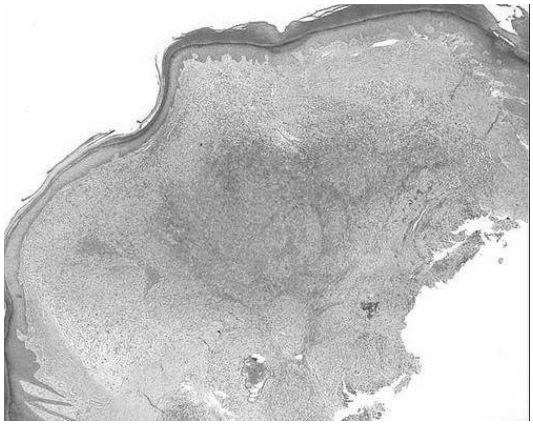
Key radiology findings

X-ray	**calcification 10–20%** · adjacent **bone erosion**
MRI	indeterminate · **T1 dark, T2 bright** · tendon-sheath nodule

Histology

Pattern	**nodular with central necrosis** in granulomatous-looking areas → mimics **necrotizing granuloma** · chronic inflammatory cells at margins

Cells epithelioid **ovoid/polygonal cells, eosinophilic cytoplasm** + spindle cells · dense hyalinized collagen · pleomorphism rare



Nodules with central necrosis

Differential

Lesion	Distinguishing feature
Epithelioid sarcoma	Young adult, nodules with **central necrosis**, keratin +
Rheumatoid nodule	Similar histology · older, RA
Squamous cell carcinoma	**Keratin pearls**
Angiosarcoma (soft tissue)	Well-formed **vascular channels**
Epidermal inclusion cyst	Benign appearance, low recurrence
Clear cell sarcoma / carcinoma	**Clear cells**
BPOP (Nora lesion)	Reactive HO, hands/feet, 3rd decade

5 Management

Wide excision + adjuvant radiotherapy all operable tumours

Sentinel lymph node biopsy at excision → regional nodal staging

Inadequate excision high rate of **multiple recurrences → re-excision of tumour bed**

Amputation multiple recurrences, to stop spread

Unplanned excision – re-resection

Unplanned ("whoops") marginal resection residual tumour in **50–60% → repeat staging + re-resection of tumour bed** (± pre/post-op RT)
✗ Wrong options: PT · AFO · tendon transfer · nerve exploration first

6 Complications

Prognosis **extremely poor** · lymph node + lung metastasis · recurrence

– – Active space – –

How it's asked

"28-yr-old, recurrent ulcerating thumb mass, nodules with necrosis?"	→ **epithelioid sarcoma** ✗ Wrong options: rheumatoid nodule · SCC (keratin pearls) · angiosarcoma · inclusion cyst
"Slowly growing finger nodule, nodular necrosis, no clear cells?"	→ **epithelioid sarcoma** ✗ Wrong options: clear cell sarcoma/ carcinoma · BPOP
"Commonest soft-tissue sarcoma of the hand?"	→ **epithelioid sarcoma**
"Sarcoma most likely to spread via lymphatics?"	→ **epithelioid sarcoma** (RACES)
"Site of metastasis more common than in other sarcomas?"	→ **regional lymph nodes** → **sentinel node biopsy**
"Foot drop after outside marginal excision of popliteal epithelioid sarcoma?"	→ **staging + re-resection of tumour bed**

EWING'S SARCOMA

– Pathology · 18 / 66 –

1 Basics

Epidemiology

Rank — **2nd commonest primary malignant bone tumour in children** (OS 1st) · 3% of paediatric malignancies, 10% of primary malignant bone tumours · ~3/million · commonest malignant bone tumour of the **1st decade** · rhabdomyosarcoma = commonest paediatric soft-tissue sarcoma

Age — **5–25** · 80–90% <20 · peak **10–15**

Sex / race — **M:F 1.5:1** · **rare in African Americans** and Asians

Site — even split axial vs appendicular · **metadiaphysis of long bones ~50%** (femur, tibia, humerus) · **pelvis ~25%** · scapula, ribs · soft tissue rare

Diaphysis myth (TO) — only **~50% diaphyseal**, 50% metaphyseal → "predominantly diaphyseal" is false

Origin — unknown, probably **neuroectodermal** (PNET family) · cell of origin likely in **marrow**

Genes & lab markers

t(11;22)(q24;q12) — **EWS–FLI1** fusion (aberrant transcription factor: EWS activation domain + FLI ETS DNA-binding domain) · **85–95%** · detect by **PCR / FISH**

Variants — **t(21;22) EWS–ERG** (10–15%) · t(7;22)

t(11;22) presence — does **not** alter clinical course **(TO)**

Translocation / fusion	Tumour
t(11;22) EWS–FLI1	✓ **Ewing**
t(X;18) SYT–SSX	Synovial sarcoma
t(12;16) TLS–CHOP	Myxoid liposarcoma
t(12;22) EWS–ATF1	Clear cell sarcoma
t(9;22) EWS–CHN	Extraskeletal myxoid chondrosarcoma
t(2;13) PAX3–FKHR	Alveolar rhabdomyosarcoma
t(X;17) ASPL–TFE3	Alveolar soft part sarcoma
RB, p53	Osteosarcoma (✗ Wrong options: Ewing)

Labs — **↑ ESR, ↑ WBC, anaemia, ↑ LDH** → mimics infection, marks extensive disease

Immunostains — **CD99 / MIC2 + (95%)** · vimentin + · **PAS +** (glycogen) · NSE +, S-100 +, Leu7 (CD57) + · **cytokeratin −**, **reticulin −** (+ in lymphoma), neurofilament − (+ neuroblastoma), myoglobin − (+ embryonal rhabdo)

NKX2.2 + — **highly specific** for Ewing (TO)

Lymphoma vs Ewing (TO) — lymphoma **PAS −, reticulin +** · Ewing PAS +, reticulin −

– – Active space – –

CD45 (LCA)	listed **+** in Orthobullets chart · TO/OB: **lymphoma is LCA +**, Ewing LCA – (OB calls LCA "CD34") – read the options

3 Presentation

Pain	commonest, worse at night · **>50% symptomatic >6 months** (delay worse in pelvis/axial)
Infection mimic	**swelling, erythema, warmth, fever (~20–25%)**, weight loss, ↑ WBC/ESR
Mass	may be impalpable until large · limp, ↓ ROM

4 Diagnosis & Classification

Key radiology findings

X-ray	large destructive **diaphyseal/metadiaphyseal** lesion · ill-defined **permeative, moth-eaten** · lytic ± reactive bone · **"onion skin"** (lamellated) or **"sunburst" / "hair-on-end"** periosteal reaction · Codman triangle · **soft-tissue mass >80%**



Onion-skin periosteal reaction, tibial diaphysis

MRI	**next step after suspicious X-ray** · soft-tissue extent, marrow, NV relation · T1 low–intermediate, **T2 high heterogeneous**, heterogeneous enhancement · monitors chemo response
Bone scan	very hot · skip/distant mets (10%) · **PET better than bone scan for Ewing mets** (vs OS)
CT chest	required for lung mets

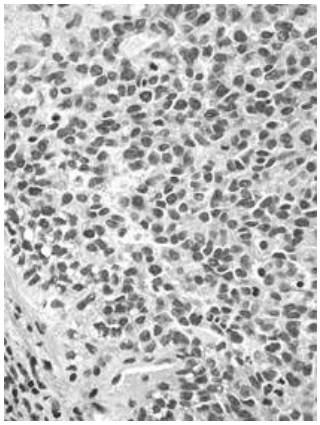
Work-up (MRI, bone marrow biopsy)

Bone marrow biopsy	**required for Ewing** (marrow mets), **not routine for OS** ✗ Wrong options: MRI · CT · bone scan · protein electrophoresis (done for both or neither)
PET-CT (TO)	staging + response · **SUV fall >50% = good response** · if **bone scan + CT chest** used instead → **must add bone marrow aspiration**
Order	X-ray → **MRI** → staging (CT chest, bone scan, **bone marrow biopsy**) → **biopsy by treating surgeon**
Not appropriate	no further work-up · arthrogram · pelvic US · chiropractor · cast / rest and repeat X-ray · IV antibiotics / I&D when ESR/CRP normal

Histology

Gross grey-white, necrosis, haemorrhage · **liquid, "pus-like"**

Micro **sheets of monotonous small round blue cells**, **high N:C ratio**, scant cytoplasm · **pseudorosettes (Homer-Wright)** · no stroma



Small round blue cells

Small round blue cell tumours – by age

Age	Likeliest
<5 yr	**Metastatic neuroblastoma**, leukaemia
5–10 yr	Eosinophilic granuloma
5–30 yr (peak 5–15)	**Ewing**
>30 (15–40)	**Lymphoma**
>40–50	**Myeloma**, metastatic small cell lung Ca
2nd–3rd decade (rare)	Small-cell OS, mesenchymal chondrosarcoma

Round-cell tumours Ewing, PNET, neuroblastoma, embryonal rhabdomyosarcoma, EG, lymphoma, myeloma, round-cell liposarcoma · not **leiomyosarcoma = spindle cells, cigar nuclei**

TB small round cells (lymphocytes) but periarticular, joint

Staging

MSTS almost all **IIB or III** · **26–28% macrometastases at presentation** · sites **lung 50%, bone 25%, marrow 20%**

– – Active space – –

Differential

	Destructive, young	Small round cells	Wide resection + chemo
Ewing	✓	✓	✓
Osteosarcoma	✓	–	✓
Lymphoma · leukaemia · EG	✓	✓	–
Osteomyelitis · desmoplastic fibroma	✓	–	–
Neuroblastoma · myeloma	–	✓	–
Dedifferentiated CS	–	–	✓

Osteomyelitis vs Ewing mixed inflammatory infiltrate (neutrophils, plasma cells, lymphocytes, histiocytes) → **debridement + antibiotics** · Ewing = uniform round cells, no inflammation

5 Management

Treatment – chemotherapy with surgery or radiation

Ewing sarcoma (chemo needed in ALL – presume micromets)

- Resectable / expendable site → NEOADJUVANT CHEMO → WIDE RESECTION → **ADJUVANT CHEMO**
  - Positive margin / poor necrosis / mets → **+ ADJUVANT RT**
- Unresectable (spine, pelvis) or unacceptable morbidity → **CHEMO + RT**
- Huge tumour on NV structures, displaced path fracture, foot/ankle → **AMPUTATION**

Chemo **VAC: vincristine, Adriamycin (doxorubicin), cyclophosphamide** (± dactinomycin) **alternating with ifosfamide + etoposide (IE)** · neoadjuvant **8–12 wk** → adjuvant **6–12 mo**
✗ Wrong options: methotrexate-based = OS

Surgery preferred local control when wide margins achievable · **negative margins → 5-yr survival ↑ >10%** · positive margin → re-resect ± RT · recurrence **5–10% with surgery** vs **35% RT alone**

Reconstruction vascularized/non-vascularized autograft (fibula, scapula, iliac crest, rib, clavicle), allograft, APC, endoprosthesis, rotationplasty

Radiation **radiosensitive** (most radio- + chemosensitive bone tumour) but trend **away** from RT · field = tumour + **2–3 cm**, **56–60 Gy** · standard = hyperfractionated · indications: **unresectable**, positive margins, poor chemo response, **all pulmonary mets**

RT indications (TO) **positive margins** · **poor chemo response (<90% necrosis)** · **unresectable** · surgery with **major morbidity (e.g. sacrum)**

Pathological fracture **not an absolute contraindication** to limb salvage (TO)

Extensive pelvic disease + lung mets **chemo + radiation**

Spine with neuro deficit, resection too morbid	local control by **radiation**
Outcomes	5-yr **82% (no mets) vs 39% (mets)** · 10-yr 63% vs 32%

Extraskeletal Ewing sarcoma (TO)

Histology same as skeletal ✓ · adolescents/young adults (✗ Wrong options: <10) · **paravertebral muscles, chest wall** (✗ Wrong options: thigh) · t(11;22), MIC2 + · **chemosensitive** (✗ Wrong options: resistant) · same treatment · better prognosis than osseous (OB)

6 Complications

Radiation complications (>60%)	**LLD**, joint contracture, muscle atrophy, **pathologic fracture**, **secondary sarcoma**

Radiation-induced sarcoma

Secondary bone sarcoma	~**20% at 20 yr** · in radiation field · dose dependent (**<60 Gy → <5%**) · e.g. lytic lesion 20 yr after RT at age 10 ✗ Wrong options: recurrence · infarct · adenocarcinoma
AML / myelodysplasia	**1–2%**, 2–5 yr after chemo (dose-intensive regimens)

DVT risk

↑ VTE in paediatric sarcoma (Ewing) — tumour activates **factor X → Xa** · no consensus on prophylaxis
✗ Wrong options: haemangioma · MSSA osteomyelitis (MRSA ↑) · NOF · PVNS

Recurrence	~20% localized, >60% metastatic · after recurrence **<10% 5-yr**
Mets cure	lung only **30%** · bone only 20% · lung + bone <15%

Prognostic factors

Most important	**metastasis at diagnosis** · lung mets better than bone/marrow · skip mets better than distant

Poor prognosis	Threshold
Age	**>10** (TO) · **>14** (Orthobullets)
Size	**>8 cm** / ≥200 mL
Site	**Pelvis/spine (worst)** > proximal > distal (best)
Chemo response	**<90% necrosis** · surgically treated: necrosis = most important (TO)
Labs	**↑ LDH (>200 IU/L)**, ↑ WBC, ↑ ESR, anaemia
Clinical	**Fever**, male sex
Marrow	Amount of marrow involvement
Molecular	**p53 mutation** + t(11;22), Ki-67, HER-2/neu

Not prognostic (TO)	**histological grade** (all Ewing high grade) · t(11;22) presence · female sex is not a bad sign
Good prognostic sign (TO)	**age 7 yr** ✗ Wrong options: 12 cm · fever · TLC 14,000

– – Active space – –

X vs Y

	Ewing	Osteosarcoma	Lymphoma of bone
Age	5–25	10–20, >65	35–55
Site	Diaphysis/metadiaphysis, pelvis	Metaphysis	Diaphysis
X-ray	Permeative, **onion skin**	Blastic, sunburst, Codman	Moth-eaten, little periosteal reaction
Histology	Round blue cells	Lacy osteoid	Mixed round cells
Markers	**CD99, PAS +, reticulin –**	SATB2	**LCA (CD45) +, reticulin +, PAS –**
Marrow biopsy	**Yes**	No	Yes
Chemo	**VAC/IE**	**MAP**	CHOP ± rituximab
RT	Radiosensitive	Radioresistant	Radiosensitive

How it's asked

"13-yr-old, mid-femoral lesion, large soft-tissue mass, onion skin?"	→ **Ewing sarcoma (TO)**
"8-yr-old, night thigh pain, ↑ WBC + ESR, onion-skin, round blue cells?"	→ **Ewing** ✗ Wrong options: osteomyelitis · OS
"Round cells EXCEPT?"	→ **leiomyosarcoma** (spindle cells)
"15-yr-old, thoracic small round blue cell tumour, RT for local control considered?"	→ **Ewing** ✗ Wrong options: chordoma · OS · chondrosarcoma (RT inadequate)
"Immunostain confirming diagnosis?"	→ **MIC-2 (CD99) +** ✗ Wrong options: cytokeratin + · PAS – · reticulin + · vimentin –
"9-yr-old, monotonous small round blue cells?"	→ **Ewing** ✗ Wrong options: neuroblastoma <5 · lymphoma · small cell lung Ca
"t(11;22) on FISH – histology?"	→ **sheets of small round blue cells, CD99 +**
"Translocation?"	→ **t(11;22)** ✗ Wrong options: 9;22 · 12;16 · X;18 · 12;22 · 2;13
"Fusion protein?"	→ **EWS–FLI1** ✗ Wrong options: SYT-SSX · ASPL-TFE3 · EWS-ATF1 · TLS-CHOP
"Neoplastic properties due to?"	→ **chromosomal translocation**
"Test required for Ewing but not OS?"	→ **bone marrow biopsy**

"Which patient needs marrow biopsy in staging?"	→ **5-yr-old, lytic diaphyseal femoral lesion, EWS-FLI1 +**
"Teen runner, pelvic lesion after failed PT?"	→ **MRI of the pelvis**
"3-yr-old, febrile, normal ESR/CRP, permeative femur?"	→ **MRI of the femur** ✗ Wrong options: biopsy + culture first · antibiotics
"Stage IIB proximal tibia Ewing?"	→ **neoadjuvant chemo → wide resection → adjuvant chemo** ✗ Wrong options: chemo + RT without surgery
"Fibula Ewing?"	→ **chemo + surgical resection** ✗ Wrong options: resection alone · RT alone
"Pelvic Ewing with sciatic symptoms + lung mets?"	→ **chemo + radiation**
"Best-prognosis regimen?"	→ **VAC (+ dactinomycin) + wide resection ± RT** ✗ Wrong options: methotrexate-based
"Distal fibula Ewing, no other lesions?"	→ **surgery + chemo ± RT** – chemo required in all cases
"14-yr-old, fever, ↑ WBC/ESR, mixed inflammatory infiltrate?"	→ **debridement + antibiotics** (osteomyelitis, not Ewing)
"Lytic lesion 20 yr after RT for childhood Ewing?"	→ **radiation-induced sarcoma**
"Second commonest primary bone malignancy in children?"	→ **Ewing**
"Paediatric condition with ↑ DVT risk?"	→ **bone sarcoma with t(11;22)**
"NOT a bad prognostic sign?"	→ **high grade** (all Ewing high grade) (✓ fever, ✓ ↑ LDH, ✓ age >10–12) **(TO)**
"Good prognostic sign?"	→ **age 7 yr (TO)**
"True of Ewing in a 10-yr-old?"	→ **raised ESR = worse prognosis (TO)** ✗ Wrong options: predominantly diaphyseal · grade prognostic · appendicular worse
"True of extraskeletal Ewing?"	→ **histology same as skeletal (TO)**
"Ewing with pathological fracture – limb salvage contraindicated?"	→ **No, not an absolute contraindication** (TO)

– – Active space – –

EXTRA-ABDOMINAL DESMOID TUMOR

– Pathology · 19 / 66 –

1 Basics

Epidemiology

Nature	benign but **most locally invasive benign soft-tissue tumour** (aggressive fibromatosis) · **no metastasis**, no malignant change (unless radiation-related)
Demographics	**25–35 yr** (15–40) · **F:M 3:1** · ~900/yr in US
Site	**shoulder (20%) > chest wall/back (17%) > thigh (13%)** · >50% extra-abdominal · **sites of previous surgery/scars** · multiple lesions in same limb → **next one appears more proximally**

Association with FAP

Familial adenomatous polyposis (FAP) / Gardner syndrome	desmoid in **4–30%** (prevalence ~21%) · **10,000× risk** ✗ Wrong options: Sinding-Larsen-Johansson · Ollier · Fanconi · McCune-Albright
Gardner features	colonic polyps + **jaw osteomas** + **epidermoid cysts**
FAP	**autosomal dominant**, **APC** tumour-suppressor gene · **~100% lifetime colon cancer → screen for colon cancer** ✗ Wrong options: lung, breast, thyroid, renal
Other associations	**Dupuytren** contracture · **Ledderhose** disease

Genes & lab markers

APC mutation	15% of tumours (APC controls **β-catenin**)
β-catenin ↑	**all** tumours
Trisomy 8 or 20	
IHC +	**oestrogen receptor-β (100%)**, somatostatin, cathepsin D, Ki-67, c-Kit · **IHC –** – ER-α, PR, HER2

Cortical desmoid

Cortical desmoid	self-limiting benign fibro-osseous lesion of **active adolescents (10–15)** · **chronic tendon traction** · **posteromedial distal femur** at **medial gastrocnemius / adductor magnus** insertion · also humerus (pectoralis, deltoid insertions)
Imaging	geographic central lucency, sclerotic margin, **"bite-like" cortical irregularity** · MRI T1 low–intermediate, T2 high, heterogeneous enhancement
Treatment	**activity modification + PT** (self-limiting)

3 Presentation

– – Active space – –

Mass	enlarging, painless or painful · **"rock hard"**, fixed, poorly circumscribed · limits joint motion
Nerve compression	numbness, tingling, weakness, stabbing pain

4 Diagnosis & Classification

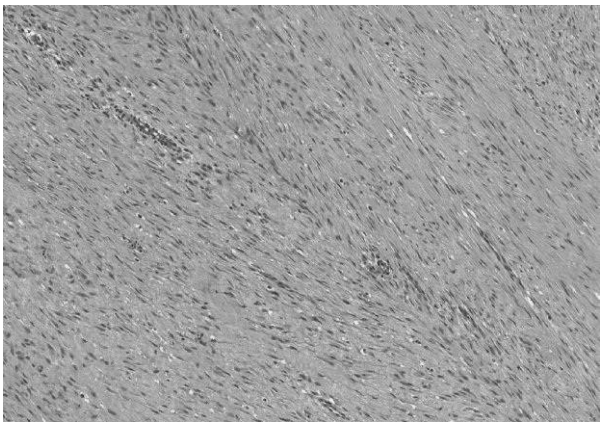
Key radiology findings

MRI – best means to diagnose · excludes multicentric lesions · **T1 low**, **T2 low–medium** (collagen) · gadolinium enhances · infiltrates muscle · 5–10 cm · may erode bone

X-ray	minimal role · bone erosion, **"frond-like" periosteal spicules** into tumour

Histology

Gross	**white, gritty**, poorly encapsulated
Micro	**well-differentiated fibroblasts/myofibroblasts**, **uniform spindle cells with elongated nuclei**, occasional mitoses, **abundant collagen**, infiltrates adjacent tissue



Uniform spindle cells in abundant collagen

Histology	Diagnosis
Uniform spindle fibroblasts, elongated nuclei, collagen	✓ **Desmoid**
Fibroblasts + Schwann cells, mast cells, wavy nuclei	**Neurofibroma**
Stromal cells, foam cells, haemosiderin giant cells	PVNS
Small round blue cells	Ewing
Storiform fibroblasts, lipophages, giant cells	NOF

– – Active space – –

5 Management

Extra-abdominal desmoid

- Asymptomatic / minimally symptomatic → **ACTIVE SURVEILLANCE, SERIAL MRI (first line)**
- Progressive, inoperable, or can't tolerate chemo → **TAMOXIFEN / LOW-DOSE CHEMO**
- Symptomatic / recurrent / failed surveillance → **WIDE RESECTION + RT (50–60 Gy)**

Active surveillance ("watch and wait")	primary strategy (Desmoid Tumor Working Group) · may stay stable or **regress spontaneously**
Tamoxifen	**oestrogen-receptor blockade** · moderate short-term success with less morbidity · for inoperable lesions / poor chemo candidates · works as lesions commoner + more aggressive in (premenopausal) women
Wide resection + radiotherapy	symptomatic or recurrent lesions · **50–60 Gy** EBRT · abdominal wall desmoid → surgery if surveillance fails

6 Complications

Recurrence

Margin	Surgery alone	Surgery + RT
Negative	**28%**	**6%**
Positive	**39%**	**25%**

Highest local recurrence of soft-tissue tumours after wide resection without adjuvant

Prognosis unpredictable · occasional spontaneous regression · **no metastasis**

How it's asked

"20-yr-old woman, rock-hard calf mass, ER-β +?"	→ **extra-abdominal fibromatosis (desmoid)** ✗ Wrong options: OS · leiomyosarcoma · angiosarcoma
"Shoulder mass, family with colonic polyps, jaw osteomas, epidermoid cysts – histology?"	→ **uniform spindle fibroblasts with elongated nuclei** ✗ Wrong options: neurofibroma · PVNS · Ewing · NOF
"Adolescent lacrosse player, posteromedial distal femoral cortical lesion from tendon traction?"	→ **cortical desmoid** → activity modification + PT
"48-yr-old, 5 cm thigh mass, biopsy → active surveillance with serial MRI?"	→ **extra-abdominal desmoid** ✗ Wrong options: lipoma · GCT · osteoid osteoma
"Inherited condition associated with desmoids?"	→ **FAP**
"Desmoid patient – screen for?"	→ **colon cancer**

"Tumour treated by oestrogen-receptor blockade?"	→ **desmoid** (tamoxifen) ✗ Wrong options: Ewing · synovial sarcoma · myeloma · GCT	– – Active space – –

– – Active space – –

FIBROUS DYSPLASIA

– Pathology · 20 / 66 –

1 Basics

Epidemiology

Incidence	1 in 5,000–10,000 · ~2.5% of bone lesions, 5% of benign bone lesions
Age	**75% onset <30** (5–30) · monostotic 10–30 · polyostotic earlier (<10; 60% symptomatic)
Sex	**M = F**
Site	any bone · **proximal femur commonest** · monostotic: **ribs 28%**, **proximal femur 23%**, skull/craniofacial, tibia, humerus · polyostotic: often **unilateral**, femur, skull/face, pelvis, spine · craniofacial: zygomatic-maxillary complex · skull + face generalized = **"leontiasis ossea" (TO)**
Location in bone	**medullary** (metaphysis/diaphysis) – not cortical (NOF, OFD, adamantinoma are cortical) **(TO)**

Pathophysiology

Defect	primitive woven bone **fails to remodel to lamellar bone** → immature trabeculae in dysplastic fibrous tissue
Natural history	active in childhood → **inactive at skeletal maturity** (monostotic grows with skeleton; polyostotic with disease severity)

Genes & lab markers – GNAS / Gs-alpha

Mutation	**activating missense GNAS (GNAS1) → Gs-α subunit (20q13)** · **somatic mosaicism**, spontaneous · **no vertical transmission** · timing → extent
Pathway	loss of intrinsic GTPase → **constitutive adenylate cyclase → ↑ cAMP** → c-fos ↑ · ↑ **WNT/β-catenin** → osteoprogenitors fail to mature · ↑ NF-κB, IL-6 → resorption
FGF-23 ↑	**renal phosphate wasting → hypophosphataemia → osteomalacia** (Looser zones)
cAMP pathway question	✓ **FD** ✗ Wrong options: achondroplasia FGFR3 · hypochondroplasia · MHE EXT · NF1 · enchondroma

Gene / defect	Condition
GNAS / Gs-α (↑ cAMP)	✓ **Fibrous dysplasia, McCune-Albright**
DTD (sulfate transporter)	Diastrophic dysplasia
CBFA1 / RUNX2	Cleidocranial dysostosis
COL1A1 / COL1A2	Osteogenesis imperfecta
FGFR3	Achondroplasia
Ring chromosome 12 (MDM2)	Low-grade osteosarcoma
t(11;22)	Ewing

Labs monostotic: none · polyostotic: **thyroid panel**, **IGF-1, GH, prolactin** (MAS screen) · **ALP** to monitor bisphosphonate; ↑ ALP with malignant change

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Associated syndromes

Syndrome	Components
McCune-Albright	**Polyostotic FD + café-au-lait ("coast of Maine", jagged, don't cross midline, trunk) + endocrinopathy** (**precocious puberty** – 85% of girls; thyroid 66%; GH excess; ovarian cysts; neonatal hypercortisolism) · **unilateral**
Mazabraud	Polyostotic FD + **intramuscular myxomas**
Jaffe-Lichtenstein	Polyostotic FD + café-au-lait, **no endocrinopathy**
Jaffe-Campanacci	Multiple **NOFs** + café-au-lait, no neurofibromas
NF1	Café-au-lait **"coast of California"** (smooth) + neurofibromas, 17q11.2

3 Presentation

Usually asymptomatic incidental

Pain high-stress sites · ↑ in **pregnancy / menses** (oestrogen receptors)

Deformity **LLD (commonest skeletal deformity)** · **shepherd's crook** (coxa vara, lateral bowing, shortening) · **saber shin** (anterior tibial bowing) · **scoliosis**

Craniofacial cosmetic deformity, blindness, hearing loss, airway obstruction, proptosis, sinus occlusion

4 Diagnosis & Classification

Key radiology findings

X-ray **expansile, central intramedullary**, well-defined, smooth cortices · lytic, sclerotic or mixed → **"ground glass"** · cortical thinning, endosteal scalloping · **no periosteal reaction** · fills bone over length



Ground-glass lesion of the proximal femur

"Rind sign" punched-out / ground-glass lesion with **thick sclerotic rim**

Parrot beak sign **fatigue fracture of the femoral neck** in FD (TO)

Looser zones pseudofractures (osteomalacia)

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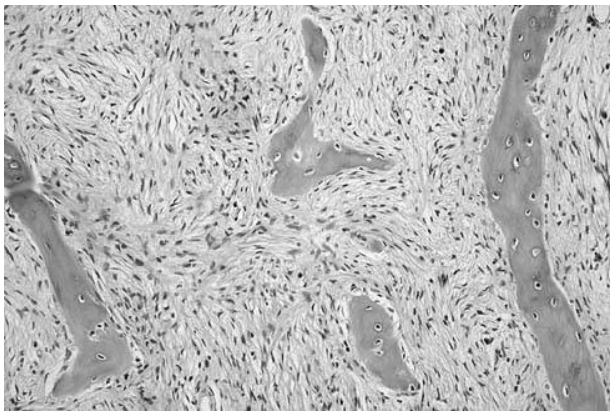
Shepherd's crook	AP + lateral hip
Spine X-rays (AP/ lateral)	in **polyostotic / MAS** (60% spinal involvement, posterior elements; scoliosis 40%) → **next step after hot spinal lesions on bone scan** ✗ Wrong options: CT · pelvis X-ray · bisphosphonates · endocrine screen
CT	ground glass 56%, sclerosis 23%, cystic 21% · enhances
MRI	**T1 dark**, T2 variable, moderate heterogeneous enhancement · separates from malignancy
Bone scan	warm until adulthood · screens for **polyostotic** disease / MAS in children
Diagnosis	history + exam + X-ray usually enough · biopsy if polyostotic/MAS doubt or malignancy

Histology

"Chinese letters" / "alphabet soup"	curvilinear irregular **woven bone trabeculae** in **fibrous stroma** (fibroblast proliferation)

No osteoblastic rimming (✓ rimming = osteofibrous dysplasia)

Gross	yellow/tan-white, **gritty/sandy** · cartilage in ~10% · cysts in older lesions



Woven bone without osteoblastic rimming in fibrous stroma

Classification

Type	Definition	Share
Monostotic	Single bone	**70%**
Polyostotic	Multiple bones	**30%**

"Touch-me-not" lesions (TO)

Definition	incidental, X-ray-diagnostic, need **no biopsy or intervention** · myositis ossificans, cortical desmoid, enchondroma, osteochondroma, **FD**, NOF, Paget, fibrous cortical defect, intraosseous lipoma, bone island, bone infarct · not **osteoblastoma** (painful, needs treatment)

Differential

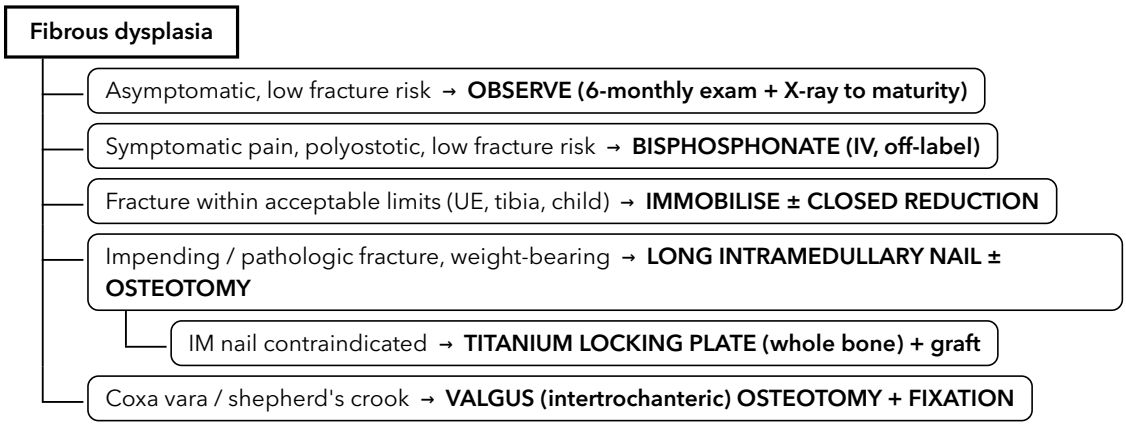
	Multiple lesions, young	Observation alone	Bisphosphonates help
FD	✓	✓	✓
EG	✓	✓	–
Ollier/Maffucci · MHE · NOF/Jaffe-Campanacci	✓	✓	–
Lymphoma · leukaemia · haemangioendothelioma	✓	–	–
Paget	–	✓	✓ (>50 yr, coarse trabeculae)
Metastasis · myeloma	–	–	✓

Osteofibrous dysplasia — **cortical**, anterior tibia, **osteoblastic rimming**

Adamantinoma — anterior tibial cortex, **epithelial nests**

Not seen in FD (TO) — **Harrison groove** (rickets) · ✓ rind sign, Chinese letters, café-au-lait

5 Management



Treatment (pathologic fracture, shepherd's crook, bisphosphonates)

IM nail — **gold standard** (load-sharing) · **long** nail to protect whole bone · elastic nails if canal small · for recalcitrant pain, impending/actual fracture, progressive deformity

Plate — **titanium** (modulus closer to bone) **locking**, **longest plate**, maximum screws · IM device > plate in lower limb

Graft choice — **calcium phosphate cement** (longest resorption) · **cortical allograft** > cancellous (slower resorption) · not **autograft** → resorbed, **converts to dysplastic bone**

Isolated curettage + graft — poor (resorption, fracture, reoperation) · only low-risk monostotic upper-limb lesions

Polyostotic + femoral neck path fracture (>2/3 diameter) — **biopsy, curettage, bone grafting + lesion-spanning internal fixation**
✗ Wrong options: observation · RT · percutaneous screws (soft lateral cortex)

– – Active space – –

5-yr-old, painful coxa vara	**valgus osteotomy + internal fixation** ✗ Wrong options: curettage + allograft · observation · spica
Symptomatic adult, no fracture risk	**bisphosphonate** ✗ Wrong options: RFA · HIFU (osteoid osteoma) · steroid injection (UBC) · curettage + autograft
Craniofacial	recontouring / curettage · **radiotherapy contraindicated** (malignant transformation) · calcitonin unsuccessful

6 Complications

Pathologic fracture — **50% of humeral lesions ≥1 fracture** · 50% of monostotic patients · humerus risk: **>50% cortical involvement, >30 mm, cystic degeneration** · femur: large neck/trochanteric lesion

Coxa vara classification

Type	Region	Neck-shaft angle	Share
1	Neck + trochanter	Normal 120–140°	24%
2	Neck + trochanter	Valgus >140°	6%
3	Neck + trochanter	**Varus <120°**	7%
4	**Lateral bowing** of proximal shaft	Normal	20%
5	Lateral bowing	Valgus	14%
6	Lateral bowing	**Varus**	**29%**

Course — **types 1–2 stable**, **types 3–6 progress** · goal: NSA 120–140°, correct bowing, apex-anterior + rotation, total-bone load-sharing fixation

Malignant transformation

Risk ~1%	higher in **polyostotic** (MAS, Mazabraud), after **radiotherapy**
Tumours	**osteosarcoma (commonest)** > fibrosarcoma > MFH · usually **diaphysis**, decades later
Clinical	new pain, swelling, mass, **↑ ALP** · **>50% mortality** (worse than primary sarcoma)
Scoliosis	40% in polyostotic

X vs Y

	Fibrous dysplasia	Osteofibrous dysplasia	Adamantinoma
Location	**Medullary**, proximal femur	**Cortical, anterior tibia**	Cortical, anterior mid-tibia
Age	<30	<10	20–40
Histology	Woven bone **without rimming**	Woven bone **with osteoblastic rimming**	Epithelial nests, keratin +
Treatment	Observe / bisphosphonate / IM nail	Observe / brace	Wide resection

	McCune-Albright café-au-lait	NF1 café-au-lait
Border	**Jagged – "coast of Maine"**	**Smooth – "coast of California"**
Midline	Don't cross	–

How it's asked

"Genetic abnormality in cAMP signalling?"	→ **fibrous dysplasia (TO)** ✗ Wrong options: achondroplasia · MHE · enchondroma
"Gs-α point mutation – which X-ray?"	→ **polyostotic FD with ground-glass proximal femur** ✗ Wrong options: adamantinoma · enchondroma · GCT · melorheostosis
"Gs-α (↑ cAMP) mutation causes?"	→ **FD** ✗ Wrong options: diastrophic dysplasia · cleidocranial · OI · achondroplasia
"Aetiology of ground-glass lesion + Chinese-letter histology?"	→ **Gs-α mutation → constitutive cAMP** ✗ Wrong options: ring chromosome 12 · parathyroid · t(11;22)
"GNAS mutation associated with?"	→ **FD** ✗ Wrong options: osteofibrous dysplasia · chondroblastoma
"McCune-Albright EXCEPT?"	→ **multiple neurofibromas**
"Jagged café-au-lait + hip lesion in a 6-yr-old?"	→ **McCune-Albright** ✗ Wrong options: NF · Holt-Oram · Ellis-van Creveld
"MAS, 3 hot spinal lesions – next?"	→ **AP/lateral spine radiographs**
"Polyostotic FD, serrated café-au-lait, endocrine, unilateral?"	→ **McCune-Albright** ✗ Wrong options: Jaffe-Campanacci
"Coxa vara, ground glass, jagged trunk pigmentation – risk of?"	→ **other bones + precocious puberty**
"NOT seen in FD?"	→ **Harrison groove (TO)**
"Lesion NOT primarily cortical?"	→ **fibrous dysplasia** (✓ NOF, OFD, adamantinoma cortical) **(TO)**
"Touch-me-not lesions EXCEPT?"	→ **osteoblastoma (TO)**
"Incidental ground-glass proximal femur / humerus?"	→ **FD** ✗ Wrong options: enchondroma (rings/arcs) · UBC · infarct
"Woven bone without osteoblastic rimming in bland stroma?"	→ **FD** ✗ Wrong options: ossifying fibroma · low-grade OS
"16-yr-old MAS, undisplaced femoral neck path fracture?"	→ **biopsy, curettage, grafting + internal fixation**

– – Active space – –

"5-yr-old, painful coxa vara with ground glass?"	→ **valgus osteotomy + internal fixation**
"38-yr-old, symptomatic distal femoral FD, biopsy-proven?"	→ **bisphosphonate therapy**
"Polyostotic FD, new pain at 39?"	→ **osteosarcoma** (malignant transformation)
"14-yr-old, huge craniofacial ground-glass lesion on CT?"	→ **maxillary fibrous dysplasia (TO)** ✗ Wrong options: EG · plasmacytoma · spina ventosa

FOCAL FIBROCARTILAGINOUS DYSPLASIA

– Pathology · 21 / 66 –

1 Basics

Nature	rare, benign tumour-like bone dysplasia · aetiology unknown
Age	**infants and toddlers**
Site	**tibia** commonest · also humerus, ulna/forearm, phalanx, femur
Association	a variant of **infantile tibia vara** that can **resolve without surgery** (vs Blount)

3 Presentation

Unilateral tibia vara	painless bowing in a walking toddler
Gait	knee hyperextension with **lateral thrust**

4 Diagnosis & Classification

Key radiology findings

X-ray (pathognomonic)	**abrupt varus at the metaphyseal–diaphyseal junction** · **medial cortical sclerosis** · **lucency in the medial cortex** just proximal to it (fibrocartilaginous tissue)



Focal fibrocartilaginous dysplasia of the tibia

Bone scan	shows deformity, **not "hot"** (not aggressive)
MRI	only in ambiguous cases

Histology

Dense bland fibrous tissue	layers of collagenous fibrous tissue + thick dense fibrocartilage · no malignant features

– – Active space – –

5 Management

Focal fibrocartilaginous dysplasia

- Most (tibia) → **OBSERVATION (± orthosis) — spontaneous correction with physeal growth**
- Persistent / progressive deformity, joint function compromised → **CORRECTIVE OSTEOTOMY (or curettage)**

Resolution — **~half of tibial cases** resolve spontaneously · much less often in femur and humerus

Osteotomy risks — secondary tibia valgus, neurologic injury

X vs Y

	FFCD	Infantile Blount
Side	**Unilateral**	Often bilateral
Apex	**Metaphyseal–diaphyseal junction**, medial cortical lucency + sclerosis	Proximal medial physis / metaphyseal beak
Natural history	**Often resolves**	Often progresses

How it's asked

"18-month-old, unilateral bowing, medial metaphyseal cortical lucency, bland fibrous histology?"	→ **focal fibrocartilaginous dysplasia** ✗ Wrong options: NF · toddler's fracture · NOF · ABC
"Treatment?"	→ **observation** · osteotomy only if persistent/progressive

GIANT CELL TUMOR

– Pathology · 22 / 66 –

1 Basics

Epidemiology

Nature	**benign but locally aggressive** · 3–5% of primary bone tumours · 1.7/million
Demographics	**females** (unlike most bone tumours) · **30–50** (Orthobullets) · 20–40 / 22–40 (OB, TO) · **skeletally mature, closed physes** (immature → metaphyseal)
Multicentric	**1–2% (TO)** · synchronous or metachronous (TO)
Population	more common in **Indian and Chinese** populations (TO)

Anatomic distribution

Order	**distal femur > proximal tibia > distal radius > sacral ala** · **50% around the knee**
Axial	**sacrum** commonest axial site (10% sacrum + spine) · spine → **vertebral body** ✗ Wrong options: **posterior elements**
Others	proximal humerus · **phalanges of hand** · apophysis (like chondroblastoma) · proximal fibula
Not typical	**diaphysis** (e.g. tibial shaft)
"Extra-articular GCT" (TO)	**GCT of tendon sheath** (giant cell synovioma) → **extremities** (hand/wrist flexor tendons) · firm painless nodule · excision

Cells

Cell	Identity
Type I – mononuclear stromal cell	**Neoplastic cell** · fibroblast-like, mesenchymal stem-cell features · **overexpresses RANKL** · carries **H3F3A (H3.3 G34W)** mutation
Type II – monocyte/macrophage	Recruited from blood · **giant-cell precursor**
Type III – multinucleated giant cell	**Hallmark** · osteoclast-like (TRAP, carbonic anhydrase II, cathepsin K, vacuolar ATPase) · **non-neoplastic bystander** · express **RANK**

Genes & lab markers

H3F3A (H3.3 G34W)	driver mutation in stromal cells **(TO)**
RANKL overexpression	by stromal cells – **key pathogenic pathway** → denosumab target
c-myc / p53	alterations – in **metastatic** lesions (lung mets overexpress c-myc)
Telomeric associations	80% · IGF-I/II activity in type II/III cells
Brown tumour	must check **serum calcium / PTH** (hyperparathyroidism, multiple lesions)

– – Active space – –

3 Presentation

Pain	insidious, activity/night/rest · joint-referred · difficulty walking
Signs	mass, tenderness, swelling, **effusion** (periarticular), ↓ ROM, antalgic gait
Pathologic fracture	common (subchondral collapse)

4 Diagnosis & Classification

Key radiology findings

X-ray	**eccentric, purely lytic, epiphyseal–metaphyseal**, extends to **subchondral bone** · **no sclerotic rim** (or partial) · poorly defined transition · expands/ breaks cortex · "neocortex" · no matrix · intra-articular extension rare · **soap-bubble** appearance **(TO – FNB 2021)**

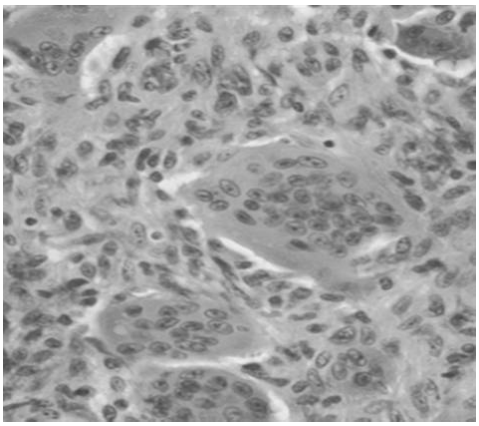


Eccentric lytic subchondral lesion

MRI	extent · **dark T1 (clear demarcation from fatty marrow)**, bright T2, avid enhancement · intermediate T2 signal (like lymphoma, Ewing) · fluid levels = **secondary ABC**
Bone scan	variable / hot
Chest X-ray or CT chest	**mandatory** staging (lung mets) ✗ Wrong options: bone scan · skeletal survey · SPEP/UPEP · MRI brain/ spine · mammogram
Biopsy first	even when imaging looks classic → **incisional biopsy** before reconstruction (malignancy can mimic)

Histology

Diagnostic	numerous **multinucleated giant cells (40–60 nuclei)** in a sea of mononuclear stromal cells · **giant-cell nuclei identical to stromal nuclei** · storiform areas, reactive bone, foam cells, haemosiderin



Giant cells with nuclei identical to stromal cells

Secondary ABC 10–14% (Orthobullets/OB) · 20% (TO) · GCT has **enhancing solid component** (primary ABC doesn't)

Campanacci grading (TO)

Grade	Behaviour	Radiographic features
I	Latent	Well-marginated, **thin rim of mature bone**, cortex intact or slightly thinned
II	Active	Well-defined, **no radiopaque rim**, cortex thinned + **moderately expanded** but present · (II + fracture graded separately)
III	Aggressive	**Fuzzy borders**, permeative · **soft-tissue mass** not contained by bone shell

Enneking (benign) most GCT **stage 2**; aggressive = **3**

Differential

	Epiphyseal	Usually curettage + graft
GCT	✓	✓
Chondroblastoma (epiphyseal)	✓	✓
ABC · osteoblastoma · CMF	–	✓

Chondroblastoma **open physes**, chicken-wire calcification, marked oedema, S-100

Brown tumour multiple lesions, abnormal calcium

Giant-cell-rich / telangiectatic / fibroblastic OS **atypia + osteoid**

Chordoma sacral **midline**, physaliferous cells

ABC fluid levels, USP6 · **UBC** – no cortical destruction

ABC vs GCT (TO) ABC: **younger**, fluid levels **without solid component** · GCT: **solid component** ± secondary ABC (fluid levels)

Clear cell chondrosarcoma epiphyseal GCT mimic · sclerotic rim possible (TO)

MRI (TO) epi-metaphyseal, eccentric · **T1 hypointense**, **T2 hyper / mixed** · solid component · **whole-bone MRI** for skip lesions

"Giant-cell-rich lesion" report (TO) **not = GCT** · insist on definite diagnosis (exclude **giant-cell-rich OS**, ABC, brown tumour) · **biopsy before curettage**, target solid part

- - Active space - -

5 Management

Giant cell tumour of bone

- Joint salvageable (most; Campanacci I–II, many III) → **EXTENDED CURETTAGE + BURR + ADJUVANT + PMMA/GRAFT ± FIXATION**
  (pathologic fracture is NOT a contraindication)
- Structural loss / collapsed subchondral bone → **EN BLOC RESECTION + RECONSTRUCTION**
  - Expendable bone (proximal fibula, distal ulna) → **RESECTION ± LCL/biceps repair**
  - Vertebral body → **TOTAL EN BLOC SPONDYLECTOMY**
- Unresectable (sacrum, spine, pelvis) / multiply recurrent → **DENOSUMAB ± EMBOLIZATION (RT last resort)**
- Hand with cortical breakthrough → **INTERCALARY RESECTION or RAY AMPUTATION**

Treatment — extended curettage with adjuvants

Preferred treatment — **extended curettage + adjuvants + reconstruction** (TO FNB 2024 — key not given; sources' standard)
✗ Wrong options: curettage alone · curettage + graft · primary RT

Technique — **large cortical window (≥ lesion size)** — no curetting under overhanging ridges · head lamp, dental mirror, angled curettes · **high-speed burr 1–2 cm** in all directions · pulsatile lavage · don't perforate subchondral plate

Most important step — **aggressive curettage with high-speed burr** (no adjuvant proven superior)

Burr depth — **1–2 cm** (Orthobullets) · **1–2 mm** of peripheral bone (TO) — read the options

Burr (TO) — **prerequisite, not an adjuvant** · "never curette without a burr" · adjuvants do not significantly improve control when curettage + burr done well

Exposure (TO) — muscle-preserving (lift **vastus lateralis off septum**) · isolate with **H_2O_2-soaked mops** · remove soft-tissue component **en bloc** · **start small window, enlarge** after debulking (avoid spillage) · headlight + loupes

C-arm (TO) — before curettage, after gross curettage, after burring · **pulsed lavage before + after adjuvants**

Adjuvant (TO)	Penetration	Note
Liquid nitrogen	**14 mm** (maximal)	**Fracture, nerve injury**
Phenol	**< 1 mm**	Wash with **ethanol** · soft-tissue irritation
Argon beam	**4 mm**	Thermal coagulation · expensive
Hydrogen peroxide	Contact only	Always used, cheap

Workflow before reconstruction (TO) **change gloves, drapes, instruments** · separate trolley · **harvest graft before curettage** (iatrogenic donor-site implantation reported)

Filling (TO)	For	Against
Bone graft	Biological · subchondral protection	Donor morbidity · delayed WB · **hard to detect recurrence**
PMMA cement	Immediate · **early WB** · **easier recurrence detection** · heat kills residual cells	–
Sandwich (subchondral graft + cement)	Minimal subchondral bone	**Subchondral collapse**

Cement and OA (TO) cement **not proven to cause arthritis** · **loss of subchondral bone** is the risk factor

Recurrence on X-ray (TO) **lysis around cement / graft** · progressive lucency · new soft-tissue shadow

Adjuvants **phenol**, **liquid nitrogen (cryo)**, **hydrogen peroxide**, **argon beam**, electrocautery, ethanol (= phenol) · burr alone recurrence **12%**

Filling **PMMA** (thermal necrosis; recurrence seen as **lucency at bone–cement interface**) · graft · ± internal fixation · not DBM + marrow aspirate (higher recurrence)

Treatment	Local recurrence
Curettage + graft alone	**20–40%** (OB: 15–25%)
+ adjuvant	**3–10%**
High-speed burr only	12%
Overall	~20%

Distal radius, cortices + joint intact **curettage + liquid nitrogen + PMMA**
✗ Wrong options: en bloc + allograft · fusion with fibula · RT

Hand no breakthrough → **curettage + cementing** · breakthrough → **intercalary resection** (free fibula) **or amputation** (lowest recurrence)

En bloc resection

Lateral condyle with collapsed subchondral bone + fracture **condylar resection + osteoarticular allograft**
✗ Wrong options: cement injection · chemo · RT · amputation

Proximal fibula **en bloc resection + repair LCL + biceps femoris to lateral tibia** → lowest recurrence, low instability
✗ Wrong options: curettage · structural allograft · denosumab (high recurrence) · observation

Knee hemicondylar osteoarticular allograft or rotating-hinge endoprosthesis · **distal radius** aggressive → resection + proximal fibular autograft (arthroplasty or arthrodesis) · distal ulna/clavicle → resection alone

Resection indications (TO) **path fracture with deformity** · **recurrence with joint destruction** · **insufficient residual / subchondral bone** · **expendable bones** (distal ulna, proximal fibula, proximal radius) · massive soft-tissue extension · en bloc in **one piece**

– – Active space – –

Path fracture / recurrence (TO) fracture **not an absolute indication** for resection (curettage + cement in many) · recurrent GCT → curettage often still possible; resect if joint destroyed

Distal radius reconstruction (TO)	Advantages	Disadvantages
Fibular arthroplasty	**Wrist motion**, cosmesis	**Unstable**, incongruent joint, donor morbidity
Ulnar translocation	**Stable**, vascularised, no donor morbidity → preferred when stability needed	Cosmetic deformity (**hour-glass wrist**)

Spine total en bloc spondylectomy (± staged anterior/posterior, cage/cement) → 10-yr disease-free up to 100%

Denosumab

Drug **fully human IgG2 monoclonal antibody against RANKL (receptor activator of nuclear factor kappa-B ligand)** (mimics OPG) · FDA approved for unresectable GCT / when resection → severe morbidity, adults + skeletally mature

Option	Identity
Denosumab	✓ **Human mAb → RANKL**
Osteoprotegerin	Decoy receptor → RANKL
Infliximab	Chimeric mAb → TNF-α
Rituximab	→ CD20
Trastuzumab	→ HER2
(none)	mAb or decoy → RANK itself

Mechanism (TO) kills/prevents **osteoclast-like giant cells** · **stromal (neoplastic) H3F3A clone survives**, quiescent
✗ Wrong options: directly kills stromal cells · halts stromal proliferation · OPG antagonist

Effect **85–90% giant-cell destruction** · sclerosis, cortical reconstitution, pain relief

Histology after denosumab (TO) **loss of giant cells + dense fibro-osseous tissue, new woven bone, stromal spindling** (mimics low-grade fibro-osseous lesion)

Dosing (TO) **120 mg SC days 1, 8, 15, 29 → every 4 weeks** · neoadjuvant 3–6 mo · **calcium + vitamin D** mandatory

Neoadjuvant downside (TO) **sclerotic, ossified margin obscures tumour boundary → ↑ recurrence after curettage** (20–30% vs 10–15%)

Stopping (TO) **rebound RANKL surge** → rapid recurrence + **severe rebound hypercalcaemia** (3–9 mo) · recurrence 20–60% · mitigate with **zoledronic acid**, tapering, calcium monitoring

Long-term (TO) **1–6% malignant transformation**, ↑ with prolonged therapy · ONJ 1–5% · atypical femoral fracture (>2 yr)

Side effects hypocalcaemia (contraindication: severe hypocalcaemia), nasopharyngitis, arthralgia

Denosumab extras (TO)	**hard shell** eases dissection · **inform pathologist** (mimics low-grade sarcoma) · **not an adjuvant** after curettage · **single preop dose → ↓** vascularity where no tourniquet (pelvis, proximal humerus) · "masks, does not kill stromal cells"
Zoledronic acid (TO)	**4 mg monthly** · osteoclast apoptosis · may ↓ recurrence · inoperable tumours
Bisphosphonates	osteoclast inhibitors · postop → **↓ recurrence 25–30% (TO: used to prevent recurrence)** · zoledronic acid stabilises inoperable disease · denosumab > bisphosphonate ✗ Wrong options: calcitonin · teriparatide · methotrexate

Radiation & embolization

Radiation	only inoperable / multiply recurrent (sacrum, large vertebral) · **~15% malignant transformation** (megavoltage)
Embolization	unresectable sacral/pelvic GCT, monthly until palliation · preop to ↓ blood loss · **definitive for inoperable** pelvis/sacrum/spine, combined with **zoledronic acid** (TO) · **pelvic GCT → embolization + limited resection + curettage** (TO)
Not used	chemotherapy · RFA

6 Complications

Lung metastasis & chest CT

"Benign" metastasizing tumour	lung mets **histologically identical** to primary (also chondroblastoma)

Source	Lung metastasis rate
Orthobullets	**2–4%** (1–6% on chest CT)
OB	**~2% overall, 6% if recurrent → counsel ~4%** · 1–9% · ~5%
TO	**~3%**
Spine GCT series	13.7%

Risk factors	**local recurrence** · **aggressive / stage 3 (Campanacci III) (TO)** · **wrist/hand, distal radius**, proximal femur, sacrum/spine · younger · immunocompromised
Timing	18–24 mo, most within 3 yr
Treatment	**resect (metastasectomy)** → ~76–80% disease-free · denosumab · RT
Mortality with lung mets	**~15% (TO)** · 17% (Orthobullets) · 5-yr disease-free 76%
Recurrence	usually **benign GCT again** (identical histology) · diagnose with CT-guided biopsy · lucency at cement interface

Malignant transformation

Rates	<1% (Orthobullets) · **1.4–6.6%** (OB) · <5% overall malignancy · radiation → ~15%
Primary vs secondary malignant GCT	secondary after **radiation** or multiple resections · latency **9 yr after RT**, **19 yr spontaneous**

– – Active space – –

Late lytic change years after curettage + biopsy shows high-grade sarcoma	**wide resection + reconstruction** (+ medical oncology) ✗ Wrong options: repeat curettage · denosumab
Pathologic fracture	periarticular lesions · **cryotherapy (liquid nitrogen)** → fracture + **vascular injury**, wound/nerve problems (phenol too)

Chemotherapy response grading (Huvos)

(osteosarcoma necrosis grading, filed under this topic's TO set)

Huvos grade	Tumour necrosis	Response
I	0–50%	Poor (little/none)
II	51–90%	**Poor** (partial)
III	91–99%	**Good**
IV	100%	Complete

88% necrosis → Huvos II, poor responder → continue standard MAP (EURAMOS-1: changing regimen doesn't help) **(TO)**

X vs Y

	GCT	Chondroblastoma	Brown tumour
Age	20–50, closed physis	10–25, open physis	>50
Site	Epiphysis–metaphysis, subchondral	Epiphysis	Multiple
Giant cells	Nuclei = stromal nuclei	Scattered + chondroblasts	Haemosiderin, osteoclasts
Labs	Normal	Normal	**↑ Ca, ↑ PTH**
Treatment	Curettage + adjuvant / denosumab	Curettage + graft / RFA	Treat hyperparathyroidism

How it's asked

"30-yr-old woman, wrist lesion, giant cells in mononuclear stroma?"	→ **GCT (TO)** ✗ Wrong options: OS · myeloma · Ewing
"Soap-bubble appearance on X-ray (FNB)?"	→ **GCT (TO)**
"Distal radius lesion, homogeneous stromal + giant-cell nuclei?"	→ **GCT** ✗ Wrong options: telangiectatic OS (malignant cells in blood spaces)
"Sacral lesion in a 25–29-yr-old with giant cells + haemosiderin?"	→ **GCT** ✗ Wrong options: chordoma (physaliferous) · lymphoma (Reed-Sternberg) · Ewing

"L3 vertebral body expansile lesion with giant cells, 55 yr?"	→ **GCT** ✗ Wrong options: osteoblastoma (posterior elements)
"Recurrence 2 yr after curettage — histology?"	→ **benign GCT** ✗ Wrong options: malignant GCT · ABC
"MCP subchondral expansile lytic lesion in a 26-yr-old?"	→ **GCT**
"Mononuclear cells overexpressing RANKL?"	→ **GCT** ✗ Wrong options: ABC (USP6)
"Common sites EXCEPT?"	→ **posterior elements of spine / tibial diaphysis**
"Sites in order?"	→ **distal femur > proximal tibia > distal radius > sacral ala**
"Common site of extra-articular GCT?"	→ **extremities** (GCT tendon sheath) **(TO)**
"Treatment of GCT — all true EXCEPT?"	→ **"pathologic fracture contraindicates curettage + cement"** (✓ bisphosphonates ↓ recurrence, ✓ denosumab inoperable, ✓ cryo → fracture) **(TO)**
"Distal radius GCT — FALSE statement (FNB 2024)?"	→ key not given · **"primidone reduces tumour size preoperatively"** fits (denosumab does) · ✓ Campanacci I → curettage, ✓ sandwich technique for periarticular III, ✓ extended curettage when joint salvageable **(TO)**
"Best treatment of knee GCT?"	→ **intralesional (extended) curettage + local adjuvant** ✗ Wrong options: chemo · RT · wide resection · RFA
"Which X-ray is best treated by burr curettage + cement?"	→ **GCT** ✗ Wrong options: adamantinoma · FD · osteochondroma · parosteal OS
"Lateral condyle GCT with subchondral collapse + fracture?"	→ **condylar resection + osteoarticular allograft**
"Cement after GCT curettage causes arthritis?"	→ **No — loss of subchondral bone does** (TO)
"Adjuvant with deepest penetration?"	→ **liquid nitrogen (14 mm)** (TO) ✗ Wrong options: phenol (<1 mm) · argon beam (4 mm) · H_2O_2
"Distal radius resection — stable reconstruction?"	→ **ulnar translocation** (TO) ✗ Wrong options: fibular arthroplasty (mobile but unstable)
"Proximal fibula GCT with peroneal palsy?"	→ **en bloc resection + LCL/biceps repair to tibia**

– – Active space – –

"Late lytic recurrence, biopsy high-grade sarcoma?"	**→ wide resection + reconstruction**
"Medication with most benefit?"	**→ denosumab** ✗ Wrong options: alendronate · calcitonin · teriparatide
"Denosumab — classification + mechanism?"	**→ human mAb against RANKL** ✗ Wrong options: against RANK · decoy receptor = OPG
"Accurate statement on denosumab in GCT?"	**→ binds RANKL, eliminates giant cells, stromal cells persist (TO)**
"Histology after 6 months of denosumab?"	**→ giant cells gone, fibro-osseous tissue, woven bone, stromal spindling (TO)**
"Neoadjuvant denosumab before curettage — main drawback?"	**→ sclerotic margin obscures tumour → ↑ recurrence (TO)**
"After stopping long-term denosumab?"	**→ rebound RANKL surge → recurrence + hypercalcaemia (TO)**
"Long-term denosumab and malignancy?"	**→ 1–6% transformation, ↑ with prolonged use (TO)**
"True about GCT (FNB 2021)?"	**→ radiographic stage 3 → higher risk of lung mets (TO)** ✗ Wrong options: 1% mets · 5% mortality · 5% multicentric
"Benign bone tumour most likely to metastasize to lung?"	**→ GCT** ✗ Wrong options: NOF · ABC · osteoid osteoma · haemangioma
"Staging for suspected metastasis?"	**→ CT chest / chest radiograph**
"Risk of lung mets after intralesional surgery?"	**→ ~4%**
"Spine GCT metastasizes to?"	**→ lung**
"Imaging shows classic GCT of the wrist — first surgery?"	**→ incisional biopsy**
"88% necrosis after MAP — Huvos?"	**→ grade II, poor responder, continue MAP (TO)**

GLOMUS TUMOR

– Pathology · 23 / 66 –

1 Basics

Epidemiology

Nature	rare **benign** tumour (hamartoma) of the **glomus body** · formerly "paraganglioma" · soft tissue and/or bone · malignant cases rare
Age / sex	**20–40** (3rd–4th decades) · females
Site	**hand 75%** · **subungual 50%** · palm, wrist, forearm, foot less often
Bone	**erosion of distal phalanx in 50%** · primary intraosseous rare

Anatomy

Glomus body	**perivascular temperature-regulating** structure (arteriovenous shunt) at **digit tip / beneath nail**

3 Presentation

Classic triad	**paroxysmal pain** · **exquisite point tenderness** · **cold intolerance** (cold water worsens pain)
Signs	small **bluish nodule** (often invisible subungually) · **nail ridging / discoloration** · may show no mass or colour change
Delay	diagnosis often delayed for years

Clinical tests

Test	Method	Accuracy
Love test	**Pinhead pressure on the spot → excruciating pain**, none just beside it	**100% sensitive**, 78% accurate
Hildreth test	**Tourniquet inflation → pain + Love-test tenderness abolished**	**92% sensitive, 91% specific**
Cold sensitivity	Cold water exacerbates pain	–

Love test diagnostic of	**glomus tumour (TO)** ✗ Wrong options: trochanteric bursitis · hydatid cyst · SLE

4 Diagnosis & Classification

Key radiology findings

X-ray	**pressure erosion** of distal phalanx, cortical deformity · intraosseous → cystic erosive change · often **normal**

– – Active space – –

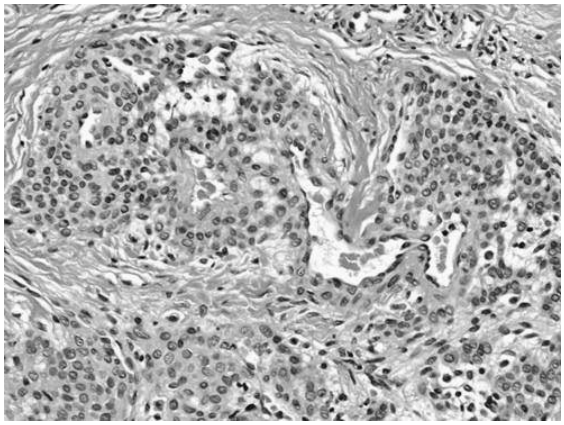
MRI establishes diagnosis · **T1 low, T2 high**, small enhancing subungual nodule



Glomus erosion into distal phalanx

Histology

Well-defined, **no atypia or mitoses** · uniform **small round cells with dark nuclei** and pink cytoplasm (epithelioid glomus cells) surrounding **small vascular spaces** in hyaline/myxoid stroma · gland-like nests



Glomus cells around vascular spaces

Differential

Lesion	Distinguishing
Glomus tumour	Triad, MRI nodule, glomus cells
Raynaud phenomenon	Cold sensitivity, **no mass**
Epidermal inclusion cyst	Prior trauma, less point tenderness / cold pain
Osteomyelitis	Labs, systemic, X-ray changes
Osteoid osteoma	Cortical nidus + sclerosis
Melanoma · SCC · synovial / epithelioid sarcoma	Different histology

5 Management

Marginal excision **curative** · for symptoms affecting quality of life · complete pain relief

Nail bed reconstruction	autologous **fat graft** for large defects
Locally aggressive / intraosseous destructive	aggressive resection, even **amputation** suggested

6 Complications

Recurrence	**~20%** (Orthobullets) · "uncommon" (benign) — usually incomplete excision

How it's asked

"20-yr-old, painful ecchymotic ring fingertip, eroded phalanx, dark-nucleated round cells?"	→ **glomus tumour** ✗ Wrong options: melanoma · SCC · synovial/epithelioid sarcoma
"43-yr-old, cold-sensitive fingertip pain, ulnar nail-bed tenderness, normal X-ray, MRI nodule?"	→ **glomus tumour** ✗ Wrong options: Raynaud · inclusion cyst · osteomyelitis · osteoid osteoma
"Little-finger tip pain worse with cold — histology vascular spaces with epithelioid cells?"	→ **glomus tumour** ✗ Wrong options: schwannoma · scleroderma · paronychia
"Love test is diagnostic of?"	→ **glomus tumour (TO)**
"Treatment?"	→ **marginal excision**

– – Active space – –

HEMANGIOMA

– Pathology · 24 / 66 –

1 Basics

Nature	**benign vascular tumour of bone**
Site	**vertebral bodies** and **craniofacial bones** · pelvis (ilium) · may be **multifocal**

3 Presentation

Usually asymptomatic	incidental
Symptomatic	pain or **pathologic fracture**

4 Diagnosis & Classification

Key radiology findings – vertebral haemangioma

Modality	Sign
Plain X-ray	**"Jailhouse" / "jail-bar" / "honeycomb"** – lytic body with **vertical striations** (thickened vertical trabeculae) **(TO)**
Sagittal / coronal CT	**"Corduroy"** appearance **(TO)**
Axial CT	**"Polka dot"** (salt-and-pepper) – white dots on black fatty background **(TO)**
MRI	**"Mottled"** – **high T1** (fat) + **high T2** (vascular), black punctate/linear signal voids · retains signal on STIR/fat-sat (unlike fatty marrow) **(TO)**
Bone scan	Warm to hot

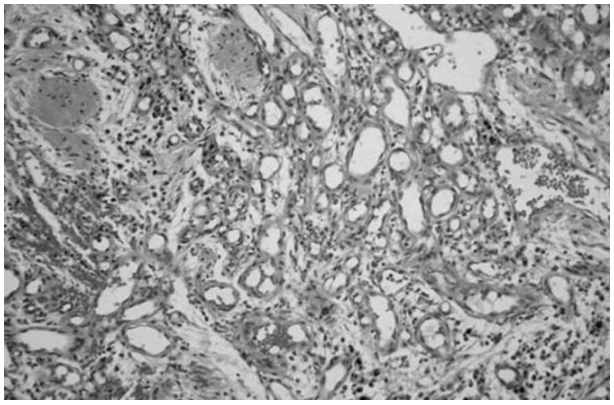


Vertical striations – jailhouse vertebra

Biopsy	almost **never required** (imaging characteristic) **(TO)**
Why these signs	normal cancellous bone replaced by **thickened vertical trabeculae** surrounded by **fat marrow / vascular lacunae**

Histology

Cavernous lesion with numerous **thin-walled blood vessels** · small vascular channels lined by **single layer of endothelium** · no giant cells (vs ABC, Paget), no granulomas (vs sarcoid)



Cavernous thin-walled vessels

Differential of the spine/ilium lesion

Lesion	Distinguishing
Haemangioma	**Organised vertical striations**, polka dots
Paget disease	Square, **enlarged** vertebra, thick cortex ("picture frame"), disorganised mineral
GCT	Lytic, destructive
Metastasis · myeloma	Lytic/sclerotic, no organisation
Chordoma (sacral)	Sacrum, midline, physaliferous cells
ABC	Fluid levels, giant cells
Sarcoid	Granulomas

Soft-tissue haemangioma (forearm)

Phleboliths on X-ray + intramuscular mass, **↑ T1 (fat)**, heterogeneous T2 + enhancement → **soft-tissue haemangioma**
✗ Wrong options: synovial sarcoma · liposarcoma · rhabdomyosarcoma

5 Management

Haemangioma of bone

- Asymptomatic → **OBSERVATION**
- Symptomatic + accessible → **CURETTAGE + BONE GRAFT (intralesional excision)**
- Symptomatic + inaccessible → **LOW-DOSE RADIATION 25–40 Gy**

How it's asked

"Polka dot sign on CT?"

→ **vertebral haemangioma (TO)**
✗ Wrong options: clavicle osteomyelitis · cortical OS · lymphoma

– – Active space – –

"Jailhouse appearance is seen on?"	→ **plain X-ray** of vertebral haemangioma **(TO)**
"Corduroy appearance is seen on?"	→ **sagittal CT** of vertebral haemangioma **(TO)**
"Mottled appearance on MRI?"	→ **vertebral haemangioma (TO)** ✗ Wrong options: marrow oedema · CML · metastasis
"Axial CT with mottled/polka-dot vertebra?"	→ **vertebral haemangioma (TO)** ✗ Wrong options: myeloma · metastasis · LCH
"D12 corduroy on CT (FNB 2024)?"	→ key not given · **vertebral haemangioma** per the sign **(TO)** ✗ Wrong options: blastic mets · myeloma · Paget
"L3 vertical striations on sagittal CT?"	→ **haemangioma** ✗ Wrong options: Paget · GCT · metastasis · chordoma
"30-yr-old, iliac lytic lesion with coarse trabeculae, thin-walled endothelial channels?"	→ **haemangioma of bone** ✗ Wrong options: ABC · sarcoid · Paget
"7-yr-old, forearm mass with phleboliths, fat on T1?"	→ **haemangioma**

HEMANGIOMA OF SOFT TISSUE

– Pathology · 25 / 66 –

1 Basics

Epidemiology

Nature	benign vascular neoplasm · **error in new formation of a vascular segment** · hormones modulate · **20% trauma-associated**
Demographics	**<30 yr** · M = F
Site	**cutaneous, subcutaneous, intramuscular** (deep, lower limb) · **hand** common · usually solitary · **diffuse haemangioma** = rare childhood form involving a whole limb
Types	**capillary (superficial)** vs **cavernous (deep)**
Tissue of origin	**vascular** ✗ Wrong options: cartilage, bone, smooth or skeletal muscle

3 Presentation

Pain varies with activity	engorgement → **aching, heaviness, swelling** · worse when **dependent**, standing, exercising, swimming · relieved by rest, elevation, ice
Mass size varies	fills after minutes of standing → examine **supine and standing** · may be warm · birthmark that enlarges

4 Diagnosis & Classification

Key radiology findings

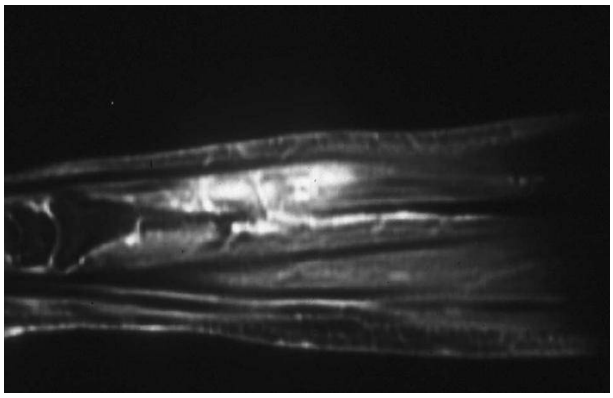
X-ray — **phleboliths** (calcified thrombi in vessels) · bone erosion · may be normal



Phleboliths in a cavernous haemangioma

MRI with gadolinium — **diagnostic study** · separates from **AVM and angiosarcoma** · **T1 iso- to hyperintense (fat)** · **T2 hyperintense** · low-signal foci = flow voids / calcification · heterogeneous vessels + **fatty infiltration → "bag of worms"** · no necrosis, no rim enhancement, no mass effect

– – Active space – –



Intramuscular haemangioma on MRI

Fat on T1 + calcification on X-ray → diagnostic of haemangioma

Histology

No malignant cells, no pleomorphism · multiple **dilated endothelial-lined vascular channels filled with erythrocytes** · **intramuscular type: lumens infiltrated by (striated) muscle fibres** · cavernous: large vessels + fat

Gross red to tan to yellow

Differential

Lesion	Distinguishing
Haemangioma	Activity-related pain/size, phleboliths, fat on T1
Low-flow venous malformation	Congenital, **multifocal, infiltrative**, adolescence (hormonal), compressible, phleboliths, interspersed fat
AVM	High flow · diagnose by exam + imaging (phleboliths, Doppler flow) · **multidisciplinary / IR**
Synovial sarcoma	Painless, calcification, different T1
Lipoma	Homogeneous bright T1, suppresses
Desmoid	Dark T1 + T2, no fat
PVNS	Similar imaging, different pain · haemosiderin giant cells
Parosteal OS	Stuck on posterior distal femur
Muscle tear haematoma	Should resolve over 3 months

5 Management

Soft-tissue haemangioma / vascular malformation

- First line (incl. children, incidental) → **OBSERVE + NSAIDs + COMPRESSION STOCKINGS + ACTIVITY MODIFICATION**
- Large, painful, failed conservative → **IR: SCLEROTHERAPY (ethanol, US-guided) or EMBOLIZATION**
- Small hand lesion failing conservative → **MARGINAL EXCISION**
- Recalcitrant, large → **WIDE RESECTION (high local recurrence)**

AVM failing conservative care	**consult interventional radiology** (coil embolization / sclerotherapy) ✗ Wrong options: CT CAP staging · biopsy · chemo · RT
Not used	chemotherapy · radiation
Recurrence after excision	high · deep lesions abut NV structures → often intralesional

6 Complications

Kasabach-Merritt syndrome	**platelet trapping → consumptive coagulopathy**, potentially fatal
Infantile haemangiomas	**involute by age 7**
Malignant transformation	**none**

How it's asked

"30-yr-old, painful posterior knee mass enlarging with activity, calcification + fat on T1?"	→ **haemangioma** ✗ Wrong options: synovial sarcoma · lipoma · parosteal OS · PVNS
"17-yr-old, enlarging hand birthmark, pain when dependent/swimming, phleboliths?"	→ **benign haemangioma** ✗ Wrong options: angiosarcoma · epithelioid sarcoma · GCT tendon sheath
"Shoulder mass painful + warm after activity, normal X-ray, fat on MRI?"	→ **intramuscular haemangioma** ✗ Wrong options: synovial sarcoma · metastasis · haematoma · abscess
"Tissue of origin?"	→ **vascular**
"Multiple infiltrative 'bag of worms' masses with fat in a teen?"	→ **low-flow venous malformation** ✗ Wrong options: desmoid · lipoma · MPNST · schwannoma
"Calf mass enlarging at end of day – histology + treatment?"	→ **vascular dilations with erythrocytes + striated muscle · observation**
"Foot AVM failing stockings + activity change?"	→ **consult interventional radiology**
"Incidental intramuscular gastrocnemius haemangioma?"	→ **observation** ✗ Wrong options: wide/marginal resection · RT · chemo

– – Active space – –

HETEROTOPIC OSSIFICATION

– Pathology · 26 / 66 –

1 Basics

Epidemiology

Definition bone in atypical **extraskeletal** soft tissue · after **local trauma**, **neurologic injury** (within **2 months** of brain/spinal cord injury), **THA/TKA**, or spontaneous

Demographics **M:F 2:1** · men with **hypertrophic OA**, women >65

Commonest location **between muscle and joint capsule**

Injury	Order of joints
TBI / stroke	**Hip > elbow > shoulder > knee** · **spastic side** · elbow HO commoner after brain injury · knee rare
Spinal cord injury	**Hip > knee > elbow > shoulder** · hip flexors + abductors · medial knee

Subtypes neurogenic HO · traumatic myositis ossificans · **fibrodysplasia ossificans progressiva** (Münchmeyer)

Pathophysiology

Mechanism activation/proliferation of **mesenchymal stem cells** → osteoprogenitors · genetic predisposition · experimental HO linked to **BMP** expression

Risk factors

Risk factor	Detail	Incidence
Injury Severity Score	**Highest direct correlation** with severe HO after knee dislocation (✗ Wrong options: GCS · timing · number of ligaments · open reconstruction)	11%
TBI	Spastic limbs · prolonged coma (20–30 yr), **prolonged ventilation**	11%
Spinal cord injury	**Complete > incomplete**	20%
Decubitus ulcers + SCI/TBI	–	**70%**
Burns	Local and remote	–
Amputation through zone of injury	**Blast** mechanism worst	63%
Antegrade femoral nail	**Piriformis** entry	25%
Distal femoral traction pin	Large Steinmann pin → **quadriceps HO** (rare) (✗ Wrong options: proximal tibia, calcaneus, radius, metacarpal)	Rare
Acetabular approach	**Extended iliofemoral > Kocher-Langenbeck > ilioinguinal** · combined approaches → HO is the commonest complication	25%
THA	**Psoas tenotomy**, **cementless** (debris, marrow spillage) · **Smith-Petersen and Hardinge > transtrochanteric > posterior (lowest)** · DA "**bikini**" incision ↓ high-grade HO (4% vs 12%)	53% (significant 5%) · symptomatic ~7%
TKA	**Anterior femoral notching**, quadriceps trauma, periosteal stripping, **MUA**, high lumbar BMD · **ankylosing spondylitis** (~20%)	–
Other	**DISH**, hypertrophic OA, previous HO	–
Not proven	**Open fractures** · **fluorosis (TO)**	–

THA risk groups (TO) **high**: previous HO, hypertrophic post-traumatic arthritis, **men with hypertrophic OA** · **moderate**: AS, DISH, Paget, Parkinson, RA, unilateral hypertrophic OA

Genes & lab markers

Marker	Use
ALP >250 IU/L	↑ early, nonspecific · **↑ at 12 wk post-op predicts** HO · **cannot determine maturity** (OB) · normalization used in timing (Orthobullets)
CRP	Tracks inflammatory activity better than ESR · **normalization ≈ maturity**
ESR >35 mm/h	At 12 wk after THA predicts HO
CK	Extent of muscle involvement

– – Active space – –

3 Presentation

Painless loss of ROM → ADL interference · ankylosis

Early **warm, painful, swollen** joint, effusion, fever (mimics infection/DVT) · CRPS association

Complications **nerve impingement** (ulnar neuropathy at elbow) · contractures → **decubitus ulcers** · skin maceration · pathologic fracture of osteoporotic bone · TKA: quadriceps snapping, patellar instability

4 Diagnosis & Classification

Key radiology findings

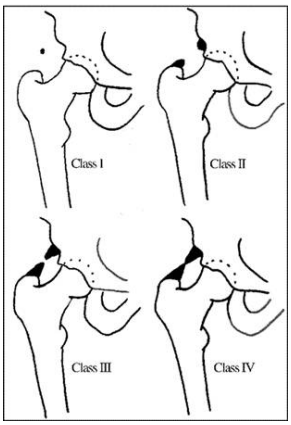
Modality	Role
Triphasic bone scan	**Best for early diagnosis**, most used · decreasing activity = maturing
Ultrasound	Early hip HO – echogenic surface + acoustic shadow
X-ray	Not early (calcium appears **7–10 days**; may be normal ~3 wk) · **maturity = cortex + sharp demarcation + trabeculae** · **Judet views** for hip
CT (3D)	**Preoperative planning** · characterize synostosis

Histology mature trabecular (lamellar) bone + fatty marrow

Brooker classification (HO after THA)

Class	Description
1	**Islands of bone** in soft tissue
2	Spurs from pelvis/femur, **≥1 cm** between opposing surfaces
3	Spurs, gap **<1 cm**
4	Apparent **bony ankylosis**

Brooker 1–2 don't affect THA outcome · **3–4** → pain, ↓ ROM



Brooker classification

5 Management

Prophylaxis (radiation, indomethacin)

Indomethacin	**75 mg/day for 10 days to 6 weeks** · proven for **distal biceps repair** (radioulnar synostosis) ✗ Wrong options: 7.5 mg
Radiation	single **700 cGy** · **4 h preop or within 72 h postop** · **<550 cGy ineffective** · blocks mesenchymal → osteoprogenitor differentiation ✗ Wrong options: 500 cGy · 2 wk preop · fractionated after
Bisphosphonates	not effective: only delay mineralization of osteoid · HO forms when stopped
Distal biceps	**single-incision** technique → lower synostosis rate than two-incision
Not prophylaxis	PTH 1-34, VAC chemo

Timing of excision & maturity

Established, symptomatic HO

- Immature (<6 mo, bone scan active, ALP/CRP raised) → **OBSERVE, PT, re-evaluate**
- Mature + severe ROM loss / function → **WIDE EXPOSURE + EXCISION (CT 3D first)**
  - Post-op → **INDOMETHACIN (5 d) and/or RT + EARLY GENTLE ROM**
- HO after THA/TKA, Brooker 3–4 → **EXCISION (± revision)**

Setting	Wait before excision
General trauma	**6 months**
Spinal cord injury	**1 year**
TBI	**1.5 years**
THA	6–18 months to mature

Maturity	**radiographs: sharp cortical margins** ✓ · marked ↓ bone scan activity + normal ALP (Orthobullets) · (✗ Wrong options: time alone · bone scan always negative · ALP (OB)) · some data: early = late results
Elbow HO 2 months after TBI with mild ulnar symptoms	**continue, re-evaluate in 3 months** ✗ Wrong options: excision ± transposition · RT · isolated release
Terrible triad, synostosis at 3 months	**await maturation → CT 3D in 3 months** ✗ Wrong options: indomethacin/RT now · radial head excision
Revision THA, 18 months, painful restricted hip	**excision** ✗ Wrong options: PT · alendronate · indomethacin · RT (none treat established HO)
Observation case	**HO 4 months after THA** (still maturing) ✗ Wrong options: mature lesions with functional loss

– – Active space – –

6 Complications

Surgical	haematoma/bleeding · **infection** (↑ in arthroplasty with HO) · fracture of osteoporotic bone · **AVN** with extensive stripping
Recurrence	∝ severity of neurological injury

How it's asked

"Highest correlation with severe HO after knee dislocation?"	→ **Injury Severity Score**
"Amputee at highest HO risk?"	→ **through the zone of injury after blast** (propane explosion, transfemoral)
"Traction pin causing HO?"	→ **distal femur** (quadriceps)
"NOT shown to increase HO?"	→ **open fractures** (✓ ventilation, SCI, blast amputation, burns)
"NOT a risk factor after THR?"	→ **fluorosis** (✓ previous HO, hypertrophic OA, DISH) **(TO)**
"Commonest complication of combined-approach acetabular fixation?"	→ **HO**
"Ankylosing spondylitis TKA – likely complication?"	→ **HO**
"DA THA 'bikini' vs longitudinal incision?"	→ **lower HO** ✗ Wrong options: less LFCN injury · more wound problems
"What could have prevented hip HO after head injury + fracture fixation?"	→ **700 cGy within 72 h of surgery**
"Prevent radioulnar synostosis after distal biceps repair?"	→ **indomethacin 75 mg daily × 6 wk**
"True about elbow HO 4 months post-injury?"	→ **radiographs assess maturity** ✗ Wrong options: early excision ↓ recurrence · ALP for maturity
"Painful Brooker 3–4 HO 18 months after revision THA?"	→ **excision**
"Elbow HO 2 months after TBI with mild ulnar paraesthesia?"	→ **re-evaluate in 3 months**
"Terrible triad, forearm stuck at 3 months?"	→ **allow maturation, CT 3D in 3 months**
"Which HO patient can simply be observed?"	→ **4 months after THA**
"Brooker classification is for?"	→ **HO around hip after THR (TO)**

– – Active space – –

IMPENDING FRACTURE & PROPHYLACTIC FIXATION

– Pathology · 27 / 66 –

1 Basics

Epidemiology & prognosis

Commonest site of all bone metastases	**spine**
Commonest site of pathologic fracture from metastasis	**proximal femur** (femur = commonest long bone; stress risers) ✗ Wrong options: distal femur · humerus · thoracic spine
Upper limb	20% of skeletal mets · **50% of those in the humerus**
Shortest survival after pathologic fracture	**lung** (and melanoma): median **<6 months**, 5-yr <5% ✗ Wrong options: thyroid · prostate · breast · renal

Why fix prophylactically

Prophylactic > fixing an actual fracture	shorter operation · less morbidity · quicker recovery
Fracture-risk predictors	significant **functional pain** · **>50% cortical destruction** · formal systems: **Harrington** and **Mirels**

4 Diagnosis & Classification

Mirels scoring

Purpose — **quantify risk of impending pathologic fracture** in a neoplastic bone lesion **(TO)**
✗ Wrong options: spinal operability · Scheuermann

Score	1	2	3
Site	Upper limb	Lower limb	**Peritrochanteric**
Pain	Mild	Moderate	**Functional**
Lesion	Blastic	Mixed	**Lytic**
Size (fraction of cortical thickness)	<1/3	1/3–2/3	**>2/3**

Threshold — **>8 → prophylactic fixation** (Orthobullets, TO) · "**8 or higher**" (OB) · one OB explanation says ">9" – read the options · **≤8 painful → radiation**

– – Active space – –

Worked examples

Case	Score	Action
Breast mets, proximal humerus, moderate pain, blastic, 1/3–2/3	**1 + 2 + 1 + 2 = 6**	**Radiation**
RCC, humerus, severe (functional) pain, lytic, >2/3	**1 + 3 + 3 + 3 = 10**	Fix
Myeloma, peritrochanteric, functional pain, lytic, >2/3	**3 + 3 + 3 + 3 = 12**	**Prophylactic fixation + RT**

Variable	Score		
	1	2	3
Pain	Mild	Moderate	Functional
Location	Upper extremity	Lower extremity	Peritrochanteric
Size	Less than ⅓	⅓ to ⅔	Greater than ⅔
Nature	Blastic	Mixed	Lytic

Mirels scoring system

Harrington criteria

Use prophylactic fixation of impending pathologic fracture of the **proximal femur (TO)**
✗ Wrong options: scoliosis · Blount stapling · proximal humeral fracture

Criterion
>50% destruction of diaphyseal cortices
>50–75% destruction of metaphysis (>2.5 cm)
Permeative destruction of subtrochanteric femur
Persistent pain after irradiation

Limitation (TO) not validated · proximal femur only

5 Management

Treatment algorithm

- **Impending pathologic fracture**
  - No proven primary with biopsy-proven bone mets → **BIOPSY FIRST**
    - Primary bone sarcoma → **CLOSE, REFER TO SARCOMA CENTRE (no nailing)**
  - Mirels ≤8, painful → **RADIATION**
  - Mirels >8 → PROPHYLACTIC FIXATION (protect whole bone) → **POST-OP RADIATION**

Biopsy may need a separate incision from nail incision · nailing a sarcoma **contaminates the whole bone** → loses limb salvage

Goals of fixation **immediate mobilization + weight bearing** · **protect the entire bone** · implant matched to prognosis

Prophylactic fixation by site

Site	Fixation
Proximal humerus (Bickel I)	**Endoprosthesis** (✗ Wrong options: IM nail (poor hold in metaphyseal flare) · plate (thin cortex))
Humeral diaphysis (Bickel II)	**IM nail** · resection + intercalary spacer · plate + screws (less preferred)
Distal humerus (Bickel III)	**Flexible nails** · elbow replacement
Femoral neck / head	**Cemented hemiarthroplasty** (fails by dislocation, infection)
Peri-/ subtrochanteric	**Cephalomedullary (IM) nail** (fails by nonunion, AVN) (✗ Wrong options: proximal femoral replacement (dissection, weakness, infection, dislocation) · curettage alone (stress riser))

Post-operative radiation ↓ pain, slows progression, treats residual tumour (✓ for myeloma lesion after fixation)

How it's asked

"Mirels criteria are used for?"	→ **quantifying impending fracture risk in a neoplastic lesion (TO)**
"Breast mets, proximal humerus, moderate pain, blastic, 1/3–2/3 — Mirels + next step?"	→ **6 → radiation** ✗ Wrong options: plate at 8 · nail at 7 · replacement
"RCC humeral lytic lesion >2/3, severe pain — Mirels?"	→ **10**
"Myeloma subtrochanteric lytic lesion, functional pain?"	→ **prophylactic internal fixation + radiation** (Mirels 12) ✗ Wrong options: RT alone · curettage · proximal femoral replacement
"Commonest site of pathologic fracture from lung mets?"	→ **proximal femur**
"Primary with shortest survival after pathologic fracture?"	→ **lung**
"Harrington criteria are used for?"	→ **prophylactic fixation of impending proximal femoral fracture (TO)**

– – Active space – –

INTRAMUSCULAR MYXOMAS

– Pathology · 28 / 66 –

1 Basics

Nature	**benign** soft-tissue tumour (mesenchymal) · **no malignant potential** · slow-growing, deep, **confined within skeletal muscle**
Demographics	**40–60** (40–70) · slight **female** predilection
Site	**thigh**, shoulder, buttock, upper arm, calf
Origin	primitive mesenchymal cells → benign fibroblasts that **lose the capacity to secrete collagen**

Associations

Mazabraud syndrome	**multiple intramuscular myxomas + (monostotic or polyostotic) fibrous dysplasia**
Myxoid liposarcoma	main differential

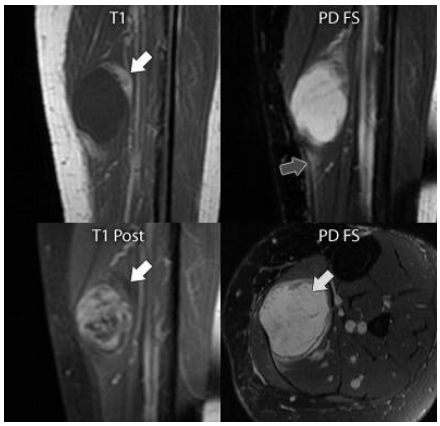
3 Presentation

Slowly growing mass · painful or painless · firm, mobile · deep to fascia

4 Diagnosis & Classification

Key radiology findings

MRI — **homogeneous** · **T1 dark**, **T2 bright** (myxoid) · **intramuscular** location · well circumscribed, no oedema or NV encasement · **rind of fat** at ≥1 pole (bright T1)



Homogeneous intramuscular myxoma on MRI

Histology

Bland, hypocellular myxoid stroma · undifferentiated **stellate cells** with thin collagen fibres · **no atypia**, low N:C ratio, **no atypical mitoses**

Differential

Lesion	Distinguishing
Intramuscular myxoma	**Intra**muscular, homogeneous, fat rind, little enhancement
Myxoid liposarcoma	**Inter**muscular · **intense diffuse enhancement** · "**pulmonary oedema**" pattern, lipoblasts, chicken-wire vessels
Neurofibroma	Follows nerve — **entering-and-exiting nerve sign** · ovoid/spindle cells in myxoid stroma
Ganglion	Dense fibrous wall around myxoid material, joint origin
Metastasis	Soft-tissue mets unusual

5 Management

Observation asymptomatic

Marginal excision **symptomatic** lesions · **no neoadjuvant chemo/RT** · recurrence + metastasis uncommon
✗ Wrong options: observation · PT when symptomatic

Excision margin terms

Margin	Plane	Example
Intralesional	Within pseudocapsule — highest recurrence	Select benign lesions
Marginal	Through **reactive zone / capsule**	✓ **Intramuscular myxoma**
Wide	Cuff of normal tissue	Sarcomas (**myxofibrosarcoma**), **solitary fibrous tumour**
Radical	Whole compartment	Highly aggressive, no limb-sparing margin

Not marginal excision ABC (curettage + adjuvant) · desmoid (often non-operative, infiltrative)

How it's asked

"40-yr-old, hip/thigh pain, homogeneous intramuscular T2-bright mass, bland myxoid histology?"	→ **surgical (marginal) excision** ✗ Wrong options: observation · neoadjuvant chemo · adjuvant RT
"56-yr-old, slowly enlarging medial thigh mass, history of thyroid cancer?"	→ **intramuscular myxoma** ✗ Wrong options: myxoid liposarcoma · neurofibroma · ganglion · metastasis
"Benign tumour treated with marginal excision?"	→ **intramuscular myxoma** ✗ Wrong options: desmoid · myxofibrosarcoma · solitary fibrous tumour
"Multiple myxomas + fibrous dysplasia?"	→ **Mazabraud syndrome**

– – Active space – –

LEIOMYOSARCOMA

– Pathology · 29 / 66 –

1 Basics

Nature	aggressive sarcoma from **smooth muscle cells lining small blood vessels** ("leio-" = smooth muscle)
Demographics	**5th–6th decades**
Bone	<100 reported cases (extra-facial) · **metaphysis of long bones** (also diaphysis) · femur, tibia, **ilium**, humerus

Genes & lab markers

Immunostains	**smooth muscle actin +**, **vimentin +**

3 Presentation

Symptoms	bone pain, palpable firm mass · ~6 months before diagnosis · weight loss
Pelvic masses	hard to see on inspection

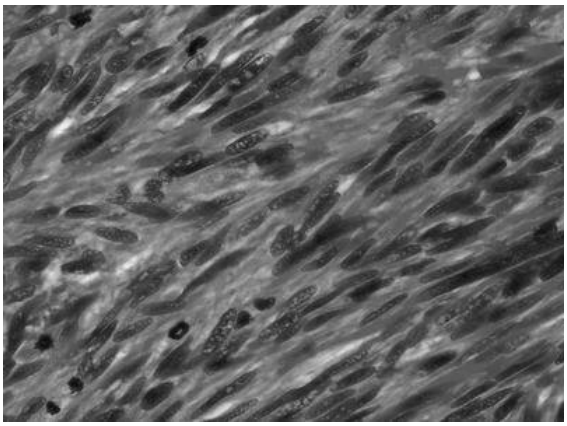
4 Diagnosis & Classification

Key radiology findings

X-ray	**purely osteolytic**, ill-defined, **moth-eaten / permeative** · intramedullary ± soft-tissue extension
MRI	**T1 dark (like muscle)** · T2 heterogeneous bright · diffuse enhancement
Staging	**chest X-ray + chest CT** (lung mets) · bone scan uptake

Histology

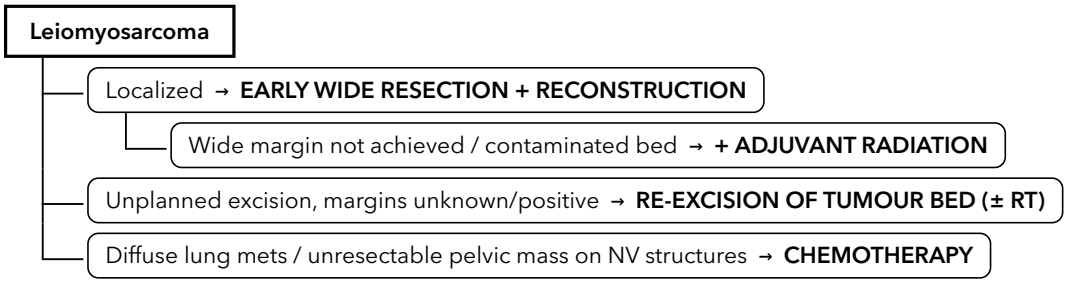
Spindle cell neoplasm (osseous = soft tissue form) · **cigar-shaped (blunt-ended) nuclei** · **interlacing fascicles** with parallel myofibrils · high N:C, mitoses



Spindle cells in fascicles with cigar-shaped nuclei

Round cells?	no: leiomyosarcoma is the **spindle-cell** exception among round-cell tumour lists

5 Management



Chemo/RT	variable response · chemo + surgery better than surgery alone · neoadjuvant/adjuvant role controversial
Unplanned excision of 3 cm superficial LMS	**repeat wide excision of tumour bed** · residual disease in up to **35%** · local recurrence **22% vs 7%** ✗ Wrong options: observation · RT alone · chemo

6 Complications

Prognosis (bone LMS)	**25% recurrence**, **25% metastasis** · **75% survival at 3 yr**

How it's asked

"35-yr-old, aggressive lytic pelvic lesion, smooth muscle actin +?"	→ **leiomyosarcoma** histology – interlacing spindle cells ✗ Wrong options: GCT · myeloma · OS · hyaline cartilage
"Outside simple excision of 3 cm thigh mass = leiomyosarcoma, margins unknown?"	→ **repeat wide excision of the tumour bed**
"Neoplasms with round cells EXCEPT?"	→ **leiomyosarcoma** (spindle cells, cigar nuclei)

– – Active space – –

LEUKEMIA

– Pathology · 30 / 66 –

1 Basics

Definition malignant proliferation of lymphocytes / precursors (or myeloid cells)

Commonest malignancy of childhood **leukaemia (ALL)**

Forms

Type	Cell	Age	Key features
ALL	Lymphoblasts	**Peak 4 yr** · **80%** of (childhood) leukaemia	**TdT +** · bone demineralization, **periostitis**, lytic lesions · **T-ALL** teens · **B-ALL**: **t(12;21) children**, **t(9;22) adults**
AML	Myeloblasts	**Older adults (5th–6th decade)**	**Auer rods** · marrow failure · **APL t(15;17)** disrupts retinoic acid receptor → **all-trans-retinoic acid** · **megakaryoblastic → Down syndrome**, <5 yr · **monocytic → gum infiltration**
CML	Mature granulocytes	5th–6th decade	**t(9;22) Philadelphia chromosome → BCR-ABL tyrosine kinase** (↑ division, ↓ apoptosis) · smear: immature granulocytes, basophilia, eosinophilia · **imatinib**
CLL	Naive B cells	5th–6th decade	Insidious · **smudge cells** · lymphadenopathy

Genes & lab markers

Marker	Leukaemia
TdT +	ALL
t(12;21)	B-ALL, children
t(9;22) BCR-ABL	CML · B-ALL adults
t(15;17) RARα	APL (AML)
Auer rods	AML
Smudge cells	CLL

3 Presentation

Symptoms **recurrent infections**, **bleeding**, **fatigue**, low-grade fever, **bone pain / limp**

Signs **hepatosplenomegaly**, **lymphadenopathy** (commonest in CLL)

4 Diagnosis & Classification

– – Active space – –

Key radiology findings

X-ray	diffuse osteopenia · **metaphyseal lucent bands** · periosteal reaction · lytic / mixed sclerotic-lytic · permeative destruction
MRI	scattered ill-defined medullary lesions · mimics **LCH and chronic osteomyelitis**
Vertebral fractures	**thoracic** commonest, then lumbar **(TO)**
Diagnosis	**peripheral smear + bone marrow biopsy** (lymphoblasts)

Differential

Lesion	Smear / histology
ALL	**Lymphoblasts**
AML	Myeloblasts + Auer rods, older
Chronic osteomyelitis	Neutrophilia, left shift
LCH	Eosinophilia, **CD1a, CD207 (langerin)**, Birbeck granules

5 Management

Chemotherapy	ALL (T and B) · may predispose to **pathologic fracture**
Targeted	**ATRA** for APL · **imatinib** for CML

How it's asked

"4-yr-old, fatigue, fevers, recurrent ear infections, limp, ill-defined medullary lesions, lymphoblasts on smear?"	→ **ALL** ✗ Wrong options: AML · CML · chronic osteomyelitis · LCH
"Most common location of vertebral fracture in leukaemia?"	→ **thoracic (TO)**
"Philadelphia chromosome?"	→ **t(9;22) BCR-ABL – CML** (imatinib)

– – Active space – –

LIPOMAS

– Pathology · 31 / 66 –

1 Basics

Epidemiology

Common benign tumour of **mature fat** · ~50% of soft-tissue neoplasms (OB)

Demographics **40–60** · slightly **male** · sedentary individuals

Site **subcutaneous** (upper back, thighs, buttocks, shoulders, arms) · deep/intramuscular (thigh, shoulder, arm) · **dorsum of foot** · **multiple in 5–10%**

Variants

Variant	Key features	Treatment
Spindle cell lipoma	Men **45–65**, neck/back/shoulder · mature fat + bland spindle cells, collagen, mast cells, myxoid · **CD34 +, S-100 –**	**Marginal excision** · rarely recurs
Pleomorphic lipoma	Middle age · lipocytes, spindle cells, **atypical giant cells** → mistaken for liposarcoma	Marginal excision
Angiolipoma	**Painful on palpation** · small nodules, upper limb · fat + nests of small arborizing vessels	Excision
Intramuscular lipoma	Often symptomatic · **lipoblasts + muscle infiltration** · may have infiltrative margins (✗ Wrong options: not malignancy) · small risk → liposarcoma	**Marginal resection**
Hibernoma	**Brown fat** · younger (**20–40**) · reddish-brown (vascular, mitochondria)	Excision

Genes & lab markers

Spindle cell lipoma **CD34 +, S-100 –**

3 Presentation

Painless, mobile, soft mass, long-standing, size plateaus · symptoms from pressure (sitting, shoe wear, tool use) · **angiolipoma painful**

4 Diagnosis & Classification

Key radiology findings

MRI (diagnostic) well-demarcated, **homogeneous, signal = subcutaneous fat on ALL sequences** · **T1 bright**, T2 bright, **fully suppressed on STIR / fat-sat** · **no enhancement**



Lipoma — homogeneous fat signal on T1

X-ray	radiolucent soft-tissue shadow · **mineralization → think synovial sarcoma**
CT	well-demarcated, like subcutaneous fat
Biopsy	usually **unnecessary** · core/incisional prone to **sampling error** in fatty tumours
Ganglion	fluid-bright on T2 (not fat signal) · commonest soft-tissue foot/ankle lesion

Histology

Mature adipocytes, eccentric nuclei, **bland acellular stroma**, no atypia · gross soft, lobular, yellow/white, ± encapsulated

Atypical lipomatous tumour

Deep (subfascial) or retroperitoneal lipomas	higher chance of being **atypical lipomatous tumour** · retroperitoneum = **well-differentiated liposarcoma** · extremity = **atypical lipoma**

MRI iso-intense with fat on all sequences → lipoma / ALT / low-grade liposarcoma → **no biopsy, marginal excision** · recurrence 25–50% at 10 yr · dedifferentiation rare

Intraosseous (calcaneal) lipoma

"Cockade sign"	well-defined lytic **calcaneal** lesion with **central calcification** → **pathognomonic** of intraosseous lipoma **(TO)** ✗ Wrong options: chondroblastoma · NOF · EG
Site	**calcaneus ~30%** (commonest) · subtrochanteric femur · around knee · ~40 yr **(TO)**
MRI	**high T1 and T2**, suppressed on fat-sat **(TO)**
DD	SBC/UBC, ABC, GCT, osteoblastoma, calcaneal pseudotumour **(TO)**

Milgram stage	Histology
I	Viable mature lipocytes
II	Necrotic foci, foamy macrophages, reactive bone
III	Complete necrosis, **calcification**, cysts (infarct-like)

– – Active space – –

5 Management

Fatty soft-tissue mass, MRI = fat on all sequences

- Painless, determinate benign → **OBSERVATION**
- Symptomatic, rapidly growing, deep/retroperitoneal → **MARGINAL EXCISION (no biopsy needed)**

Not indicated core / incisional biopsy · wide/radical resection · RT · chemo · amputation

6 Complications

Local recurrence **<5%** (Orthobullets) · 15–60% (OB review, lipomatous masses) · 25–50% at 10 yr for ALT

X vs Y

	Lipoma	Well-diff. liposarcoma / ALT	Myxoid liposarcoma
Depth	Subcutaneous mostly	**Deep**, retroperitoneal	Deep, intermuscular
MRI	= fat all sequences	Fat + thick septa/nodules	T2 bright, myxoid, enhances
Histology	Mature fat, no atypia	Atypical stromal cells, lipoblasts	Lipoblasts, chicken-wire vessels
Treatment	Observe / marginal	Marginal excision (extremity)	Wide resection + RT

How it's asked

"55-yr-old, posterior thigh fullness worse sitting, gluteal mass T1 bright, STIR dark, lipoblasts + muscle?"	→ **intramuscular lipoma** ✗ Wrong options: dedifferentiated liposarcoma · fibrosarcoma
"Forearm mass, T1 = fat, STIR not bright — histology?"	→ **lipoma** (mature fat) ✗ Wrong options: myxoid liposarcoma · chondrosarcoma · Ewing
"Symptomatic intramuscular arm lipoma, CT CAP negative?"	→ **surgical excision** ✗ Wrong options: core biopsy · RT
"Painful dorsal foot fatty mass, no enhancement?"	→ **lipoma, marginal excision** ✗ Wrong options: wide resection
"Enlarging painful forearm intramuscular lipoma?"	→ **marginal resection** ✗ Wrong options: core/incisional biopsy
"Large symptomatic axillary/chest wall mass iso-intense to fat?"	→ **marginal excision** ✗ Wrong options: biopsy · wide + RT
"Shoulder mass, CD34 + spindle cells in fat?"	→ **spindle cell lipoma → marginal excision**
"Cockade sign?"	→ **calcaneal (intraosseous) lipoma (TO)**

"Heel pain, lucent calcaneal lesion with central calcification, T1 + T2 bright?"	→ **intraosseous lipoma (TO)** ✗ Wrong options: ABC · SBC · osteoblastoma	– – Active space – –

– – Active space – –

LIPOSARCOMA

– Pathology · 32 / 66 –

1 Basics

Epidemiology

Rank	**2nd commonest soft-tissue sarcoma in adults** (UPS 1st) · ~20% of adult STS
Demographics	**50–80** · myxoid in younger adults (30–60, peak 5th decade) · **male** > female
Site	**deep to fascia** (almost never subcutaneous) · **75% extremities** (lower > upper) · **retroperitoneum**

Lipomas do not predispose to liposarcoma

Hallmark cell

Lipoblast	**signet-ring** cell, indented nucleus, vacuolated cytoplasm · distinguishes from lipoma and UPS

Subtypes & grade

Subtype	Grade	Share	Metastasis
Well-differentiated / ALT	**Low**	40–45% (OB)	**<1%** (ALT ~0%)
Myxoid	Low–**intermediate**	**~50% – commonest** (Orthobullets/OB)	**10–30%**
Round cell	High	(spectrum with myxoid)	>50%
Pleomorphic	High	**<5%**, most aggressive	>50%
Dedifferentiated	High	<10% · "commonest" per one OB explanation	**~50%** (highest)

Genes & lab markers

Marker	Subtype
MDM2 + CDK4 amplification (12q13–15, **ring / giant marker chromosome 12**)	**ALT / WDL** and **dedifferentiated** · CDK4 sens 86%/ spec 89%, MDM2 86%/74%
t(12;16) FUS(TLS)–DDIT3(CHOP)	**Myxoid** (± round cell) · rarely t(12;22)
p53 mutation, complex karyotype	**Pleomorphic**

Best test to separate ALT from lipoma	**FISH for MDM2 amplification** ✗ Wrong options: MRI · S-100 (MPNST/melanoma) · CD138 (myeloma) · CD20 (lymphoma)

MDM2 negative regulator of p53

Genetic finding	Tumour (distractors)
MDM2/CDK4	✓ ALT/WDL
t(12;16)	✓ Myxoid liposarcoma
t(2;13)	Alveolar rhabdomyosarcoma
t(X;18) SYT–SSX	Synovial sarcoma
ETV6–NTRK3	Infantile fibrosarcoma
t(12;22)	Clear cell sarcoma
t(11;22)	Ewing

3 Presentation

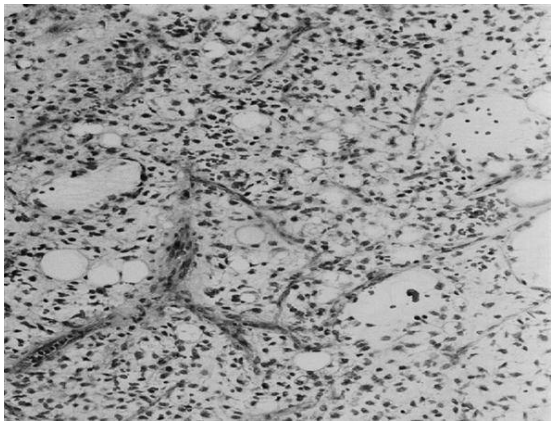
Slow-growing, painless, deep mass · often 10–20 cm · incidental · mass effect (sciatic paraesthesia, retroperitoneal)

4 Diagnosis & Classification

Key radiology findings

MRI with contrast best characterization (sens 100%, spec 71%, PPV 100%)

Subtype	MRI
ALT / WDL	Like lipoma – fat signal (Orthobullets: **bright T1, dark T2**) · **thick (>2 mm), irregular septa / stranding**, nodules, mild enhancement, oedema → dedifferentiation focus
Myxoid	Heterogeneous, T2 bright, little fat (~10%) · **intense diffuse enhancement** (vs myxoma)
High grade	Like other sarcomas – **T1 dark, T2 bright** (only ~10% fat)



Myxoid liposarcoma histology

X-ray soft-tissue mass · calcification/ossification in WDL

Biopsy areas **not consistent with fat** on MRI must be biopsied

– – Active space – –

Histology

Subtype	Histology
ALT / WDL	Mature fat with **atypical stromal cells**, scattered lipoblasts, minimally cellular · lipoma-like, sclerotic, inflammatory variants
Myxoid	Proliferating lipoblasts (signet ring at lobule edges) in **myxoid stroma** + **delicate plexiform "chicken-wire" capillaries** · "**pulmonary oedema**" pattern
Round cell	Small round blue cells
Pleomorphic	**Giant lipoblasts, bizarre nuclei**, eosinophilic giant cells (UPS-like + lipoblasts)
Dedifferentiated	ALT/WDL transformed to high-grade sarcoma

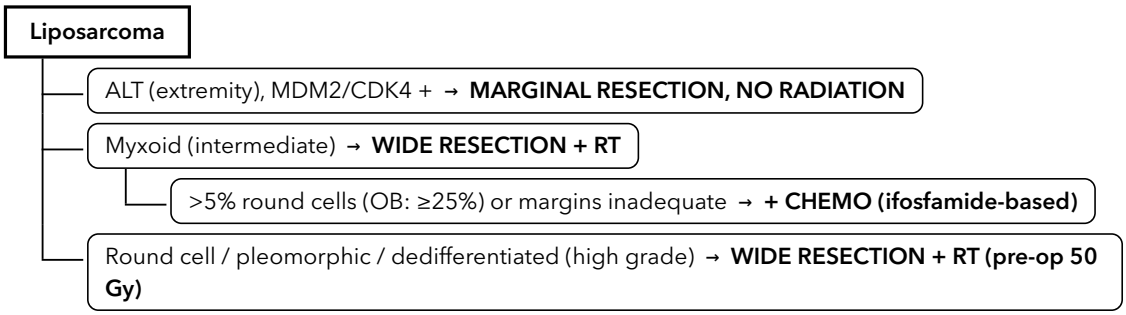
Staging

Myxoid liposarcoma – extrapulmonary spread lungs + **abdomen, retroperitoneum, spine**, other extremities' soft tissue, bone marrow

Myxoid staging/ surveillance **CT chest/abdomen/pelvis + whole-spine MRI** + MRI of primary site
✗ Wrong options: chest-only · PET (misses spine) · brain MRI · sentinel node · marrow biopsy

Other STS chest CT

5 Management



ALT/WDL **marginal resection without RT** (OB) · Orthobullets lists WDL for wide resection + RT (vs ALT marginal) – read the options ·
✗ Wrong options: amputation · chemo · RT alone · wide resection for extremity ALT

Myxoid **most chemo- and radiosensitive** liposarcoma

Radiation **50 Gy preop**, **66 Gy postop** standard for STS · ↓ local recurrence

Technique tourniquet without exsanguination · frozen section margins · NV resection only if bypass possible and morbidity acceptable

Soft tissue sarcoma treatment (margins, radiation, staging)

Most important factor preventing local recurrence **adequacy of surgical margin**
✗ Wrong options: chemo · RT alone

Large (>5 cm), deep, high-grade STS **wide excision + radiation** (pre- or post-op)

Pleomorphic liposarcoma, no mets — **neoadjuvant RT 50 Gy → wide resection**
✗ Wrong options: RFA · resection alone · chemo-based · disarticulation

– – Active space – –

6 Complications

Mass effect NV bundles, retroperitoneum, peritoneum

Subtype	Local recurrence	Dedifferentiation	Survival
ALT	**10–50% over 10 yr**	**3–7%** (2% extremity)	~100%
WDL (central)	High	**20%**	~80% 5-yr
Intermediate/high	~20%	–	50–80% 5-yr

X vs Y

	ALT	Lipoma
MDM2/CDK4	**+**	–
Septa	Thick (>2 mm), nodular	Thin
Depth	Deep	Usually subcutaneous
Treatment	Marginal excision	Observe / marginal

	Myxoid liposarcoma	Intramuscular myxoma
Location	**Inter**muscular	**Intra**muscular
Cellularity	Cellular, **lipoblasts**, chicken-wire vessels	**Hypocellular**, bland
Enhancement	Intense	Little

How it's asked

"Thigh fatty tumour, MDM2 + CDK4 +?"	→ **atypical lipomatous tumour** ✗ Wrong options: lipoma · RMS · LMS · myxofibrosarcoma
"Genetic basis + treatment of thigh WDL?"	→ **MDM2 amplification; marginal resection without radiation**
"Gluteal ALT, MDM2 +?"	→ **marginal resection** ✗ Wrong options: wide · RT · chemo
"Best test to separate ALT from lipoma?"	→ **FISH for MDM2**
"Feature allowing marginal excision alone?"	→ **MDM2 + CDK4 amplification** ✗ Wrong options: t(12;16) · t(2;13) · ETV6-NTRK3 · SYT-SSX
"Lipomatous tumour with highest metastatic rate?"	→ **dedifferentiated liposarcoma**
"Risk of ALT dedifferentiating?"	→ **3–7%**
"Thigh mass, signet-ring lipoblasts in myxoid stroma — translocation?"	→ **myxoid liposarcoma, t(12;16)** ✗ Wrong options: alveolar RMS · synovial t(X;18)

– – Active space – –

"33-yr-old, thigh mass, 12;16 translocation?"	→ **myxoid liposarcoma** ✗ Wrong options: WDL · round cell · fibrolipomatous hamartoma
"Posterior thigh mass with sciatic symptoms, chicken-wire vessels?"	→ **myxoid liposarcoma** ✗ Wrong options: synovial sarcoma · MFH · fibrosarcoma
"Myxoid liposarcoma after resection — extra staging?"	→ **whole-spine MRI**
"Myxoid liposarcoma surveillance?"	→ **MRI of thigh + CT chest/abdomen/pelvis**
"Slide warranting marginal resection alone?"	→ **well-differentiated liposarcoma / ALT** ✗ Wrong options: myxoid · dedifferentiated · UPS · pleomorphic
"Thigh mass following fat with stranding + enhancement?"	→ **liposarcoma** ✗ Wrong options: synovial sarcoma (cystic, no fat)
"Most important factor against local recurrence in UPS?"	→ **adequate surgical margin**
"Large, deep, high-grade dedifferentiated liposarcoma?"	→ **wide excision + radiation**
"15 cm pleomorphic liposarcoma, p53 +, t(12;16) −?"	→ **neoadjuvant RT (50 Gy) → wide resection**

LYMPHOMA

– Pathology · 33 / 66 –

1 Basics

Epidemiology

Primary lymphoma of bone (PBL)	**very rare** · 2–3% of primary bone malignancies (OB) · 3–7% (TO, osteosarcoma chapter)
Extranodal	10–35% of NHL patients
Demographics	**males** · all ages, **35–55** commonest (median 45–60)
Site	bones with **persistent red marrow**: **pelvis, spine, ribs** · knee (distal femur, proximal tibia), **proximal femur**, shoulder girdle · **diaphysis** of long bones
Risk factors	immunodeficiency (HIV, hepatitis) · viral/bacterial infection

Forms

Form	Definition
Primary lymphoma of bone	Solitary site, no other disease for **6 months** · mostly **non-Hodgkin B-cell** · **better prognosis** than secondary bone involvement
Multiple bony sites	No visceral disease
Bone + soft tissue	–

Subtype **diffuse large B-cell lymphoma (DLBCL)** = **70–80%** of bone lymphomas

Genes & lab markers

Immunostains + **CD20**, **CD45 (leukocyte common antigen)**, CD79a, BCL-6 (B cell) · **CD3 –**

Negative **CD99 –**, **no t(11;22)** (vs Ewing) · reticulin + (vs Ewing –)

Stain	Tumour
CD20 + CD45	✓ **Lymphoma**
CD99 + FLI-1 (EWSR1)	Ewing
CD138	Myeloma
CD31 + CD34	Vascular tumours
Desmin + MyoD1 (myogenin)	**Rhabdomyosarcoma**

3 Presentation

Pain unrelieved by rest · **B symptoms** – fever, **night sweats**, weight loss, fatigue

Pathologic fracture **25%**

Mass warm, swollen, large **soft-tissue mass** · spinal cord compression

Labs may mimic infection (mild ↑ ESR/CRP, lymphocytosis)

– – Active space – –

4 Diagnosis & Classification

Key radiology findings

X-ray — **large ill-defined, diffuse, permeative/moth-eaten lytic** lesion, subtle mottling · **diaphysis** · cortical thickening 25% · may be **sclerotic** · **"ivory" vertebra** · multiple sites · often **subtle/unremarkable**



Diffuse lytic mottled lesion of the femur

MRI — **extensive marrow involvement + large soft-tissue mass out of proportion to bone destruction** · T1 dark, T2 bright

Bone scan — intensely positive · **PET/CT** – staging + follow-up

CT chest/abdomen/pelvis — staging

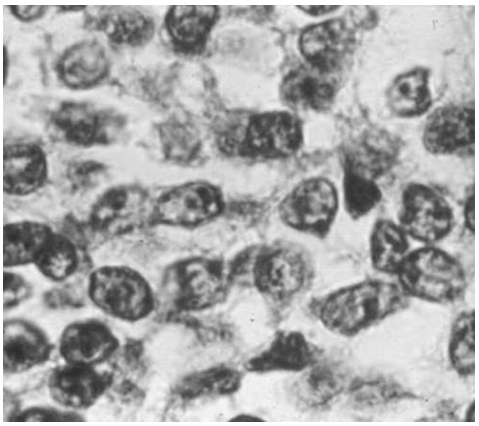
Bone marrow aspiration + biopsy — **required for staging** · **before any definitive bony stabilization** when no known primary
✗ Wrong options: chemo first · resection · prophylactic nail first

Moth-eaten (type 2) margins (TO) — myeloma, mets, infection, EG, OS, chondrosarcoma, **lymphoma** · 45-yr-old, proximal humeral path fracture, moth-eaten → **lymphoma**

Histology

Mixed small round blue cells of **different sizes and shapes** · **diffuse infiltration** of trabeculae (not nodular) · little stroma

Needle biopsy — often non-diagnostic (**crush artefact**)



Mixed small round blue cell infiltrate

Differential

	Destructive, young	Malignant, older	Small cells	Chemo alone
Lymphoma	✓	✓	✓	✓
Leukaemia · Ewing	✓	–	✓	–
OS · EG · osteomyelitis · desmoplastic fibroma	✓	–	–	–
Metastasis	–	✓	✓	–
Myeloma	–	✓	–	✓
Chondrosarcoma · MFH · secondary sarcoma	–	✓	–	–
RMS · neuroblastoma	–	–	✓	–

Ewing vs lymphoma Ewing younger, CD99 +, t(11;22) · lymphoma older, CD20/CD45 + · TB → caseating granulomas · osteomyelitis → mixed inflammatory cells

5 Management

Treatment — chemotherapy (R-CHOP) & fixation

Primary lymphoma of bone

- First → **BONE MARROW BIOPSY + PET/CT (staging)**
- All → **MULTI-AGENT CHEMO (R-CHOP) ± LOCAL RT (persistent disease)**
- Impending / actual pathologic fracture → **STABILISE (IM nail / hemiarthroplasty) then chemo ± RT**

Chemo **CHOP: cyclophosphamide, doxorubicin (hydroxydaunorubicin), vincristine (Oncovin), prednisone ± rituximab (R-CHOP)** · chemo alone effective for most · **70% 5-yr survival** in disseminated disease

Radiation adjunct for local control of persistent disease · RT alone insufficient · routine RT in R-CHOP era uncertain

Surgery **only to stabilise** pathologic / impending fractures (prophylactic IM nail) · surgery mustn't delay chemo

Femoral neck lesion found at hemiarthroplasty (frozen section = lymphoma) **complete hemiarthroplasty, then staging + chemoradiotherapy**
✗ Wrong options: close wound + chemo · radical resection + megaprosthesis · marrow transplant · disarticulation

Not indicated wide resection / limb salvage · amputation · hemipelvectomy · marrow transplant (relapse only) · neoadjuvant chemo + resection (OS/Ewing protocol)

6 Complications

Prognosis **primary bone lymphoma better** than secondary bone involvement

– – Active space – –

X vs Y

	Lymphoma of bone	Ewing	Myeloma
Age	35–55	5–25	>50
X-ray	Permeative, mottled, ± sclerotic, big soft-tissue mass	Permeative, onion skin	**Punched-out**, cold bone scan
Marker	**CD20, CD45**	**CD99**	**CD138**
Treatment	**R-CHOP ± RT**	VAC/IE + surgery ± RT	Chemo ± RT

How it's asked

"45-yr-old, destructive lesion + soft-tissue mass, round blue cells, CD99 –?"	→ **lymphoma of bone** ✗ Wrong options: Ewing · chondrosarcoma · adenocarcinoma · OS
"32-yr-old distal femur, sheets of blue cells, no t(11;22)?"	→ **lymphoma** ✗ Wrong options: Ewing · bacterial/TB osteomyelitis · OS
"50-yr-old, tibial lesion, B symptoms, large soft-tissue mass – stains?"	→ **CD20 + CD45** ✗ Wrong options: CD31/CD34 · CD99/FLI-1 · CD138 · desmin/MyoD1
"48-yr-old, fevers, weight loss, CD20/CD45 – histology?"	→ **small round blue cells**
"76-yr-old proximal femur, PET shows single site, prophylactic fixation + chemo – supporting pathology?"	→ **large atypical lymphoid cells CD20 +, CD79a +, BCL-6 +, CD3 –**
"Pathologic femoral neck fracture through lymphoma at hemiarthroplasty?"	→ **hemiarthroplasty, then staging + chemoradiotherapy**
"Severe thigh pain, extensive femoral lymphoma?"	→ **IM nail + multi-agent chemo ± RT** ✗ Wrong options: AKA · wide resection + prosthesis
"Knee pain, failed antibiotics, biopsy = lymphoma?"	→ **multi-agent chemo ± local RT**
"Iliac DLBCL, PET no other lesions?"	→ **CHOP chemotherapy** ✗ Wrong options: marrow transplant · CT CAP for primary · hemipelvectomy · RT alone
"40-yr-old, distal thigh lytic lesions, weight loss, no known primary – next?"	→ **bone marrow aspiration + biopsy** ✗ Wrong options: chemo · resection · nail first

MALIGNANT PERIPHERAL NERVE SHEATH TUMOR

– Pathology · 34 / 66 –

1 Basics

Synonyms	**neurofibrosarcoma**, malignant schwannoma · high-grade STS from a **nerve or pre-existing neurofibroma**
Age	solitary **30–55** · with **NF1 20–40**

Diagnosis & NF1 association

NF1	most cases · **4% of NF1 patients** develop MPNST · **autosomal dominant** (17q11.2) ✗ Wrong options: AR · X-linked

Genes & lab markers

S-100 + (diffuse) · **keratin –**

S-100 positive tumours	**melanoma, nerve sheath tumours, clear cell sarcoma** (± synovial sarcoma) · chondroblastoma, chondroid lesions

3 Presentation

Mass on a **large nerve** – **sciatic, brachial plexus**, femoral

Neurologic	motor + sensory deficit, radicular pain · **pressure → dysaesthesia** distally
NF1 red flag	a neurofibroma that **enlarges** and becomes **constantly painful**

4 Diagnosis & Classification

Key radiology findings

X-ray	normal / non-specific soft-tissue mass
MRI	**T1 low, T2 high** (like other high-grade sarcomas) · traced to a major nerve · oedema, **rim enhancement (necrosis)** · serial MRI: enlargement of a benign nerve sheath lesion = malignant change
PET	**SUV >2.0** → malignant

– – Active space – –



MPNST arising from the sciatic nerve

Bone scan mildly positive

Histology

Atypical pleomorphic spindle cells with wavy nuclei resembles **fibrosarcoma** · hypocellular stroma areas

	MPNST	Schwannoma	Neurofibroma
Cells	**Pleomorphic, atypical** spindle cells	Bland, Antoni A/B	Bland, wavy nuclei, mast cells
MRI	Necrosis, oedema, rim enhancement	**Target sign**	Entering-exiting nerve, target sign
S-100	+	+	+

5 Management

Wide resection including the entire affected nerve + **preoperative (neoadjuvant) radiation** – treated as high-grade sarcoma

Chemotherapy not useful

6 Complications

5-yr survival **solitary 75%** · **NF1-associated 30%** (Orthobullets) · one OB-cited series: NF1 not independently worse

Poor prognosis size, mitotic rate, need for amputation

How it's asked

"45-yr-old, radicular buttock pain, S-100 + pleomorphic spindle cells?"	→ **MPNST** ✗ Wrong options: schwannoma (bland, target sign) · liposarcoma · GCT tendon sheath
"NF patient, enlarging painful posterior thigh mass on sciatic nerve, rim enhancement, PET SUV >2?"	→ **MPNST** ✗ Wrong options: benign PNST · MFH · liposarcoma · synovial sarcoma

"S-100 + spindle-cell tumour of femoral nerve — associated condition's inheritance?"	→ **autosomal dominant** (NF1)
"Treatment?"	→ **wide resection of tumour + entire nerve + preop RT** ✗ Wrong options: chemo

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MELANOMA

– Pathology · 35 / 66 –

1 Basics

Epidemiology

Nature	aggressive skin malignancy of **melanocytes** (from **neural crest**, deepest epidermal layer)
Demographics	**50–70** · **M:F 1.2:1**
Hand/foot sites	**thumb > great toe > index finger** · sun-exposed areas
Subtype	**acral lentiginous melanoma** (palms, subungual) · **subungual melanoma** = ALM subtype

Risk factors

Sun exposure (UV suppresses immunity, DNA damage) · **family history** · fair skin, blue eyes, freckling · **xeroderma pigmentosum** · FAMM syndrome · multiple / **dysplastic naevi (precursor)** · immunosuppression

Prevention	sunscreen, avoid sun

Genes & lab markers

S-100 + (melanoma)

LDH ↑	predicts poor prognosis · **AST/ALT** – liver mets · CBC

3 Presentation

Pigmented lesion with recent change in shape or size · itching, bleeding, ulceration

ABCDE	**A**symmetry · **B**order irregularity · **C**olour variation · **D**iameter (>6 mm suspicious; <6 mm benign) · **E**levation / enlargement
Subungual	dystrophic nail prone to trauma → often blamed on **injury**
Hutchinson sign	brown-black pigment extends from nail bed/plate onto **cuticle and nail folds** ± nail lifting
Longitudinal melanonychia – malignant features	band **>6 mm**, **single nail**, wider proximally, colour heterogeneity, blurry borders, pigment into nail fold

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Subungual melanoma with Hutchinson sign

4 Diagnosis & Classification

Key radiology findings

CXR lungs often first metastatic site · **US** – nodes · **PET/CT** – metastases

Biopsy **full-thickness biopsy of the nail matrix** after nail plate removal
✗ Wrong options: trephination · antibiotics · no work-up

Histology

Melanocytes with marked atypia + dermal invasion · vacuolated cytoplasm · hyperchromatic nuclei, prominent nucleoli

Breslow thickness

Stage	Thickness
1	**≤0.75 mm**
2	0.76–1.5 mm
3	1.51–4 mm
4	**>4 mm**

Clark level

Level	Invasion
I	Epidermis only (in situ)
II	Papillary dermis
III	Papillary dermis to papillary–reticular interface
IV	Reticular dermis
V	Subcutaneous tissue

Differential

General naevi · seborrhoeic keratosis · basal cell carcinoma

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Subungual	**trauma / subungual haematoma** (contained under plate, needs trauma) · **onychomycosis** (thickening, yellow-brown) · psoriasis (pitting) · SCC (no pigment) · splinter haemorrhages (endocarditis → echo) · glomus tumour (painful, blue) · DFSP (trunk)

5 Management

Margins by thickness

Lesion	Treatment
<1 mm	Local resection, **1 cm** margin
1–2 mm	Resection **1–2 cm** + **sentinel node biopsy** → radical node dissection if positive
Thick	**>3 cm** margin (OB)
Metastatic	Resection + node dissection + chemo

Subungual melanoma

Amputation	**thumb: IP joint disarticulation / through proximal phalanx** · finger: PIP level · **+ sentinel node biopsy**
Distal = proximal	distal amputation with adequate margins has **same recurrence and survival** as MCP/ray/CMC amputation → preserves function
Not used	neoadjuvant chemo to downsize · 1–2 **mm** margin · Mohs (controversial; for SCC/BCC) · isolated limb perfusion routinely

6 Complications

Recurrence	usually **regional lymph nodes**
Prognosis	**depth = most important factor** · <0.7 mm → **96%** survival · >4 mm → **47%**
Poor prognosis	deep lesion · **male** · **neck/scalp** · positive nodes/mets · **ulceration** · ↑ LDH
Subungual	**5-yr survival 40–60%** (late diagnosis, thicker, ulcerated)
Bone metastasis	melanoma + lung: worst survival after pathologic fracture (<6 months)

How it's asked

"62-yr-old, enlarging brown nail streak on ring finger – biopsy to exclude?"	→ **melanoma** ✗ Wrong options: DFSP · endocarditis · glomus · epithelioid sarcoma
"Persistent thumbnail pigment after hammer blow, Hutchinson sign – next step?"	→ **full-thickness nail matrix biopsy** ✗ Wrong options: trephination · antibiotics · IPJ disarticulation first
"65-yr-old, thumbnail discoloration onto folds, no trauma?"	→ **melanoma** ✗ Wrong options: SCC · onychomycosis · haematoma · psoriasis

"Thin (0.6 mm) subungual thumb melanoma — best cure with function?"	**→ IP joint disarticulation + sentinel node biopsy** ✗ Wrong options: MCP disarticulation · ray amputation + limb perfusion · Mohs · chemo + 1–2 mm margin	– – Active space – –
"Most important prognostic factor?"	**→ depth (thickness)**	

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MELORHEOSTOSIS

– Pathology · 36 / 66 –

1 Basics

Nature	rare, **benign, painful** developmental dysplasia of cortical bone · formation of **periosteal (or endosteal) new bone** – "flowing hyperostosis"
Demographics	**before age 40** · **no sex predilection**
Genetics	**non-hereditary**
Site	**lower extremities** commoner · any bone (long bones, ulna, hand, foot/metatarsals)

Genes & lab markers

ALP	often **elevated**

3 Presentation

Symptoms	**pain**, **reduced ROM**, **joint contractures**, swelling
Skin	**fibrosis with induration and erythema** over the lesion
Incidental	may be found on unrelated imaging (asymptomatic)

4 Diagnosis & Classification

Key radiology findings

X-ray (diagnostic)	**"dripping candle wax"** – dense **cortical hyperostosis flowing along the cortex** · may **cross joints** → mechanical block



Dripping candle-wax hyperostosis

CT	non-aggressive sclerosis, may **obliterate the medullary canal**
Bone scan	**hot** (↑ osteoblastic activity) – not a sign of malignancy here

Histology

Normal Haversian systems, enlarged trabeculae · bland osteoid · **no atypia, no mitoses**

Differential

Lesion	Distinguishing
Melorheostosis	Flowing candle-wax cortical hyperostosis, bland bone
Metastatic prostate cancer	Blastic lesions, carcinoma on histology (↑ PSA is a distractor)
Periosteal OS	Surface sunburst, malignant osteoid
Spindle cell sarcoma	Malignant spindle cells
Stress fracture	Fracture line, callus

Association melorheostosis can underlie **secondary osteosarcoma** (rare)

5 Management

Melorheostosis

- Asymptomatic / incidental → **OBSERVATION (X-ray diagnostic, no biopsy)**
- Mild symptoms, adequate motion → **SYMPTOMATIC CARE + BISPHOSPHONATES (zoledronate)**
- Severe contracture, immobility, pain → **HYPEROSTOTIC BONE RESECTION + CONTRACTURE RELEASE (± osteotomy)**

Bisphosphonates help **pain and swelling**
✗ Wrong options: calcitonin · rituximab · tamoxifen (desmoid) · stereotactic radiation

How it's asked

"55-yr-old, foot pain, sclerotic 2nd metatarsal, hot scan, bland bone, ↑ PSA?"	→ **melorheostosis** ✗ Wrong options: metastatic prostate · periosteal OS · spindle cell sarcoma · stress fracture
"21-yr-old, incidental flowing hyperostosis of proximal femur?"	→ **observation** ✗ Wrong options: needle/incisional biopsy · curettage · resection
"Elbow pain, swelling, ROM 30–90°, candle-wax hyperostosis — beneficial drug?"	→ **zoledronate**

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METASTATIC DISEASE OF EXTREMITY

1 Basics

Epidemiology

Commonest cause of a destructive bone lesion in adults · commonest tumour of bone · aggressive lytic lesion >40 yr: **metastasis > myeloma > lymphoma > Paget sarcoma > post-radiation sarcoma**

Solitary bone lesion >40 yr	**500×** more likely metastatic carcinoma than primary bone tumour
Bone = 3rd commonest metastatic site (after lung, liver)	
Age	>50
Primaries (BLT + Kosher Pickle)	**Breast** (commonest in women; overall), **Lung**, **Thyroid**, **Kidney**, **Prostate** (commonest in men) · overall **breast > prostate > lung (TO)** · men: prostate > lung · women: breast > lung · children: **neuroblastoma**
Breast	bone is the first metastatic site in 40–75% **(TO)**

Sites

Commonest site of bone metastasis	**spine (thoracic) (TO)** > pelvis > proximal femur > humerus · then ribs, skull · red marrow
Pathologic fracture	**proximal femur** commonest (then proximal humerus) · femoral neck 50%, subtrochanteric 30%, peritrochanteric 20% (OB: 50% neck, 30% subtroch, 20% intertroch) · TO: femoral neck commonest, transverse lines
Acral (distal to elbow/ knee)	rare · **lung** commonest (then kidney) → **arterial tree** spread · **tibia** commonest bone **(TO)** · poor prognosis
Occult primary	biopsy shows carcinoma but CT CAP + bone scan negative → **lung** ✗ Wrong options: colon, kidney, prostate, breast
Batson vertebral plexus	**valveless** venous plexus → lung, kidney, breast, prostate, thyroid to **vertebral bodies, pelvis, skull, proximal girdles** · breast → azygos → Batson in **thoracic** spine
Spine	vertebral body involved before pedicle, but **pedicle destruction ("winking owl")** is commonest first plain-film sign **(TO)**

Biology of metastasis (integrins, MMPs, RANKL, endothelin)

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Step	Molecule
Release from primary	↓ **E-cadherin** (cell adhesion lost) · **PDGF** → migration
Intravasation, avoid immune surveillance	–
Homing to bone	**CXCL12** (marrow stromal chemokine)
Attachment to endothelium / bone	**Integrin** (tumour cell CAM)
Extravasation, ECM degradation	**MMPs** (collagenases, gelatinases, stromelysins)
Angiogenesis	**VEGF**
Survival	Genomic instability, ↓ **apoptosis** · **thrombospondin inhibits** tumour growth

Necessary steps intravasation, immune evasion, target localization, angiogenesis
✗ Wrong options: **direct osteoclast stimulation** (indirect, via RANKL) · **increased apoptosis**

Lytic "vicious cycle" tumour **PTHrP** (+ IL-6) → **osteoblasts release RANKL** → RANK on osteoclast precursors → osteolysis → **TGF-β, IGF-1, Ca^{2+}** released → more PTHrP · blocked by bisphosphonates, denosumab, OPG

Blastic **endothelin-1** → ET-A receptor on osteoblasts, ↓ **DKK-1** → WNT ↑ · also TGF-β, BMPs **(TO)**

Lytic vs blastic

Pattern	Primaries
Lytic	**Kidney, thyroid, lung** (also GI, melanoma)
Blastic	**Prostate (90%)**, seminoma, carcinoid, medulloblastoma · breast 60% (OB)
Sclerotic (TO list)	Prostate, **bladder**, carcinoid, **medullary thyroid carcinoma** · lytic: **myeloma, RCC, thyroid, lung** (TO)
Mixed	**Breast** – lytic, blastic or mixed ("all of the above") **(TO)** · commoner lytic (TO)

Pulsatile metastases **follicular thyroid + RCC** (most pulsatile tumours are mets > ABC > GCT) **(TO)**

Bone-to-bone metastasis Ewing > osteosarcoma **(TO)**

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Genes & lab markers

Test	Looks for
CBC, ESR, BMP, LFTs, PT/PTT, Ca, phosphate	Baseline, hypercalcaemia
SPEP / UPEP	**Myeloma**
PSA	Prostate
LDH	Lymphoma of bone
ALP	Paget, bone turnover
Urinalysis	Renal cell
Acid phosphatase · urine hydroxyproline	Prostate · massive destruction **(TO)**
β2-microglobulin, immunoglobulins (TO)	Myeloma
Calcitonin (TO)	**Medullary thyroid** carcinoma
CEA (TO)	Breast, pancreas, gallbladder
Thyroglobulin (TO)	Thyroid cancer
LDH (TO)	Myeloma, lymphoma

Histology IHC — **keratin**, **CK7** (breast, lung), **TTF-1** (lung) · RCC: **clear cell change, pseudoalveolar** pattern

Breast receptor status — **ER, PR, HER2/neu** guide therapy (tamoxifen/letrozole, trastuzumab)
✗ Wrong options: RANKL · BRCA · CA-125 · t(11;22)

3 Presentation

Pain — dull, gradual; **night pain** (tumoral) or **weight-bearing** (mechanical) · weight loss, mass

Pathologic fracture — **8–30% at presentation** · **90% need surgery** (rarely heal)

Atraumatic lesser trochanter avulsion in an adult — **malignancy until proven otherwise**

Metastatic hypercalcaemia (emergency) — confusion, weakness, **polyuria/polydipsia**, N/V, dehydration → **IV fluids**, loop diuretic, **bisphosphonate**

Complications (Coleman) — hypercalcaemia 19%, pathologic fracture 19%, cord compression 10%, marrow failure 9%

4 Diagnosis & Classification

Workup of solitary lytic lesion – staging & biopsy before fixation

New painful/destructive bone lesion, >40 yr

- 1. History + exam (breast, thyroid, prostate) + LABS (SPEP/UPEP, PSA…)
- 2. X-ray whole bone + CXR → **CT CHEST/ABDOMEN/PELVIS + WHOLE-BODY BONE SCAN (± MRI lesion)**
- 3. Primary found + multiple lesions → **biopsy most accessible lesion if needed**
- 4. Solitary lesion (even with remote/known cancer) → **BIOPSY (treating team, intramuscular tract)**
  - Carcinoma confirmed → FIX (whole bone) + RT · Sarcoma → **wide resection protocol**

Never treat a bone lesion without a tissue diagnosis
✗ Wrong options: IM nail with **reamings sent as biopsy** (contaminates femur, abductors, skin) · empirical chemo/RT · ORIF

Remote cancer + solitary lesion	**biopsy** (could be primary sarcoma) – breast cancer 10–26 yr earlier, lung cancer resected, colon cancer, Paget patient
Nailed "metastasis" proves high-grade chondrosarcoma	whole femur contaminated → **wide resection incl. hip disarticulation** (or total femoral resection)
Open biopsy	**intramuscular** interval ✗ Wrong options: intermuscular

Lab + CT + MRI before biopsy for atraumatic lesser trochanter fracture
✗ Wrong options: suture anchor repair · IM nail first

Pathologic humeral shaft fracture, no cancer history	**coaptation splint, then bone scan + CT CAP** (then biopsy) ✗ Wrong options: immediate nail with reamings · ORIF + biopsy first
Painful TKA with new tibial osteolysis, normal infection labs	**CT CAP / PET-CT staging** ✗ Wrong options: revision
Primary identified	**~85%** by history + exam + X-ray + CT CAP (OB: CT CAP **90%**, >90% sensitive) · bone scan + CT ~50%

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Key radiology findings

Modality	Fact
X-ray	Two planes, joint above + below · lytic / mixed / blastic · **sensitivity ~50%** – needs **30–50% mineral loss**, 1–2 cm · lung → **cortical** mets
CT CAP	**Find primary in all >40 yr with unknown lytic lesion** · CT of lesion for cortical containment · sens 71–100%
MRI	**Most sensitive (95%) + specific (90%)** · marrow change before blastic change · T1 low, T2 high · local staging
Bone scan	Other skeletal lesions · **cold in myeloma + thyroid → skeletal survey** · **investigation of choice for occult blastic mets (TO)**
PET-CT	Investigation of choice for **occult lytic** mets **(TO)** · PET alone 30% specificity
Indium scan	Infection (✗ Wrong options: tumour)

Bone scan pitfalls (TO) — **false negative: RCC, myeloma** · **false positive: trauma, infection** · PET-CT also used for **response assessment**

Clinical hunt for primary (TO) — tobacco, urinary symptoms, non-healing ulcer, weight loss, bleeding from orifices · exam: **nodes, breast, thyroid, per-rectal (prostate)**

"Never assume metastasis – prove it" (TO) — biopsy only after **complete systemic evaluation** · core needle · **reaming biopsy only ~60% diagnostic** (crushed tissue)



Acral metastasis to the hand from lung carcinoma

Histology

Epithelial cells in clumps / glands in fibrous stroma · RCC: **clear cells, pseudoalveolar** · metastatic gastric adenocarcinoma: glands + bone



Metastatic carcinoma — epithelial glands in stroma

Fix-without-wide-resection histology **metastatic carcinoma (breast)**
✗ Wrong options: OS lacy osteoid · dedifferentiated CS · fibrosarcoma herringbone · chordoma physaliferous cells

Differential

	Malignant, older	Multiple, older	Epithelial glands	Bisphosphonates help	Wide resection + RT
Metastasis	✓	✓	✓	✓	(RCC solitary)
Myeloma	✓	✓	–	✓	–
Lymphoma of bone	✓	✓	–	–	–
Chondrosarcoma · MFH · secondary sarcoma	✓	–	–	–	–
Paget	–	✓	–	✓	–
FD	–	–	–	✓	–
Synovial sarcoma · glomus	–	–	✓	–	–
Hyperparathyroidism	–	✓	–	–	–

Mirels criteria

Score	Site	Size (of shaft diameter)	Appearance	Pain
1	Upper limb	<1/3	Blastic	Mild
2	Lower limb	1/3–2/3	Mixed	Moderate
3	**Peritrochanteric**	**>2/3**	**Lytic**	**Functional (weight-bearing)**

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Recommendation (Orthobullets) **<8 RT + observe** · **8 judgement** · **>8 prophylactic fixation** · OB explanations vary: "≥8", "≥9", ">9" — read the options

Mirels	Fracture risk (6 mo)
4–6	0%
7	4%
8	15%
9	**33%**
10	72%
11	96%
12	100%

Sensitivity 80–90%, specificity 30–35%

Harrington (prophylactic fixation) **>50% diaphyseal cortex** destruction · **>50–75% metaphysis** · **permeative subtrochanteric** destruction · **persistent pain after RT** · pain with weight bearing · (OB: proximal femur lesion >2.5 cm, avulsion of lesser trochanter)

Spinal metastasis prognostic scores (Tokuhashi, Tomita)

Tokuhashi **prognosis after spinal metastasis** to guide treatment **(TO)** · ✓ includes **general condition** · **≤8 → <6 months** (conservative) · **9–11** individual · **≥12 → >1 yr** (excisional)

Modified Tokuhashi parameter	Scores
General condition (KPS)	Poor 10–40% = 0 · moderate 50–70% = 1 · good 80–100% = 2
Extraspinal bone mets	>2 = 0 · 1–2 = 1 · 0 = 2
Vertebral body mets (original score)	0–2
Visceral (major organ) mets	Non-removable 0 · removable 1 · none 2
Primary	**Lung, OS, stomach, bladder, oesophagus, pancreas 0** · liver, gallbladder, unidentified 1 · other 2 · **kidney, uterus 3** · rectum 4 · **thyroid, breast, prostate, carcinoid 5**
Palsy	Frankel A–B 0 · C–D 1 · E 2

Tomita (2001) **3 parameters**: primary growth, visceral mets, bone mets
✗ Wrong options: **general condition** (that's Tokuhashi) **(TO)**

Tomita parameter	Score
Primary: slow (breast, thyroid) / moderate (kidney, uterus) / rapid (lung, stomach)	1 / 2 / 4
Visceral mets: none / treatable / untreatable	0 / 2 / 4
Bone mets: solitary / multiple	1 / 2

Tomita action	**2–4 aggressive surgery** · **5–7 palliative surgery** · **8–10 non-operative / supportive**
Others	Bauer, Linden, Rades, Douglas, Katagiri scores **(TO)**
Katagiri (TO)	**primary growth rate** · **visceral mets** · **ECOG** · **prior chemo** · **multiple bone mets** → higher score = poorer survival
Modified Bauer (TO)	visceral mets · primary site · **palsy** · bone-only / multiple sites

5 Management

General principles

Goals	pain control, independence, prevent fracture · **implant should outlive the patient**, allow **immediate weight bearing**, **protect the whole bone**, not rely on healing
Nonoperative	asymptomatic · non-displaced fractures of humeral shaft, pelvis, scapula · non-weight-bearing bones · **short life expectancy** · systemically sensitive tumours
Life expectancy for surgery	lower-limb fixation if ≥6–12 weeks (OB) · arthroplasty if >3 months (OB) · **>6 months = best predictor of fracture healing** (Orthobullets) · RT alone for pain if <6 months (OB)
Bisphosphonates	lytic, blastic, mixed · ↓ skeletal-related events + hypercalcaemia · oral clodronate, IV pamidronate/**zoledronic acid** · not for acute impending fracture · complication **ONJ**
Denosumab	solid-tumour mets + myeloma · **superior to zoledronic acid** for SRE prevention · SC · ONJ
Doses (TO)	**denosumab 120 mg SC monthly** · **zoledronic acid 4 mg IV monthly**
RT protocols (TO)	**8 Gy single fraction** or **30 Gy in 10 fractions**
Surgery indications (TO)	pathological fracture · impending fracture · **solitary metastasis** (resectable) · **IM nail + cement augmentation** common
Radiation	palliation + local control · **pain ↓ in 70% at 2 wk**, complete relief ~30% · **RCC + thyroid radioresistant** (✓ breast, prostate, lymphoma, myeloma radiosensitive) · post-RT bone weakens first → fracture risk ~1 yr · radiating a high-risk lesion first ↑ fracture risk
Other	RFA (periosteal nerves), **cementoplasty** (pain, not structure), cryoablation, HIFU · radionuclide therapy (TO)

Postoperative radiation

All patients after fixation (unless death imminent or previously irradiated) · start **2–3 weeks post-op** · **field covers entire implant** (e.g. whole femur after nailing)
✗ Wrong options: tumour site only

Preoperative RT	↑ wound complications

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Femoral lesions — cephalomedullary nailing & radiation

Location	Treatment
Femoral head / neck	**Arthroplasty (cemented)** · hemiarthroplasty if acetabulum free · **THA if acetabular lesions** / OA (✗ Wrong options: screws · DHS · pinning (high failure))
Intertrochanteric	Calcar-replacing hemi/THA **or** cephalomedullary nail
Peri-/ subtrochanteric	**Long, statically locked cephalomedullary nail** ± PMMA · or proximal femoral replacement (✗ Wrong options: **DHS / sliding hip screw** · plate · dynamic locking (shortens) · standard antegrade nail · Enders + cement)
Femoral diaphysis	**Statically locked cephalomedullary nail**

Arthroplasty vs nail/ ORIF — arthroplasty **lower failure** · THA more dislocation than hemi · arthroplasty more infection than nails

Cemented stems — immediate stability, **RT inhibits ingrowth** · intra-op metastatic surprise during THA → **convert to cemented stem + cup**

PMMA — fills defects, immediate structural support
✗ Wrong options: autograft (unreliable)

Post-op rehab — **full weight bearing** with cane (lung mets, Mirels 12) · no axillary crutches with axillary nodes

Periarticular lesions — arthroplasty & endoprosthesis

Site	Treatment
Proximal humerus	Arthroplasty / **endoprosthesis** (ORIF option) · RCC → **proximal humerus replacement** (✗ Wrong options: locking plate · nail · standard reverse)
Humeral diaphysis	**IM nail**
Distal humerus	Flexible nails / elbow replacement
Tibia	**IM nail** (pathologic fractures progress, don't heal in cast/brace)

Solitary renal cell metastasis — embolisation & wide resection

Preoperative embolization — **RCC + thyroid** (hypervascular, VEGF) → ↓ blood loss, no effect on healing · palliative pain control

Solitary RCC metastasis — **wide (en bloc) resection + endoprosthetic reconstruction** (cemented long stem) → ↓ revision, ↑ survival (median **1.3 → 4.8 yr** with complete resection), ↓ complications, ↓ local recurrence
✗ Wrong options: IM nail / plate (progression, hardware failure)

RCC = strongest predictor of local progression after nailing

Polymetastatic RCC — en bloc resection only gives **lower local recurrence**
✗ Wrong options: survival benefit · cheaper · less blood loss, reoperation equivalent

RCC features — lytic, **acral** possible, **radio- and chemo-resistant**, hypervascular, high complication rate with intralesional treatment, survival variable (6 mo – 4–5 yr) · fracture healing **44%** · IL-2 immunotherapy response 18–35%
✗ Wrong options: sensitive

Thyroid met (subtrochanteric)	**curettage + IM nailing** · long survival → stabilise even tibial fracture

6 Complications

Pathologic fracture healing	**65% nonunion** overall · prostate **42%**, RCC **44%** (comparable across carcinomas)

Fracture healing by primary (TO)	Rate
Myeloma	**67%**
RCC	**44%**
Breast	**27%**
Lung	**0%**

Highest pathological fracture risk (TO)	**lung, kidney, myeloma**
Prostate mets	**blastic (90%)**, **radiosensitive**, **fewer SREs**, **better survival**, comparable fracture healing
1-yr survival after surgery for metastatic disease	~30% (prostate series)

Prognosis

Primary	Median survival with bone mets
Thyroid	**48 months**
Prostate	**40 months** (Coleman 53)
Breast	**24 months** (Coleman 20)
Kidney	Variable: **6 months – 4–5 years**
Myeloma	2–3 years
Lung	**6–7 months** (worst) · <5% 5-yr

How it's asked

"Commonest cause of an aggressive lytic lesion >40 yr?"	→ **metastatic bone disease**
"Solitary supracondylar humeral lesion, breast cancer 10 yr ago, staging negative?"	→ **biopsy** ✗ Wrong options: TEA · curettage · cement · RFA
"Paget patient with THA, lytic pelvic lesion?"	→ **bone scan + CT CAP** ✗ Wrong options: revision · ORIF
"Knee pain, clear cells, pseudoalveolar pattern?"	→ **metastatic RCC** ✗ Wrong options: chordoma
"IM nail placed through 'metastasis' that is high-grade chondrosarcoma?"	→ **wide resection incl. hip disarticulation**

– – Active space – –

"Heavy smoker, new painful femoral diaphyseal lytic lesion?"	→ **biopsy, then treat by result** ✗ Wrong options: nail + reamings
"Lung cancer history, new femoral lesion, no previous biopsy-proven bone mets?"	→ **biopsy** ✗ Wrong options: nail + reamings
"Solitary lesion after remote colon cancer + MoM THA?"	→ **percutaneous biopsy**
"45-yr-old, knee lesion, no cancer history?"	→ **CT CAP then biopsy**
"68-yr-old smoker, atraumatic lesser trochanter fracture?"	→ **labs (CBC, chemistry, SPEP/UPEP) + CT + MRI**
"59-yr-old, thigh pain, unknown primary?"	→ **CT chest/abdomen/pelvis** before biopsy
"64-yr-old, 3 mo groin pain, lytic proximal femur?"	→ **bone scan + CT CAP + labs**
"Lytic humeral lesion, hot bone scan, 55 yr?"	→ **MRI of lesion + CT CAP** ✗ Wrong options: indium · skeletal survey · CT needle biopsy first · LDH
"Pathologic humeral shaft fracture, no history?"	→ **coaptation splint, then bone scan + CT CAP**
"Post-TKA tibial osteolysis, aspirate negative?"	→ **CT CAP**
"Smoker, solitary femoral lesion, staging negative — biopsy how?"	→ **open biopsy via intramuscular interval**
"Isolated femoral carcinoma, primary not found?"	→ **lung**
"Humeral pathologic fracture, CT CAP negative — next?"	→ **core needle biopsy**
"NOT a necessary step in bony metastasis?"	→ **direct osteoclast stimulation**
"Endothelin-1 is involved in?"	→ **osteoblastic metastases**
"NOT a step in becoming metastatic?"	→ **increased apoptosis**
"Tumour cells attach to host endothelium via?"	→ **integrin** ✗ Wrong options: E-cadherin · MMP · VEGF · CXCL12
"PTHrP + breast cancer cells cause osteolysis via?"	→ **osteoblastic RANKL production**
"ECM degradation in metastasis?"	→ **MMPs** ✗ Wrong options: thrombospondin (inhibits) · VEGF · PDGF
"Role of MMPs in a pathologic fracture?"	→ **ECM degradation for invasion/ metastasis**
"Osteoblastic spinal mets in a 65-yr-old man — primary?"	→ **prostate (TO)**
"Commonest site of bony metastasis?"	→ **thoracic spine (TO)**
"Lytic spinal secondaries from?"	→ **renal carcinoma (TO)** ✗ Wrong options: medulloblastoma, prostate, carcinoid (blastic)
"Breast metastases are?"	→ **lytic, blastic, or mixed — all of the above (TO)**

"Distal to elbow/knee / acral mets from?"	→ **lung**
"Metastatic pathological fracture least likely to heal?"	→ **lung (0%)** (TO) ✗ Wrong options: myeloma (67%) · RCC (44%) · breast (27%)
"Bone scan false negative?"	→ **RCC, myeloma** (TO)
"Breast cancer pathology guiding medical therapy?"	→ **HER2/neu** (ER, PR)
"Least radiosensitive metastasis?"	→ **renal** ✗ Wrong options: breast, prostate, lymphoma, myeloma
"Prostate mets vs others?"	→ **comparable fracture healing** ✗ Wrong options: radioresistant · lytic · more SREs · worse survival
"RCC — LEAST associated?"	→ **sensitivity to radiation + cytotoxic chemo**
"Median survival, lung with bone mets?"	→ **~7 months**
"Subtrochanteric pathologic fracture, lung cancer?"	→ **statically locked cephalomedullary nail** ✗ Wrong options: THA · hemi · SHS · dynamic locking
"Esophageal cancer, <6 mo life expectancy, subtrochanteric lesion?"	→ **cephalomedullary nail** ✗ Wrong options: SHS · plate · PFR
"Breast mets, pathologic femoral fracture, biopsy-proven?"	→ **IM nail + RT to entire femur + chemo**
"Breast mets, diaphyseal femur lesion needing cane?"	→ **closed IM nail + post-op RT**
"Lung mets, Mirels 12 peritrochanteric, active 12–18 mo prognosis?"	→ **prophylactic IM nail + full weight bearing with cane**
"Subtrochanteric impending fracture, breast cancer?"	→ **long cephalomedullary nail**
"Thigh pain despite chemo + RT, multiple mets?"	→ **prophylactic rodding** ✗ Wrong options: bisphosphonates acutely · megaprosthesis
"Lung history, peritrochanteric lytic lesion, Mirels 11?"	→ **cephalomedullary nail** ✗ Wrong options: cementoplasty · uncemented PFR
"Thyroid papillary mets, peritrochanteric, functional pain?"	→ **Mirels 12 → IM nail**
"Myeloma, femoral head/neck lesion, 1-yr life expectancy?"	→ **cemented hemiarthroplasty + RT** ✗ Wrong options: DHS · APC · RT alone
"Thyroid mets, tibial pathologic fracture, 1-yr survival?"	→ **surgery** (IM nail) ✗ Wrong options: brace · cast · RT alone
"Breast mets, peritrochanteric lesion 40% circumference, non-ambulatory?"	→ **Mirels 11 → hemiarthroplasty** (nail or THA also acceptable)

– – Active space – –

"Solitary RCC metastasis proximal femur, 5 yr after nephrectomy?"	→ **proximal femur replacement** ✗ Wrong options: nail after curettage · chemo/RT
"Breast carcinoma found intra-op after neck cut in THA?"	→ **convert to cemented stem + cup**
"RCC pathologic proximal humerus fracture?"	→ **proximal humerus replacement**
"Breast mets, femoral neck fracture + acetabular lesions, good prognosis?"	→ **THA**
"RCC femoral lesion — preoperative step?"	→ **embolization**
"Isolated RCC proximal humerus metastasis 7 yr after nephrectomy?"	→ **wide resection + prosthetic reconstruction**
"Why EPR over intralesional nailing for solitary RCC?"	→ **lower revision + longer survival + fewer complications**
"Procedure improving survival in solitary RCC femoral met?"	→ **wide resection + proximal femoral replacement**
"Best candidate for en bloc resection?"	→ **solitary RCC proximal femoral lesion** ✗ Wrong options: multifocal breast spine · thyroid (curettage + nail) · myeloma · lymphoma
"Polymetastatic RCC — en bloc vs nailing benefit?"	→ **lower local recurrence**
"Highest risk of local progression after IM nail?"	→ **solitary RCC metastasis**
"Histology treatable by fixation without wide resection?"	→ **metastatic breast carcinoma**
"Histology benefiting from preop embolization?"	→ **RCC (clear cells)** ✗ Wrong options: myeloma · chondrosarcoma · chordoma · prostate
"Tokuhashi score is for?"	→ **prognosis after spinal metastasis (TO)**
"NOT a Tomita component?"	→ **general condition** (✓ visceral mets, ✓ number of bone mets, ✓ primary type) **(TO)**

METASTATIC DISEASE OF SPINE

– Pathology · 38 / 66 –

1 Basics

Epidemiology

Commonest destructive bone lesion in adults	metastasis · bone = **3rd** site (after lung, liver) · >40 yr
Spine = commonest site of bone metastasis	**thoracic** commonest (symptomatic: thoracic **68–70%**, lumbosacral 16–22%, cervical 8–15%) · spine > proximal femur > humerus
Spinal disease	up to **90%** of metastatic cancer patients, only **10–20% symptomatic** · **first sign of malignancy in 20%** · >90% of spinal tumours are metastatic

Primary tumours metastasising to spine

Primary	% of patients developing spine mets
Breast	**16–37%**
Lung	12–15%
Prostate	9–15%
Kidney	3–6%
Thyroid	4%

Order of spinal secondary spread	breast, prostate, lung, kidney ≈ **80%** · plus thyroid, myeloma (~75% of bone mets)
Rarely to spine	**colon** · hepatic, uterine, skin, "osseous" are distractors · haemangioma doesn't metastasize ✗ Wrong options: breast, lung, renal, thyroid
Risk clues	**BRCA1** (breast) · smoking (lung) · **von Hippel–Lindau / VHL** (renal) · hypercalcaemia ✗ Wrong options: **APC + positive FOBT** (colon — least likely)

Mechanism of metastasis

Theories	**seed and soil** (compatible end-organ) · **circulation** (blood flow from primary)
Steps	↓ **E-cadherin** → release · immune evasion · **integrin** → endothelial attachment · extravasation · **VEGF** angiogenesis · genomic instability · ↓ apoptosis ✗ Wrong options: osteocalcin
Batson plexus	**valveless** venous plexus → vertebral bodies, pelvis, skull, proximal girdles

– – Active space – –

Osteoclast-mediated bone resorption

Lysis	**tumour-induced activation of osteoclasts** · tumour releases TNF-α, IL-6, IL-11, **PTHrP**, TGF-β → **osteoblasts secrete RANKL** → RANK on osteoclasts · **OPG** binds RANKL, blocks resorption ✗ Wrong options: direct resorption by tumour cells (only osteoclasts resorb bone) · macrophages · necrosis · angiogenesis
Sclerosis	prostate (± breast) → **endothelin-1** → ET-A receptor on osteoblasts, ↓ **DKK-1** → WNT ↑ ✗ Wrong options: ET-1 causes lysis

Genes & lab markers

Labs (unknown primary)	CBC + diff, **ESR**, BMP, LFTs, Ca, phosphate, ALP, **SPEP/UPEP**, TFTs, **PTH**, **PSA**, urinalysis (± CEA)
Red flags → first tests	CBC, ESR, urinalysis (+ PSA, CEA with cancer history)
IHC	keratin · **CK7** (breast, lung) · **TTF-1** (lung) · **vimentin** (renal) · **CDX2** (colon)
Associated	**hypercalcaemia 10–30%** (lung, breast) → IV fluids, loop diuretic, bisphosphonate · anaemia · hypercoagulability / VTE

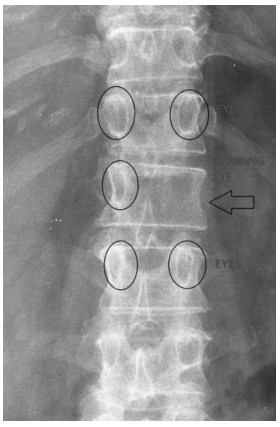
3 Presentation

Pain	**axial night pain** / rest pain · mechanical pain (instability) · **pain worse supine**, relieved standing
Red flags (tumour)	age >50, cancer history (most significant), weight loss, night pain, pain lying down
Neurologic	cord compression → **UMN signs: hyperreflexia**, clonus, gait ataxia, weakness, bowel/bladder · distinguish **root vs cord**
Chemo-related	peripheral neuropathy

4 Diagnosis & Classification

Key radiology findings

X-ray (AP + lateral)	**"winking owl" sign** = **unilateral absent pedicle** (pedicle destruction) · lytic or mixed · **up to 40% invisible** on X-ray · discs preserved (vs infection)



Winking owl sign — absent pedicle

MRI with contrast	**next step** for winking owl / suspected mets · dedicated region + **limited sagittal whole spine** · soft-tissue mass, ↑ T2, cortical permeation, **cord compression** · **98%** sensitive + specific
CT	sensitivity 66%, specificity 99% · **CT myelography if MRI contraindicated** · **CT CAP** for unknown primary
Bone scan	extent · **cold in myeloma + thyroid → skeletal survey**

Blastic pelvis/spine in older man → prostate (→ **prostate biopsy** confirms) · lung blastic only 25–30%

Workup & diagnosis

Known cancer, no prior mets / no cancer history	evaluate before treating · history, exam, X-ray, labs, **CT CAP + bone scan** → identifies primary in **85%** · biopsy adds **8%**
Staging before biopsy	✗ Wrong options: CT-guided biopsy first · excision · RFA · NSAIDs + PT
Winking owl in IV drug user, normal WBC/ESR/CRP	**CT CAP** ✗ Wrong options: cervical MRI · antibiotics · kyphoplasty
Multiple spinal lesions + very high PSA	staging + diagnostic tests together (prostate biopsy, CT CAP, bone scan, CT-guided spine biopsy) ✗ Wrong options: **bone marrow biopsy** (prostate rarely to marrow)
Spine biopsy indications (TO)	**life expectancy > 3 months** · **neurological deficit** · **uncertain diagnosis**
Known metastatic disease	biopsy not required
Biopsy technique	**posterior transpedicular**, wide-gauge **Jamshidi**, coring needle, image guidance · posterior (cervical) → vertebral artery risk · occult adenocarcinoma → **small lung primary**

Histology

Epithelial cells in clumps/glands in fibrous stroma

Tokuhashi (revised) score

Six elements general condition · extraspinal bone mets · number of vertebral mets · visceral mets · primary tumour (0–5) · palsy · max **15**

Total	Survival / approach
0–8	**<6 months** – conservative / palliative
9–11 (9–12)	**>6 months** – individualise
≥12 (12–15)	**>1 year** – excisional surgery

Worked example (OB) good health 2 + no extraspinal bone mets 2 + 2 spinal lesions 1 + removable visceral 1 + **rectal primary 4** + incomplete palsy 1 = **11** → surgery

– – Active space – –

Tomita primary growth + visceral mets + bone mets
✗ Wrong options: **general condition (TO)**

Tomita (TO)	Points
Primary growth	Slow 1 · moderate 2 · rapid 4
Visceral metastasis	None 0 · treatable 2 · untreatable 4
Bone metastasis	Solitary 1 · multiple 2

Tomita score → surgery (TO) **2–3: wide / marginal excision (long-term control) · 4–5: palliative intralesional · 6–10: non-operative / supportive**

WBB (Weinstein-Boriani-Biagini) (TO) vertebra in **12 radial zones** + layers **A** extraosseous soft tissue · **B** intraosseous superficial · **C** intraosseous deep · **D** extraosseous extradural · **E** extraosseous intradural → decides **en bloc** feasibility

Age (TO)	Likely lesion
10–30 yrs	Benign – osteoid osteoma, osteoblastoma, ABC, osteochondroma
30–50 yrs	Chordoma, chondrosarcoma, Hodgkin
> 50 yrs	**Metastasis, myeloma**

Strategy by primary (TO) slow (breast, thyroid, prostate) → aggressive / en bloc · moderate (renal, uterine) → marginal · rapid (lung, stomach) → palliative fixation · **surgery only if expected survival > 6 weeks** (TO) – conflicts with "life > 3 months / > 6 months" thresholds in chapter – read the options

NOMS framework

NOMS **N**eurologic (ESCC grade) · **O**ncologic (**radiosensitivity**) · **M**echanical instability (**SINS**) · **S**ystemic illness (disease burden, comorbidity, ECOG) · most comprehensive current decision framework (FNB 2024 key not given – NOMS fits; ✗ Wrong options: Bigliani · DeWald; Tomita is a score) **(TO)**

ESCC (epidural cord compression) **0–1 low grade**, **2–3 high grade**

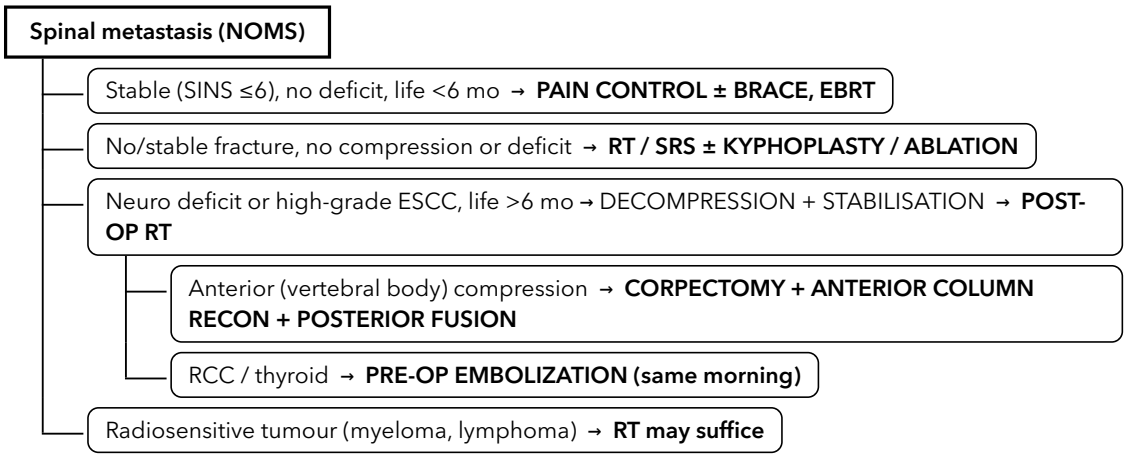
SINS (Spinal Instability Neoplastic Score)

Element	Scores
Location	**Junctional** (occiput–C2, C7–T2, T11–L1, L5–S1) **3** · mobile (C3–C6, L2–L4) 2 · semi-rigid (T3–T10) 1 · rigid (S2–S5) 0
Pain (mechanical – relief lying down / pain on loading)	Yes **3** · occasional non-mechanical 1 · none 0
Lesion	**Lytic 2** · mixed 1 · blastic 0
Alignment	**Subluxation/translation 4** · de novo deformity (kyphosis/scoliosis) 2 · normal 0
Body collapse	>50% 3 · <50% 2 · none but >50% body involved 1 · none 0
Posterolateral involvement (facet, pedicle, CV joint)	Bilateral 3 · unilateral 1 · none 0

SINS	Meaning
0–6	**Stable** – no surgical consult
7–12	**Potentially unstable** → surgical referral
13–18	**Unstable** → surgical referral

Worked example (OB) T6 (1) + occasional non-mechanical pain (1) + mixed (1) + **focal kyphosis (2)** + 30% collapse (2) + no posterolateral (0) = **7** → surgical referral
✗ Wrong options: normal-alignment combinations scoring 4–6

5 Management



TO rule **radioresistant → decompression + stabilisation** · **radiosensitive → RT first, then stabilisation** · vertebroplasty / kyphoplasty for pain from instability (TO)

Nonoperative & radiation

Pain + brace stable fracture, no deficit, **life <6 months** (Tokuhashi)

External beam RT painful lesions **without deficit or cord compression**
✗ Wrong options: embolization · laminectomy · corpectomy

SRS **88% local control** if no instability or compression

Kyphoplasty / vertebroplasty / RFA no or stable fracture, **no compression, deficit, or instability** · transpedicular, fluoroscopy · kyphoplasty = balloon cavity then PMMA (partial reduction) · not with **posterior cortex disruption** or neurologic impairment

Surgery – decompression, stabilisation & radiation

Patchell RCT **direct decompression + post-op RT > RT alone** for metastatic cord compression → best for **regaining/maintaining ambulation**

Indications neurologic deficit + life **>6 months** · intractable pain · instability

Sequence **surgery first, then radiation** · chemo too slow for acute deficit
✗ Wrong options: RT before surgery → **3× wound complications**

– – Active space – –

Anterior compression (vertebral body)	**corpectomy + anterior column reconstruction + posterior instrumented fusion** · posterior-only via **costotransversectomy / transpedicular** with expandable cage ✗ Wrong options: **laminectomy alone** → doesn't decompress, progressive **kyphosis** · **posterior fusion alone** → continued compression
Approach	posterior for thoracic, lumbar, sacral · cervical anterior or posterior · cement augments screws
Separation surgery	decompress + stabilise to create space for **SRS** · no en bloc resection for metastases ✗ Wrong options: goal of total resection; margin usually intralesional
Outcomes	**95% local control** with surgery + RT · **16%** of ASIA E with high-grade ESCC decline even with surgery ✗ Wrong options: laminectomy + steroids alone
Breast mets, failed chemo, T9 ventral compression, hyperreflexia	**thoracic corpectomy + instrumented fusion + post-op RT** ✗ Wrong options: CHOP · posterior decompression after kyphoplasty · RT alone

Preoperative embolisation (renal, thyroid)

Hypervascular	**renal + thyroid → preop embolization** (✗ Wrong options: breast, colon, prostate, lung, liver) · ideally **morning of surgery** (✗ Wrong options: post-op) · target the **tumour capillary bed**, not main feeder (collaterals) · agents: glue, alcohol, PVA particles, Gelfoam
RCC spine with deficit	**embolization → decompression + stabilization → RT** · RCC radioresistant → RT alone won't relieve compression ✗ Wrong options: kyphoplasty · steroids · cryotherapy
Colon cancer	no embolization needed

Bisphosphonates

Zoledronic acid	**↓ skeletal-related events 30–40%** (fracture, need for RT/surgery, cord compression, hypercalcaemia) · **pain ↓ in ≥50%** · **ONJ ~2.5%** · **renal toxicity** → dose by creatinine clearance · **zoledronic acid > pamidronate** ✗ Wrong options: 15% · 25%

6 Complications

Recurrence · **hardware failure / instability** (RT, chemo, frailty; **>6 levels** → instrumentation failure) · nonunion

Revision surgery – soft tissue reconstruction

Wound complications	risk: smoking, **prior RT**, instrumentation, revision · **immediate soft-tissue reconstruction** → better outcomes (delayed = **2×** wound complications) ✗ Wrong options: preop RT · tobacco · posterior approach

Prognosis

Timing	cancer → spinal met **32 months** · spinal met → fracture **27 months**
1-yr survival after spine-met surgery	**50%** · epidural disease worsens

Survival factors	histology + prior treatment, function + neurology, age, extent, nutrition
Median survival	thyroid **48 mo** · prostate **40** · breast **24** · kidney variable (≥6) · **lung 6**

How it's asked

"Smoker, osteoblastic pelvic/lumbar lesions — confirm diagnosis?"	→ **prostate biopsy** ✗ Wrong options: UPEP · thyroid biopsy · renal US
"Winking owl — LEAST likely?"	→ **APC mutation + positive FOBT (colon)** (✓ BRCA1, smoking, VHL, hypercalcaemia)
"Thoracic lesion on MRI, weight loss, no history — next?"	→ **CT CAP + whole-body bone scan** ✗ Wrong options: CT biopsy first · excision · RFA
"Multiple spinal mets + ↑ PSA — NOT in initial work-up?"	→ **bone marrow biopsy**
"Arrow at absent pedicle?"	→ **metastatic disease** ✗ Wrong options: haemangioma · infection
"Back pain worse lying down, weight loss, prostate cancer history, normal X-ray?"	→ **CBC, ESR, urinalysis, PSA, CEA** ✗ Wrong options: PT · manipulation
"68-yr-old, thoracic pain, winking owl?"	→ **MRI with gadolinium**
"Diffuse hot bone scan + blastic CT lesions, 85-yr-old man?"	→ **metastatic prostate carcinoma** ✗ Wrong options: myeloma · RCC · thyroid (lytic)
"Colon mets with cord compression, incontinence — true?"	→ **hyperreflexic patellar reflexes** ✗ Wrong options: osteocalcin binding · ET-1 lysis · RT before surgery · embolization
"IV drug user, winking owl, normal inflammatory markers?"	→ **CT CAP**
"Isolated T9 breast met, ventral compression, worsening gait?"	→ **corpectomy + instrumented fusion + post-op RT**
"L1 destroyed, breast mets, leg weakness — best function?"	→ **decompression + instrumentation + post-op RT**
"ASIA C, single thoracic met from rectal primary, Tokuhashi 11?"	→ **decompression + stabilisation → RT** ✗ Wrong options: RT first (3× wound problems) · RT alone · chemo
"Lung mets, can't walk 48 h, life 6–12 mo?"	→ **circumferential decompression + stabilisation + post-op RT** ✗ Wrong options: RT alone · laminectomy + steroids · en bloc
"Breast mets, anterior compression, life >3 yr on targeted therapy?"	→ **corpectomy + anterior column reconstruction + posterior fusion** ✗ Wrong options: laminectomy + PSF · kyphoplasty

– – Active space – –

"RCC thoracic met with compression — true?"	→ **posterior fusion alone leaves cord compressed** ✗ Wrong options: laminectomy → lordosis · kyphoplasty
"T6 met — which features warrant surgical referral?"	→ **mixed lesion + 30% collapse + focal kyphosis (SINS 7)**
"RCC spinal met with clonus — before surgery?"	→ **arterial embolization**
"Metastases needing preop embolization?"	→ **renal + thyroid**
"Highly vascular tumour?"	→ **thyroid**
"Isolated RCC met, knee-extension weakness?"	→ **embolization → decompression/ stabilization → RT**
"RCC spinal biopsy, failed nonoperative care?"	→ **embolization the morning of corpectomy + stabilization**
"Zoledronic acid in breast mets — true?"	→ **↓ fractures, RT, surgery, cord compression, hypercalcaemia (SRE ↓ 30–40%)** ✗ Wrong options: 25% ONJ · no pain benefit · less effective than pamidronate · renal function irrelevant
"Tumours most likely causing spinal mets?"	→ **breast, renal, prostate, lung** (+ thyroid)
"Rarely metastasizes to spine?"	→ **colon**
"Lytic T12 lesion resorption mediated by?"	→ **tumour-induced osteoclast activation**
"Painful lumbar prostate mets, no deficit?"	→ **external beam radiation**
"Improves outcome in revision metastatic spine surgery?"	→ **immediate soft-tissue reconstruction**
"Tokuhashi score is used for?"	→ **prognosis after spinal metastasis (TO)**
"Not a Tomita component?"	→ **general condition (TO)**
"Most comprehensive current decision framework for spinal mets (FNB 2024)?"	→ key not given · **NOMS** fits **(TO)**
"Tomita 6–10 points?"	→ **non-operative / supportive care** (TO)

MULTIPLE MYELOMA

– Pathology · 39 / 66 –

1 Basics

Epidemiology & eponym (Kahler disease)

Kahler disease = multiple myeloma **(TO)**
✗ Wrong options: OS · MFH · benign fibrous histiocytoma

Most common primary (malignant) bone tumour (TO) · 3–4 per 100,000 · 1% of all malignancies

Demographics **>40 yr (median 60; 65 in OB)** · **male** · **2× African Americans**

Forms

Form	Key features
Multiple myeloma	Commonest · marrow + multiple lesions · CRAB
Solitary plasmacytoma	Single skeletal site, negative marrow · **radiosensitive** · **>50% progress to MM** · MRI + FDG-PET find more lesions in 33% · **best prognosis**
Osteosclerotic myeloma (POEMS)	**P**olyneuropathy (symmetric, distal → proximal, doesn't improve) · **O**rganomegaly · **E**ndocrinopathy · **M** protein · **S**kin changes (trunk) · **sclerotic** lesions
MGUS / asymptomatic myeloma	Annual surveillance

Plasma cell dyscrasia (TO)	Criteria	Treatment
MGUS	M protein **< 30 g/L** · plasma cells **< 10%** · no CRAB	Observe
Smouldering myeloma	M protein **> 30 g/L** or plasma cells **10–60%** · no CRAB	**Surveillance only**
Multiple myeloma	Plasma cells **> 60%**, or **10–60% + ≥1 CRAB**	Treat
Solitary plasmacytoma	One biopsy-proven bone lesion · no systemic disease · marrow **< 60%** (TO – verify: usual criterion is normal marrow / < 10% clonal plasma cells)	**RT**

Pathophysiology (RANKL, osteoblast suppression)

Osteolysis myeloma cells bind marrow stromal cells → **↑ RANKL**, M-CSF, **IL-6, IL-11**, MIP-1α → osteoclast activation · **OPG ↓** (decoy receptor for RANKL)

Osteoblast suppression **TNF** and **DKK-1** ↓ osteoblast function → **no sclerotic rim**, **cold bone scan** · **ALP normal** despite severe destruction – key clue (TO)

Renal failure (TO) **light-chain (Bence Jones) nephropathy** → tubular obstruction → glomerular damage · Bence Jones = early renal clue

Lytic lesions (mets + myeloma) **RANK ligand acting on osteoclasts** (RANKL secreted by osteoblasts → RANK receptor; OPG = decoy)
✗ Wrong options: RANKL on neoplastic cells · OPG on osteoclasts

– – Active space – –

Genes & lab markers

Lab	Finding
SPEP	**M spike** · **IgG ~50%, IgA ~25%** (heavy chains IgG 52%, IgA 21%, IgM 12%)
UPEP	**Bence Jones proteins** = **monoclonal light chains (κ, λ)** · proteinuria
Blood	**Anaemia**, ↑ **creatinine**, **hypercalcaemia (30%)**, ↑ ESR
β2-microglobulin	Tumour burden, **prognosis** (ISS)
IHC	**CD138 +**

Poor prognosis genetics — **del 13**, **t(4;14)**, t(14;16)

Genetics list (TO) — hyperdiploidy · **13q14 deletion** · **t(11;14)**, t(4;14), t(14;16) · **N-RAS, K-RAS, B-RAF** mutations

Blood (TO) — **rouleaux** on smear · circulating plasma cells **> 2,000/mm³ → plasma cell leukaemia** · **reversed A:G ratio** · ↑ LDH, ↑ uric acid · M spike in **gamma** region

Marrow cytology (TO) — perinuclear halo, clock-face chromatin · **Mott cells, Russell bodies** (immunoglobulin inclusions)

3 Presentation

Uncommon features (TO) — hyperviscosity · hepatosplenomegaly · **cord compression (plasmacytoma)** · pleural effusion

Localized bone pain (**spine, ribs**) · **pathologic fracture** (low-energy vertebral compression) · **fatigue** (anaemia, renal failure, hypercalcaemia) · radiculopathy, fevers

4 Diagnosis & Classification

Key radiology findings

X-ray — **multiple "punched-out" lytic lesions**, **no sclerotic border** · visible only after **>50% destruction** · diffuse osteopenia, moth-eaten · marrow-packing process



Punched-out lytic lesions

Skull lesion (TO)	Disease
Punched out, no bevelling	**Myeloma**
Double bevelled	**EG / LCH**
Salt and pepper	**Hyperparathyroidism**
Cotton wool	**Paget**

Spine **vertebra plana** may occur (also plasmacytoma) (TO)

Skeletal survey **preferred screen** · **bone scan cold in ~30%** (30–50%) – no osteoblastic activity
✗ Wrong options: bone scan ideal

MRI more sensitive than X-ray (spine, pelvis) · T1 dark, T2 bright · marrow replacement · focal lesion ≥5 mm = MDE

FDG-PET **93% sensitive** (> X-ray) · finds more lesions in "solitary" plasmacytoma

Diagnostic criteria (IMWG)

Marrow **clonal plasma cells ≥10%** (or biopsy-proven plasmacytoma) **+ ≥1 CRAB feature or MDE**

CRAB	Threshold
Calcium	>0.25 mmol/L (>1 mg/dL) above ULN or **>2.75 mmol/L (>11 mg/dL)**
Renal	CrCl **<40 mL/min** or creatinine **>177 µmol/L (>2 mg/dL)**
Anaemia	Hb >20 g/L below LLN or **<100 g/L**
Bone	**≥1 lytic lesion** (X-ray, CT, PET-CT) · if marrow <10%, >1 lesion needed

Myeloma-defining event (MDE)
≥60% clonal marrow plasma cells
Involved:uninvolved **free light chain ratio ≥100** (involved ≥100 mg/L)
≥1 focal MRI lesion ≥5 mm

Plasma cell % – conflict IMWG/Orthobullets/TO: **≥10%** · OB: **plasmacytoma 10–30% vs myeloma >30%** (keyed **35%** as minimum) – read the options · normal marrow <2%

Next test with anaemia + serum M protein **bone marrow aspiration + biopsy (TO)**
✗ Wrong options: UPEP (already have serum M protein) · CT head · EMG · lumbar puncture · vertebral CT biopsy

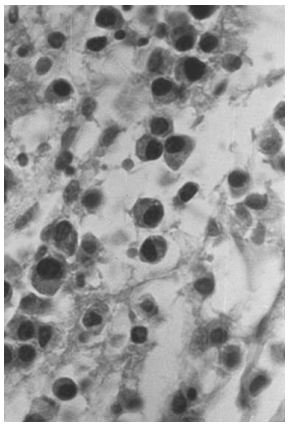
International Staging System (ISS)

Stage	Criteria	Median survival
I	**β2-M <3.5 mg/L** + albumin ≥3.5 g/dL	62 months
II	β2-M 3.5–5.4 mg/L	44 months
III	**β2-M ≥5.5 mg/L**	29 months

– – Active space – –

Histology

Plasma cells round, **eccentric nucleus**, **"clock-face" chromatin**, prominent nucleolus, basophilic cytoplasm · **perinuclear clear zone (Hof)** = Golgi (immunoglobulin production)



Plasma cells with eccentric clock-face nuclei

Differential

	Older patient	Multiple lesions	Bisphosphonates help	Chemo + RT
Myeloma	✓	✓	✓	(chemo ± RT)
Metastasis	✓	✓	✓	–
Lymphoma	✓	✓	–	✓
Paget	–	✓	✓	–
Hyperparathyroidism	–	✓	–	–

5 Management

Plasma cell disorder

- MGUS / asymptomatic → **ANNUAL SURVEILLANCE**
- Solitary plasmacytoma → **EBRT 45–50 Gy (fix if fracture/impending)**
- Osteosclerotic (POEMS) → **CHEMO + RT + PLASMAPHERESIS**
- Multiple myeloma → **MULTIAGENT CHEMO ± STEM-CELL TRANSPLANT + BISPHOSPHONATES**
  - Impending / complete fracture, life >3 mo → **STABILISE (whole bone, cement) + EBRT**
  - Painful VCF / instability or compression resistant to RT → **KYPHOPLASTY**

Treatment (fixation, radiation, bisphosphonates)

Chemo non-transplant (>65, comorbid): **melphalan + prednisone + thalidomide/bortezomib**, lenalidomide/thalidomide + dexamethasone · transplant candidates: **bortezomib/lenalidomide + dexamethasone** (± thalidomide) → autologous SCT (not curative, +2–3 yr DFS)

Very radio- and chemosensitive RT may be primary for impending lesions · surgery shouldn't delay induction / transplant

Bisphosphonates ↓ skeletal events · **zoledronic acid IV q4 wk × 2 yr**, **avoid in renal failure** (TO)

Denosumab (TO) **120 mg SC** (TO lists weeks 0, 1, 2 then q4 wk) · **safe in renal failure** → preferred

Systemic agents (TO) **melphalan + prednisolone/dexamethasone** · IMiDs: thalidomide, lenalidomide, pomalidomide · **proteasome inhibitors: bortezomib**, carfilzomib, ixazomib · anti-CD38: **daratumumab**, isatuximab · elotuzumab · **BCMA**: belantamab (ADC), **teclistamab** (bispecific CD3–BCMA), **CAR-T** (idecabtagene, ciltacabtagene) · **autologous SCT** best < 65 yr, not curative

Ortho rules (TO) **IM nail spanning whole bone** · periarticular → arthroplasty / megaprosthesis · **post-op RT at ~3 weeks** (radiosensitive) · plasmacytoma → **RT**, fix if **Mirels > 8**, kyphoplasty for vertebrae · treat hypercalcaemia aggressively

Surgery complete or impending fracture, **life expectancy >3 months** · whole-bone protection, immediate stability · **post-op RT**

Lesion	Fixation
Femoral neck fracture	**Cemented hemiarthroplasty or THA** (✗ Wrong options: IM nail · percutaneous screws · DHS · radical resection)
Proximal femur (stem)	**Full-length cemented** stem (✗ Wrong options: press-fit (RT/chemo impair ingrowth))
Peritrochanteric, Mirels 12	**Stabilisation + RT + bisphosphonates** (✗ Wrong options: wide resection (never for myeloma) · hormone therapy · curettage + graft)
Distal humerus path fracture	**Internal fixation without resection** (IM rod) → RT (✗ Wrong options: megaprosthesis · chemo first)
Periacetabular, intact subchondral plate	**Curettage + PMMA + modified Harrington pins**
Periacetabular, subchondral plate violated	**Complex THA + acetabular cage**
Painful VCF	**Kyphoplasty/vertebroplasty** – restores height, corrects sagittal deformity

Prophylactic fixation indicated before fracture so **systemic therapy isn't delayed** (bed rest, immunocompromise)

6 Complications

Fracture healing myeloma has **among the best healing (67%)** of metastatic-type lesions · MM vertebral fractures heal poorly, progressive height loss

Prognosis **median 3 yr** · 5-yr **30%**, 10-yr 11% · **renal failure → shortest survival**

Poor prognosis **del 13 / t(4;14) / t(14;16)** · circulating plasma cells · **↑ β2-microglobulin** · **↓ albumin** · ↑ marrow microvessels

– – Active space – –

X vs Y

	Myeloma	Metastatic carcinoma
X-ray	**Punched-out**, no sclerotic rim	Lytic / blastic / mixed
Bone scan	**Cold ~30%** → skeletal survey	Hot (except thyroid, RCC lytic)
Labs	M spike, Bence Jones, CRAB	Primary-specific (PSA…)
Marker	**CD138**	Keratin, CK7, TTF-1
Wide resection	**Never**	Solitary RCC

How it's asked

"Anaemia + serum monoclonal protein, radiculopathy – next investigation?"	→ **bone marrow biopsy (TO)** ✗ Wrong options: UPEP · CT vertebral biopsy · LP
"Spinal biopsy plasma cells – UPEP shows?"	→ **monoclonal light chains** ✗ Wrong options: polyclonal · heavy chains
"Cell type in T6 biopsy (asterisk)?"	→ **plasma cells**
"Lumbar/thoracic tenderness, L5 weakness, anaemia + serum M protein?"	→ **bone marrow aspiration + biopsy** ✗ Wrong options: urine electrophoresis · CT head · EMG
"Chronic anaemia, moth-eaten femora, marrow-packing MRI?"	→ **multiple myeloma** ✗ Wrong options: FD · osteoporosis · infarcts
"Minimum marrow plasmacytosis for myeloma (options 5/10/35/50/75%)?"	→ **35%** (OB: >30%) · IMWG uses ≥10%
"Low-energy VCF + punched-out lesions + M protein – histology?"	→ **sheets of clock-face plasma cells**
"Ca 12, creatinine 2.1, Hb 9.3 – SPEP/UPEP?"	→ **IgG 50%, IgA 25%, M spike; κ and λ light chains**
"Mechanism of lytic lesions in mets/myeloma?"	→ **RANK ligand acting on osteoclasts**
"True of myeloma bone turnover?"	→ **TNF suppresses osteoblasts** ✗ Wrong options: DKK-1 promotes osteoclasts · OPG ↑ · IL-6/IL-11 counteract · bone scan reliable
"Myeloma hip lesion, IgG/IgA ↑ – osseous care?"	→ **surgical stabilization + radiation + bisphosphonates**
"Myeloma femoral neck lesion?"	→ **cemented hemiarthroplasty or THA**
"Distal humeral pathologic fracture, myeloma?"	→ **internal fixation without resection**

"Myeloma femoral neck fracture after debating prophylaxis — true?"	→ **prophylactic fixation was indicated to avoid delaying systemic therapy** ✗ Wrong options: press-fit stem · radio/chemo-insensitive · poor healing · bone scan ideal
"Bone-protecting agent in myeloma with renal failure?"	→ **denosumab** (TO) ✗ Wrong options: zoledronic acid
"Normal ALP despite extensive lytic lesions?"	→ **multiple myeloma** (TO)
"Kahler disease?"	→ **multiple myeloma (TO)**
"Most common primary malignant bone tumour?"	→ **multiple myeloma (TO)**

– – Active space – –

– – Active space – –

MYOSITIS OSSIFICANS

– Pathology · 40 / 66 –

1 Basics

Diagnosis & traumatic aetiology

Nature	**reactive, benign** soft-tissue bone formation (heterotopic ossification) · aka **hematoma ossificans**
Cause	**direct trauma / contusion → intramuscular haematoma** · near diaphysis of long bones · ⅓ idiopathic · also paraplegia, GBS, AIDS encephalopathy, head injury, burns, coma ✗ Wrong options: surgery · parathyroid tumour · bisphosphonates · steroids
Demographics	**young active males 15–35**
Muscles	**quadriceps, brachialis, glutei** · elbow = commonest joint · **massage / vigorous passive physio after elbow injury provokes it (TO)**
Genetics	post-traumatic, not genetic ✗ Wrong options: COMP · type X collagen · t(9;22) · p53

Genes & lab markers

Early	↑ CRP, ↑ **CPK** (inflammation, muscle)
Mature phase	↑ **ALP** · falling ALP + mature bone on X-ray = maturity **(TO)**
FOP	**ACVR1** mutation (see below)

3 Presentation

Phases (TO)	**acute** (days–weeks): pain, swelling, joint stiffness · **pseudotumoral** (2–3 wk): painless hard mass · **resolving** (3–6 mo)
Mass	grows over months to **3–6 cm**, then firm · palpable · ↓ ROM (e.g. TKA stiffness after thigh blow)

4 Diagnosis & Classification

Key radiology findings

X-ray	initially **normal** · mineralization at **2–3 weeks**, matures **3–6 months** · **peripheral ossification with central lucency** ("dotted veil") · ossified, smooth · **next to but not stuck on / not sharing the cortex**



Myositis ossificans – peripheral ossification

CT	**"eggshell"** peripheral rim
MRI with gadolinium	**rim enhancement in first 3 weeks** · intramuscular · surrounding oedema early
Zonation phenomenon (Ackerman)	matures **periphery → centre** (✓ MO) vs osteosarcoma **centre → periphery**
Variants (TO)	stalked (peduncle to bone) · sessile (broad periosteal base) · muscle-only

Histology – zonal pattern

Zone	Histology
Periphery	**Mature lamellar + woven bone trabeculae** (osteoblastic rimming) · **mature fibroblasts** → calcified on X-ray
Centre	**Immature, proliferating fibroblasts** (mitoses, **atypia may be seen**), ± cartilage · not calcified

TO key

mature fibroblasts at periphery, proliferating at centre (TO) ·
Orthobullets: no cellular atypia · biopsy may mimic sarcoma → **clinical + radiographic correlation critical**
✗ Wrong options: reverse

Differential

Lesion	Distinguishing
Myositis ossificans	**Peripheral** maturation, separate from cortex, trauma history
Extraskeletal OS	**Older**, central ossification, no zonation
Parosteal OS	**Stuck on** posterior distal femur, linear radiolucent gap, 30–40 F
Osteochondroma (sessile)	**Cortex + medulla continuous** with bone
Synovial sarcoma	Wispy irregular calcification within mass
Periosteal chondrosarcoma	On the bone, cartilage mineral
Ossifying lipoma · ossifying mets · extraskeletal chondrosarcoma	Different histology

- - Active space - -

5 Management

Management – observation

Myositis ossificans

- Acute → **REST, ICE, NSAIDs, COMPRESSION, SPLINT, ACTIVE ROM (no passive stretching)**
- Asymptomatic / maturing → **OBSERVE + SERIAL X-RAYS (no biopsy if imaging diagnostic)**
- Symptomatic after maturity (≥6 mo; ~1 yr) → **SURGICAL EXCISION**

Passive stretching / massage	**contraindicated** (worsens) · PT for active ROM
Excision	only if still a problem **after maturation** · **wait ≥6 months** · excision at **6–12 months** → recurrence (Orthobullets) · maturation ~1 yr (TO)
Asymptomatic 13-yr-old with peripheral mature ossification after injury	**observation** ✗ Wrong options: NSAIDs · rheumatology · biopsy · resection

6 Complications

Prognosis	self-limiting · mass shrinks **after ~1 year**

Fibrodysplasia ossificans progressiva (ACVR1)

Gene	**ACVR1** (activin A type I receptor = **BMP type-1 receptor**) · **R206H** gain-of-function → uninhibited SMAD signalling (Activin A, **BMP 2/4**/6/7) · sporadic or **AD**
Hallmarks	**progressive HO** of muscles, fascia, tendons, ligaments, capsules after minor trauma ("**Stone Man**") + **congenital bilateral monophalangeal hallux valgus** (pathognomonic) · cervical fusions, short broad femoral necks, osteochondromas · stooped posture, trismus, recurrent "bony prominences"
Treatment	conservative · surgery only for refractory pain, motion block, nerve compression (excision provokes more HO)

Gene / protein	Condition (distractors)
ACVR1	✓ **FOP**
LMX1B	Nail-patella
GNAS (Gs-α)	Fibrous dysplasia
DTDST (sulfate transporter)	Diastrophic dysplasia
RUNX2	Cleidocranial dysplasia
EXT1/2/3	MHE
HLA-B27	Ankylosing spondylitis

How it's asked

"Groin mass a year after helmet blow, dense ossification adjacent to bone, peripheral enhancement?"	→ **myositis ossificans** ✗ Wrong options: ossifying lipoma · synovial sarcoma · periosteal chondrosarcoma
"Stiff TKA 6 mo after baseball hit to thigh, prostate cancer history?"	→ **myositis ossificans** ✗ Wrong options: ossifying mets · parosteal OS · sessile osteochondroma
"Most likely cause?"	→ **contusion**
"Medial arm mass after arm wrestling, peripheral mineral on CT?"	→ **myositis ossificans** (zonation of Ackerman) ✗ Wrong options: extraskeletal OS · parosteal OS
"Thigh mass after jogging?"	→ **myositis ossificans** ✗ Wrong options: sarcoma · pseudoaneurysm · abscess
"Quadriceps ossified mass in footballer — linked to?"	→ **history of thigh trauma**
"Asymptomatic ossified hip 'knot' 4 mo after injury?"	→ **observation**
"Stone man, trismus, bony prominences, sister affected?"	→ **ACVR1 mutation** ✗ Wrong options: EXT · HLA-B27
"Thigh deformity since youth, ACVR1 mutation?"	→ **FOP** ✗ Wrong options: melorheostosis · Paget · FD · MHE
"Congenital bilateral hallux valgus + stiffness — mutated protein?"	→ **activin receptor type 1A**
"True about myositis ossificans?"	→ **mature fibroblasts at periphery, proliferating at centre (TO)**

– – Active space – –

NEURILEMMOMA

– Pathology · 41 / 66 –

1 Basics

Synonym	**schwannoma** · benign, **encapsulated** nerve sheath tumour of **Schwann cells** on the **surface (epineurium) of a peripheral nerve** · motor or sensory nerves
Age / sex	**3rd–6th decades** (tested ages 40, 45; 20–50) · M = F
Site	**flexor surfaces** of extremities, head and neck · large ones in pelvis · tibial nerve (posterior ankle, tarsal tunnel), median, superficial peroneal
Malignant transformation	**extremely rare** (vs neurofibroma)

Genes & lab markers

NF2 gene mutations
✗ Wrong options: NF1 (neurofibroma)

S-100	**strong, uniform/diffuse** staining

3 Presentation

Usually asymptomatic · slow-growing mass · **paraesthesia** in nerve distribution · discomfort with shoe wear

Percussion (Tinel) sign	**radiating pain distally on tapping** = most reliable sign · sensory deficit possible, **motor weakness rare** · **rest pain = worrying for MPNST**

4 Diagnosis & Classification

Key radiology findings

MRI	**T1 dark**, **T2 bright**, **diffuse gadolinium enhancement** · fusiform, homogeneous, in continuity with a named nerve · **eccentric** to nerve
String sign	nerve tapering into tumour above and below · **target sign** – central low signal



String sign of a schwannoma

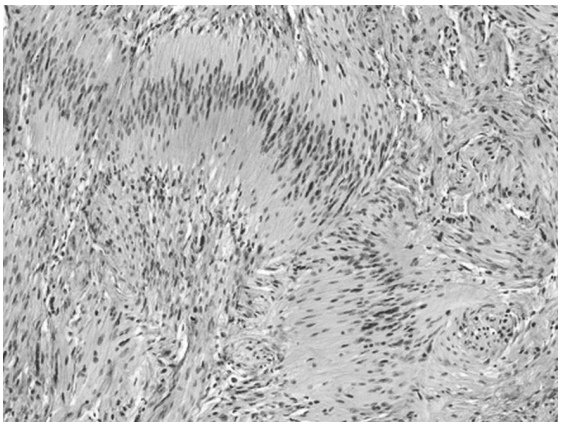
Benign vs MPNST on MRI **well-defined capsule** favours benign
✗ Wrong options: >2 cm peritumoral oedema · lack of enhancement · ≥2 deep lobulations · limb location irrelevant

Hard to separate from neurofibroma on MRI

Histology

Gross grey, **well encapsulated** in nerve sheath

Feature	Description
Antoni A	**Dense, cellular** spindle cells in intersecting bundles ("spindle cell battle formation")
Antoni B	**Hypocellular, loose myxoid** areas
Verocay bodies	**Two rows of palisading nuclei – pathognomonic**



Verocay bodies – palisading nuclei

– – Active space – –

Differential

	Schwannoma	Neurofibroma	MPNST
Relation to nerve	**Eccentric**, on surface, encapsulated	**Within** nerve, fusiform	From nerve/ neurofibroma
Gene	**NF2**	**NF1**	NF1
Histology	Antoni A/B, Verocay	Wavy nuclei, mast cells	Pleomorphic spindle cells
Excision	**Marginal, shell out, nerve preserved**	Requires nerve sacrifice	Wide + nerve
Recurrence / malignancy	Rare / extremely rare	Higher / can transform	–

Others ganglion (mucin-filled synovial-lined sac, near joint) · nodular fasciitis (deep fascia) · lipoma (bright T1 + T2) · haemangioma · synovial sarcoma (periarticular, heterogeneous)

5 Management

Treatment – marginal excision

Observation asymptomatic

Marginal (intracapsular) excision symptomatic / affecting QoL · **excise parallel to nerve fascicles** → tumour extrudes, nerve preserved · recurrence **rare (<1%)** · small risk of sensory deficit

Not indicated wide resection / nerve reconstruction · radiation · chemotherapy · RFA

Fibrolipomatous hamartoma

Intraneural lipoma congenital, benign · **median nerve** commonest · MRI: **"coaxial cable"** (axial) / **"spaghetti"** (longitudinal) – fat between thickened fascicles (T1 bright) · asymptomatic → observe · symptomatic → **carpal tunnel decompression ± epineurolysis**

How it's asked

"Posterior ankle mass, percussion radiates into foot – true?"	→ **Verocay bodies on histology** ✗ Wrong options: NF1 · mucin sac (ganglion) · MPNST transformation · high recurrence
"Tender shoulder mass, Antoni A/B, uniform S-100?"	→ **schwannoma** ✗ Wrong options: MFH · lymphoma · synovial sarcoma · neurofibroma
"String + target signs, palisading nuclei?"	→ **schwannoma** ✗ Wrong options: nodular fasciitis · lipoma · haemangioma
"Medial ankle mass, Tinel over tarsal tunnel?"	→ **neurilemmoma** ✗ Wrong options: neurofibroma · ganglion

"MRI feature favouring benign nerve sheath tumour?"	→ **well-defined capsule**
"Symptomatic arm schwannoma — treatment?"	→ **marginal resection only** ✗ Wrong options: + RT · wide + chemo · RFA
"Which patient needs only marginal resection?"	→ **62-yr-old with nerve-sheath mass + Tinel sign** ✗ Wrong options: Ewing · OS · liposarcoma · impending fracture
"Median-nerve schwannoma, frozen section Verocay bodies?"	→ **marginal resection parallel to fascicles**
"Superficial peroneal schwannoma with paraesthesia?"	→ **marginal excision** ✗ Wrong options: observation · wide resection
"Fibrolipomatous hamartoma — commonest nerve?"	→ **median**

– – Active space – –

NEUROBLASTOMA

– Pathology · 42 / 66 –

1 Basics

Nature	malignant tumour of **sympathetic neural tissue** (neural crest) · near **adrenal gland** or spinal cord (neck, chest, abdomen, pelvis)
Incidence	~1/100,000 children · slight male predominance · majority **<2 yr**
Rank	**commonest (malignant) solid tumour of childhood** (Orthobullets, OB) · TO: commonest **extracranial / abdominal / infancy** tumour; commonest **solid** tumour overall = **brain tumour** – read the options · leukaemia = commonest childhood malignancy (not solid)
Small round blue cell tumour by age	**<5 yr → metastatic neuroblastoma** (also leukaemia) · bone mets common in children

3 Presentation

Systemic	fever, malaise, weight loss, diarrhoea · **tachycardia** (adrenal)
Exam	**abdominal mass**, hepatosplenomegaly

4 Diagnosis & Classification

Key radiology findings

X-ray	bone mets: **permeative, lytic**, anywhere · may cause vertebra plana
CT chest/abdomen/ pelvis	initial study, visceral extent · **MRI** – anatomy · **bone scan** – bony mets



Metastatic neuroblastoma

Biopsy	needle or open incisional required

Histology

Small round blue cells forming rosettes (Homer-Wright) · neurofilament + (vs Ewing –)

Other small round blue cell tumours of childhood	rhabdomyosarcoma, non-Hodgkin lymphoma, Ewing/PNET, blastemic Wilms

– – Active space – –

5 Management

Neuroblastoma

- Very young infant, low risk → **OBSERVE (spontaneous regression)**
- Low grade, resectable with wide margin → **SURGICAL EXCISION**
- High grade / metastatic, unresectable → **CHEMO (cisplatin, ifosfamide) + STEM-CELL TRANSPLANT**
- High grade near NV structures → **CHEMO + SCT + EXCISION**

6 Complications

Prognosis	**majority cured** · **bone metastasis = poor prognosis** · spontaneous regression in very young

How it's asked

"Most common malignant solid tumour of childhood?"	→ **neuroblastoma** ✗ Wrong options: leukaemia (not solid) · lymphoma · Ewing · EG
"Most common tumour in children (FNB)?"	→ **brain tumour** (neuroblastoma = commonest extracranial) **(TO, OS chapter)**
"Small round blue cell tumour in a child <5?"	→ **metastatic neuroblastoma**

– – Active space – –

NEUROFIBROMA

– Pathology · 43 / 66 –

1 Basics

Nature benign peripheral nerve sheath tumour · mostly with **neurofibromatosis (NF1)**

Age sporadic **20–40** · NF1 **<20**

Site sporadic: peripheral nerves of extremities · NF1: near **spinal cord/brain**, unusual sites (parotid, tongue)

Cells **fibroblasts predominate** + **non-myelinating Schwann cells (cell of origin)** + perineural cells, mast cells (vs schwannoma = Schwann cells only)

Genes & lab markers

NF1 **autosomal dominant**, 17q11.2 · **biallelic inactivation** of NF1 tumour-suppressor in non-myelinating Schwann cells → loss of **neurofibromin** → **↑ RAS** (RAS-MAPK) activity

MPNST transformation + loss of **CDKN2A or TP53**

S-100 **variable** (vs uniform in schwannoma)

Schwann cell	Axons	Disease
Myelinating	Large-diameter, myelin	–
Non-myelinating	**Small-diameter**, cytoplasmic processes	**Neurofibromatosis**

Subtypes

	Dermal	Plexiform
Origin	**Single** nerve	**Multiple** nerve bundles
Look	**Fusiform** swelling	**"Bag of worms"**
Association	90% sporadic, 10% NF1	**Pathognomonic of NF1**
Onset	Puberty	Early childhood
Malignancy	Never	**10% → MPNST**

3 Presentation

Symptoms asymptomatic · stinging, itching, pain, disfigurement · **exquisite tenderness** (benign nerve sheath tumour hallmark) · reduced sensation

Cutaneous "**button-holes**" through skin defect with finger pressure

NF1 diagnostic criteria (NIH – 2 of 7)

Criterion
≥2 neurofibromas or 1 plexiform
Axillary/groin freckling (Crowe sign)
≥6 café-au-lait spots ("coast of California") – >5 mm prepubertal, >15 mm postpubertal
Sphenoid wing dysplasia or long-bone cortical thinning (**anterolateral tibial bowing**)
≥2 Lisch nodules (iris hamartomas, slit lamp)
Optic glioma
First-degree relative with NF1

NF1 associations **dystrophic kyphoscoliosis**, **rib penciling**, intraspinal **dumbbell** neurofibromas, **dural ectasia**, meningiomas · cancer risk ~4×

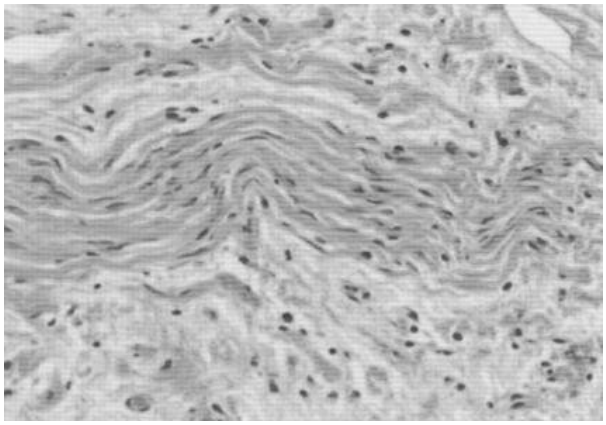
4 Diagnosis & Classification

Key radiology findings

MRI ± contrast T1 low, T2 very high (myxoid-type)

Sign	Meaning
Nerve continuity (entering-exiting)	Fusiform tumour in continuity with NV bundle
Split fat	Thin rim of fat around mass in NV bundle
Target	High-signal periphery, low centre (STIR)

Neurofibroma vs schwannoma neurofibroma **central** to nerve (fascicles run through it, no capsule) · schwannoma **eccentric**, encapsulated



Neurofibroma — wavy collagen

PET-CT **distinguishes MPNST from benign neurofibroma** (↑ SUV) in NF1 with multiple nodules
✗ Wrong options: UPS vs MPNST · infection vs tumour

– – Active space – –

Histology

Hypocellular, **fibroblasts** + Schwann cells, mast cells, lymphocytes · **elongated, wavy nuclei** · **"wire-like"/rope-like wavy collagen** · may be myxoid · corpuscle-like (Pacini/Meissner) structures

Histology	Diagnosis
Wavy collagen, elongated cells, hypocellular	✓ **Neurofibroma**
Herringbone, atypia	Fibrosarcoma
Stromal cells, haemosiderin macrophages, giant cells	GCT of tendon sheath
Mono/biphasic epithelioid cells	Synovial sarcoma

MPNST risk

MPNST in NF — **5% (10–25% lifetime)** · usually from **plexiform** · **painful, enlarging** mass · poor prognosis

5 Management

Observe — asymptomatic

Surgical excision — symptomatic · tumour within nerve → **may need nerve sacrifice / grafting** (vs schwannoma shelled out)

How it's asked

"Forearm mass 6 yr, episodic sharp pain, fusiform on median nerve?"	→ **nerve sheath tumour** ✗ Wrong options: sarcoma · metastasis · haematoma · abscess
"Buttock pain sitting, sciatic paraesthesia, wavy collagen?"	→ **neurofibroma** ✗ Wrong options: myxoid liposarcoma · myxoma · MFH · fibromatosis
"Exquisitely tender long-standing leg mass in tibial nerve, rope-like collagen?"	→ **neurofibroma** ✗ Wrong options: fibrosarcoma · GCT tendon sheath · synovial sarcoma
"NF1 with multiple nodules – PET-CT distinguishes?"	→ **MPNST vs benign neurofibroma**

NODULAR FASCIITIS

– Pathology · 44 / 66 –

1 Basics

Nature	**commonest fibrous soft-tissue lesion** · **pseudosarcomatous, self-limiting reactive** proliferation of **fibroblasts + myofibroblasts**
Demographics	M = F · adults **20–40** · young people
Site	**half in upper limb** · **volar forearm**, back, chest wall, head and neck
Trigger	often a recalled trauma

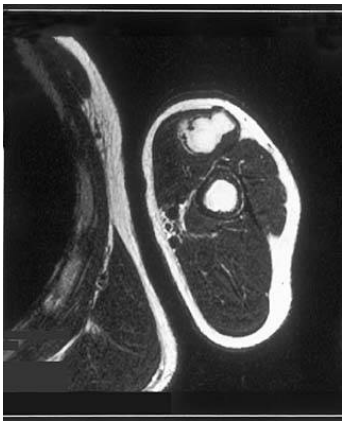
3 Presentation

Rapidly enlarging mass over 1–2 weeks (present only weeks) · **1–2 cm** · painful in half

4 Diagnosis & Classification

Key radiology findings

X-ray	normal
MRI	deep-seated, **inhomogeneous** mass · **nodularity extending along fascial planes** · **avid gadolinium enhancement** · usually superficial, also intramuscular · T2 hyperintense, T1 iso/hypo



Nodular fasciitis on MRI

Histology

Plump, immature-looking fibroblasts (like granulation tissue) in **short, irregular bundles and fascicles** · **dense reticulin network** · small amounts of mature collagen

Evolution	short duration → **myxoid** · longer → **hyaline fibrosis**

– – Active space – –

5 Management

Marginal resection

Treated with marginal excision
Nodular fasciitis
Periosteal chondroma
Neurilemmoma
Epidermal inclusion cyst
Glomus tumour

How it's asked

"40-yr-old, rapidly enlarging shoulder mass, short irregular fibroblast fascicles in reticulin?"	→ **nodular fasciitis** ✗ Wrong options: fibrosarcoma · intramuscular haemangioma · schwannoma · rhabdomyoma
"Deep fascia lesion, homogeneous T2-bright (vs schwannoma)?"	→ **nodular fasciitis** arises in deep fascia

NON-OSSIFYING FIBROMA

– Pathology · 45 / 66 –

1 Basics

Epidemiology

Commonest benign bone lesion of childhood (with osteochondroma) · in **30–40% of skeletally immature children** · multiple in **8%**

Demographics	**males 2:1** · **5–15 yr**
Site	**metaphysis** of long bones · **80% lower limb**: **distal femur > proximal tibia > distal tibia** · uncommon proximal femur/humerus
Synonyms	metaphyseal fibrous defect, non-osteogenic fibroma, **fibrous cortical defect** (<2 cm), fibroxanthoma

Pathophysiology

Benign fibrogenic lesion from **dysfunctional ossification** – abnormal subperiosteal osteoclastic resorption during metaphyseal remodelling

Genetics	somatic **RAS-MAPK activating mutations** → considered a neoplasm (RASopathy family) · mutation in a subset of cells; histiocytes + giant cells reactive

Associated conditions

Jaffe-Campanacci syndrome	**multiple NOFs + café-au-lait + mental retardation/cognitive delay + heart, eye, gonadal defects + giant cell lesions of the mandible** ✗ Wrong options: Jaffe-Lichtenstein (FD + café-au-lait) · Maffucci · Mazabraud · McCune-Albright
Others	**neurofibromatosis** · familial multifocal NOF · ABC

Genes & lab markers

RAS-MAPK somatic mutations · no lab abnormality

3 Presentation

Asymptomatic, incidental	found after ankle/knee sprain or fall · pain from the injury, not the lesion
Pathologic fracture	± stress fractures with cortical involvement

4 Diagnosis & Classification

Key radiology findings

X-ray (diagnostic)	**eccentric, metaphyseal, cortically based "bubbly"/scalloped lytic lesion** with **thin sclerotic rim** · narrow zone of transition · cortex may be thinned/expanded · **length > width** · no periosteal reaction, no soft-tissue mass

– – Active space – –

TO descriptors **"bunch of grapes"**, sclerotic / **crenated margins** · **no marrow oedema** · can be **multifocal** (TO)



Eccentric bubbly metaphyseal lesion with sclerotic rim

With growth migrates to diaphysis, enlarges (1–7 cm) · near maturity → **sclerotic**, reossifies

Quantitative CT predicts **fracture risk**

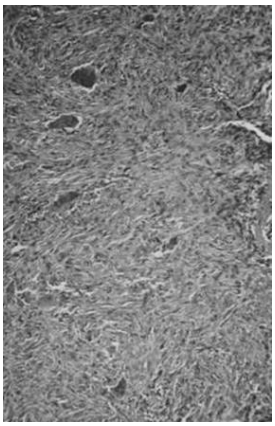
MRI T1 low · T2 varies (higher when developing) · thin enhancing rim · rarely needed

Mimic **adductor magnus avulsion / cortical desmoid** at posteromedial distal femur

No biopsy, MRI, bone scan, staging for classic lesions

Histology

Fibroblastic spindle cells in whorled / storiform pattern ("helicopters in a wheat field") · fibroblastic background · **lipophages (foamy histiocytes)**, **giant cells** · **haemosiderin** · occasional ABC component



Storiform fibroblasts with giant cells

Differential

	Bubbly lytic X-ray	Haemosiderin	Observation alone
NOF	✓	✓	✓
ABC · UBC	✓	–	–
PVNS	–	✓	–
FD · enchondroma · osteochondroma · EG · Paget	–	–	✓

Lesion	Distinguishing
NOF	**Eccentric, scalloped, metaphyseal**, sclerotic rim
UBC	**Central**, expands to physis width, **fallen leaf / fallen fragment** sign
FD	Central, **ground glass**
GCT	Painful, skeletally mature, subchondral, no mineral, destructive
Osteofibrous dysplasia	**Diaphyseal, anterior tibial cortex**
Enchondroma	Central, mineralized, not sclerotic rim
Osteosarcoma	Painful, permeative, periosteal reaction, mineralized mass
Chondroblastoma	Epiphyseal

Cortical desmoid (TO)

Posteromedial distal femoral metaphysis (**adductor magnus** traction), boys 10–15 · cortical erosion with sclerotic base, no lytic lesion · **no biopsy**, observe · "posterolateral" is wrong **(TO)**

5 Management

Non-ossifying fibroma

- Incidental, asymptomatic → **OBSERVE (X-ray 6 + 12 mo, then yearly; or 2–3/3–6 mo check)**
- Pathologic fracture, non-displaced → **CAST / SLING, let fracture heal, then observe**
- Symptomatic + large (>50–75% cortex / >50% diameter) or high QCT risk → **CURETTAGE + GRAFT ± PROPHYLACTIC FIXATION**

Observation — first line · **reossifies spontaneously** in 2nd–3rd decade · non-urgent referral
✗ Wrong options: biopsy · MRI · bone scan · CT staging · cast · CPS

Pathologic fracture — treat the fracture (casting · humerus → **sling + analgesia**) · curettage only after healing unless ORIF needed
✗ Wrong options: IM nail without diagnosis · aspiration + injection (UBC)

Curettage + bone grafting — **symptomatic + large** (>50–75% cortical involvement / just over half diameter in weight-bearing bone) · persistent activity pain → **+ prophylactic internal fixation**
✗ Wrong options: RFA · doxycycline · steroid injection · sclerotherapy · wide excision

– – Active space – –

6 Complications

Pathologic fracture

90% lower limb · 50% distal tibia

Risk factors	**>50% of transverse diameter · >33 mm length** in weight-bearing bones
Prognosis	spontaneous resolution · **no malignant or metastatic potential** ✗ Wrong options: 5–10% (that's MHE)

How it's asked

"20-yr-old skier, incidental lateral distal femoral sclerotic-rimmed lesion?"	→ **reassurance + WBAT** ✗ Wrong options: biopsy · resection
"Distal fibula lesion after sprain, asymptomatic 16-yr-old?"	→ **activities as tolerated, X-ray in 3–6 mo** ✗ Wrong options: staging · cast · curettage
"6-yr-old, knee abrasion, tibial lesion?"	→ **repeat X-ray in 3 months**
"14-yr-old tibial lesion after ankle injury?"	→ **routine follow-up** ✗ Wrong options: endocrine (McCune-Albright) · GI cancer (Gardner) · open biopsy
"18-yr-old soccer player — diagnosis + treatment?"	→ **NOF, observation** ✗ Wrong options: enchondroma · ABC curettage
"Associated with NOF EXCEPT?"	→ **fallen leaf sign** (UBC) (✓ stress fractures, ✓ commonest childhood lesion, ✓ Jaffe-Campanacci, ✓ haemosiderin)
"9-yr-old, mild ankle pain, family doctor calls?"	→ **symptom management, non-urgent referral** ✗ Wrong options: urgent CT/labs · RFA · curettage
"Incidental distal tibial / proximal tibial lesion?"	→ **observation** ✗ Wrong options: MRI · bone scan · biopsy
"15-yr-old 'shadow on the bone', eccentric sclerotic metaphyseal?"	→ **NOF** ✗ Wrong options: GCT · infection · OFD (diaphyseal) · enchondroma
"NOF best characterized as?"	→ **eccentric scalloped metaphyseal radiolucency** ✗ Wrong options: fallen leaf · central fusiform to physis width (UBC) · ground glass (FD)

"True of NOF?"	→ **typically ossifies spontaneously** ✗ Wrong options: trauma-related · cast first-line · high fracture risk · 5–10% malignancy
"Trampoline ankle injury, bubbly lesion — histology?"	→ **storiform fibroblasts**
"Pathologic fracture through which lesion (lobulated, eccentric)?"	→ **NOF**
"Humeral pathologic fracture through eccentric soap-bubble lesion?"	→ **sling + pain medication**
"Outside core biopsy, no treatment needed — histology?"	→ **whorled/storiform fibroblastic spindle cells**
"11-yr-old gymnast, limp, NOF >50% diameter, daily pain?"	→ **curettage + bone grafting + prophylactic internal fixation**
"Persistent pain, 4 cm NOF over half the diameter?"	→ **curettage + bone grafting** ✗ Wrong options: observation · cast · sclerotherapy
"Multiple NOFs, café-au-lait, cognitive delay, eye problems, mandibular lesion?"	→ **Jaffe-Campanacci syndrome**
"Cortical desmoid — all true EXCEPT?"	→ **"posterolateral distal femur"** (it is posteromedial) **(TO)**

– – Active space – –

OSTEOBLASTOMA

– Pathology · 46 / 66 –

1 Basics

Epidemiology

Rare	**1% of primary bone tumours**, 1–5% (3%) of benign · **10% of osseous spinal neoplasms** · less common than osteoid osteoma
Demographics	**M:F 2:1** · **10–30 yr**
Site	**posterior elements of spine + sacrum (30–40%; >35–40%)** · long bones (lower > upper), diaphysis/metadiaphysis (femur commonest), never epiphyseal · **mandible ("cementoblastoma")**
Spinal level (TO)	**cervical > lumbar** > thoracic > sacrum (17%) – no firm consensus; keyed answer **cervical**
Posterior element tumours (TO)	**osteoid osteoma > osteoblastoma > ABC** · GCT → vertebral body
Nature	**benign, locally aggressive, progressive** ✗ Wrong options: self-limiting

Genes & lab markers

FOS (chr 14) / FOSB (chr 19) rearrangements

Nuclear β-catenin staining → osteoblastoma · **cytoplasmic/membranous** → osteosarcoma

Associations	**secondary ABC 10–40%** · **oncogenic osteomalacia**

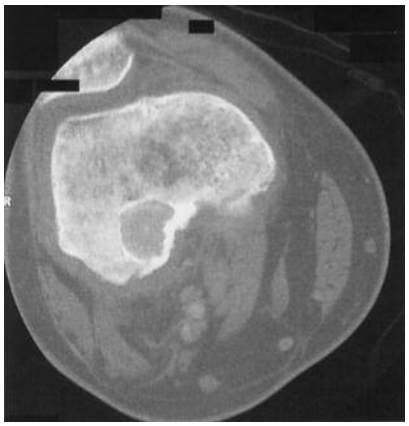
3 Presentation

Pain	slowly progressive **dull ache** · night pain **not typical** · **partial NSAID response** (TO FNB stem: night pain improving with NSAIDs, posterior element, 28 yr → keyed osteoblastoma)
Spine	**neurologic symptoms** (root compression, weakness, rarely paraplegia), scoliosis
Signs	swelling, muscle atrophy, limp · may be incidental

4 Diagnosis & Classification

Key radiology findings

X-ray	**lytic or mixed lytic-blastic**, **radiolucent nidus 2–10 cm** (>2 cm), intralesional densities · reactive sclerosis · expansile · 66% cortical, 33% medullary · **25% look aggressive** (mistaken for malignancy)

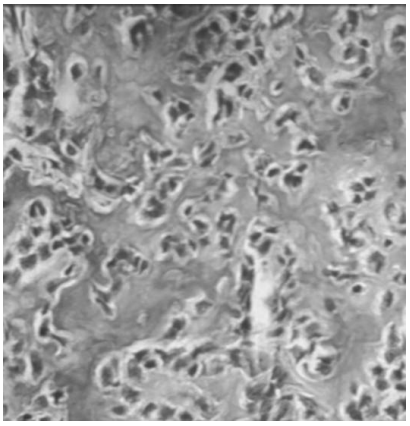


Osteoblastoma on CT

CT	full characterization: size, location, cortical breach, soft-tissue component
MRI	reactive soft-tissue/marrow **oedema** · T1 low–intermediate, T2 intermediate–high
Bone scan	**hot**, intense focal uptake

Histology

Like osteoid osteoma, more giant cells · **sharp demarcation nidus–reactive bone** · nidus of **immature osteoid + plump osteoblasts** (normal nuclei) · irregular osteoid/woven spicules rimmed by osteoblasts in **loose fibrovascular stroma** · mitoses but **no atypical** ones · rich vascularity, extravasated RBCs · **no permeation of host lamellar bone**



Osteoblastoma histology

Osteoid osteoma vs osteoblastoma

	Osteoid osteoma	Osteoblastoma
Incidence	10% of benign	3% of benign
Size	**<2 cm** (<1.5)	**>2 cm** (3–3.5)
Site	>50% long-bone diaphysis (proximal femur > tibia > phalanges > spine)	**>35% posterior spine**
Natural history	**Self-limited**	**Progressive**
Symptoms	**Night pain, relieved by NSAIDs** · painful scoliosis, lesion on **concave** side	Dull ache, **partial NSAID response**, neuro symptoms
Spine management	Nonoperative first → RFA/resection	**Always surgery**

– – Active space – –

Differential

Lesion	Distinguishing
Osteosarcoma	Periostitis, soft-tissue mass, **permeates lamellar bone**, malignant cells, cytoplasmic β-catenin
ABC	Fluid levels, septations, blood-filled sinusoids
Osteoid osteoma	<2 cm, thick sclerosis around small nidus
GCT	Epiphysis–metaphysis, giant cells with stromal nuclei
Chondroblastoma	Epiphyseal, chondroid
NOF	Eccentric metaphyseal, storiform
FD	Ground glass, Chinese letters
Osteomyelitis · stress fracture	Sinus/skin changes · no large lucent lesion

5 Management

Osteoblastoma (doesn't respond to nonoperative care)

- Most → **CURETTAGE / MARGINAL EXCISION + BONE GRAFT (± phenol, cryo)**
- Large, locally aggressive, recurrent, some spinal → **EN BLOC RESECTION (lower recurrence)**
- Selected small lesions → **ABLATION (RFA / MRgFUS)**

Observation rarely (keeps growing)

Treatment — ablation (MRgFUS vs RFA) (TO)

MRgFUS **less invasive** (needleless, no ionizing radiation), real-time thermal map · does not penetrate deeper (deep lesions → RFA needle reaches better)

RFA CT-guided, ~17G needle, single-day stay

Both equally safe/effective · equal resolution of **bone oedema** (oedema loss = inactive lesion) · post-ablation **"halo/ring"** sign (necrotic centre + sublethal rim) · not "RFA less chondral damage" (claim not supported)

6 Complications

Local recurrence **15–25%**, mostly within **2 years**

Prognosis excellent, most disease-free after first surgery

How it's asked

"Most common site of osteoblastoma (spinal levels)?"	→ **cervical spine** (keyed) **(TO)**
"28-yr-old, posterior element tumour, night pain improves with NSAIDs, no deficit?"	→ **osteoblastoma** (keyed; ✗ Wrong options: ABC · GCT · lymphoma) **(TO)**

"17-yr-old, distal humeral bone-forming lesion extending beyond cortex, large osteoblasts, no atypia?"	→ **osteoblastoma** ✗ Wrong options: osteoid osteoma (smaller) · OS · chondroblastoma · GCT
"18-yr-old knee, well-marginated lytic/blastic, benign osteoblasts?"	→ **osteoblastoma** ✗ Wrong options: FD · OS · NOF · osteomyelitis
"23-yr-old, back pain radiating down leg relieved by NSAIDs, mixed lesion >2 cm?"	→ **osteoblastoma** ✗ Wrong options: OS · GCT · ABC · stress fracture
"True about osteoblastoma ablation?"	→ **MRgFUS less invasive than RFA (TO)**
"Touch-me-not lesions EXCEPT?"	→ **osteoblastoma** (painful, needs intervention) **(TO, FD chapter)**

– – Active space – –

OSTEOCHONDROMA & MULTIPLE HEREDITARY EXOSTOSIS

– Pathology · 47 / 66 –

1 Basics

Epidemiology

Commonest benign bone tumour (20–50%) · commonest benign cartilage tumour · tested ages 9–24 (adolescents, young adults)

Site	**bone surface**, metaphysis, often at **tendon insertions** · **knee** (distal femur > proximal tibia) · proximal femur, proximal humerus · **femur 30%**, tibia + humerus 10–20% each · **2nd commonest site: pelvis (iliac crest) (TO)** · spine rare (posterior elements of **cervical** spine) · **never facial bones** (intramembranous) · subungual exostosis (hallux)
Polyostotic conditions (TO)	FD (McCune-Albright), **MHE** ("diaphyseal aclasis"), enchondroma (Ollier/ Maffucci) ✗ Wrong options: **haemangioma** (non-polyostotic)

Pathophysiology

Hamartomatous bone + cartilage · physeal cartilage escapes through a **perichondral ring (node/ zone of Ranvier) defect** and grows by endochondral ossification under the periosteum → **grows away from the joint**

Secondary causes	Salter-Harris fracture · surgery · **radiation** (**commonest benign radiation-induced bone tumour**, 6–24%, latency 3–17 yr)

MHE genetics (EXT1/EXT2, heparan sulfate, AD)

Inheritance	**autosomal dominant** · penetrance **96% females, 100% males** (>95%) · child of affected parent ≥50% risk · ~15% of osteochondroma patients
Genes	**EXT1 (8q24), EXT2 (11p13)**, EXT3 (19) – **tumour suppressors** (exostosins) → **↓ heparan sulfate** in physeal chondrocytes · affect **prehypertrophic chondrocytes** · dysregulated **Indian hedgehog** signalling

EXT1 vs EXT2 EXT1 **more severe**: more exostoses, more malalignment, shorter limbs/height, less forearm + knee ROM, more pelvic/flat-bone lesions, **higher chondrosarcoma rate**
✗ Wrong options: lower

Gene	Condition (distractors)
EXT1/2/3	✓ **MHE**
PHEX	X-linked hypophosphataemic rickets
PMP-22	Charcot-Marie-Tooth
SYT-SSX	Synovial sarcoma
GNAS	FD / McCune-Albright
IDH1	Ollier / Maffucci
DTDST	Diastrophic dysplasia (cauliflower ear, hitchhiker thumb) – AR
FGFR2 · FGFR3	Apert · achondroplasia
RUNX2/CBFA1	Cleidocranial dysplasia

Other inheritance contrasts Ollier sporadic · fibular hemimelia none · sickle (infarcts) AR

3 Presentation

Solitary **painless mass** · mechanical symptoms, irritation (bursitis), NV compression · grows until skeletal maturity

MHE family history · multiple masses (knees, wrists, shoulders)

Forearm & ankle deformity in MHE

Deformities **short stature**, **LLD**, femoral shortening · **coxa valga** (lesser trochanter lesions) · **knee valgus** + patellar dislocation and **ankle valgus** (short fibula) · **ulnar shortening + radial bowing**, **ulnar deviation of wrist**, **radial head (radiocapitellar) dislocation** · pectoral/pelvic asymmetry · premature OA · no absent thumb

Masada	Description	Frequency
I	**Distal ulnar exostosis → short ulna + radial bowing, radial head NOT dislocated**	**Commonest (55–61%) (TO)**
II	Short ulna + **radial head dislocated** · IIa proximal radial exostosis · IIb no proximal radial exostosis	22–33%
III	**Distal radial** exostosis → relative **radial shortening**	–

Masada treatment (TO) I: conservative / excise painful lesion · II: wait to ~13 yr, then levelling ± radial head resection or ring-fixator reduction · III: excise painful lesion, ulnar shortening for positive variance

Forearm rotation loss **osteochondroma excision** most effective · deformity → excision + ulnar lengthening + radial closing-wedge osteotomy
✗ Wrong options: osteotomies · radial head resection

Ankle valgus in a 7-yr-old **medial (distal tibial) hemiepiphysiodesis**
✗ Wrong options: full epiphysiodesis · cast · curettage · resection alone

– – Active space – –

4 Diagnosis & Classification

Key radiology findings

X-ray (diagnostic) — **sessile (broad) or pedunculated (stalk)** surface lesion · **pedunculated points away from the joint** (in both solitary and MHE) · **cortex continuous** with host · **medullary cavity continuous** with host (**pathognomonic**) · cartilage cap radiolucent, involutes at maturity



Medullary continuity of osteochondroma on CT

Cartilage cap — **MRI best** (T1 low, T2 high) · US and CT also measure it

Cap thickness	Meaning
Child (immature)	**1–3 cm** normal – thick cap = growth, **not** malignancy
Adult (mature)	**2–3 mm** normal
Adult >2 cm (≥2 cm)	**Secondary chondrosarcoma** concern (Orthobullets, TO) · OB variants: **>1.5 cm**, ">1–2 cm", ">2–3 cm" · **thickening** in adulthood (✗ Wrong options: thinning)

Biopsy — not recommended for diagnosis
✗ Wrong options: needle biopsy · bone scan · PET

Histology

Like a normal physis — **hyaline cartilage cap** + **well-defined perichondrium** + linear columns of active chondrocytes + normal primary trabeculae · stalk = cortical + cancellous bone

Differential (bony bumps)

Lesion	Key feature
Osteochondroma	**Cortical + medullary continuity**
Subungual exostosis (Dupuytren)	Dorsal distal phalanx (hallux), **no medullary continuity**
Trevor disease (dysplasia epiphysealis hemimelica)	**Epiphyseal** osteochondromas, lower limb, M:F 3:1, no malignancy
Turret exostosis	Dome on dorsal proximal/middle phalanx, post-traumatic, no continuity
BPOP (Nora lesion)	Hand phalanges/metacarpals, bone + cartilage + fibrous, high recurrence
Parosteal OS	Posterior distal femur, dense lobulated, no continuity
Juxtacortical chondroma	Saucerized cortex, 20–40
Subperiosteal haematoma	Smooth elliptical, trauma
Myositis ossificans	Doesn't share cortex, zonation

"Benign bone tumour presenting as a bump" → **osteochondroma** (turret exostosis + Nora lesion = non-tumours) **(TO)**

Cartilage tumours (TO) chondroblastoma, osteochondroma, CMF all cartilage → "EXCEPT" = **none**

5 Management

Treatment – observation vs marginal excision

Osteochondroma

- Asymptomatic / minimal (incl. MHE child) → **OBSERVATION**
- Symptomatic, cosmetic, mechanical → **MARGINAL RESECTION at stalk base + whole cartilage cap**
  (delay to skeletal maturity if possible → ↓ recurrence)
- MHE deformity → **excision, hemiepiphysiodesis, osteotomy**
- Growth/pain after maturity, cap >2 cm → MRI/biopsy → secondary chondrosarcoma → **WIDE RESECTION**

Marginal resection resect at **base of stalk including entire cartilage cap** (↓ regrowth) · graft/fix large defect
✗ Wrong options: chemo · RT · curettage/liquid N_2 · radical resection

No further tests before excising a classic symptomatic lesion (MRI already done)

Incidental humeral / femoral sessile lesion, asymptomatic **observation**
✗ Wrong options: MRI, CT, bone scan, biopsy, long-leg films if alignment normal

MHE child **observe** · excision for radial head dislocation, loss of rotation, symptoms
✗ Wrong options: biopsy largest · PET · remove all · bisphosphonates

– – Active space – –

Subungual exostosis

Hallux, adolescents · reactive metaplasia (microtrauma) · painful, may ulcerate · mistaken for wart · histology: mature trabecular bone capped by **fibrocartilage**

Treatment — **nail removal + excision / osteotomy through base**, local wound care (protect nail matrix) · recurrence possible
✗ Wrong options: observation if painful · amputation · chemo/RT

6 Complications

Malignant transformation to chondrosarcoma

Setting	Risk
Solitary osteochondroma	**<1%** (Orthobullets) · **1%** (TO)
MHE	**5–10%** (Orthobullets) · 3–10% · 0.57–8.3% · ~10% · "1 in 10,000" (one OB) – read options

MHE (TO) — **5%** malignant transformation

Risk factors — **pelvic/central (proximal) location** (highest), **sessile**, MHE, **EXT1**, **older age**, cap >2 cm in adult

Secondary chondrosarcoma — **low grade (~90%; 67–85%)** · 3rd–4th decade (tested 50) · **pelvis** commonest · <1% in children · presents as **new pain / growth after maturity** · pelvic mass → urinary retention, perineal numbness

Treatment — **wide resection** (same as chondrosarcoma), curative 70–90% · pelvic → multidisciplinary (general + plastic)
✗ Wrong options: chemo · RT · local excision

Other complications

Vascular — **popliteal artery pseudoaneurysm** (rare), compression, true aneurysm, arterial/venous thrombosis

Nerve compression (up to 25%) — sciatic, **common peroneal** (anterior + lateral compartment atrophy), radial

Tendon — shoulder: cuff impingement, subscapularis tear, bicipital tendinitis

Bursa formation (± metaplastic cartilage nodules) · **recurrence 2–5%** after resection

How it's asked

"HME – EXCEPT?"	→ **"exostoses grow towards the physis"** (they grow away) (✓ AD, ✓ prehypertrophic chondrocytes, ✓ EXT1-3, ✓ medullary continuity)
"MHE – EXCEPT?"	→ **"grow towards joint in MHE but away in solitary"** (both grow away)
"NOT associated with MHE?"	→ **absence of the thumb**
"EXT1 vs EXT2 – all EXCEPT?"	→ **lower sarcomatous transformation** (EXT1 is higher)

"AD gene defect with ↑ chondrosarcoma risk?"	→ **EXT-1** ✗ Wrong options: PHEX, PMP-22, SYT-SSX, GNAS-1
"EXT-1 defect — diagnosis + tumour risk?"	→ **MHE, chondrosarcoma**
"X-ray for EXT1/2/3 mutation?"	→ **multiple osteochondromas** ✗ Wrong options: rickets · cleidocranial · NF bowing · Madelung
"Which X-ray shows an AD disorder?"	→ **MHE** ✗ Wrong options: infarct/sickle AR · diastrophic AR · fibular hemimelia · Ollier sporadic
"MHE with EXT — true?"	→ **↓ heparan sulfate in physis** ✗ Wrong options: cap thinning = malignancy · EXT2 worse · 25% malignancy · distal lesions riskier
"Father with exostoses + hemiepiphysiodesis, child with bumps — gene?"	→ **EXT1**
"Incidental lesion — true in children?"	→ **secondary transformation usually low grade, cured by surgery alone** ✗ Wrong options: cap <1 mm · 20%
"62-yr-old, enlarging lifelong medial thigh bump, multiple bumps?"	→ **EXT; secondary low-grade chondrosarcoma**
"MHE, pelvic grade 2 chondrosarcoma with urinary retention?"	→ **wide resection with multidisciplinary team**
"Highest risk of malignant transformation?"	→ **pelvic location** ✗ Wrong options: sessile · isolated · young age · 2 mm cap
"Non-polyostotic tumour?"	→ **haemangioma (TO)**
"2nd commonest site of osteochondroma?"	→ **pelvis (TO)**
"Benign tumour presenting as bump on long bones?"	→ **osteochondroma (TO)**
"Sessile medial distal femoral lesion sharing cortex + medulla?"	→ **osteochondroma** ✗ Wrong options: myositis ossificans · parosteal OS
"Thoracic transverse-process lesion on trauma CT — aetiology?"	→ **aberrant cartilaginous growth**
"Symptomatic pedunculated tibial osteochondroma?"	→ **marginal resection** ✗ Wrong options: chemo · radical · curettage + liquid N_2
"14-yr-old, sessile anterior femoral exostosis, painless?"	→ **observation**
"Painful, cosmetic distal femoral osteochondroma, skeletally mature?"	→ **marginal resection with complete cap removal**
"Tests before resecting classic symptomatic lesion?"	→ **none**
"7-yr-old with multiple exostoses?"	→ **observation**

– – Active space – –

"Most common forearm deformity in HME?"	**→ ulnar exostosis with radial bowing (Masada I) (TO)**
"Limited pronation/supination in MHE?"	**→ osteochondroma excision**
"7-yr-old, ankle valgus from fibular exostosis?"	**→ medial hemiepiphysiodesis**
"Painful hallux lesion misdiagnosed as wart?"	**→ nail removal + excision / osteotomy through base**

OSTEOFIBROUS DYSPLASIA

– Pathology · 48 / 66 –

1 Basics

Nature	rare, **benign fibro-osseous** tumour-like lesion of the **tibial (± fibular) cortex** · aka **ossifying fibroma**
Age	**<10 yr** (peak 1–5) · **M > F**
Site	**anterior tibial cortex**, diaphysis (± metaphysis)

Genes & lab markers

Trisomies 7, 8, 12, 22 reported

No Gs-α (GNAS) mutation (vs fibrous dysplasia)

Keratin – (vs adamantinoma: keratin + epithelial nests)

3 Presentation

Painless swelling of the shin · may be tender when bumped

Deformity	**anterior / anterolateral tibial bowing (procurvatum)**, limp
Pseudarthrosis	**10–30%**

4 Diagnosis & Classification

Key radiology findings

X-ray (AP + lateral)	**eccentric, intracortical, multiloculated lytic lesion of the anterior tibial cortex** · well-marginated lucencies, sclerotic borders · expands/thins anterior cortex · **no periosteal reaction** · confined to anterior cortex



Anterior tibial cortical lesion with bowing

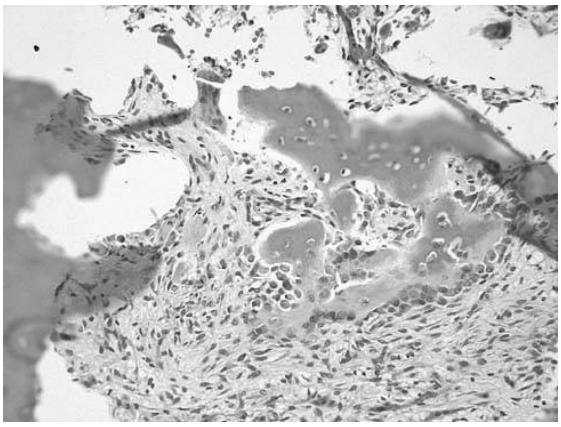
CT	cortical expansion, no periosteal reaction or matrix · **bone scan** – focally hot
Biopsy	needed when **young adult (15–35)** or red flags (growth, change, aggressive features) → exclude **adamantinoma**

– – Active space – –

Histology

Like fibrous dysplasia EXCEPT osteoblastic rimming

fibroblast proliferation (bland spindle stroma) around **woven bone islands lined by osteoblasts** (lamellar rim) · mitoses common · ± giant cells · **no epithelial nests**



Osteoblastic rimming around woven bone

Differential

	Tibial diaphysis	Observation alone
OFD	Yes	Yes
Adamantinoma	Yes	No (wide resection)
FD · EG · enchondroma · osteochondroma · NOF · Paget	–	Yes

	OFD	Adamantinoma	Fibrous dysplasia
Age	**<10**	20–40	<30
Location	**Anterior tibial cortex**	Anterior mid-tibial cortex	**Medullary**, femur/ribs/skull
Histology	Woven bone **with osteoblastic rimming**	**Epithelial nests, keratin +**	Woven bone **without rimming**
Gene	Trisomies	–	**GNAS**
Treatment	**Observe**	**Wide resection**	Observe / fix

Others

NOF (doesn't erode cortex, storiform) · UBC/ABC (metaphyseal, femur/humerus) · Ewing (onion skin, round cells) · EG

5 Management

Osteofibrous dysplasia

- Most → **OBSERVATION (annual follow-up) – regresses by maturity**
- Significant deformity interfering with walking → **BRACE**
- Impending pathologic fracture / pain → **CURETTAGE + GRAFT**
- Persistent deformity after skeletal maturity → **CORRECTIVE OSTEOTOMY**

Avoid surgery before ~15 yr	**very high local recurrence**

No curettage + grafting for routine OFD · (✗ Wrong options: casting · wide resection · chemo)
✗ Wrong options: "accepted for all EXCEPT OFD" question (see ABC chapter)

6 Complications

Prognosis	lesions **regress**, no problems in adulthood · link to adamantinoma debated (continuum?)

How it's asked

"3-yr-old, bowing + limp, lucent anterior proximal tibial diaphyseal lesion?"	→ **osteofibrous dysplasia** ✗ Wrong options: FD (ground glass) · UBC/ABC (metaphyseal) · NOF
"12-yr-old, tender bump below knee, anterior cortical lucencies, osteoblastic rimming?"	→ **OFD** ✗ Wrong options: adamantinoma · NOF · ABC · low-grade CS
"9-yr-old shin mass, trabeculae lined by osteoblasts?"	→ **OFD** ✗ Wrong options: NOF · osteoblastoma · chondroblastoma · Ewing
"6-yr-old, tender tibial shaft lucency, hot bone scan?"	→ **OFD** ✗ Wrong options: GCT · NOF · FD · EG
"10-yr-old painless lower-leg deformity, OFD histology – next?"	→ **observation** ✗ Wrong options: brace · cast · curettage · osteotomy
"7-yr-old, incidental anterior tibial lesion, keratin-negative biopsy?"	→ **observation with annual follow-up** ✗ Wrong options: wide resection (adamantinoma)

– – Active space – –

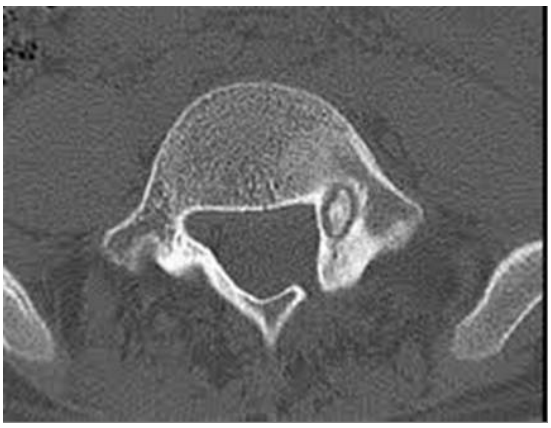
OSTEOID OSTEOMA

– Pathology · 49 / 66 –

1 Basics

Epidemiology

Incidence	**10–14% of benign** bone tumours · **2–3% of primary** bone tumours · **most common true benign bone tumour (TO)**
Demographics	**M:F 2.5:1** (Orthobullets) · **2–4:1** (TO, spine) · **5–25 yr**, **70% before 20** · peak 2nd decade
Site (Orthobullets)	lower limb >50%: **proximal femur > tibial diaphysis**, **cortical** · spine 10–15% (thoracolumbar > cervical/sacral, **posterior elements**, **concave side** of scoliosis) · hand 5–10% (**scaphoid, proximal phalanx**) · foot <5% (**talar neck**)
Site (TO)	**femur > tibia, cortical, diaphysis** (keyed "femur, cortical diaphysis") · vertebrae **7–20%** · usually cortical, can be intramedullary, subperiosteal, intra-articular
Spinal OO (TO)	**lumbar** · **posterior elements 75%** · **pars interarticularis** commonest · CT: lesion at **pedicle–lamina junction**



CT – nidus at pedicle–lamina junction (TO)

Pathophysiology (COX, PGE2)

Nidus	central nodule of **woven bone + osteoid with osteoblastic rimming** · **reactive zone** – thickened bone + fibrovascular tissue
Pain	↑ **prostaglandin E2** (+ PGI2) and **COX-1 / COX-2** expression · ↑ number and size of **unmyelinated nerve fibres** in nidus
NSAIDs	**cyclooxygenase inhibition** → ↓ PGE2 · ibuprofen/naproxen competitive · aspirin irreversible acetylation ✗ Wrong options: mu-opioid agonist · 30S · transpeptidase · ACh agonist
Most important pain mediator	**PGE2** · first shown by **Makley and Dunn** ✗ Wrong options: PGI2 (endothelial COX-2, anti-platelet) · IGF-1 · glutamate · ATP
Associations	**scoliosis, lesion on concave side** (paravertebral spasm) · **LLD** (epiphyseal → overgrowth) · flexion contractures

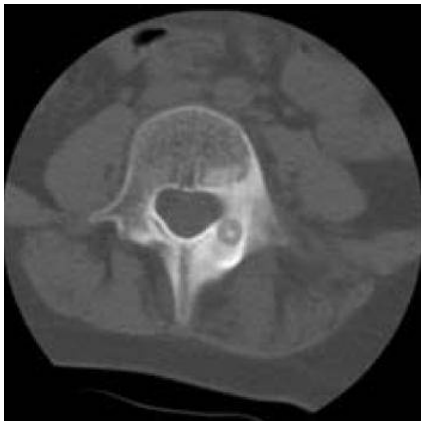
3 Presentation

Pain	constant, progressive, **worse at night** and with **alcohol**, **not activity-related**, **relieved by NSAIDs/aspirin** · near joint → mimics arthritis
Spine	**night pain 80–100%** · **painful scoliosis 63–70% (TO)** · 75% (Orthobullets) · backache + paravertebral spasm + scoliosis in adolescent → suspect OO
Hand	painless swelling possible
Exam	swelling, tenderness, effusion, contracture, limp, muscle atrophy
Delay	mean ~**14 months** to diagnosis · often misdiagnosed as **stress fracture** in runners
Labs	**ESR, CRP, WBC normal** ✗ Wrong options: osteomyelitis

4 Diagnosis & Classification

Key radiology findings

X-ray (orthogonal views)	intense **reactive sclerosis around a radiolucent nidus** · may be normal early · spine: subtle pedicle sclerosis, typical only in half · **obliques** help (pedicle, facet, pars) **(TO)**
CT (modality of choice)	**nidus <1.5 cm** (>1.5 cm → osteoblastoma) · **vascular groove sign** in **80%** (1 mm serpentine cortical lucencies) · bull's-eye calcified centre (50%) · sclerosis may exceed nidus · **fine-cut CT** separates OO from **stress fracture**



CT – radiolucent nidus with sclerosis

Bone scan	**most sensitive for localization (TO)** · **double-density sign** – intense nidus uptake + lower surrounding uptake

– – Active space – –

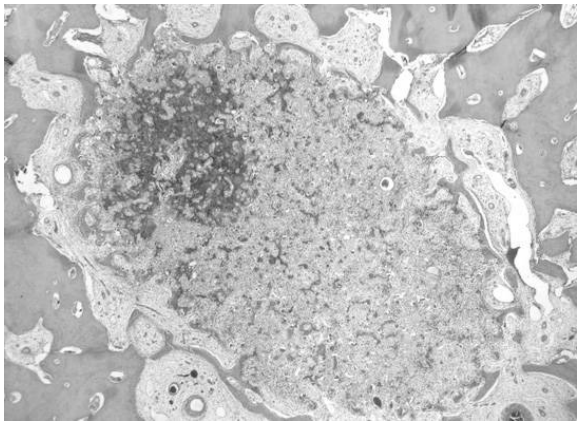


Double-density sign

MRI	**not recommended** – marrow **oedema masks features** · useful for intra-articular nidus · not needed for diagnosis
Diagnosis	clinical history + NSAID response + X-ray/CT → **no biopsy, no staging, no labs needed**
Femoral neck OO in young athlete (TO)	marrow oedema + effusion but **no fracture line**, thickened neck cortex → **CT for the nidus** · **intracapsular** location explains the large effusion · can occur in the 20s

Histology

Distinct demarcation nidus vs reactive bone · uniform **immature osteoid trabeculae (woven bone)** with **osteoblastic rimming** · plump, regular osteoblasts · vascular stroma · **no infiltration** of host bone · no neutrophils/dead bone
✗ Wrong options: Brodie



Nidus – woven bone with osteoblastic rimming

Enneking staging of benign lesions

Stage	Grade	Examples
1	**Latent**	Enchondroma, NOF
2	**Active**	**Osteoid osteoma**, UBC, ABC, chondroblastoma, CMF, GCT
3	**Aggressive**	GCT, ABC

Differential

Lesion	Distinguishing
Brodie's abscess	Neutrophils, dead bone · similar cortical X-ray
Stress fracture	Activity-related · **fine-cut CT** separates
Osteoblastoma	**>2 cm**, intramedullary, aggressive, dull ache
ABC	Posterior elements, fluid levels
Ewing	Permeative, onion skin
Low-grade intramedullary OS	Intramedullary sclerosis, ~1% of OS
Enchondroma	Intramedullary, painless, cartilage
Osteoma	**Dense juxtacortical** bony mass
Chondroblastoma	**Epiphyseal** – OO is diaphyseal **(TO)**

5 Management

Osteoid osteoma

- First line (most extremity; painful spine without scoliosis) → **NSAIDs + OBSERVATION**
  (50% burn out over several years)
- Failed NSAIDs / GI side effects → **RFA (standard of care) or MR-HIFU**
- Near skin / nerve / cord, digit, recurrence after RFA, fixed spinal deformity → **SURGICAL EXCISION**

NSAIDs & observation

First line
NSAIDs + observation
✗ Wrong options: RFA/HIFU/resection before trial · MRI

Incidental OO
unrelated acute symptoms (e.g. contralateral radiculopathy after injury) → **continued observation**

Radiofrequency ablation & MR-HIFU

RFA
CT-guided electrode into nidus · **90°C for 4–6 min → 1 cm** necrosis zone · **90% success in 1–2 sessions** (95% one procedure – OB) · **10–15% recurrence** · least invasive, fastest resolution, least morbidity

CT-guided percutaneous drilling
alternative to RFA (TO) · **"vessel sign"** around nidus on CT (TO)

RFA distances
>1 cm from skin · **>1 cm from articular surface** · **1–1.5 cm from neurovascular** structures · spine safe with motor response testing

RFA after failed NSAIDs
hip, tibia, femur → **RFA**
✗ Wrong options: biopsy · CT chest · en bloc · curettage + cement

MR-HIFU
non-invasive, **non-ionizing** · thermal ablation of periosteal nerves + nidus · comparable to RFA · for failed medical management (e.g. NSAID GI side effects)

– – Active space – –

Surgical excision (spine, hand)

Indications	not amenable to RFA (**close to skin, nerve, cord/dura**) · **digital lesions** (RFA → thermal skin necrosis, digital NV injury) · **recurrence after RFA** · **spine with fixed deformity**
Spine near neural elements	**surgical resection / en bloc excision of nidus** ✗ Wrong options: RFA · vertebrectomy/corpectomy · fusion
Digit	**curettage with power burr** (<10% local recurrence) ✗ Wrong options: RFA · wide resection
Principle	complete marginal removal of **nidus** (sclerotic bone can stay) · **94% success**
Spinal deformity	**resolves if excised within 15 months** of onset · no TLSO for mild curve

6 Complications

RFA	recurrence 10–15% · cellulitis, thrombophlebitis, **skin necrosis** (<1 cm from skin) · NV damage (<1–1.5 cm)
LLD	epiphyseal lesion → limb overgrowth
Open curettage/en bloc near hip	late joint morbidity
Prognosis	pain resolves ~**3 yr** (2–4 yr, hand) · lesion resolves **5–7 yr** · malignant transformation extremely rare

X vs Y

	Osteoid osteoma	**Osteoblastoma**
Incidence	10% of benign	3% of benign
Size	**<2 cm (typically <1.5)**	**>2 cm** (3–3.5)
Site	>50% long-bone diaphysis · proximal femur > tibia > phalanges > spine	>35% posterior spine
Natural history	**Self-limited**	**Progressive**
Symptoms	**Night pain, relieved by NSAIDs** · painful scoliosis (concave side)	Dull ache, partial NSAID response, neuro symptoms
Spine treatment	Nonoperative first → RFA/resection	**Surgery always**

Bone-tumour one-liners (TO)

Fact	Answer
Location	Most metaphyseal · **epiphyseal**: GCT, chondroblastoma · **diaphyseal**: Ewing, OO/osteoblastoma, FD sometimes
Age	Most <20 · **40–60**: haemangioma, chondrosarcoma · **20–40**: GCT · **bimodal**: OS (10–20 primary, 40–60 secondary) · Ewing 5–15
Most common tumour of bone	**Metastasis**
Most common primary	**Myeloma > OS > chondrosarcoma**
Most common benign	**Osteochondroma** · true benign: **OO** · lesion: **fibrous cortical defect**
Malignant, 1st decade	**Ewing**
Radiation-induced	**OS > fibrosarcoma > MFH** · soft tissue: **MFH**
Hand	**Enchondroma** · malignant bone: chondrosarcoma · malignant overall: SCC
Chest wall malignant bone	**Chondrosarcoma**
Night pain + NSAID response	**OO**
Pulsatile	ABC, GCT, telangiectatic OS, **thyroid follicular / renal** mets (renal most)
Egg-shell calcification	**GCT**
Codman tumour	**Chondroblastoma**
OS virus	**FBJ murine virus**
Lowest recurrence after extended curettage	**Liquid nitrogen**
HME inheritance	**Autosomal dominant**

Most common site (TO)	
FD	**Femoral neck = craniofacial**
UBC · ABC	**Proximal humerus** · **proximal femur**
Enchondroma · chondroblastoma	**Hand** · **distal femur**
CMF	**Proximal tibial metaphysis**
Osteochondroma	Distal femur > proximal tibia
OO · osteoblastoma	Femur > tibia (diaphysis) · posterior vertebra > femoral diaphysis
Haemangioma	**Spine (T4–L4)** > skull
GCT	Distal femur > proximal tibia > distal radius
OS · chondrosarcoma · Ewing	Distal femur · proximal femur · femoral diaphysis > flat bones
Chordoma · ameloblastoma · adamantinoma	**Sacrum** · **mandible** · **tibia**
EG · plasmacytoma/myeloma	**Skull** · **spine**
Glomus · synovial sarcoma · PVNS	**Subungual** · **knee** · **knee**

– – Active space – –

Sign / pattern (TO)	Lesion
Ground glass, rind sign, shepherd's crook	FD (crook also Paget, OI)
Fallen leaf, trap door	UBC
Corduroy / jail-house, polka dot	Haemangioma
Nidus <1.5 cm / >1.5 cm	OO / osteoblastoma
Soap bubble	GCT · also ABC, chondroblastoma, enchondroma, CMF, FD, ameloblastoma, plasmacytoma, chordoma
Codman triangle, sunray	OS (Codman also Ewing)
Popcorn calcification	**Chondrosarcoma**
Onion peel	Ewing
Punched-out, double contour (skull)	EG
Punched-out, no sclerotic rim	Myeloma
Chinese letters · chicken wire	FD · chondroblastoma
Storiform · herringbone	MFH · fibrosarcoma
Small round cells	Ewing, lymphoma, neuroblastoma, PNETs
Biphasic	Synovial sarcoma

Marker / syndrome (TO)	
Ewing	**t(11;22)** (85%), **CD99, MIC-2**
Synovial sarcoma	**t(X;18)**
LCH	**S-100, CD1a**, NSE, Birbeck granules
McCune-Albright	Polyostotic FD + precocious puberty + café-au-lait
Mazabraud	Polyostotic FD + intramuscular myxomas
Diaphyseal aclasia	HME · Masada – forearm
Ollier · Maffucci	Enchondromas · + haemangiomas
Vertebra plana	**EG (commonest)**, Ewing, lymphoma/leukaemia, Gaucher, ABC, infection

Metastasis & sarcoma (TO)	
Mets primary	Males: prostate > lung · females: breast > lung · children: **neuroblastoma** · overall breast > prostate > lung
Site	**Thoracic spine** · pathological fracture: **femoral neck** · distal to knee/elbow → lung primary, tibia
Occult mets	**Bone scan** (blastic) · **PET-CT** (lytic)
STS adults · young adults · children	**MFH** > liposarcoma · **synovial sarcoma** · **rhabdomyosarcoma** (GU > extremity)
STS hand/UL · foot	**Epithelioid sarcoma** · **synovial sarcoma**
STS prognosis	**Histological grade**
Bone resorption markers	N/C-telopeptides, hydroxyproline, deoxypyridinoline, hydroxylysine glycosides, **TRAP**, bone sialoprotein
Bone formation markers	**Bone-specific ALP**, **osteocalcin**, procollagen-1 extension peptides

Spinal tumours (TO) most common benign primary bony: **vertebral haemangioma** · most common overall: **epidural metastasis**

How it's asked

"Axial CT spine, lesion at pedicle–lamina junction — diagnosis?"	→ **osteoid osteoma (TO)** ✗ Wrong options: OS · EG · Brodie
"Most common site of osteoid osteoma?"	→ **femur, cortical, diaphysis (TO)** ✗ Wrong options: metaphysis · tibia intramedullary
"17-yr-old runner, shin pain eased by NSAIDs — confirmatory histology?"	→ **woven bone nidus with osteoblastic rimming** ✗ Wrong options: hyaline cartilage (enchondroma) · physaliphorous (chordoma)
"17-yr-old, failed scoliosis bracing, normal labs, hot spine lesion?"	→ **osteoid osteoma** ✗ Wrong options: idiopathic scoliosis · osteomyelitis
"23-yr-old fibular cortical lytic lesion, oedema, plump osteoblasts, no inflammation?"	→ **osteoid osteoma** ✗ Wrong options: Brodie · osteoblastoma (>2 cm)
"18-yr-old night thigh pain, intracortical nidus, woven bone spicules?"	→ **osteoid osteoma** ✗ Wrong options: low-grade intramedullary OS
"31-yr-old shin pain, NSAIDs help, intracortical lucent nidus?"	→ **osteoid osteoma** ✗ Wrong options: osteoma (dense juxtacortical)
"15-yr-old athlete, tibial diaphysis pain worse at night — next step?"	→ **fine-cut CT** ✗ Wrong options: biopsy · referral · chest X-ray

– – Active space – –

"Mechanism of the drug that relieves OO pain?"	→ **cyclooxygenase inhibition**
"True of this painful finger lesion?"	→ **↑ cyclooxygenase activity** ✗ Wrong options: needs MRI · no spontaneous resolution · day pain
"Most important pain mediator?"	→ **PGE2** ✗ Wrong options: PGI2 · glutamate · ATP
"Failed NSAIDs, normal labs, tibial/hip OO — next?"	→ **percutaneous / CT-guided RFA** ✗ Wrong options: biopsy · en bloc · antibiotics
"Hip OO — fastest relief, least morbidity?"	→ **RFA** ✗ Wrong options: steroid injection · curettage with hip dislocation
"Runner with groin pain, ibuprofen helps, CT nidus?"	→ **RFA** ✗ Wrong options: MR arthrogram · intra-articular steroid
"NSAID GI side effects, anterior tibial OO?"	→ **MR-HIFU** ✗ Wrong options: radical resection · chemo
"Cervical OO near cord, failed NSAIDs?"	→ **surgical resection / en bloc excision** ✗ Wrong options: RFA · corpectomy · C2–3 fusion
"Little-finger OO, can't take NSAIDs?"	→ **curettage** ✗ Wrong options: RFA (skin/NV necrosis) · wide resection
"12-yr-old, mild curve, OO abutting canal, NSAIDs failed?"	→ **surgical excision** ✗ Wrong options: TLSO · phenol
"Water-skier, right leg pain, left L5 OO on CT?"	→ **continued observation** (incidental)
"Night pain worse with alcohol, nidus on CT, no treatment yet?"	→ **NSAID trial + observation** ✗ Wrong options: MRI · RFA first
"15-yr-old, epiphyseal lytic soap-bubble lesion?"	→ **chondroblastoma (TO)** ✗ Wrong options: OO (diaphyseal) · GCT (20–40)

PAGET'S DISEASE

– Pathology · 50 / 66 –

1 Basics

Epidemiology

Nature	idiopathic **abnormal bone remodelling** – excessive resorption + abnormal new bone formation (osteitis deformans) · **2nd commonest chronic bone-remodelling disorder** in the US after osteoporosis · **3–4% of US population >40**
Demographics	**>40 yr**, peak **5th decade** · **M = F** · **Caucasians** (northern European / **Anglo-Saxon**)
Distribution	**monostotic or polyostotic**
Sites	**femur > pelvis > tibia > skull > spine** (Orthobullets) · **pelvis > tibia** (TO) · also clavicle · **sacrum** typically involved

Pathophysiology – osteoclast abnormality

Primary abnormality	↑ **osteoclastic bone resorption** (✗ Wrong options: osteoblast · osteocyte · chondroblast · synoviocyte) → **focally increased remodelling** (✗ Wrong options: ↓ vascularity · sclerotic bone replacing marrow)
Cause	**slow virus** (intranuclear nucleocapsid-like inclusions) – **paramyxovirus**, **RSV**
Osteoclast diseases	Paget · **osteopetrosis** (defective osteoclastic resorption) · pycnodysostosis – share **nuclear inclusion bodies** ✗ Wrong options: OI (osteoblast) · osteochondroma (chondrocyte) · myeloma · osteoporosis (uncoupling, not inherited)
Osteopetrosis genes	carbonic anhydrase II · vacuolar proton pump · chloride channel 7

Genes & lab markers

Inheritance	mostly spontaneous · familial clusters, **~40% autosomal dominant** · family history in **15%** (OB)
SQSTM1 (p62/ sequestosome 1), 5q35-QTER	most important · severe disease
TNFRSF11A	insertion mutation in the gene encoding **RANK**
Labs	↑ **serum ALP** (most useful test; tracks extent + activity) · ↑ **urinary collagen cross-links** · ↑ **urinary hydroxyproline** · ↑ **urinary N-telopeptide, α-C-telopeptide, deoxypyridinoline** · **calcium + phosphate normal** · PTH normal (↑ in small subset with very active disease)
α-CTX	rises more than β-CTX → ↑ α/β CTX ratio **(TO)** · **urinary NTX** – sensitive, monitors treatment response **(TO)**
Hypercalcaemia	only with **immobilization** (↑ osteocyte **sclerostin**), fracture, predominant lytic phase, or sarcoma **(TO)** · ambulatory → normal calcium
vs myeloma	**ALP ↑ in Paget**, normal in myeloma **(TO)**

– – Active space – –

Phases

Phase	Process	X-ray
Lytic	Intense osteoclastic resorption, hypervascular	Lucent, expanded, thin intact cortex · **blade of grass / flame** advancing edge · skull **osteoporosis circumscripta**
Mixed	Resorption + compensatory formation (weak brittle bone)	Lysis + sclerosis, coarse trabeculae
Sclerotic (burnt-out)	Osteoblastic formation dominates · dense avascular **mosaic** bone	**Bone enlargement, cortical thickening** – commonest presentation

All three phases can coexist in one bone

3 Presentation

Asymptomatic commonest · found incidentally (↑ ALP, X-ray for another complaint)

Pain commonest presenting symptom · dull ache · stress fractures, hypervascularity, warmth · **cyclic** recurring leg pain

Deformity **long-bone bowing** (femur, **tibia varus**) · kyphosis · short neck (flattened skull base, basilar invagination) · skull enlargement

Fractures brittle bone · **transverse** – **banana / incremental fracture (TO)** · subtrochanteric

Arthritis hip + knee OA (83% have a rheumatic syndrome – OB)

Neuro **deafness** (CN VIII compression, otosclerosis, cochlear BMD loss) · impaired vision, facial palsy, trigeminal neuralgia · **spinal stenosis** · **vascular steal** → cerebral ischaemia, spinal claudication **(TO)**

Cardiac **high-output heart failure** – large/multiple lesions + poor cardiac reserve

Red flag **new intense pain + swelling → Paget sarcoma**

4 Diagnosis & Classification

Key radiology findings

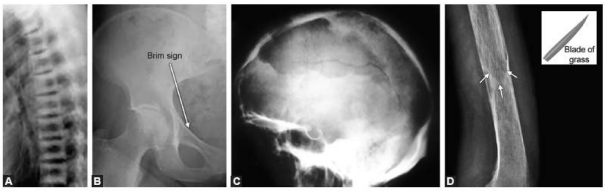
X-ray **coarsened trabeculae, blastic** appearance · **remodelled cortices** – loss of cortex–medulla distinction · **bone enlargement + cortical thickening** · bowing · fractures · OA



Coarsened trabeculae, cortical thickening

– – Active space – –

Sign (TO)	Finding
Picture-frame vertebra	Peripheral cortical thickening (✗ Wrong options: VDRR · fluorosis · MED)
Ivory vertebra	Diffusely sclerotic vertebra
Cotton-wool spots	Fluffy sclerotic skull patches
Osteoporosis circumscripta	Focal lytic skull lesion, lytic phase
Brim sign	Sclerotic iliopectineal line
Blade of grass	V-shaped diaphyseal lucency
Banana fracture	Transverse pathological fracture of bowed bone



Picture frame · brim sign · osteoporosis circumscripta · blade of grass

Skull sign (TO)	Disease
Salt and pepper	Hyperparathyroidism
Bevelled edge, double contour	Eosinophilic granuloma (LCH)
Osteoporosis circumscripta · cotton wool	Paget (lytic · sclerotic)
Rain-drop skull (punched-out)	Myeloma

Spine sign (TO)	Disease
Picture frame · ivory	Paget
Rugger jersey	Hyperparathyroidism / renal osteodystrophy · osteopetrosis
Cod-fish vertebra	Osteoporosis (biconcave)

Bone scan marks disease sites accurately · **intensely hot in lytic + mixed**, less in sclerotic · pagetoid bone hot even without sarcoma

MRI thick trabeculae with **normal marrow signal** · lumbar stenosis

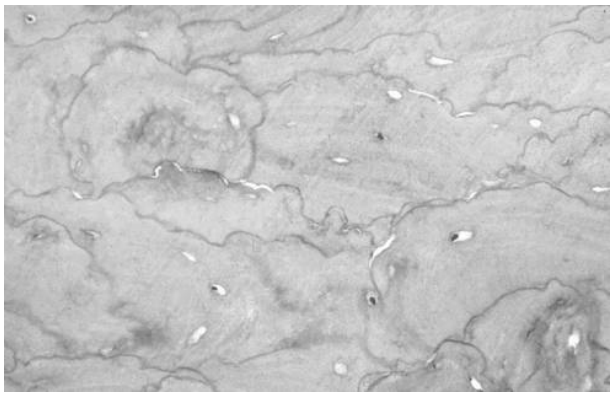
CT cortical thickening, coarse trabeculae

Sarcoma cortical destruction + soft-tissue mass

Histology

Mosaic pattern woven + lamellar bone, **broad irregular trabeculae**, **disorganized cement lines** · **large osteoclasts, many nuclei (up to 100)**, **virus-like inclusion bodies** · **fibrovascular tissue** between trabeculae · biopsy rarely needed **(TO)**

– – Active space – –



Mosaic cement lines

Differential

	Benign, older (40–80)	Multiple lesions, older	Observation only	Bisphosphonate helps
Paget's disease	✓	✓	✓	✓
Enchondroma (older adult)	✓	–	✓	–
Bone island	✓	–	–	–
Bone infarct	✓	✓	–	–
Hyperparathyroidism (brown tumour)	✓	✓	–	–
Metastatic bone disease · myeloma	–	✓	–	✓
Lymphoma of bone	–	✓	–	–
NOF · osteochondroma · EG	–	–	✓	–
FD	–	–	–	✓

Distractors osteoporosis (compression fractures, neck/IT) · renal osteomalacia (erosions, amyloid) · lymphoma (blastic but no enlargement/coarse trabeculae) · melorheostosis (periosteal new bone) · infarct

5 Management

Paget's disease

- Asymptomatic → **OBSERVATION + SUPPORTIVE (physio, orthoses, NSAIDs, analgesics)**
  (treat if active disease at sites where complications likely)
- Symptomatic → BISPHOSPHONATES (1st line) → **CALCITONIN (2nd line)**
- Hip/knee OA → **THA / TKA after medical pre-treatment**
- Bowing with fracture / impending fracture / failed brace → **METAPHYSEAL OSTEOTOMY + PLATE**

Medical treatment – bisphosphonates

Indications **bone pain** · **pre-op** (lower ALP to **50% of pretreatment** → less bleeding) · hearing loss, spinal stenosis, HOHF · prevention of fracture/deformity

Mechanism **osteoclast inhibition**
✗ Wrong options: TNF-α · IL-1R · ER blockade · COX

Drug	Notes
Alendronate, risedronate	Oral
Etidronate	Old, inhibits osteoclasts **and** osteoblasts · **≤6 months** at a time
Pamidronate, zoledronic acid	IV, osteoclasts only · **single 5 mg IV zoledronic acid** — most effective, normalizes ALP up to **6.5 yr**
Calcitonin	**2nd line** (bisphosphonate intolerance) · osteoclasts shrink within minutes · SC/IM
Calcium + vitamin D	Prevent hypocalcaemia on bisphosphonate/calcitonin · not as treatment
Teriparatide	**Contraindicated** — secondary OS risk

Teriparatide contraindications Paget · hypercalcaemic disorders · OS · bone mets · pregnancy · skeletal radiation

NSAIDs pain relief, no change in course
✗ Wrong options: methotrexate · monoclonal antibody · chemo · radiation · wide resection

Benign biopsy, lytic sacral lesion in old melanoma patient Paget → **bisphosphonate**

Arthroplasty & osteotomy in Paget disease

Procedure	Commonest complication
THA	**Bleeding** (rich blood supply) · loosening ~15%
TKA	**Malalignment**

Before arthroplasty **pharmacological pre-treatment** to cut bleeding · activity by **ALP + urine hydroxyproline**

Bowing + persistent pain after 6 mo brace **corrective osteotomy + plate fixation**
✗ Wrong options: IMN without osteotomy · amputation · knee arthrodesis

6 Complications

Paget sarcoma

Incidence **<1%** · **5–10%** in advanced polyostotic disease **(TO)**

Type **osteosarcoma > fibrosarcoma, chondrosarcoma** (spindle-cell sarcomas) · pleomorphic spindle cells, giant cells, atypia

Patient older (**55–80**, 7th decade) · polyostotic · **pelvis, femur, humerus**, proximal skeleton

Treatment **chemotherapy + wide resection + reconstruction** · palliative RT if too infirm
✗ Wrong options: surgery alone

Prognosis **worst of bone sarcomas** · metastatic 5-yr survival **<10%** (Orthobullets/OB) · **<5%** (OB2004) · non-metastatic **>50%** · secondary OS in >60 yr 7.5% vs primary 37% (Huvos)

– – Active space – –

Premalignant for secondary OS (TO)	FD · Paget · irradiation · multiple osteochondromatosis · bone infarct
Others	fractures · OA · HOHF · deafness · stenosis

How it's asked

"Picture-frame vertebrae seen in?"	→ **Paget's (TO)** ✗ Wrong options: VDRR · fluorosis · MED
"NOT true of Paget's?"	→ **decreased urinary N-telopeptide** (also false: ↑ Ca/PO_4 in lytic phase — better pick is NTX) **(TO)**
"60-yr-old, back/hip pain, cotton-wool skull, ↑↑ ALP?"	→ **Paget's** (one copy keyed "hyperparathyroidism" but its own explanation says Paget's) **(TO)**
"Subtrochanteric fracture, ↑ ALP, normal Ca/PO_4?"	→ **Paget's** ✗ Wrong options: osteoporosis · renal osteomalacia
"↑ urinary hydroxyproline, NTX, α-CTX — which X-ray?"	→ **Paget's** ✗ Wrong options: infarct · myeloma · melorheostosis
"Paget's — which cell?"	→ **osteoclast**
"Excess resorption + disordered formation?"	→ **Paget's** ✗ Wrong options: osteopetrosis · OI
"Expected lab?"	→ **↑ urinary NTX + α-CTX · ↑ serum ALP** ✗ Wrong options: ↓ calcium · ↑ calcium · ↓ cross-links
"Ambulatory 40-yr-old, macrophage-lineage cell disease — lab?"	→ **normal calcium** ✗ Wrong options: ↑ phosphate · low ALP · normal hydroxyproline
"Enlarged sclerotic hemipelvis with coarse trabeculae?"	→ **Paget's** ✗ Wrong options: lymphoma
"Another disease of the same abnormal cell?"	→ **osteopetrosis**
"Paget's = osteoclastic abnormality causing?"	→ **focally increased remodelling**
"Asymptomatic Paget's?"	→ **supportive therapy** ✗ Wrong options: bisphosphonate · calcitonin
"Symptomatic hip, ↑ ALP, normal Ca?"	→ **bisphosphonates** ✗ Wrong options: antibiotics · chemo
"Contraindicated drug in Paget's?"	→ **teriparatide**
"Bowed leg, mixed lesion — drug mechanism?"	→ **osteoclast inhibition**
"Benign Paget biopsy, hemipelvis — next?"	→ **observation + evaluate for bisphosphonate**

"Painful pagetoid femur, normal marrow on MRI?"	→ **medical management** ✗ Wrong options: prophylactic fixation · RT
"Enlarged sclerotic bowed tibia, cyclic pain?"	→ **bisphosphonate** ✗ Wrong options: NSAIDs alone · vit D · MTX
"Paget sarcoma — EXCEPT?"	→ **treated by surgery alone**
"Humeral Paget sarcoma — treatment?"	→ **chemo + wide resection + reconstruction**
"Commonest malignant transformation?"	→ **osteosarcoma** ✗ Wrong options: fibrosarcoma · myeloma · GCT
"Paget's predisposes to?"	→ **all** — chondrosarcoma, OS, spindle-cell sarcoma
"Worst-prognosis bone sarcoma?"	→ **sarcoma in Paget's** ✗ Wrong options: MFH · OS · Ewing · clear-cell CS
"Bowed tibia, brace failed?"	→ **corrective osteotomy + plate**
"Paget THA — commonest complication?"	→ **increased blood loss** ✗ Wrong options: malalignment (TKA)
"Paget TKA — commonest complication?"	→ **malalignment**

– – Active space – –

PAROSTEAL OSTEOSARCOMA

– Pathology · 51 / 66 –

1 Basics

Nature	**low-grade**, well-differentiated **surface** osteosarcoma · **<5%** (OB7521) / **~5%** (OB8282) of osteosarcomas
Demographics	**females > males** · **30–40 yr** (3rd–4th decades) · range in stems 20–42
Site	**surface of metaphysis** · **posterior distal femur** > proximal tibia > proximal humerus · **80% femur** (Orthobullets) · posterior distal femur **~75%** (OB7488) / **80%** (OB4407)
Marrow invasion	**25%** · does not affect long-term survival

Genes & lab markers

MDM2 amplification + **ring chromosome** – high rate

3 Presentation

Painless mass · slow-growing, firm, mild/no pain · long history (>3 yr)

↓ Knee flexion / tightness	large posterior distal femoral lesion · tenderness on deep posterior palpation · effusion
Labs	ESR, CBC normal

4 Diagnosis & Classification

Key radiology findings

X-ray — **heavily ossified, lobulated mass arising from cortex**, **"stuck-on"** the metaphyseal surface over a broad area · **cloud-like** ossification · may be multifocal



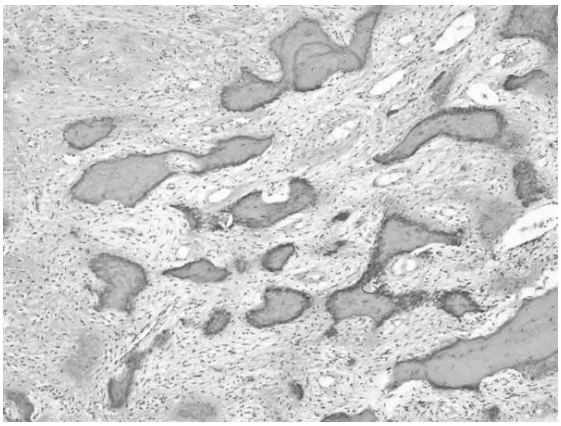
Lobulated ossified mass stuck on posterior distal femur

MRI — **mandatory**, whole bone · soft-tissue/marrow involvement, skip lesions · **very dark** signal (heavy mineralization)

CT chest	**mandatory** staging (lung mets)
Bone scan	**mandatory** · **always hot**
Angiography	finds **hypervascular dedifferentiated areas** → target biopsy
Next step	**biopsy** after staging ✗ Wrong options: arthroscopy · bisphosphonate · chemo/RT first

Histology

Low grade	**regularly arranged normal osseous trabeculae** · **atypical spindle cells between trabeculae** = most prominent feature · bland fibroblastic stroma · some atypia needed for diagnosis
Pitfall	mistaken for **fibrous dysplasia** histologically (X-ray differs)



Spindle cells between regular trabeculae

Differential

	Bone surface lesion	Similar histology	Surgery alone
Parosteal OS	✓	–	✓
Periosteal OS	✓	–	–
Periosteal chondroma	✓	–	✓
Osteochondroma / MHE	✓	–	✓
Developmental defect (adductor magnus insertion)	✓	–	–
Myositis ossificans	✓	–	–
Chondrosarcoma · **adamantinoma**	–	–	✓
Chordoma	–	–	✓ (controversial)
FD	–	✓	–

– – Active space – –

Lesion	Key distinction
Osteochondroma	**Medullary continuity** with host bone on cross-section
Myositis ossificans	Thin **egg-shell peripheral** calcification, often **not touching bone**
Surface chondrosarcoma	More cortical destruction, **rings and arcs**
Dedifferentiated parosteal	Hypervascular areas on angiogram

5 Management

Parosteal osteosarcoma

- Low grade → **WIDE RESECTION ALONE**
  (geometric osteotomy to keep native joint; no chemo or RT)
- Dedifferentiated (high-grade component) → **NEOADJUVANT CHEMO + WIDE RESECTION**

Treatment – wide resection alone

Only OS subtype treated with wide resection alone
✗ Wrong options: conventional · periosteal · telangiectatic · secondary (all chemo + wide + chemo)

Low-grade intramedullary (central) OS	also **surgery with wide margins only** · "lacey" osteoid, minimal atypia, long premorbid history
✗ Wrong options	curettage (high recurrence) · marginal resection · observation
Conventional OS for contrast	pre-op chemo **8–12 weeks**, maintenance **6–12 months** after surgery

6 Complications

Prognosis	**95% long-term survival** with local control · better than other OS
Dedifferentiation	**poor prognostic factor** → chemo

How it's asked

"21-yr-old, dense bone 'stuck-on' proximal humerus, mature bone in fibroblastic stroma?"	→ **parosteal OS** ✗ Wrong options: periosteal OS · osteochondroma · MO
"33-yr-old knee, surface lesion, hypervascular on angiogram, high-grade areas?"	→ **dedifferentiated parosteal OS** – neoadjuvant chemo + wide resection
"34-yr-old woman, ↓ knee flexion, lobulated posterior distal femoral surface mineralization?"	→ **parosteal OS** ✗ Wrong options: chondrosarcoma (rings and arcs) · lymphoma
"Field hockey player, ↓ knee flexion, posterior femoral surface lesion – next?"	→ **surgical biopsy** ✗ Wrong options: chemo · RT · arthroscopy
"Low-grade intramedullary OS, no mets – treatment?"	→ **surgery only**

"Parosteal OS (biopsy) — definitive management?"	→ **wide surgical resection** ✗ Wrong options: neoadjuvant chemo (conventional OS)
"24-yr-old, tight knee, surface OS — plan?"	→ **surgical resection only** ✗ Wrong options: chemo · RT · observation
"42-yr-old slowly enlarging thigh mass, surface OS?"	→ **wide resection** ✗ Wrong options: curettage (recurrence)
"Which OS subtype gets wide resection alone?"	→ **parosteal**

– – Active space – –

– – Active space – –

PERIOSTEAL CHONDROMAS

– Pathology · 52 / 66 –

1 Basics

Nature	rare, **benign chondrogenic surface** tumour (under periosteum)
Age	**10–20 yr**
Site	**proximal humerus (59%)** · distal femur, proximal femur · **hand** (metacarpals, phalanges)

3 Presentation

Pain	many painful from **irritation of overlying tendons** · night pain possible · palpable, mildly tender mass

4 Diagnosis & Classification

Key radiology findings

X-ray	**well-demarcated, shallow cortical defect** · **saucerization** of underlying cortex · punctate mineralization in **1/3** · cortically based, eroding cortex ✗ Wrong options: medullary epicentre (enchondroma)

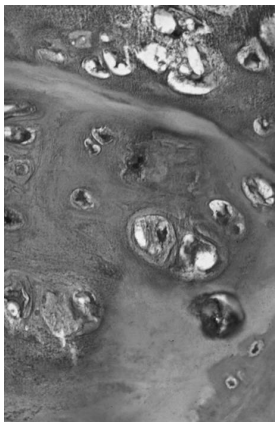


Saucerized cortical defect, proximal tibia

X-ray is the key	histology can look like **chondrosarcoma**

Histology

Like enchondroma, but more cellular with more malignant-looking cells · **bland hyaline cartilage** · small **chondroid cells in lacunae**



Bland hyaline cartilage

Differential

	Surface lesion	Similar chondroid histology	Marginal excision
Periosteal chondroma	✓	✓	✓
Osteochondroma / MHE	✓	✓	–
Parosteal OS · periosteal OS	✓	–	–
Enchondroma / Ollier / Maffucci	–	✓	–
Chondrosarcoma	–	✓	–
Neurilemmoma · nodular fasciitis (soft tissue)	–	–	✓
Epidermal inclusion cyst · **glomus tumour**	–	–	✓

Lesion	Key distinction
Enchondroma	Same histology, **central medullary** epicentre
Surface chondrosarcoma	Malignant histology
CMF	Fibrous + myxoid elements
Chondroblastoma	Polyhedral cells, **chicken-wire** calcification
Periosteal OS / CS	Similar location – malignant

5 Management

Periosteal chondroma

- Mild symptoms → **OBSERVATION**
- Severe / progressive pain, function impaired → **MARGINAL EXCISION INCLUDING UNDERLYING CORTEX**
  - Large defect → **BONE GRAFT**

Recurs if cartilage left behind

✗ Wrong options: wide excision · RT · chemo

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How it's asked

"19-yr-old shoulder night pain, surface cartilage lesion on CT + biopsy — treatment?"	→ **marginal excision including underlying cortex** ✗ Wrong options: wide excision · chemo · RT 60 Gy
"10-yr-old wrist, cortical saucer-shaped defect, benign cartilage?"	→ **periosteal chondroma** ✗ Wrong options: enchondroma (medullary) · surface CS · CMF · chondroblastoma

PERIOSTEAL OSTEOSARCOMA

– Pathology · 53 / 66 –

1 Basics

Nature	extremely rare **intermediate-grade surface** osteosarcoma
Demographics	**15–25 yr · F > M**
Site	**diaphysis** of long bones · **femur, tibia**

Genetics (p53, RB, Li-Fraumeni)

p53 germ-line mutation	**15–20%** of periosteal OS
Li-Fraumeni	**AD**, germ-line **p53** inactivation · breast, colon cancers, sarcomas – **50% are osteosarcoma → OS = commonest bone tumour with p53** ✗ Wrong options: chondrosarcoma (p53 = aggressive behaviour) · Ewing (EWSR1–ETS) · myeloma
Sporadic OS	p53 mutation in **15–30%**
Retinoblastoma (RB1)	hereditary (bilateral) retinoblastoma → OS · RB1 linkage analysis identifies at-risk relatives

Genes & lab markers

Finding	Tumour
RB + p53 alterations	**Conventional high-grade OS**
Supernumerary ring chromosomes	Parosteal OS
t(11;22)	Ewing
t(X;18)	Synovial sarcoma
Telomeric associations	GCT of bone

3 Presentation

Pain	commonest symptom · **pathologic fracture 25%**
Exam	regional swelling, tenderness · may present after a minor twist

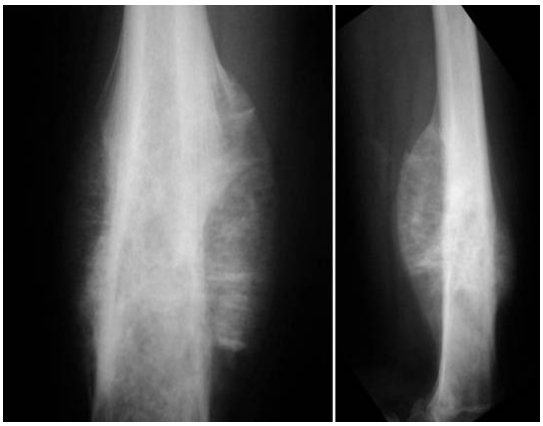
4 Diagnosis & Classification

Key radiology findings

X-ray	**"sunburst" / "hair-on-end"** periosteal reaction, often in a **saucerized cortical depression** · **no medullary involvement** typically

Pathology

– – Active space – –



Sunburst periosteal reaction

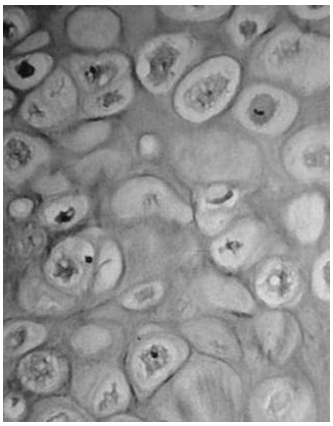
Staging **CT chest** (lung mets) + **bone scan** (very hot) required

Workup — MRI of entire bone

Bone-forming lesion, indistinct margins → MRI of the entire bone ASAP before biopsy · biopsy by the treating surgeon
✗ Wrong options: biopsy first · amputation · observation · NSAIDs

Histology

Grossly lobular, cartilaginous · osteoid + chondroblastic matrix · no osteoid → chondrosarcoma



Chondroblastic matrix with osteoid

Surface OS compared

	Parosteal OS	**Periosteal OS**
Grade	Low	**Intermediate**
Age · sex	30–40 · F	**15–25** · F
Site	Posterior distal femoral **metaphysis**	Femur/tibia **diaphysis**
X-ray	**Stuck-on**, lobulated, ossified	**Sunburst** in saucerized cortex
Histology	Spindle cells between regular trabeculae	**Chondroblastic** + osteoid
Genes	**MDM2**, ring chromosome	**p53** (15–20%)
Treatment	**Wide resection alone**	**Chemo + wide resection + chemo**

Other distractors osteochondroma (medullary continuity, not aggressive) · dedifferentiated CS (from prior cartilage tumour) · synovial sarcoma (soft tissue, spindle cells, may calcify)

Conventional OS distractors (same module)

Lesion	Histology
Osteosarcoma	**Osteoid from pleomorphic / malignant spindle cells** · Codman triangle, soft-tissue mass
Ewing	Uniform **small blue round cells** (reactive osteoid not prominent)
Osteomyelitis	Inflammatory cells, dead bone, no mass
EG	Histiocytes + eosinophils, rarely mass
MFH of bone	Lytic, **storiform**, no osteoid
Osteoblastoma	Benign stromal cells
Stress fracture	**T1 MRI fracture line**, no mass

5 Management

Treatment — chemotherapy & resection

Same as intramedullary OS **neoadjuvant multi-agent chemo (8–12 wk) → wide resection (limb salvage) → maintenance chemo (6–12 mo)**

Limb salvage allograft / megaprosthesis · rotationplasty · amputation if needed

✗ Wrong options radiation (OS is not radiosensitive) · chemo alone · surgery alone · resection before chemo

6 Complications

Pulmonary mets **20–35%**

Prognosis **intermediate** between parosteal and intramedullary OS

Good sign **90–98% necrosis** after neoadjuvant chemo

Bad sign **MDR gene** (pumps chemo out) · in **25% of primaries, 50% of mets**

How it's asked

"47-yr-old knee pain/swelling, surface lesion, osteoid + cartilage?"	→ **periosteal OS** ✗ Wrong options: chondrosarcoma · Ewing · GCT · FD
"24-yr-old softball twist, sunburst surface ossification?"	→ **periosteal OS** ✗ Wrong options: parosteal (stuck-on) · osteochondroma · dedifferentiated CS · synovial sarcoma
"16-yr-old distal femoral bone-forming lesion, pleomorphic osteoid?"	→ **osteosarcoma** ✗ Wrong options: Ewing · rhabdomyosarcoma · MFH

– – Active space – –

"10-yr-old distal femur, destructive bone-forming, malignant spindle cells?"	→ **osteosarcoma** ✗ Wrong options: osteoblastoma · Ewing
"10-yr-old distal tibia, Codman triangle, soft-tissue mass?"	→ **osteosarcoma** ✗ Wrong options: EG · osteomyelitis · trauma
"Li-Fraumeni (p53) — commonest primary bone tumour?"	→ **osteosarcoma**
"9-yr-old, retinoblastoma, mother bilateral RB, humeral OS — treatment?"	→ **chemotherapy + wide resection** ✗ Wrong options: RT
"16-yr-old conventional OS — genetic feature?"	→ **RB + p53 alterations** ✗ Wrong options: ring chromosomes · t(11;22) · t(X;18) · telomeric
"13-yr-old high-grade OS — treatment?"	→ **neoadjuvant chemo → resection + reconstruction** ✗ Wrong options: resection then chemo · RT
"10-yr-old proximal tibial OS?"	→ **surgery + chemotherapy**
"10-yr-old humeral OS?"	→ **chemo + limb-sparing surgery with reconstruction** ✗ Wrong options: amputation alone · RT
"15-yr-old, indistinct bone-forming knee lesion — next?"	→ **MRI of the entire bone** ✗ Wrong options: biopsy first

PIGMENTED VILLONODULAR SYNOVITIS

– Pathology · 54 / 66 –

1 Basics

Epidemiology

Name	now **tenosynovial giant cell tumour (TGCT)** · WHO 2013: **diffuse TGCT = PVNS** · **localized TGCT = GCT of tendon sheath** (commonest in **fingers**) **(TO)**
Nature	**locally aggressive neoplastic synovial disease** ("not a true neoplasm" – Orthobullets) · neoplastic: **aneuploidy, trisomy 7**, ~20 malignant/ metastatic cases **(TO)** · "non-neoplastic inflammatory" (OB7584) · rarely metastasizes
Incidence	**9.2 per million/yr** (Orthobullets) · **1.8 per million/yr**, 1 in 250 arthroscopies **(TO)**
Demographics	**30–40 yr** (any age; stems 14–58) · **M = F** (Orthobullets) · slight **F** predominance (OB9003)
Site	monoarticular · **knee > hip > ankle > shoulder > elbow** · localized: **anterior knee (80%)**, patellofemoral, infrapatellar fat pad
History of trauma	50%

CSF-1 pathogenesis & pexidartinib

Driver **overexpression of CSF1** → CSF1R-expressing cells cluster → synovial hyperplasia · chromosome **1p13** · **5q33** rearrangement

Gene distractor	Lesion
CSF1 ↑	**PVNS / TGCT**
USP6	ABC, nodular fasciitis
MDM2	Parosteal / low-grade OS, ALT
APC	Desmoid (FAP)
c-myc	Aggressive GCT of bone
EXT	MHE
RB	OS (up to 35%)
p53	Li-Fraumeni sarcomas, breast
FGFR3 · RUNX2	Achondroplasia · cleidocranial dysplasia
Ki-67 · NF1 · NTRK	Ewing · neurofibromatosis · various solid tumours

– – Active space – –

Localized vs diffuse

	Localized	Diffuse
Location	Knee > hip > ankle	Knee (75%)
Age	30–50	<40
Sex	M = F	F ≥ M
Presentation	**Painless**, swollen, long-standing · locking/ catching (meniscal mimic) **(TO)**	**Painful**, swollen, tender, ↓ motion
X-ray	Erosion from local pressure	Degenerative change **both sides of joint**
MRI	**Well-circumscribed** mass	**Ill-defined** mass
Recurrence after synovectomy	**8%**	**30%**
Treatment	**Nodule excision + partial synovectomy**	**Total synovectomy** (anterior arthroscopic + posterior open)

3 Presentation

Hallmark — **recurrent atraumatic haemarthrosis** · insidious pain, stiffness, swelling · boggy effusion, warmth, erythema, joint-line tenderness, ↓ ROM

Pain — often not the main complaint **(TO)** · vague for years

Labs — **ESR, CRP normal**

Aspirate — **grossly bloody / brown** · e.g. >50,000 RBC, 1,100 WBC

4 Diagnosis & Classification

Key radiology findings

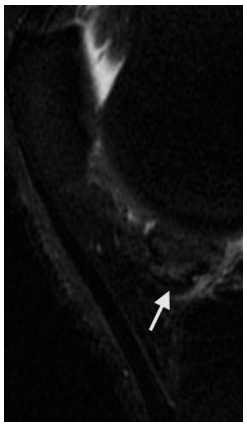
X-ray (AP + lateral) — soft-tissue swelling · **cystic erosions with sclerotic margins on both sides of the joint** · knee often shows advanced OA; erosions/cysts **commoner in hip + ankle** (tight joints) **(TO)** · **no mineralization**
✗ Wrong options: synovial chondromatosis



Cystic erosions with sclerotic margins on both sides

CT — extent of cystic bone loss

MRI (most sensitive) **low signal on T1 and T2** (haemosiderin) · **blooming artefact** on gradient echo (iron) · fat signal on T1 · effusion, synovial expansion, erosion · **posterior extra-articular extension** · modest enhancement · brighter foci = less haemosiderin **(TO)** · "somewhat bright on T2" (OB7420)



Low T2 signal of PVNS

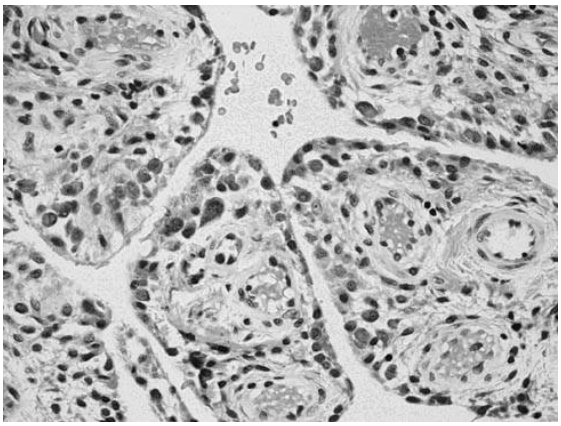
Arthroscopy (gold standard + biopsy) **brownish/reddish inflamed synovium**, **frond-like papillary projections**



Brown frond-like synovium

Histology

Haemosiderin-stained multinucleated giant cells · mononuclear stromal cells · vascular villi lined by plump hyperplastic synoviocytes · **pigmented foam cells (lipid-laden histiocytes)** · mitoses common (few in OB stems) · gross: yellow-brown mass



Haemosiderin-laden giant cells

Pathology

– – Active space – –

Histology distractor	Lesion
Biphasic spindle + epithelial	Synovial sarcoma
Lipoblasts, myxoid, signet-ring	Myxoid liposarcoma
Antoni A/B	**Neurilemmoma**
Plasma cells, lymphoid follicles	RA
Caseating granulomas	TB
Mature chondrocyte nodules	Synovial chondromatosis
Osteoclast-like cells + stroma + osteoid	OS / GCT of bone

Differential

Lesion	Key distinction
Synovial chondromatosis	Nodules **low T1, high T2**, mineralized loose bodies · no haemorrhagic effusion
Synovial sarcoma	**Periarticular, not intra-articular** · T2 bright · spindle cells
Haemophilia	Recurrent effusions, but MRI not nodular mass
RA	Pannus, uniform signal · no haemorrhage
Gout	Punched-out erosions, overhanging edges, tophi
Tumoral calcinosis	Extracapsular periarticular calcification
GCT of bone	Eccentric epiphyseal/metaphyseal lesion in bone
Baker's cyst	Fluid outpouching, semimembranosus–medial gastrocnemius
ABC · chondroblastoma	Intra-osseous (PVNS is extra-osseous)

Behaviour descriptor **locally aggressive**
✗ Wrong options: malignant · inflammatory · normal variant

5 Management

PVNS / TGCT

- Asymptomatic → **OBSERVATION**
- Localized → **NODULE EXCISION + PARTIAL SYNOVECTOMY (arthroscopic if anterior)**
- Diffuse → **TOTAL SYNOVECTOMY (arthroscopic anterior + open posterior) ± EBRT**
- End-stage joint → SYNOVECTOMY + ARTHROPLASTY (ankle → **ARTHRODESIS)**
- Recurrent / unresectable, surgery morbid → **PEXIDARTINIB**

Surgical treatment & recurrence

Mainstay **synovectomy** (open or arthroscopic) · arthroscopic → less morbidity, better ROM · open → pain, stiffness **(TO)**

Margin synovectomy = **intralesional** (OB417 — only lesion where intralesional is standard) · = **marginal** excision (Orthobullets, OB7568)

Localized	**resection of nodule only** · arthroscopic anterior excision + partial synovectomy ✗ Wrong options: complete synovectomy · RT · FNA (excisional biopsy)
Diffuse knee	lowest recurrence: **open anterior + posterior synovectomy** (morbid) · arthroscopic anterior + **open posterior** (between gastrocnemius heads — **popliteal NV bundle at risk**) · **posteromedial portal** for posterior compartment **(TO)**
Foot/ankle	**marginal excision** · radioisotopes poor (multi-joint spread, **skin necrosis** if extravasated) ✗ Wrong options: amputation · wide excision
EBRT	**30–35 Gy / 15 fractions** or **50 Gy / 25** · with total synovectomy → recurrence **10–20%** · for recurrent/destructive disease · not RT alone
Intra-articular Y-90 / radiocolloid	combined with surgery historically **(TO)** · now removed/abandoned — haematomyelopoietic disease, high complications (OB)
Not used	**chemotherapy (TO)** · intra-articular steroids · plica excision · TKA in young without end-stage cartilage

Pexidartinib

Mechanism	oral **CSF-1R antagonist** (TKI) · also KIT + FLT3-ITD · **first and only FDA-approved** systemic therapy (**2019**) **(TO)** ✗ Wrong options: imatinib · emactuzumab · nilotinib (used, not approved)
Indication	**diffuse TGCT not amenable to resection** / severe morbidity from surgery / recurrent recalcitrant ✗ Wrong options: synovial chondromatosis · GCT of bone · synovial sarcoma **(TO)**
Efficacy	once daily 24 wk → ~40% improve (ENLIVEN) · partial responses → surgical downstaging
Side effects	**cholestatic hepatotoxicity** (REMS, LFT monitoring) · **hair depigmentation** · dose taper ✗ Wrong options: alopecia

6 Complications

Recurrence	**most common** complication · **30–50%** despite complete synovectomy (Orthobullets) · **~50%** (OB219276) · 8% localized / 30% diffuse
Joint destruction	early OA → arthroplasty (↑ stiffness, infection) · arthrodesis or amputation
Radiation	skin necrosis, **radiation-induced sarcoma**

Uncommon: arthrofibrosis, chondrolysis, metastasis

How it's asked

"45-yr-old, post-trauma knee nodules, arthroscopy finds far more loose bodies than X-ray showed?"	→ key absent; sources point to **synovial chondromatosis (TO)** ✗ Wrong options: PVNS · OCD · OA
"Recurrent effusion, nodularity, deep-red fronds at arthroscopy?"	→ **PVNS** ✗ Wrong options: synovial sarcoma (not intra-articular) · chondromatosis

– – Active space – –

"24-yr-old anterior ankle mass, giant cells + haemosiderin?"	→ **PVNS** ✗ Wrong options: ABC · chondroblastoma (intraosseous)
"Ankle, low T1 + T2 lobulated synovial mass?"	→ **PVNS** ✗ Wrong options: GCT · gout · haemophilia · tumoral calcinosis
"Low T1/T2, blooming on gradient echo?"	→ **diffuse PVNS** ✗ Wrong options: synovial chondromatosis (high T2)
"Commonest site of localized PVNS?"	→ **anterior knee**
"58-yr-old, lobular low-T1 posterior knee lesion into tibia, no mineral?"	→ **PVNS** ✗ Wrong options: synovial osteochondromatosis
"Best descriptor?"	→ **locally aggressive**
"Recurrent haemorrhagic effusions, nodular low-signal masses?"	→ **PVNS** ✗ Wrong options: RA · chondromatosis
"Correct histology?"	→ **lipid-laden macrophages, haemosiderin, giant cells** ✗ Wrong options: biphasic · Antoni
"Same disease that also occurs in joints?"	→ **tenosynovial giant cell tumour** ✗ Wrong options: GCT of bone · Baker's cyst
"Brown synovium at total synovectomy — histology?"	→ **haemosiderin-stained multinucleated giant cells** ✗ Wrong options: caseating granulomas · chondrocyte nodules
"NOT part of PVNS knee treatment?"	→ **chemotherapy** (✓ arthroscopic synovectomy, ✓ RT, ✓ imatinib) **(TO)**
"Intralesional resection is standard of care for?"	→ **PVNS**
"Ankle PVNS, bloody aspirate — next?"	→ **arthroscopic or open resection** ✗ Wrong options: chemo + wide · RT + radical
"Commonest complication after synovectomy?"	→ **recurrence** ✗ Wrong options: arthrofibrosis · chondrolysis · metastasis
"Brown aspirate, parapatellar tenderness — next?"	→ **arthroscopic or open resection** ✗ Wrong options: plica excision
"Symptomatic localized knee PVNS?"	→ **resection of nodule only** ✗ Wrong options: complete synovectomy · RT
"Hindfoot mass from subtalar joint, low T1/T2?"	→ **marginal excision** ✗ Wrong options: radioisotope · amputation

"Diffuse knee PVNS — lowest recurrence?"	→ **open anterior + posterior synovectomy** ✗ Wrong options: limited arthroscopic · steroid · radioisotope
"16-yr-old, well-circumscribed anterior knee mass?"	→ **arthroscopic excision + partial synovectomy** ✗ Wrong options: open posterior (diffuse)
"Indication for pexidartinib?"	→ **PVNS (TO)**
"Only FDA-approved medical treatment for TGCT?"	→ **pexidartinib (TO)**
"Recurrent effusions, brown synovium — driver?"	→ **CSF-1 overexpression** ✗ Wrong options: USP6 · MDM2
"New drug for PVNS — mechanism?"	→ **CSF-1 receptor antagonist**
"Knee synovectomy planned — mutated gene?"	→ **CSF-1** ✗ Wrong options: APC · c-myc · EXT · RB
"Recurrent PVNS after open synovectomy — prevent recurrence?"	→ **oral pexidartinib** ✗ Wrong options: Y-90 · steroid · TKA
"CSF-1R drug treats?"	→ **TGCT** ✗ Wrong options: gout · haemophilia · RA · chondromatosis
"Pexidartinib side effect + mechanism?"	→ **hair depigmentation from CSF-1R inactivation** ✗ Wrong options: alopecia · anthracycline cardiotoxicity

– – Active space – –

PLANTAR FIBROMATOSIS (LEDDERHOSE DISEASE)

– Pathology · 55 / 66 –

1 Basics

Epidemiology

Nature	benign **myofibroblast + collagen** proliferation of the plantar fascia · **commonest soft-tissue neoplasm of the foot**
Demographics	**males** · **middle-aged / elderly** (any age) · Caucasian
Bilateral	**25–50%**
Risk factors	male · Caucasian · **epilepsy** · repeated trauma · **diabetes** · **long-term alcohol** · chronic liver disease · other fibrosing conditions
Associations	**Dupuytren's** (plantar fibromatosis in **25%** of Dupuytren's) · **Peyronie's**
Genetics	basis **unclear** (unlike Dupuytren's)

Anatomy

Plantar fascia	**3 bands** (central, medial, lateral) · **medial band most affected** · origin medial + anterior calcaneus → 5 digital slips at MTPJs → base of proximal phalanges

Phases

Phase	Feature
Proliferative	↑ Fibroblast activity, cell proliferation
Involutional / active	**Nodule** formation
Residual / resting	↓ Fibroblast activity, collagen maturation, scar/contracture

3 Presentation

Subcutaneous thickening / nodules, **medial plantar** foot · usually asymptomatic, may become painful · **digital contractures rare**

Exam	look for Dupuytren's, Peyronie's · ankle/hindfoot motion · Achilles/ gastrocnemius contracture



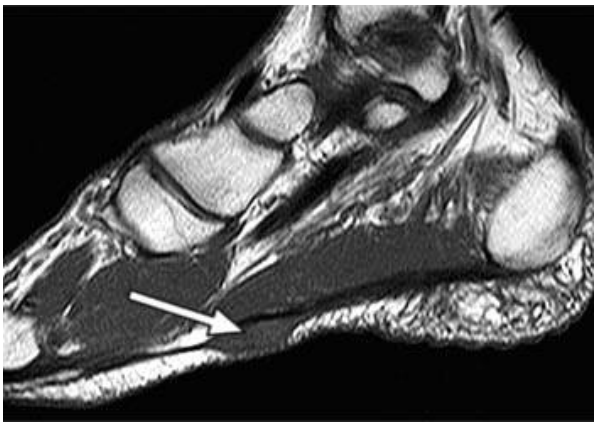
Bilateral medial plantar nodules

4 Diagnosis & Classification

Key radiology findings

X-ray (AP + lateral) usually normal

MRI (study of choice) nodular thickening on **medial** plantar aponeurosis · **low T1** (acellular, high collagen) · **low/medium T2** (high-signal areas if aggressive)



Low T1 nodule on plantar aponeurosis

US less reflective fibroma within bright fascia · **comb sign** – alternating hypo- and iso-echoic linear bands



Comb sign

– – Active space – –

Histology

Dense fibrocellular tissue	mature collagen, fibrocytes at varying maturity · **no atypia, no abnormal mitoses** · gross: lobulated firm irregular mass
Differential	post-traumatic neuroma · fibrosarcoma · clear-cell sarcoma

5 Management

Plantar fibromatosis

- Small / minimally painful → **OBSERVATION + SUPPORTIVE**
  (NSAIDs, orthotics, PT, steroid injection, radiotherapy)
- Large, painful, activity-limiting → **TOTAL FASCIECTOMY ± RADIATION**

Treatment & recurrence

Excision	Recurrence
Local excision	**57–100%**
Wide excision (2–3 cm margins)	8–80%
Total fasciectomy (preferred)	**0–50%**

Technique	avoid incisions directly over nodules · partial fasciectomy → **excise nodule + overlying skin** (± skin graft)
Surgical complications	skin necrosis · nerve entrapment (medial plantar nerve) · **loss of arch height / windlass** · wound breakdown
Radiotherapy	best evidence: **randomized double-blind sham-controlled trial (2023)**, durable efficacy at ≥1 yr · **post-op RT** for high-grade lesions to cut recurrence
Evidence grades	collagenase: insufficient (case reports) · orthotics: expert opinion (Level V) · surgery: no Level I superiority data

6 Complications

Recurrence	**60% of excised lesions** · recurs **more aggressive** · risk: **bilateral**, **multiple nodules**, **family history** · prevent with adjuvant RT
Prognosis	malignant transformation **rare**

How it's asked

"Plantar fascial lesion – excisional biopsy's likely complication?"	→ **recurrence** ✗ Wrong options: metastasis · wound breakdown, nerve injury, windlass loss = wide excision problems
"Bilateral plantar nodules, failed orthotics/NSAIDs – highest-grade evidence?"	→ **radiotherapy (RCT, double-blind sham-controlled)** ✗ Wrong options: collagenase RCTs · surgery RCTs

PLEOMORPHIC SARCOMA OF BONE (MALIGNANT FIBROUS HISTIOCYTOMA)

– Pathology · 56 / 66 –

1 Basics

Epidemiology

Incidence	**<2% of primary malignant bone tumours** · **2nd commonest bone sarcoma in adults** after chondrosarcoma
Demographics	**20–80 yr** (peak 2nd–4th decades) · **M:F 3:2** · Caucasian
Site	**75% appendicular** · **metaphysis** of long bones: **distal femur, proximal tibia, proximal humerus**
Secondary (25%)	from **bone infarct, Paget's, prior radiation**

Genes & lab markers

p53 mutations – secondary MFH of bone

Labs	↑ WBC · **eosinophilia** · abnormal LFTs · **hypoglycaemia**

3 Presentation

New, painful mass · **pathologic fracture** · constitutional (fever, chills, night sweats, weight loss) · swelling, limp, ↓ ROM

4 Diagnosis & Classification

Key radiology findings

X-ray (AP + lateral)	**purely lytic, destructive metaphyseal** lesion · **no matrix mineralization** · variable periosteal reaction · cortical destruction + soft-tissue extension · non-specific



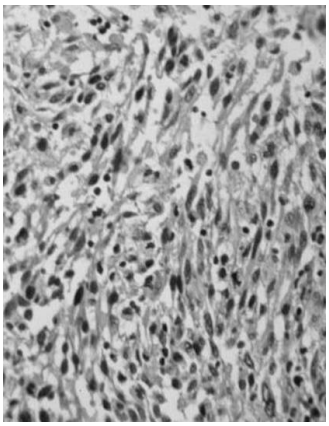
Lytic destructive femoral lesion

– – Active space – –

MRI	deep heterogeneous mass · internal haemorrhage · **T1 low, T2 high**
CT	**chest CT staging** (lung mets) · bone destruction
Bone scan	staging · ↑ uptake in adjacent cortex

Histology

Pleomorphic spindle + histiocytic cells in storiform pattern · malignant multinucleated **giant cells with grooved/indented nuclei** · chronic inflammatory cells · variable collagen · fascicles radiating from hypocellular foci · haemorrhage/necrosis → high grade · **no osteoid** (vs OS)



Storiform pleomorphic spindle cells

Differential

	Malignant destructive, older	Sacral, older	Wide resection + chemo
MFH / fibrosarcoma	✓	✓	✓
Chondrosarcoma (conventional)	✓	✓	–
Dedifferentiated chondrosarcoma	✓	✓	✓
Metastatic disease · myeloma	✓	✓	–
Lymphoma of bone	✓	✓	–
Secondary sarcoma	✓	✓	✓
Osteosarcoma · **Ewing**	–	–	✓
Chordoma (sacral)	–	✓	–
Rhabdomyosarcoma (soft tissue)	–	–	✓

Enneking staging (OB)

Stage	Meaning
I	Low grade
II	High grade
III	Distant metastasis
A / B	**Intracompartmental / extracompartmental**

Stage IIB spindle-cell sarcoma → pre-op chemo + wide excision + chemo

5 Management

Diagnosis & treatment

Treat like osteosarcoma	**neoadjuvant chemo (8–12 wk) → wide resection → adjuvant chemo (6–12 mo) ± RT** ✗ Wrong options: wide resection alone · chemo alone · RT alone · observation
Margin	**wide excision or amputation** → better 5-yr survival than intralesional/marginal
RT	incomplete/questionable margins · adjunct
Soft-tissue UPS for contrast	**radiation + wide resection** (painless, slow, T1 dark/T2 bright)

Reconstruction after resection

Limb salvage endoprosthesis · resection arthrodesis · allograft · **rotationplasty**

Proximal tibia	Megaprosthesis (MP)	Allograft-prosthetic composite (APC)
10-yr implant survival	**79%**	**94%**
Failure rate	No significant difference	No significant difference
Function	Similar	Similar · **better extensor mechanism in high-demand patient**
Leg length	Both restore	Both restore

6 Complications

Survival	**50–60% at 5 yr**
Lung mets	up to **30%**
Prognosis	**secondary worse than primary** · **>90% necrosis** after chemo → better · **age <40** → better DFS

X vs Y

Distractor (soft-tissue thigh mass)	Key feature
UPS	Painless, slow, deep, T1 dark / T2 bright
Liposarcoma	Lipoblasts, signet-ring cells
Desmoid	Upper torso, **dark T1 + T2**, fibroblasts + collagen
Synovial sarcoma (periarticular)	Biphasic, **t(X;18)**, SYT-SSX
Schwannoma	Homogeneous, **target sign**, Antoni A/B

PET ↓ glucose uptake = early response predictor (OS, Ewing, GIST)

– – Active space – –

How it's asked

"75-yr-old, slow painless 7 cm thigh mass, T1 dark/T2 bright, pleomorphic?"	→ **undifferentiated pleomorphic sarcoma** ✗ Wrong options: liposarcoma · desmoid · synovial sarcoma · schwannoma
"60-yr-old, spindle-cell sarcoma of bone — optimal treatment?"	→ **neoadjuvant chemo + wide resection + adjuvant chemo** ✗ Wrong options: wide resection alone
"12-yr-old, purely lytic distal femoral metaphyseal lesion, storiform malignant spindle cells?"	→ **chemotherapy + surgery** (MFH of bone)
"Proximal tibial resection — APC vs megaprosthesis?"	→ **APC better extensor mechanism in high-demand patient** ✗ Wrong options: MP longer survival · APC higher failure

RADIATION THERAPY

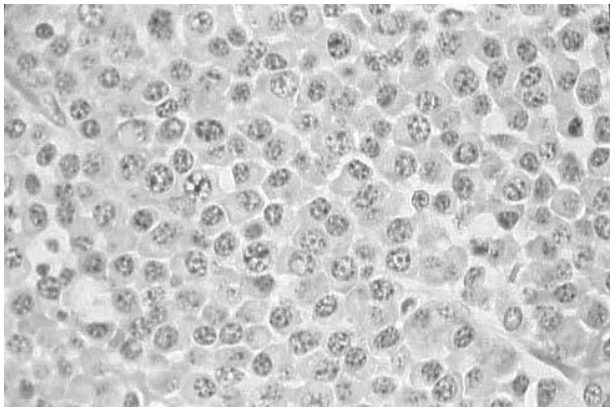
– Pathology · 57 / 66 –

1 Basics

Mechanisms **free radical production · direct genetic (DNA) damage**

Radiosensitive tumours

Use	Tumours
Definitive control (primary bone)	**Ewing / PNET · primary lymphoma of bone · haemangioendothelioma · solitary plasmacytoma**
First-line RT alone (spinal mets – OB)	**Leukaemia, lymphoma, myeloma/plasmacytoma, haemangioendothelioma, Ewing**
Adjuvant to surgery	**Soft-tissue sarcomas** (pre- or post-op)
Palliation / after fixation	**Metastatic bone disease** – after fixing impending/pathologic fractures



Solitary plasmacytoma – radiosensitive

Metastasis sensitivity	
Prostate	**Very radiosensitive**
Breast	**70% sensitive**, 30% resistant
GI, renal	**Not radiosensitive**

Radioresistant (OB) **fibrosarcoma, chondrosarcoma, melanoma, angiosarcoma** · osteosarcoma (not radiosensitive – OS chapters)

– – Active space – –

5 Management

Technique

1 rad = 1 cGy · typical **180–200 cGy/day** in **fractions** · RT is **cumulative** (total = sum of fractions)

Total dose	Tissue healing
<45 Gy	Usually uncomplicated
45–55 Gy	Heals, with problems
>60 Gy	**Unlikely to heal**

Preoperative vs postoperative radiation

	Pre-op (neoadjuvant)	Post-op (adjuvant)
Wound complications	**More (up to 40%)**	Less (<20%)
Oedema, fibrosis, stiffness	**Less (<15%)**	More (>20%, often temporary)
Lymphoedema · fracture	**Lower**	Higher
Field · dose	**Smaller field**, lower dose	Larger field, higher dose
Pseudocapsule	**Forms** → close-margin resection	No
Tumour shrinkage	**Yes**	No
Local recurrence	**Lower** (Orthobullets) · **no difference** (OB6336)	Higher (Orthobullets)
Overall survival	No difference by timing	No difference

6 Complications

Radiation complications (fracture, secondary sarcoma)

Early **delayed wound healing**, **infection**, **desquamation**

Late **fibrosis**, **joint stiffness**, **secondary sarcoma**, **fractures**

Post-radiation fracture **~25%** after STS resection + EBRT

Fracture risk factor	Yes / No
Dose ≥59 Gy	Yes
Weight-bearing bone, esp. femur	Yes
Female	Yes
Volume of bone irradiated	Yes
Anterior femoral compartment resection	Yes
Age	Yes
Periosteal stripping (→ consider prophylactic fixation)	Yes
Osteoporosis	Yes
Method of RT administration	No

Post-radiation sarcoma	sarcoma in a previously irradiated field · **~13%** · ↑ with prior chemo · **very poor prognosis**
Criteria	**≥5 yr** after RT (range **3–50 yr**) · **in the RT field** · **different histology** from the original disease



Lytic destructive clavicle lesion — radiation-associated sarcoma

How it's asked

"Pre-op vs post-op RT — higher with pre-op?"	→ **wound necrosis** ✗ Wrong options: fibrosis · lymphoedema (post-op) · local recurrence (no difference)
"Synovial sarcoma (SS18-SSX), pre-op RT (A) vs post-op RT (B)?"	→ **A: more wound complications, less fibrosis**
"NOT a risk factor for post-RT fracture?"	→ **method of RT administration** (✓ female, ✓ age, ✓ dose, ✓ periosteal resection)
"47-yr-old, clavicle lytic lesion, mantle RT 22 yr ago, atypia + bone formation?"	→ **radiation-associated sarcoma** ✗ Wrong options: osteonecrosis · osteomyelitis
"L4–5 lytic metastases — RT first-line for?"	→ **lymphoma** ✗ Wrong options: fibrosarcoma · chondrosarcoma · melanoma · angiosarcoma

– – Active space – –

RHABDOMYOSARCOMA

– Pathology · 58 / 66 –

1 Basics

Epidemiology & subtypes

Nature malignant tumour of **primitive mesenchyme** (mesoderm, skeletal-muscle precursors)

Commonest soft-tissue sarcoma in children
✗ Wrong options: foot (synovial sarcoma) · calcifies (synovial sarcoma) · Maffucci (haemangiomas) · from myxomas

Site **head/neck, genitourinary, retroperitoneum** · GU > extremities (TO, OO chapter)

Subtype	Age	Histology	5-yr survival
Embryonal	Infants, young children	**Small round blue cells**, ± cross-striations	**80%**
Alveolar	**Adolescents, young adults** (15–40)	Poorly differentiated round cells + **multinucleated giant cells**, aggregates in **dense fibrous septae**	**60%**
Botryoid ("bunch of grapes")	Infants, young children · **vagina**	–	**<30%** ("uniformly fatal")
Pleomorphic	**40–70 yr**	Multiple cell types, like other pleomorphic sarcomas	**25%**

Genetics – t(2;13) PAX3-FKHR

Alveolar **t(2;13) → PAX3-FKHR** · **t(1;13) → PAX7-FKHR** (as named in OB8296; paired the other way round in some sources) · **high metastatic risk**

Translocation	Tumour
t(2;13)	**Alveolar RMS**
t(11;22) EWS-FLI1	Ewing
t(12;16)	Myxoid liposarcoma
t(16;17)	ABC
t(9;22)	CML
t(X;18)	Synovial sarcoma
GNAS · IDH1	FD · Maffucci/Ollier
p53 · RB	Osteosarcoma

Genes & lab markers

IHC **MyoD1 +, myoglobin +, myosin +, desmin +, vimentin +**

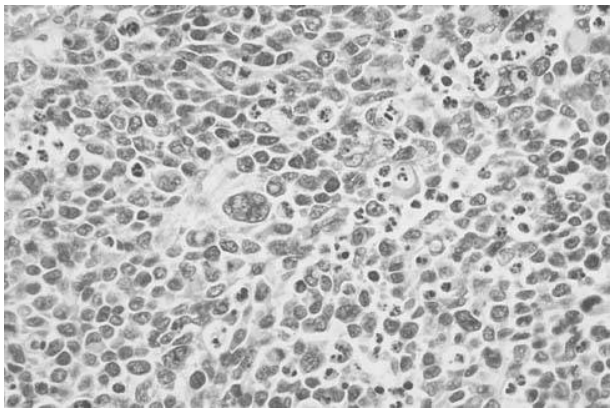
3 Presentation

Rapidly growing painless mass · ± **enlarged regional nodes** (e.g. axillary)

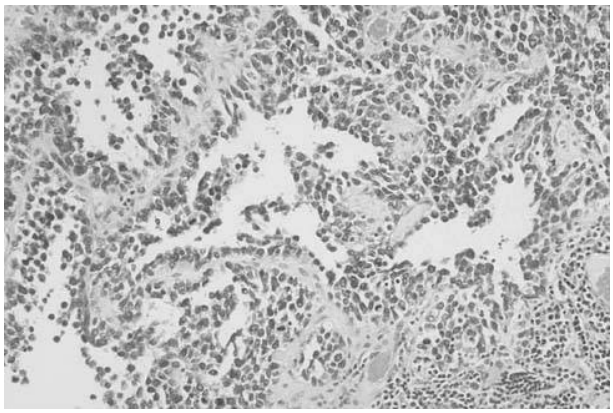
4 Diagnosis & Classification

Key radiology findings

MRI	non-diagnostic but needed for planning · **dark T1, bright T2** (like other STS)
CT chest	required for staging
Bone marrow biopsy	**required** for staging · marrow mets → worse prognosis
Nodes	**sentinel node biopsy** (esp. if nodes enlarged)



Embryonal RMS – small round blue cells



Alveolar RMS – fibrous septae

Lymph-node-metastasizing sarcomas

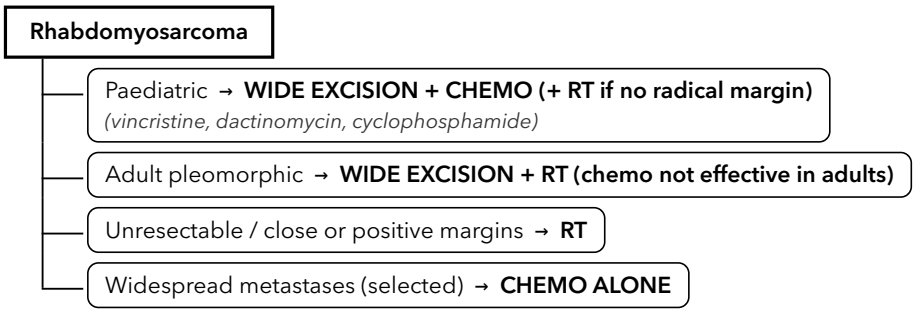
"SCARE"	**synovial sarcoma, clear-cell sarcoma, angiosarcoma, rhabdomyosarcoma, epithelioid sarcoma** ✗ Wrong options: Ewing · myxoid liposarcoma

Small round blue cell differential

RMS · neuroblastoma · lymphoma · leukaemia · Ewing · metastatic disease

– – Active space – –

5 Management



Teenager, isolated limb RMS — **wide resection + chemo + RT** · chemo has a **substantial survival benefit** · amputation rarely (only if better function)
✗ Wrong options: observation · marginal excision

How it's asked

"True of rhabdomyosarcoma?"	→ **commonest soft-tissue sarcoma in a child** ✗ Wrong options: commonest in foot · calcifies · Maffucci
"Teenager, lower-limb RMS — management?"	→ **wide resection + chemo + RT** ✗ Wrong options: hemipelvectomy · marginal
"Alveolar RMS — genetic abnormality?"	→ **PAX3-FKHR** ✗ Wrong options: EWS-FLI1 · p53 · RB
"9-yr-old, rapid painless arm mass, axillary nodes?"	→ **t(2;13)** ✗ Wrong options: t(12;16) · GNAS · t(11;22) · IDH1
"Translocation of a sarcoma with lymphatic mets?"	→ **t(2;13)** ✗ Wrong options: t(16;17) (ABC) · t(9;22) (CML)

SOFT TISSUE SARCOMA

– Pathology · 59 / 66 –

1 Basics

Epidemiology

Incidence	**12,000/yr** in US · 4,700 deaths/yr · **M > F** · **85% >15 yr**
Site	**60% extremities** · **44% thigh**
Origin	mesoderm (connective tissue, muscle, fat, nerve, vessels) · grow **centripetally with a pseudocapsule**, compress neighbours
Types	**>50** histologic types (~100 — OB) · same presentation, imaging, differential and treatment · commonest: **UPS/MFH, liposarcoma, leiomyosarcoma, synovial sarcoma**
Commonest by group (TO)	adults: **MFH > liposarcoma** · young adults: **synovial** · children: **rhabdomyosarcoma** · hand/UL: **epithelioid** · foot: **synovial**

Genes & lab markers

Translocation / association	Sarcoma
t(X;18), SYT-SSX	**Synovial sarcoma**
t(2;13)	**Alveolar rhabdomyosarcoma**
t(12;16)(q13;p11)	**Myxoid liposarcoma**
NF1	**MPNST**
Stewart-Treves / chronic lymphoedema	**Angiosarcoma**
S-100 + HMB45	**Clear-cell sarcoma** (melanoma of soft parts)

Sarcomas with lymph node spread

"RACES"	**R**habdomyosarcoma · **A**ngiosarcoma · **C**lear-cell sarcoma · **E**pithelioid sarcoma · **S**ynovial sarcoma → consider **sentinel node biopsy**
Overall	only **5%** of STS go to nodes ✗ Wrong options: liposarcoma · desmoid · GCT · Ewing · OS · chondrosarcoma · MFH · MPNST
Leiomyosarcoma	can spread via lymphatics but mainly **haematogenous** → lungs

3 Presentation

Slow-growing, painless mass · trauma draws attention · **rapid growth → higher grade**

Red flags	**>5 cm**, **deep to fascia**, **immobile**, rapid growth, pain in a previously painless mass, recurrent swelling
Mimics	lipoma, **haematoma** (no trauma, no ecchymosis → suspect sarcoma; mean **7-month delay** when misdiagnosed), desmoid, nerve sheath tumours, abscess, biceps tear

– – Active space – –

"Sarcoma until proven otherwise"	large (>5 cm), deep, heterogeneous mass

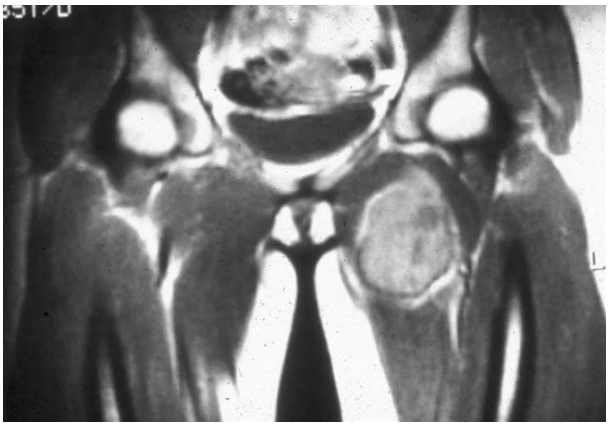
4 Diagnosis & Classification

Workup of soft tissue mass (MRI, biopsy)

Soft-tissue mass

- X-rays (2 planes) → **bone involvement, mineralization (synovial sarcoma)**
- Contrast MRI → diagnostic & benign & symptomatic → **EXCISE WITHOUT BIOPSY**
  (lipoma, ganglion, schwannoma, intramuscular myxoma)
- MRI indeterminate / sarcoma-like → CORE NEEDLE or OPEN BIOPSY → **STAGING CT**

MRI (mandatory)	**T1 low** (iso to muscle), **T2 high**, gadolinium: **peripheral enhancement + non-enhancing necrotic centre** · ~**90%** accurate benign vs malignant (ill-defined, large, heterogeneous)



Deep adductor sarcoma on MRI

CT chest	staging (lungs) · **myxoid liposarcoma → CT chest/abdomen/pelvis (± bone scan)**
PET	evolving · useful for node-spreading sarcomas · ↓ uptake predicts response
Next step order	X-ray → **MRI** (✗ Wrong options: biopsy before MRI · FNA · excisional biopsy) → **biopsy** (✗ Wrong options: RT · chemo · resection · observation before tissue diagnosis) → staging

Biopsy

Core needle	modality of choice · correct in **68–93%** (Orthobullets) · **84%** (Skrzynski) · cheaper ($1106 vs $7234)
Open biopsy	gold standard · after **non-diagnostic core** (e.g. "only blood" in a presumed haematoma)
Excisional biopsy	only small (**<3 cm**) clearly benign lesions
Who	at the treating centre, by the definitive surgeon · **2–12× more complications** otherwise
Technique	**longitudinal** incision · avoid NV structures/compartments · meticulous haemostasis · **drain distal + in line** with incision · expert pathologist ✗ Wrong options: transverse

Histology distractors

Lesion	Histology
Leiomyosarcoma	Spindle cells, smooth muscle
Myxoma	Paucicellular myxoid stroma, bland spindle cells, macrophages
Haematoma	Cystic cavity, necrotic debris, fibrin, clot
Neurofibroma (dermal)	Discoid proliferations of peripheral nerves
Fibroma	Poorly marginated hypocellular collagen
Fibrosarcoma	**Herringbone** spindle cells **(TO)**
MFH / UPS	**Storiform** pleomorphic
Clear-cell sarcoma	Nests of **clear cells** in fibrocollagenous septa, giant cells

AJCC 8th staging (extremity / trunk)

Stage	Size	Nodes	Mets	Grade
IA	<5 cm	None	None	Low
IB	>5 cm	None	None	Low
II	<5 cm	None	None	Intermediate/high
IIIA	**5–10 cm**	None	None	Intermediate/high
IIIB	**>10 cm**	None	None	Intermediate/high
IVA	Any	**Any N+**	None	Any
IVB	Any	Any	**M+**	Any

8th edition changes **separates appendicular/truncal from other sites** (head/neck, visceral, retroperitoneum) – widely accepted · **depth removed** (not independent) · new T3 (10–15 cm) + T4 (>15 cm) · **node = stage IV** (controversial, equated with lung mets)
✗ Wrong options: margins · histologic subtype · hospital type

Vanderbilt staging includes depth · **Enneking** – no size, not for marrow/RES tumours, limited for spine

5 Management

Soft tissue sarcoma (localized)

- Most → **WIDE RESECTION + RT (pre- or post-op)**
- Abutting critical structure → **PLANNED CLOSE / MICRO-POSITIVE MARGIN + RT**
  (remove adventitia / perineurium / periosteum)
- Encasing critical structure → **EN BLOC with the structure (bypass graft if possible)**
- Can't get negative margin / irreversible nerve loss / comorbid → **AMPUTATION (often foot)**
- Resectable lung mets → **WEDGE RESECTION / LOBECTOMY**

– – Active space – –

Surgical margins & resection

Most important treatment factor (subcutaneous sarcoma DFS) — **size of operative margin**
✗ Wrong options: chemo dose · RT · RFA

High grade	at least **wide** · not intralesional or marginal · dorsum of foot → **wide resection + fasciocutaneous flap** ✗ Wrong options: STSG over tendons
Abutting SFA + sciatic nerve, tumour within 1 mm, ink negative	**equivalent to wide resection removing the structures** (with RT)

Periosteal resection >10 cm → fracture risk → **prophylactic nailing**

Large midfoot synovial sarcoma	**transtibial amputation** (best local control, fastest return)
Resection technique	elliptical incision **incorporating biopsy + drain sites** · tourniquet **without exsanguination** · frozen section margins
Lung mets	first-line **surgical resection** if complete resection possible (wedge/lobectomy) · curative in up to **25%** ✗ Wrong options: RT · chemo first
Chemotherapy	**controversial** (no survival benefit in RCTs for most) · consider high grade · **standard for rhabdomyosarcoma + Ewing** · synovial sarcoma more chemo-sensitive

Radiation therapy

Effect	**↓ local recurrence regardless of margin status** · **no overall survival benefit** (Yang) · transient strength/ROM loss · does not hamper limb salvage
Dose	**50 Gy pre-op**, **66 Gy post-op** (Orthobullets, OB7588) · **60 Gy** typical adjuvant (OB1117), range 45–65 Gy · palliative bone mets **~30 Gy** · HO prophylaxis **~6 Gy**

	Pre-op	Post-op
Wound complications	↑ — 20–30% (Orthobullets) · 30% (O'Sullivan) · **35% vs 17%** (OB7588)	↓
Dose · volume	**Lower, smaller**	Higher, larger
Fibrosis, oedema, stiffness, fracture, lymphoedema	↓	↑ · ↑ radiation-induced sarcoma
Function	Trend better	–
Local recurrence · survival	**No difference**	No difference

RT not used	chondrosarcoma, osteosarcoma (as adjuvant) · **WD liposarcoma / ALT → excision alone** (recurrence 2%)

Unplanned excision — tumour bed re-excision

Commonest mistake in STS treatment · **74% residual disease**

Unplanned ("whoops") excision

- First → **MRI limb + CT chest/abdomen/pelvis (no RT or surgery before imaging)**
- Positive margins → **TUMOUR BED RE-EXCISION (wide, ≥3 cm along scar) ± RT**
- Negative margins → **OBSERVATION vs ADJUVANT RT**

Deep, positive margins (e.g. sartorius)	**wide tumour bed excision + RT** ✗ Wrong options: RT alone · amputation · chemo
High-grade superficial forearm	**wide resection of tumour bed + coverage** ✗ Wrong options: marginal to spare NV
After wide re-excision vs planned	↑ **plastic reconstruction** · less blood loss · same function · **5-yr LR-free >90%** · no extra chemo

Specific subtypes (clear cell, alveolar soft part)

Clear-cell sarcoma	**malignant melanoma of soft parts** · adolescents/young adults · foot/ankle, bound to tendons · **S-100, HMB45** · **lymph node** mets · chemo + RT resistant → **wide excision / amputation**
Alveolar soft part sarcoma	**<1%** of STS · very vascular · lung, bone, **CNS** mets at presentation · LR 11–50% · **wide resection** + metastasectomy
Fibrosarcoma (TO)	diagnosis of exclusion · **herringbone** spindle cells · 5–10 cm slow painless deep lower-limb mass, **30–50 yr**

Synovial sarcoma — histology & prognosis (TO)

Not synovial in origin	arises near bursae, tendon sheaths, capsules from multipotent stem cell rests → **biphasic = epithelial + mesenchymal (spindle)** · **lacks synovial differentiation**
Subtypes	**biphasic** (epithelial + spindle) · **monophasic** (spindle only) · poorly differentiated
Genetics	**t(X;18)(p11;q11)**, SS18 (SYT) on 18 + SSX1/2/4 on X → **SYT-SSX** · in **>90%** / **95%** · **SYT-SSX1 → biphasic** · markers (poorly differentiated): **CD56, CD99, HMW keratin** · BCL-2 under study
Epidemiology	**7%** (6–8%) of STS · commonest **foot** STS · commonest **non-RMS paediatric** STS · median **35 yr**, 15–40 · slight **M** · **knee** commonest
Imaging	X-ray: heterogeneous well-defined septate cystic mass, **calcification in 1/3** (CT best) · MRI: cystic, sharp, multiloculated → **misread as benign** · PET for mets/recurrence
Grade	**all high grade** · FNCLCC grading used

– – Active space – –

Treatment wide excision (**>4 cm** margin) + RT (all >5 cm: EBRT, brachytherapy, IMRT) ± (neo)adjuvant chemo · amputation for fungating mass, difficult fracture, infection · metastatic → **doxorubicin + ifosfamide**

Advanced-disease drug	Mechanism
Pazopanib	**Only approved targeted drug for STS** · TKI vs VEGFR, PDGFR, KIT
Trabectedin	Blocks DNA binding of the fusion oncoprotein
Tazemetostat	**EZH2 inhibitor** (INI1/SMARCB1-negative)
NY-ESO-1 T cells	Antigen in **80%**
Isolated limb perfusion	**TNF-α + melphalan**/doxorubicin, hyperthermic · limb preservation **72–96%**

Poor prognosis (TO)	Yes / No
Age >50 (younger better)	✓
Size ≥5 cm	✓
Deep	✓
Axial site (distal extremity best)	✓
Higher stage · FNCLCC grade II–III · mitoses	✓
Monophasic histology	✓
CXCR4, IGF-1R nuclear expression	✓
Inadequate margins	✓
Biphasic histology	**No (favourable)**

Metastasis most important factor · in **~50%** · lung **80%**, nodes **up to 20%**, bone 9.9%, liver 4.5% · **5-yr OS ~60%** (61–73%)

6 Complications

Metastasis **lungs commonest (40–60%)** · then **lymph nodes** (OB6341) · or bone/soft tissue (OB7545) · liver, brain rare

Local recurrence **<10%** with surgery + RT · commonest site for low-grade recurrence = **local** · most within **5 yr**, but follow indefinitely

Radiation pre-op wound problems · post-op fibrosis, fracture, secondary sarcoma

Prognosis

Most important factor **tumour stage** (Orthobullets) · **metastatic disease at presentation** (OB6333, bone sarcoma) · **histological grade** (TO one-liner)

Poor high grade · mets · **>5 cm** · **deep to fascia** · delayed diagnosis · **unplanned excision**

Stage	5-yr survival
Low grade	**90%**
II–III	**50–75%**
IV	**15%** · median 12–18 months

How it's asked

"52-yr-old, large deep thigh mass — initial step?"	→ **biopsy** ✗ Wrong options: marginal excision · RT · chemo
"Slow-growing thigh mass with ecchymosis, 'haematoma', core yields only blood?"	→ **open surgical biopsy** ✗ Wrong options: RICE · drain · I&D
"4 cm deep thigh mass, X-rays clear — next?"	→ **contrast MRI** ✗ Wrong options: FNA · core · excisional biopsy
"64-yr-old fixed posteromedial thigh mass, varicosities, spindle cells?"	→ **soft tissue sarcoma** (leiomyosarcoma) ✗ Wrong options: myxoma · haematoma · fibroma
"83-yr-old enlarging forearm mass >5 cm?"	→ **needle biopsy** ✗ Wrong options: marginal resection
"Slow gluteal mass, heterogeneous on MRI?"	→ **CT-guided biopsy** ✗ Wrong options: wide resection first
"Medial ankle mass after tripping, MRI suggestive of sarcoma?"	→ **biopsy**
"Painless subfascial arm mass after lifting boxes?"	→ **CT-guided core needle biopsy** ✗ Wrong options: ED drainage · biceps repair
"55-yr-old painless heterogeneous deep thigh mass on MRI?"	→ **core-needle biopsy** ✗ Wrong options: excisional (<3 cm benign)
"Low-grade STS shelled out, positive margins, no mets?"	→ **tumour bed re-excision**
"Incompletely excised high-grade superficial forearm sarcoma?"	→ **wide tumour-bed resection + coverage**
"'Lipoma' in sartorius → high-grade sarcoma, positive margins?"	→ **wide tumour-bed excision + RT**
"Piecemeal 'lipoma' excision → t(12;16) — next?"	→ **imaging** (CT CAP + MRI) before re-excision
"Wide re-excision after unplanned excision vs planned?"	→ **↑ plastic reconstruction** ✗ Wrong options: ↑ blood loss · worse function · LR >50%
"Most important treatment factor for subcutaneous sarcoma DFS?"	→ **size of operative margin**
"39-yr-old large midfoot synovial sarcoma — best local control + fastest function?"	→ **transtibial amputation** ✗ Wrong options: wide resection + RT · ray resection
"28-yr-old 5.7 cm high-grade dorsal foot sarcoma?"	→ **wide resection + flap coverage** ✗ Wrong options: marginal + STSG · BKA

– – Active space – –

"Liposarcoma abutting SFA + sciatic nerve, close margin, ink negative?"	**→ equivalent to wide resection removing both structures**
"60 Gy is typical for?"	**→ adjuvant RT for STS** ✗ Wrong options: HO ~6 Gy · bone mets ~30 Gy
"True about RT in STS?"	**→ reduces local recurrence** ✗ Wrong options: improves survival · pre-op fewer wound problems
"Wide excision + RT for all EXCEPT?"	**→ well-differentiated liposarcoma** ✗ Wrong options: round cell · fibrosarcoma · synovial · pleomorphic
"Pre-op vs post-op RT?"	**→ higher wound complications** ✗ Wrong options: higher dose · larger volume · worse survival
"Most significant factor for long-term survival in bone sarcoma?"	**→ metastatic disease at presentation**
"Commonest site of STS metastasis?"	**→ lungs**
"Solitary lung nodule 9 months after STS resection?"	**→ wedge resection** ✗ Wrong options: pneumonectomy · chemo · RT
"15 cm deep arm sarcoma on biopsy — next imaging?"	**→ chest CT** ✗ Wrong options: PET · bone scan
"7 cm high-grade STS, N0 M0 — AJCC?"	**→ IIIA**
"AJCC 8th major change?"	**→ appendicular separated from other sites**
"4 cm synovial sarcoma, groin node +, lungs clear — AJCC?"	**→ IVA**
"High grade, 8 cm, N0 M0?"	**→ IIIA** ✗ Wrong options: 7 cm intermediate = IB (it is IIIA)
"Five STS that go to nodes?"	**→ rhabdomyosarcoma, synovial, epithelioid, clear-cell, angiosarcoma**
"25-yr-old dorsal wrist clear-cell sarcoma — spreads to?"	**→ lymph nodes**
"6 cm medial thigh mass + inguinal nodes — differential?"	**→ clear-cell, synovial, rhabdomyosarcoma** ✗ Wrong options: GCT · liposarcoma · desmoid
"Dorsal wrist 'ganglion' excised — nests of clear cells?"	**→ clear-cell sarcoma** ✗ Wrong options: epithelioid · synovial
"Posterior leg mass, vascular nests — diagnosis + treatment?"	**→ wide resection for alveolar soft part sarcoma**

"Synovial sarcoma — all true EXCEPT?"	→ **"biphasic with synovial + epithelial differentiation"** (it lacks synovial differentiation) (✓ knee, ✓ young adults, ✓ t(X;18) >90%) **(TO)**
"NOT a poor prognostic factor in synovial sarcoma?"	→ **biphasic histology** (✓ old age, ✓ large size, ✓ axial) **(TO)**
"Herringbone pattern?"	→ **fibrosarcoma (TO)** ✗ Wrong options: Ewing · synovial · MFH (storiform)

– – Active space – –

– – Active space – –

SQUAMOUS CELL CARCINOMA

– Pathology · 60 / 66 –

1 Basics

Nature	malignant tumour of **keratinocytes** (epidermis) · elderly
Hand	**commonest malignancy of the hand** (skin cancers ≫ primary MSK malignancy) · **commonest subungual malignancy** · dorsum of hand + forearm
Skin overall	**2nd commonest skin cancer** after BCC (OB212007) · in the hand, **BCC is 2nd** (OB220030) · face/neck > hand
Risk factors	**UV exposure** (main) · **actinic keratosis** · Bowen's disease, keratin horns, leukoplakia · **solid-organ transplant** (immunosuppression) · **chronic osteomyelitis**

Marjolin ulcer

Definition	SCC in **chronic draining sinus tracts** (osteomyelitis), **burn scars**, pressure sores
Rate	**~1%** malignant change · latency **~30 yr** · many low grade
Distractors	Mooren (corneal) · chancroid (genital) · **Curling** (peptic, burns) · **Cushing** (gastric, ↑ ICP)
High suspicion	biopsy chronic draining sites

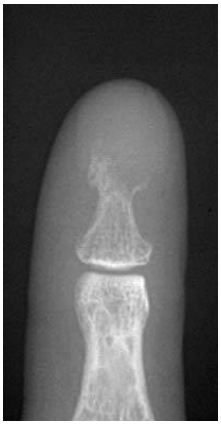
3 Presentation

Indurated, scaly, erythematous plaque/lesion · pain · subungual mass mistaken for **haematoma**

4 Diagnosis & Classification

Key radiology findings

X-ray (AP + lateral) **lytic** lesion when bone involved (e.g. distal phalanx tuft destruction)

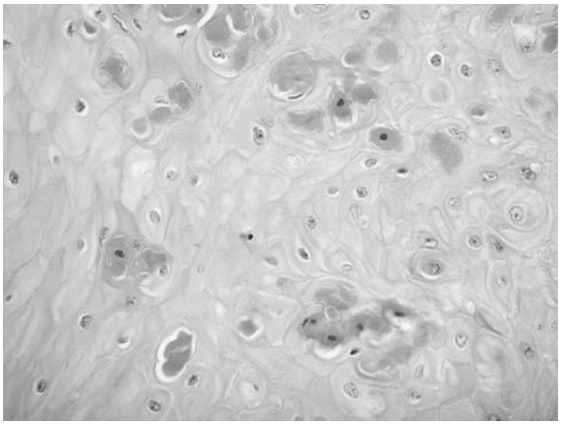


Lytic destruction of distal phalanx

Biopsy **punch, shave or incisional**

Histology

Keratin nodules / squamous pearls · tumour nests penetrating deep fascia



Keratinizing squamous cells

Skin lesion	Feature
SCC	**Keratin pearls**
BCC	Pink/flesh papule, central ulcer · nests of basal cells
Melanoma	From naevus · ABCD changes
Seborrhoeic keratosis	"Stuck-on" waxy lesion · histology like SCC
Lentigo	Small pigmented macule
M. marinum	Culture 30–32°C
Sarcoidosis	Non-caseating granulomas

5 Management

Hand SCC – diagnosis & treatment

Squamous cell carcinoma

- Most → **WIDE LOCAL RESECTION (6 mm margin to subcutaneous fat) ± SKIN GRAFT**
- Small lesions → **MOHS MICROSURGERY (highest cure rate)**
- Bone involvement (digit) → **AMPUTATION**
- Marjolin ulcer → **WIDE RESECTION / AMPUTATION + soft-tissue reconstruction**

Adjuvant RT (Orthobullets) **>2 cm** wide · **4 mm** deep · **perineural invasion** · **nodal mets** · OB: RT not needed for primary SCC, not typical for SCC ("radiation" is wrong in Marjolin and hand stems)

Lymph node biopsy may be needed

Marjolin ✗ Wrong options: IV antibiotics · flap or STSG alone

Treated with wide resection alone	
SCC · **adamantinoma** · **chordoma** · **chondrosarcoma** · **parosteal OS**	✓

– – Active space – –

6 Complications

Prognosis	**nodal metastases + high histologic grade** = poor · **more metastatic than BCC** · lymphatic → nodes, then lungs, liver, brain, bone, skin

How it's asked

"SCC from chronic sinus drainage or burn scars?"	→ **Marjolin's ulcer** ✗ Wrong options: Curling · Cushing · Mooren · chancroid
"67-yr-old, chronic calcaneal osteomyelitis sinus, squamous pearls?"	→ **wide resection** ✗ Wrong options: antibiotics · flap · STSG · RT
"Most common malignancy of the hand?"	→ **SCC** ✗ Wrong options: epithelioid sarcoma · lung mets · chondrosarcoma
"El Paso construction worker, scaly indurated hand lesion, SCC?"	→ **wide local resection** ✗ Wrong options: transradial amputation · RT
"Subungual 'haematoma', bone involved, digit amputated?"	→ **SCC** ✗ Wrong options: BCC · epithelioid sarcoma
"Kidney transplant, UV exposure, dorsal hand plaque, keratin nodules?"	→ **SCC** ✗ Wrong options: BCC · melanoma · seborrhoeic keratosis

SYNOVIAL CHONDROMATOSIS

– Pathology · 61 / 66 –

1 Basics

Diagnosis – benign metaplasia

Nature	**benign metaplastic** proliferation of synovium → **cartilage metaplasia** → **multiple intra-articular loose bodies** (soft synovial tissue → firm cartilage nodules → calcify → ossify) ✗ Wrong options: neoplasia · reactive arthritis · OA · vascular
Metaplasia	one non-cancerous phenotype → another (cf. Barrett's oesophagus – but Barrett's has high malignant risk)
Demographics	**30–50 yr** (3rd–5th decades) · **M:F 2:1**
Site	**knee** commonest: **70%** (OB219396) · **50–65%** (TO) · then **hip (20%)**, shoulder, elbow, ankle, wrist · also tendon sheaths, **bursae** (e.g. over an osteochondroma) · not purely intra-articular
Genetics	occasional **chromosome 6** abnormalities

Primary vs secondary (TO)

	Primary (Reichel syndrome)	Secondary
Frequency	Less common	**More common**
Joint	One large joint, no underlying pathology	Knee/hip/shoulder with OA, trauma, inflammation, osteonecrosis, neuropathy
Age · sex	3rd–5th decade · **M >2:1**	Older
Bodies	Many, uniform · **"ring and arc"** calcification	**Fewer, larger, variable size** · **concentric rings**
Origin	Synovial metaplasia	Shed hyaline cartilage implants in synovium

Phases (TO)

Phase	Pathology
1	Active synovial cartilage formation, **no loose bodies**
2	Synovial proliferation **+ shedding** of loose bodies (commonest MRI pattern)
3	Many loose bodies, **no active synovial disease**

3 Presentation

Slow pain + swelling (worse with activity) · **mechanical** symptoms – catching, locking, **stiffness** · ↓ ROM (e.g. 0–40°), crepitus, effusion, quadriceps atrophy · ± warmth, erythema

– – Active space – –

4 Diagnosis & Classification

Key radiology findings

X-ray (AP + lateral) **multiple stippled / spheroidal calcified loose bodies** · **ring-and-arc** chondroid calcification (pathognomonic) · extrinsic bone erosion possible · **20% radiographically occult** (mineralization is time-dependent)



Stippled calcified loose bodies, hip



Multiple calcified masses about the knee

MRI early disease: **cartilage nodules only on MRI** · lobular · nodules **low T1, high T2** · later **signal drop-out** (calcification)

Arthroscopy finds **far more nodules than the X-ray shows** (unmineralized)

Histology

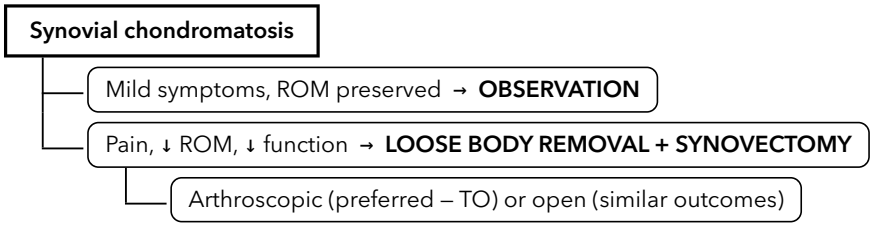
Discrete hyaline cartilage nodules at various stages of calcification/ossification · **clustered** chondrocytes with **mild atypia**, binucleate cells, occasional mitoses

Differential

Lesion	Key distinction
PVNS	**No mineralization**, bloody effusions, low T1+T2, blooming · CSF1
Synovial sarcoma (periarticular)	Rarely intra-articular, diffuse mineralization, no loose bodies
Synovial haemangioma	Diffuse mineralization (phleboliths), no loose bodies
Lipoma arborescens	No calcification
OA loose bodies	**Fewer, smaller**, mobile
Gout	Urate crystals, erosions, no calcified bodies
Septic arthritis	Acute, no cartilage on biopsy
Chondroblastoma	**Osseous**, cobblestone chondroblasts

5 Management

Treatment & recurrence



Benign metaplasia → symptomatic treatment only
✗ Wrong options: wide excision · chemo · disarticulation

Hip, young adult, joint space preserved	**arthroscopic loose body removal** (+ synovectomy, chondroplasty, labral debridement) ✗ Wrong options: THA · resurfacing · 600 cGy RT
Knee	"open, complete synovectomy" (OB7782) · complete synovectomy role **debated** (OB219396) · may prevent degenerative change
Recurrence	**up to 30%** (OB219396) · **3–23%** overall, arthroscopic hip **7.1–25%** (OB220967) · due to **incomplete synovial removal** · not joint-specific or predicted by ring-and-arc

6 Complications

Malignant transformation (synovial chondrosarcoma)

Incidence	**1%** to **6.4%** (TO) · **~5%** (OB) · long-standing disease with **multiple recurrences**

– – Active space – –

Grade mostly low (grade 1: 20%, grade 2: 70%)

Feature	Recurrence	Malignancy
Extensive calcified mass in/around joint	Yes	Yes
Soft-tissue / bursal extension	Yes	Yes
Deep **extrinsic bone erosion**	Yes	Yes
Cortical destruction + marrow invasion / permeation	No	✓ **(most reliable)**
Rapid growth, rapid clinical decline	–	Suspicious → biopsy
Lung mets	No	Definitive

Histology of malignancy marked **myxoid** matrix · **sheet-like / "lining up"** chondrocytes (loss of clustering) · **spindling** · necrosis · marrow permeation · moderate–marked atypia

Untreated articular destruction, chronic pain

How it's asked

"45-yr-old, knee trauma, nodules on X-ray, far more loose bodies at arthroscopy?"	→ **synovial chondromatosis** (key absent in source; ✗ Wrong options: PVNS · OCD · OA) **(TO)**
"47-yr-old groin pain, hip loose bodies, cartilage nodules – true statement?"	→ **benign metaplastic process, symptomatic treatment** ✗ Wrong options: wide excision · neoplastic
"30-yr-old painful ankle, calcified loose bodies?"	→ **synovial chondromatosis** ✗ Wrong options: PVNS · lipoma arborescens · haemangioma · synovial sarcoma
"55-yr-old swollen knee, multiple calcified masses, benign chondroid biopsy?"	→ **synovial chondromatosis** ✗ Wrong options: OA loose bodies · chondroblastoma
"Stiff hip, arthroscopy or open debridement – underlying process?"	→ **metaplasia**
"Knee ROM 0–40°, calcified bodies – true?"	→ **benign metaplastic process** ✗ Wrong options: hip commonest · purely intra-articular · CSF1 · urate
"Best feature separating recurrence from malignant change?"	→ **marrow invasion (TO)** ✗ Wrong options: soft-tissue extension · extrinsic erosion · calcified mass
"Painful hip chondromatosis, young adult?"	→ **arthroscopic loose body removal** ✗ Wrong options: THA · 600 cGy
"Recurrence 3 yr after arthroscopic partial synovectomy?"	→ **recognized, from incomplete synovial removal** ✗ Wrong options: pathognomonic of chondrosarcoma · hip >50%

SYNOVIAL SARCOMA

– Pathology · 62 / 66 –

1 Basics

Epidemiology

Nature malignant, usually **high-grade** STS · **unknown cell of origin** · spreads along fascial planes
✗ Wrong options: type II synoviocyte (misnomer from microscopic resemblance)

Frequency **5–10%** of STS (Orthobullets, OB5861) · **6%** (OB7586) · **7% / 6–8%** (TO) · **4th commonest** STS (OB8312)

Age **15–40 yr** · **commonest STS in young adults/adolescents** (16% of STS at that age; RMS 15%) · 30% of childhood STS · mean <30

Sex M ≈ F (**1.2:1**; 1.1:1)

Site **periarticular, rarely intra-articular** (only **10%** in a major joint) · **lower limb 60%** · **knee / popliteal fossa** commonest → ankle, foot · shoulder, elbow, axilla · **commonest sarcoma of the foot**

Genetics – t(X;18) SYT-SSX

t(X;18)(p11;q11) in **>90%** · **SYT (SS18) on 18 + SSX1 / SSX2 / SSX4 (rare) on X → SYT-SSX** fusion · detect by **RT-PCR**

Mechanism fusion binds **BAF complex**, displaces tumour suppressor **BAF47** → activates **Sox2**

	SYT-SSX1	**SYT-SSX2**
Frequency	**More common (60%)**	40%
Histology	**Biphasic**	**Monophasic**
Sex	M:F 1:1	M:F 1:2
Presentation	Larger, with mets	Smaller, without mets
Survival	**Worse** (Orthobullets, OB213030 answer)	**Better**

– – Active space – –

Conflict TO: **monophasic histology = adverse**, biphasic favourable (and OB213030's explanation says biphasic survives better) · Orthobullets: SSX1 (biphasic) worse · **fusion type = most important prognostic factor** (Orthobullets)

Fusion / translocation	Tumour
SYT-SSX, t(X;18)	**Synovial sarcoma**
EWS-FLI1, t(11;22)	Ewing
PAX3-FKHR, t(2;13)	Alveolar RMS
EWS-ATF1, t(12;22)	**Clear-cell sarcoma**
FUS-CHOP / FUS-DDIT3, t(12;16)	**Myxoid liposarcoma**
t(9;22)	Myxoid chondrosarcoma · BCR-ABL CML
COL1A1-PDGFB	DFSP
Ring chromosome 12	WD liposarcoma, MFH (ring chromosomes also parosteal OS)
Monosomy 14, 22	GIST

Genes & lab markers

IHC **vimentin +**, **epithelial membrane antigen (EMA) +**, **keratin / cytokeratin +** (nearly all biphasic, many monophasic), sporadic **S-100** · only synovial sarcoma (vs MFH) stains vimentin + EMA · TO: CD56, CD99, HMW keratin (poorly differentiated)

3 Presentation

Slow-growing mass near a joint · painless or painful · mistaken for ganglion/bursitis · large deep >5 cm

Nerve signs e.g. popliteal mass → plantar tingling, weak push-off, ↓ knee flexion

Check regional lymph nodes

4 Diagnosis & Classification

Diagnosis & features (calcification, foot, histology)

X-ray **soft-tissue calcification** – **15–20%** (OB6293) · **1/3** (TO) · irregular (vs round phleboliths of haemangioma) · may mimic HO · commonest STS with calcification
✗ Wrong options: Ewing · clear-cell · MFH

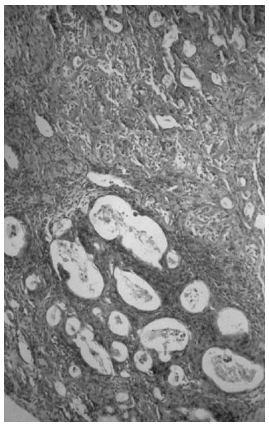


Soft-tissue calcification in axillary synovial sarcoma

US	non-specific hypoechoic mass
CT	well-circumscribed, calcification · **CT chest** staging
MRI (gold standard)	heterogeneous, **dark T1 / bright T2** · **T2 "triple sign"** (high: necrosis/cyst · intermediate · low: calcification/fibrosis) · cysts in up to 20% · well-defined → misread as benign
Biopsy	core needle or incisional · **not FNA** (no architecture)
Workup	**MRI of the limb + CT chest** · nodes examined · **no bone scan** · not US only

Histology

Biphasic (commonest) – **epithelial cells** (gland, nest, cyst-like; pale columnar) + **spindle cells** · **monophasic** – spindle only · spindle cells small, uniform, sheets/fascicles, scant cytoplasm, dark atypical nuclei



Biphasic glands + spindle cells

– – Active space – –

Histology distractor	Lesion
Round cells in fibrous septae, giant cells	Alveolar RMS
Nests of clear cells, fibrous septa, giant cells	Clear-cell sarcoma
Signet-ring lipoblasts, **chicken-wire** capillaries	Myxoid liposarcoma
Small round blue cells, rosettes	Ewing
Granulomatous look	Epithelioid sarcoma
Storiform pleomorphic	MFH
Haemosiderin macrophages, villi	PVNS
Osteoid · "blue balls of cartilage"	OS · chondrosarcoma

5 Management

Standard — **wide resection + perioperative RT** (neoadjuvant or adjuvant – no consensus) · RT ↓ local recurrence and **improves OS + DSS** (Orthobullets)

Chemotherapy — **one of the more chemosensitive STS** · may improve local control + OS · reserved for **high-risk, metastatic, younger** patients · metastatic → **doxorubicin + ifosfamide (TO)**

Large midfoot tumour — transtibial amputation (STS chapter)

6 Complications

Lymph node spread & staging

Metastasis — **30–60%** · **lung commonest** · **lymph nodes** (with angiosarcoma, clear-cell, RMS, epithelioid) → **sentinel node biopsy** · nodes worse than none but better than lung mets · ↑ with large, deep, high grade

Only RACES member with SYT-SSX1 → synovial sarcoma

vs myxoid liposarcoma t(12;16) — synovial has **fewer bony mets** · but **more nodal mets**, **more local recurrence**, commoner in foot

Liposarcoma mets by grade — WD **<1%** · myxoid **10–30%** · high grade **>50%**

Prognosis — poor · **5-yr 50–60%** · **10-yr ~25%** · TO 5-yr ~60% (61–73%) · factors: size, grade, location, age, margins, **fusion type**

How it's asked

"t(X;18) is found in?"	→ **synovial sarcoma** ✗ Wrong options: myxoid liposarcoma · Ewing · clear-cell
"Fusion protein of synovial sarcoma?"	→ **SYT-SSX** ✗ Wrong options: BCR-ABL · EWS-FLI1 · FUS-CHOP · COL1A1-PDGFB
"36-yr-old firm fixed foot mass – confirmatory cytogenetics?"	→ **X;18** ✗ Wrong options: 12;22 (clear-cell)

"16-yr-old painless posterior arm swelling, cystic, biphasic?"	→ **(X;18)(p11;q11)**
"46-yr-old medial arm mass, biphasic?"	→ **t(X;18)**
"13-yr-old thigh mass, t(X;18)?"	→ **synovial sarcoma** ✗ Wrong options: myxoid chondrosarcoma · GIST
"37-yr-old slowly enlarging foot mass — genetic defect?"	→ **SYT-SSX fusion** ✗ Wrong options: p53 · ring chromosome
"Soccer player, painless popliteal mass, plantar tingling?"	→ **SYT-SSX1** ✗ Wrong options: EWS-ATF1 · FUS-DDIT3 · PAX3-FKHR
"Great-toe mass, SYT-SSX1 vs SSX2?"	→ **biphasic** ✗ Wrong options: smaller · monophasic · less common
"t(X;18) vs t(12;16) sarcoma — lower rate of?"	→ **bony metastases**
"Painful axillary mass, vimentin + EMA +?"	→ **synovial sarcoma** ✗ Wrong options: MFH · liposarcoma
"22-yr-old painful foot mass, glands + spindle cells?"	→ **synovial sarcoma** ✗ Wrong options: Morton's neuroma · desmoid
"Synovial sarcoma EXCEPT?"	→ **"named because origin is the type II synovial cell"**
"Commonest sarcoma of the foot?"	→ **synovial sarcoma** ✗ Wrong options: epithelioid (hand/forearm)
"Which histology is t(X;18)?"	→ **biphasic epithelial + spindle** ✗ Wrong options: alveolar RMS · clear-cell · myxoid liposarcoma · Ewing
"Malignant tumour most often with soft-tissue calcification?"	→ **synovial sarcoma** ✗ Wrong options: haemangioma (benign phleboliths)
"40-yr-old painless ankle mass, biphasic?"	→ **synovial sarcoma** ✗ Wrong options: clear-cell · epithelioid · PVNS
"NOT a characteristic of synovial sarcoma?"	→ **usually intra-articular**
"29-yr-old forefoot pain, heterogeneous T2-bright mass?"	→ **synovial sarcoma** ✗ Wrong options: plantar fibromatosis · Morton's
"SYT-SSX1 tumour that spreads to nodes?"	→ **synovial sarcoma**
"Thigh SYT-SSX sarcoma — further workup?"	→ **MRI femur + CT chest** ✗ Wrong options: bone scan · US

– – Active space – –

TELANGIECTATIC INTRAMEDULLARY OSTEOSARCOMA

– Pathology · 63 / 66 –

1 Basics

Nature	rare (**4% of OS**), malignant, **high-grade** OS
Demographics	**5–25 yr · M > F**
Site	**same as ABC – proximal humerus, proximal femur, distal femur, proximal tibia** · knee commonest in stems
Risk factor	prior **radiation**

Genes & lab markers

Tumour suppressors	**Rb-1**, **p53** · **oncogenes – HER2/neu**, **c-myc**, **c-fos**
Prognostic labs	↑ **LDH**, ↑ **ALP** → poor

3 Presentation

Pain (ABC-like), night pain, poor NSAID relief, limp · swelling, tenderness

Pathologic fracture **25%**

4 Diagnosis & Classification

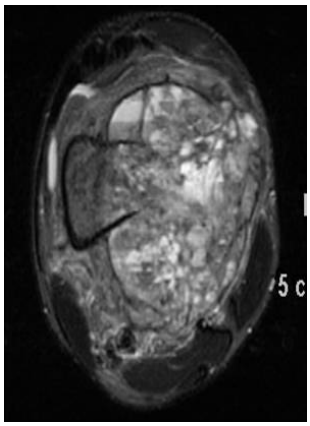
Key radiology findings

X-ray **lytic, destructive, expansile**, eccentric, septae · **cortex may be disrupted** · periosteal reaction possible



Lytic expansile proximal humeral lesion

MRI (all cases) **fluid-fluid levels** (less defined than ABC) · **extensive surrounding oedema** · soft-tissue component

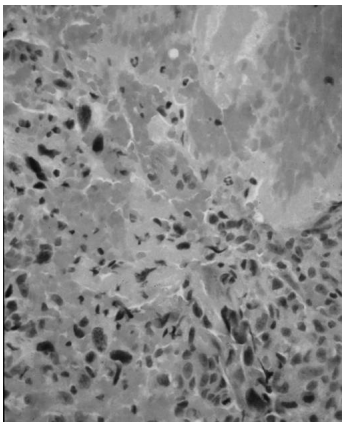


Fluid-fluid levels with oedema

Bone scan ↑ uptake

Histology

Lakes of blood mixed with malignant cells (not in ABC) · high-grade cells with **mitoses** in intervening cellular septa · **less osteoid** than conventional OS · gross: **"bag of blood"** · experienced MSK pathologist essential



Blood lakes with malignant cells

ABC vs telangiectatic OS

	ABC	Telangiectatic OS
Age · site	**75% <20** · same four sites	5–25 · same four sites
X-ray	Expansile, eccentric, lytic, septae, **cortical thinning**	Same, **possible cortical disruption**
MRI	**Clear** fluid levels, may extend to soft tissue	Fluid levels **less defined**, **surrounding oedema**
Histology	Blood spaces, spindle cells, **benign giant cells, no malignant cells**	**Blood lakes with malignant cells + mitoses**
Treatment	**Curettage + graft**	**Chemo + wide resection + chemo**

Other distractors GCT (females, 3rd–5th decade, curettage) · chondroblastoma (epiphyseal, cobblestone chondroblasts) · UBC (aspiration + steroid)

– – Active space – –

5 Management

Neoadjuvant chemo (8–12 wk) → wide resection (limb salvage) → adjuvant chemo (6–12 mo) · as conventional OS

Agents	**doxorubicin (adriamycin), cisplatin, methotrexate, ifosfamide**
Limb salvage	possible in **~90%**
✗ Wrong options	wide resection alone · curettage + graft (ABC) · RT · steroid injection · observation · hip disarticulation first

6 Complications

Local recurrence	**~5%** · poor prognosis · ↑ with **displaced pathologic fracture**
Good	**≥98% necrosis** after chemo · good response + wide margins ↓ recurrence
Poor	**advanced stage (most important)** · ↑ LDH, ↑ ALP · **MDR gene** (25% primaries, 50% mets)
Survival	**more chemosensitive, same survival** as intramedullary OS · localized **~70%** · metastatic **~20%**

How it's asked

"21-yr-old proximal tibial mass, blood lakes with malignant cells — first step after staging?"	→ **neoadjuvant chemotherapy** ✗ Wrong options: wide excision first · curettage · RT
"14-yr-old knee night pain, expansile lesion with blood lakes + atypia?"	→ **osteosarcoma — neoadjuvant chemo + wide resection** ✗ Wrong options: ABC (wide + RT) · GCT · wide resection alone
"18-yr-old femur, poorly defined fluid levels, blood with mitoses + osteoid?"	→ **neoadjuvant chemo → wide resection → adjuvant chemo** ✗ Wrong options: curettage · steroid injection

TUMORAL CALCINOSIS

– Pathology · 64 / 66 –

1 Basics

Diagnosis & phosphate metabolism (FGF23)

Nature	rare, **hereditary** dysfunction of **phosphate** regulation → massive **periarticular, extracapsular** calcinosis ✗ Wrong options: calcium · oxalate · chloride metabolism · renal tubular insufficiency
Demographics	**females** · **African-Americans** · onset childhood / early adolescence (stems ~30 yr)
Site	**hip (trochanteric bursa commonest)** > **shoulder** (Orthobullets) · extensor surfaces: hip, **elbow (olecranon)**, shoulder, foot, wrist (OB7499)

Genes & lab markers

FGF-23 defect → familial hyperphosphataemic tumoral calcinosis
✗ Wrong options: IDH1 (Maffucci) · dystrophin (DMD) · ACVR1 (FOP) · 1α-hydroxylase (rickets)

Labs	**hyperphosphataemia** (OB) · **calcium normal** (OB7499) · Orthobullets: Ca, PO_4, uric acid, ALP **normal or slightly ↑**; renal and parathyroid function normal/slightly ↑ · PTH may ↑
Associations	metabolic defects, collagen vascular disorders, trauma

3 Presentation

Palpable mass / swelling around a joint · slow-growing, rarely painful · pain from compression of neighbours · ROM usually preserved

4 Diagnosis & Classification

Key radiology findings

X-ray	**amorphous, cystic, lobular (round/oval), well-demarcated periarticular calcification** · bones/joints rarely involved directly

– – Active space – –



Lobular periarticular calcification



Hip tumoral calcinosis

CT	**fluid-fluid levels – calcium layering ("sedimentation sign")** · homogeneous → less active, less growth
MRI	diffuse low signal · or bright nodular pattern alternating with signal voids · fluid levels
Bone scan	intense uptake in the masses

Histology

Lobulated masses with defined capsules + thick septae · filled with **calcareous material (calcium phosphate, carbonate, hydroxyapatite) + fluid** · loculated calcific debris in fibrous stroma

Differential

Lesion	Key distinction
Calcinosis of renal failure (secondary)	Same X-ray + histology · **CRF / dialysis**, GFR, vit D, ↑ PO_4 · ~1% prevalence · any race · **medical treatment** (diet, binders, dialysate)
Calcific tendinitis	**Within tendon** · shoulder > hip > elbow > wrist > knee · **no sedimentation**
Synovial osteochondromatosis	**Intra-articular** / tendon sheath · **rings and arcs**
Myositis ossificans / HO	Not lobular · faint calcification → bone in **6 weeks** · mature cortex + medulla
Gout	Hyperuricaemia · **focal bone erosions**
Osteochondroma · chondrosarcoma · BPOP (Nora)	Bone-attached lesions

5 Management

Tumoral calcinosis

- Asymptomatic → **OBSERVATION**
- Symptomatic, ↓ function → **COMPLETE SURGICAL EXCISION (+ normalize phosphate)**

Recurrence risk — **incomplete excision** · **poorly circumscribed** lesion · excision while **actively growing**

How it's asked

"59-yr-old dialysis-dependent diabetic, painless calcified hip bump?"	→ **calcinosis of renal failure** ✗ Wrong options: synovial chondromatosis · MO · parosteal OS
"30-yr-old African-American woman, hip swelling 3 yr, lobulated calcific masses?"	→ **tumoral calcinosis** ✗ Wrong options: osteochondroma · chondrosarcoma · Nora's
"Tumoral calcinosis is a hereditary disorder of?"	→ **phosphate metabolism** ✗ Wrong options: calcium · oxalate
"Olecranon calcified nodule – lab + gene?"	→ **hyperphosphataemia; FGF-23** ✗ Wrong options: IDH1 · ACVR1 · dystrophin

– – Active space – –

UNDIFFERENTIATED PLEOMORPHIC SARCOMA

– Pathology · 65 / 66 –

1 Basics

Name	formerly **malignant fibrous histiocytoma (MFH)** · MFH removed from **WHO 2013** · high-grade, aggressive, malignant fibrogenic tumour · cell of origin unknown
Commonest STS	in **adults 55–80** (Orthobullets) · commonest STS overall / of the extremity in mature adults (OB) · **commonest radiation-associated STS** · also after **Paget's**
Incidence	1–9 / 100,000
Demographics	**55–80 yr** (Orthobullets) · mean **59**, range 32–80 (OB) · **M > F (1.2:1)** · Caucasian > African-American
Site	**lower limb (~50%) > upper limb (~20%) > retroperitoneum** · from bone in **1–5%** (see Pleomorphic Sarcoma of Bone chapter)

Genes & lab markers

PD-1 expression in **30%**

Labs	**hypoglycaemia**, ↑ WBC · ± fever
IHC – diagnosis of exclusion	**no lineage markers**: negative for desmin, myogenin, SMA, S-100, CD34, MDM2, EMA · FISH negative for t(X;18), t(12;16)

Positive marker	Excludes UPS →
Desmin, myogenin	**Rhabdomyosarcoma**
Desmin, SMA	**Leiomyosarcoma**
S-100	MPNST
CD34	DFSP
MDM2 amplification	WD liposarcoma
EMA, t(X;18)	Synovial sarcoma
t(12;16)	Myxoid liposarcoma

3 Presentation

Slow-growing, painless mass (can grow rapidly, fungate) · trauma draws attention · no ecchymosis (vs haematoma) · mass effect → NV symptoms · tight trousers, ↓ hip flexion

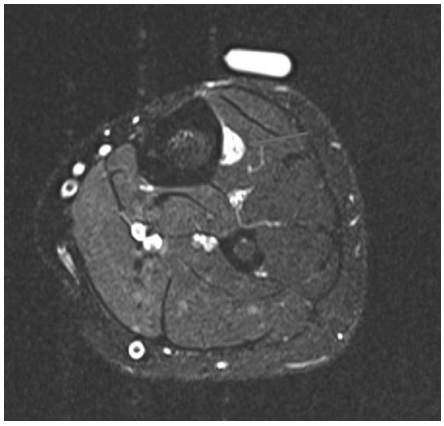
4 Diagnosis & Classification

Diagnosis (MRI, immunohistochemistry)

X-ray	orthogonal views first

MRI (mandatory) **T1 isointense to muscle (low)** · **T2 heterogeneous high** · heterogeneous enhancement, **non-enhancing necrotic centre** (poor prognosis)

– – Active space – –



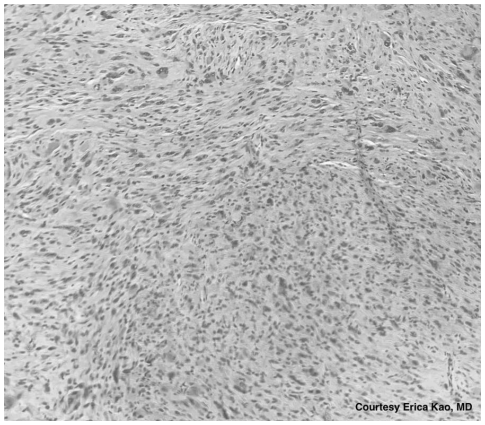
UPS — high T2 signal

MRI distractor	Lesion
Isointense to fat on all sequences	Lipoma / low-grade liposarcoma
Well-circumscribed, **homogeneous** T2 high	Myxoid lesion
Intramuscular, T2 high, **peri-lesional oedema at poles**, fat rind, thin septal enhancement	**Intramuscular myxoma** (40–70, F)

Histology

Spindle + histiocytic + multinucleated eosinophilic (bizarre) giant cells · **storiform (cartwheel)** pattern around small vessels · atypia, high mitotic index · gross: nodular grey-white

Subtype	Share
Pleomorphic	**80–85%**
Giant cell	10%
Inflammatory	<10%



Storiform pleomorphic spindle cells

– – Active space – –

AJCC staging (extremity / trunk)

Stage	Size	Nodes	Mets	Grade
IA	<5 cm	None	None	Low
IB	>5 cm	None	None	Low
II	<5 cm	None	None	Intermediate/high
IIIA	5–10 cm	None	None	Intermediate/high
IIIB	>10 cm	None	None	Intermediate/high
IVA	Any	**Regional**	None	Any
IVB	Any	Any	**Present**	Any

Surgical margins (Enneking)

Margin	Plane
Intralesional	Through tumour – pseudocapsule violated
Marginal	Through **pseudocapsule / reactive zone** (may leave satellite cells)
Wide	Tumour + **cuff of normal tissue**
Radical	Whole compartment / amputation

R classification (AJCC) — **R0** microscopically negative (goal of curative surgery) · R1 microscopic positive · R2 gross · UICC **R+1** variant

Myxofibrosarcoma — infiltrative → plan **≥3 cm** margin, frozen sections

5 Management

Treatment – wide resection & radiation

Standard — **wide resection + RT** (pre- or post-op)
✗ Wrong options: wide resection alone · RT alone · marginal ± RT · wide + chemo · debridement · intralesional debulking

RT dose — **50 Gy pre-op** · **66 Gy post-op** (Orthobullets, OB217430) · **50–60 Gy** (OB213031)

Pre-op RT — lower dose, **~30% wound complications** · post-op → more late toxicity, **radiation-induced sarcoma**

Chemo — some cases · **neoadjuvant chemo improves 5-yr RFS + OS in tumours >10 cm**

Amputation / disarticulation — only if negative margins impossible, major nerve loss, or comorbidity · limb salvage whenever reconstruction possible

Biopsy — at the treating centre · longitudinal · drain distal + in line · excised en bloc with tumour (elliptical incision)

Follow-up — close surveillance **5–10 yr**

6 Complications

Prognosis — **5-yr survival 50–60%** · aggressive

How it's asked

"64-yr-old forearm mass, FISH negative for t(X;18) + t(12;16), desmin/myogenin/CD34 negative?"	→ **high-grade pleomorphic undifferentiated sarcoma** ✗ Wrong options: alveolar RMS · myxoid liposarcoma · DFSP · synovial
"UPS MRI?"	→ **T1 isointense to muscle, T2 heterogeneous high** ✗ Wrong options: fat-isointense (lipoma) · homogeneous T2 (myxoid) · polar oedema (myxoma)
"65-yr-old deep thigh mass, pleomorphic, SMA/desmin/S100/MDM2 negative?"	→ **UPS** ✗ Wrong options: leiomyosarcoma · WD liposarcoma · MPNST · synovial
"Fungating posteromedial leg mass, necrotic centre, pleomorphic spindle cells?"	→ **wide resection + RT** ✗ Wrong options: wide resection alone · debridement
"63-yr-old enlarging thigh mass, UPS?"	→ **wide resection + radiation** ✗ Wrong options: wide + chemo · marginal + RT
"Truck driver, large anteromedial thigh UPS, no mets?"	→ **RT + wide surgical excision** ✗ Wrong options: hip disarticulation · debulking · RT alone
"Neoadjuvant RT then resection, normal tissue cuff at closest margin?"	→ **wide** ✗ Wrong options: marginal · radical

– – Active space – –

UNICAMERAL BONE CYST

– Pathology · 66 / 66 –

1 Basics

Diagnosis, location & natural history

Nature **non-neoplastic, serous (yellow) fluid-filled** cyst · aka **simple bone cyst (SBC)** · **3%** of biopsied bone tumours / of primary tumours in first 2 decades

Age **<20 yr (85%)** · peak **3–14**, mean **~9** · **M:F 2:1**

Site long bones **95%**, **central metaphysis next to physis** → migrates to diaphysis with growth

Site	Share
Proximal humerus	**50–60%** (Orthobullets) · **52%** (OB219296)
Proximal femur	**30%** (Orthobullets) · **25%** (OB219296) · 50% of femoral cysts in patients >17
Humerus + femur	**>80%** (TO) · 90% (OB5435)
Calcaneus	6% · commonest tarsal
Others	Distal tibia, ilium, spine (posterior elements), metacarpals, phalanges, distal radius · iliac wing, ribs, talus, radius → adults

Pathophysiology **venous stasis** in cancellous bone → resorption · ↑ pressure, inflammatory proteins, **↑ lysosomal enzymes** in fluid · "temporary failure of medullary bone formation near the physis" · resolves after losing contact with physis

Genes / labs none · **no labs needed**
✗ Wrong options: ALP (OS) · ESR · IgG (myeloma) · LDH (Ewing/OS)

Active vs latent

	Active	Latent
Distance to physis	**Adjacent, ~1 cm** (Orthobullets) · **<5 mm** (TO)	Normal bone between cyst and physis · **>5 mm** (TO)
Steroid injection	Indicated, but lower success	–
Curettage / fixation	**Avoid → growth arrest**	Symptomatic, failed injections, proximal femur



Latent cyst — normal bone between cyst and physis

Natural history — **shrinks near skeletal maturity, may heal after growth** · Donaldson: **none fully healed** though 87% of physes closed
✗ Wrong options: grows · malignant

3 Presentation

~80% asymptomatic · found incidentally or after fracture

Pathologic fracture — presents with fracture in **50–75%** · **63–87%** eventually fracture **(TO)** · **commonest cause of pathologic fracture in children** · minor trauma (throwing, sport)

4 Diagnosis & Classification

Key radiology findings

X-ray (diagnostic) — **central, lytic, well-demarcated metaphyseal** lesion, thin sclerotic rim · **unilocular**, **symmetric cortical thinning**, smooth endosteal scalloping · **no periosteal reaction** · **not wider than the physis** · oblong along long axis · 2–3% cross physis · multiloculated/trabeculated after fractures

Sign	Meaning
Fallen leaf / fallen fragment	Cortical fragment in dependent part after fracture — **pathognomonic** (✗ Wrong options: before fracture) **(TO)**
Trapdoor	Fragment hinged on periosteum, folds inward **(TO)**
Rising bubble	Gas in non-dependent part — pathognomonic
Doughnut (bone scan)	Cold centre, warm rim

– – Active space – –



Fallen fragment sign

CT	spine/pelvis · wall thickness, occult fracture, fracture risk · thin wall, pseudo-septa
MRI	T1 dark (**better fracture-risk predictor** than X-ray) · T2 bright · **rim enhancement** · few septae/fluid levels · use when malignancy not excluded ✗ Wrong options: double-density fluid levels (ABC)
Bone scan / PET	no role

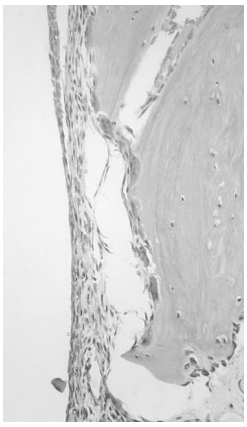
Fracture-risk predictors (TO)

Index	High risk
Bone cyst index (Kaelin & MacEwen) = cyst area ÷ diaphysis diameter²	**>4 humerus** · **>3.5 femur** (MRI better than X-ray)
Bone cyst diameter (Ahn & Park)	**>85%** on AP **and** lateral
Minimal cortical thickness	**<2 mm**

Caveat poor inter/intra-observer reliability · BCI remains best

Histology

Thin fibrous lining · fibrous tissue, **giant cells**, **haemosiderin** · few chronic inflammatory cells · **cementum spherules** (10%) · bland spindle cells · biopsy rarely needed (**fracture callus may mimic osteosarcoma**)



Thin fibrous cyst lining

Differential

	Bubbly lytic	Usually aspiration + injection	Occasionally curettage + graft
UBC	✓	✓	✓
ABC	✓	–	–
NOF	✓	–	✓
Enchondroma (solitary)	–	–	✓

Eccentric vs central (TO)	Features
SBC – central	Least likely for an **eccentric** lytic lesion
ABC – eccentric	1–6% of primary bone tumours · <20 · slight F · femur > tibia, spine, pelvis · expansile, bulging cortex · **wider than physis** · MRI fluid levels (CT shows them in 1/3) · secondary, solid (giant cell reparative granuloma), soft-tissue forms
Fibrous cortical defect – eccentric	2–15 yr · commonest benign bone lesion (with NOF up to 40%) · M 2:1 · intracortical oval lucency, posteromedial distal femur
GCT – eccentric	Epiphysis → metaphysis, **closed physes**, non-sclerotic margin, subchondral · fluid levels 10–14% (ABC component)

Other distractor	Key feature
Telangiectatic OS	Expansile, matrix, Codman, soft-tissue mass · ↑ WBC, platelets, **LDH, ALP**
FD	Ground glass
OFD	Anterior tibial cortex, osteoblastic rimming
EG	Coffee-bean nuclei, **Birbeck (tennis-racket)** granules
ABC histology	Many RBCs, giant cells, septa
Chondroblastoma · CMF · osteochondroma	Epiphyseal · chondromyxoid lobules · cartilage cap

5 Management

Pathologic fracture – sling & allow healing

UBC with pathologic fracture

- Proximal humerus / non-weight-bearing → **SLING (4–6 wk) – let fracture heal** *(no biopsy, staging, MRI or ORIF)*
  - Cyst persists → **ASPIRATION + INJECTION (steroid / marrow / DBM)**
- Proximal femur / weight-bearing, displaced, unstable, open → **CURETTAGE + GRAFT + INTERNAL FIXATION**

Immobilization sling **4–6 wk** (OB) · 3–6 wk (OB5683) · plaster cast **(TO)** · **~15% fill in** after fracture (Orthobullets) · **<10%** heal (OB6098) · **fracture healing usually does not resolve the cyst**

– – Active space – –

✗ Wrong options (initially)	biopsy, staging (CT chest, bone scan), MRI, ORIF, IM nail, steroid injection, curettage, RT/chemo, RFA

Proximal femur & calcaneus – surgical treatment

Proximal femur fracture	high refracture + malunion/nonunion nonoperatively → **ORIF + bone grafting** (hip screw + proximal femoral plate) ✗ Wrong options: bed rest · spica · traction
Femoral neck	varus malunion, AVN, growth arrest (<10%)
Calcaneus	surgery if symptomatic + **100% of ML width + ≥30% of AP length → open curettage + bone augmentation** / cannulated screws ✗ Wrong options: MRI · bone scan · CT chest · wide excision

Persistent cyst – curettage & injection

Option	Indication
Observation	Small, asymptomatic, upper limb · X-ray at 6 mo · **42%** success
Aspiration + scraping ("Q-Ropsy") ± BMA / steroid	Symptomatic cyst (TO)
Aspiration + methylprednisolone acetate	**Active** cysts, proximal humerus · several injections (young children)
Bone marrow / DBM injection	Effective, alone or combined
Curettage + bone graft (± fixation)	Symptomatic latent cyst failing injections · displaced humeral fractures · proximal femur · **recurrent / non-responding**
Curettage + ESIN + graft	**Highest success (88%)**
Decompression	IM nail, cannulated screws, Rush rod (rarely needed in upper limb)

Not recommended (TO)	**polidocanol injection** (✓ curettage, ✓ DBM injection, ✓ decompression)
Recurrence prevention	**intralesional corticosteroid + DBM + autologous marrow** ✗ Wrong options: denosumab (GCT) · BMP-2 · pexidartinib (PVNS) · bisphosphonates
Adjuvants	**liquid nitrogen / phenol → ABC**, not UBC

Healing after surgery	Criteria
Complete	**>95% opacification + cortical thickening + no pain**
Partial	80–95% opacification ± cortical thickening
Incomplete	<80% opacification, no cortical thickening

6 Complications

Recurrence	**10–30%** · risk: **age ≤10** · follow all cysts with serial X-rays
LLD / angular deformity	active cyst crossing physis or into epiphysis · surgery in active phase
Prognosis	favourable · close follow-up in active phase (recurrence, fracture, growth arrest)

How it's asked

"5-yr-old, proximal humeral SBC with pathological fracture — next?"	→ **conservative (TO)** ✗ Wrong options: curettage + graft · ORIF · nailing
"9-yr-old, painful snap throwing, humeral cyst fracture?"	→ **sling** ✗ Wrong options: staging · biopsy · IM nail
"8-yr-old, car-door injury — diagnosis + treatment?"	→ **UBC, sling and swathe** ✗ Wrong options: aspiration + cement · GCT curettage
"8-yr-old shoulder pain after collision?"	→ **sling immobilization** ✗ Wrong options: steroid injection first · RFA
"10-yr-old skateboard fall, humeral cyst fracture?"	→ **immobilization alone** ✗ Wrong options: MRI
"11-yr-old, benign-looking cyst on X-ray + CT with fracture?"	→ **immobilization** ✗ Wrong options: biopsy
"Hockey player, sling from trainer, cyst fracture?"	→ **continue sling** ✗ Wrong options: CT chest · MR arthrogram
"Lacrosse injury, non-displaced cyst fracture?"	→ **sling for comfort** ✗ Wrong options: urgent biopsy · plate
"7-yr-old pathologic humeral fracture — initial?"	→ **sling and swathe** ✗ Wrong options: needle biopsy (callus mimics OS)
"Fallen leaf sign — next?"	→ **let fracture heal, serial X-rays** ✗ Wrong options: staging + biopsy
"5-yr-old fallen flake sign?"	→ **shoulder immobilizer** ✗ Wrong options: MRI · oncology
"Gym-class throw, central lucency, thin cortex?"	→ **sling + closed fracture care**
"7-yr-old softball, UBC histology?"	→ **immobilize 4–6 wk** ✗ Wrong options: liquid nitrogen (ABC) · chemo
"7-yr-old ice-skater, proximal femoral cyst fracture?"	→ **ORIF + bone grafting** ✗ Wrong options: bed rest · spica
"12-yr-old heel pain, large calcaneal cyst?"	→ **open curettage + bone augmentation** ✗ Wrong options: MRI · wide excision
"NOT a recommended treatment of SBC?"	→ **polidocanol injection (TO)**
"12-yr-old, healed fracture, now daily pain, recurrent cyst?"	→ **curettage + bone grafting** ✗ Wrong options: IM nail · biopsy

– – Active space – –

"Adjunct to prevent UBC fracture recurrence?"	→ **steroid + DBM + autologous marrow injection** ✗ Wrong options: denosumab
"Fallen fragment sign seen in?"	→ **simple bone cyst (TO)** ✗ Wrong options: GCT · ABC · FD
"Trapdoor sign?"	→ **SBC (TO)**
"Eccentric lytic lesion — least likely?"	→ **SBC (TO)** ✗ Wrong options: ABC · GCT · FCD
"Fallen fragment sign is pathognomonic of?"	→ **fracture of UBC (TO)** ✗ Wrong options: SBC before fracture
"15-yr-old calcaneal cyst — course at maturity?"	→ **decreases, may disappear**
"Commonest site in children?"	→ **proximal humerus**
"7-yr-old, prior biopsy, metaphyseal cyst?"	→ **UBC** ✗ Wrong options: ABC (RBCs) · chondroblastoma · CMF
"13-yr-old distal femoral cyst, normal labs — next lab?"	→ **none required**
"16-yr-old proximal femoral cyst, fibrous wall?"	→ **UBC** ✗ Wrong options: ABC (double-density levels) · FD · OFD · EG
"Femoral cyst fracture fixed — commonest UBC site?"	→ **proximal humerus**
"Tennis player, fallen leaf — histology?"	→ **fluid-filled fibrous-lined cavity, occasional giant cells**

TUBERCULOSIS OF HIP

– Pathology · TO notes –

1 Basics

Epidemiology

Rank	**2nd commonest osteoarticular TB** after spine · ~**15%** of osteoarticular TB (TO)
Age	mostly **first three decades** (TO)
Always secondary	to **pulmonary / lymph node / visceral** focus · spread **haematogenous** or direct extension (TO)
Diagnostic dilemma	stages mimic many hip disorders (TO)

Initial focus

Focus (TO)	Note
Acetabular roof	**Commonest** · joint involvement **delayed**
Epiphysis	Intracapsular → **rapid spread to joint**
Metaphysis (Babcock's triangle)	Common in **children**
Greater trochanter	Spreads via **trochanteric bursa**

Pathogenesis

Sequence (TO)	synovial hypertrophy + congestion → **subchondral necrosis, sequestrum** → **kissing lesions** on opposing surfaces → **cold abscess** perforates capsule
Cold abscess exit	**weak inferior capsule** → intrapelvic · **femoral triangle** · medial / lateral / posterior thigh · **ischiorectal fossa** (TO)

3 Presentation

Systemic	fever with **evening rise**, weight + appetite loss, cough, night sweats (TO)
Early local	insidious pain **referred to medial knee / thigh** · stiffness → limp · fullness from cold abscess (TO)
Night cries	child wakes when **protective muscle spasm relaxes in sleep** (TO)
First movement lost	**extension** (TO)
Late	LLD · deformity · stiffness · **pathological dislocation** · cold abscess / sinus · untreated → **fibrous ankylosis** in deformed position (TO)
Wandering acetabulum	destruction of head + acetabulum, constant muscle pull **migrates head proximally** (TO)

– – Active space – –

Clinical stages

Stage (TO)	Pathology	Attitude	Limb length
1 Synovitis	↑ joint fluid · terminal movements restricted	**FABER** (flexion, abduction, ER)	**Apparent lengthening**
2 Early arthritis	Cartilage involved · spasm of strong flexors + adductors	**FADIR** (flexion, adduction, IR)	**Apparent shortening**
3 Advanced arthritis	Cartilage destroyed · ↓ joint space	FADIR, exaggerated	**True shortening**
4 Advanced + subluxation / dislocation	Wandering acetabulum	Gross deformity · jog of movement (fibrous ankylosis) or excess (dislocation)	**Gross shortening**

4 Diagnosis & Classification

Key radiology findings

X-ray — early **haziness**, soft-tissue swelling · later **osteopenia**, **peripheral erosions** (TO)

Phemister triad — **juxta-articular osteoporosis + peripheral erosion + gradual joint-space narrowing** → classic osteoarticular TB (TO)

MRI — **most sensitive for early diagnosis** · marrow oedema, effusion (TO)

Tissue diagnosis

Biopsy — histopathology · **AFB stain** · **PCR + culture** · failure of non-invasive diagnosis → **tissue diagnosis mandatory** (TO)

AFB positivity — synovial fluid **< 10%** · synovial tissue **< 20%** → biopsy + PCR important (TO)

Mantoux — high **false positives** in endemic areas (TO)

Shanmugasundaram radiological classification

Type (TO)	Feature	Recovery
Normal	Synovial disease only	**92%**
Wandering acetabulum	Proximal migration of head	**29%** (worst)
Dislocating	Partial dislocation, muscle spasm	**50%**
Perthes type	Head destruction, Perthes-like	**80%**
Protrusio acetabuli	Medial wall weakness	–
Atrophic	Minimal infection, small head	–
Mortar and pestle	Saucer acetabulum + small head	–

5 Management

Conservative

Old hip spica 1.5 yr → "frame knee" stiffness → **obsolete** (TO)

Modern **ATT + traction** + gradual mobilisation ± stage-wise surgery (TO)

Traction relieves spasm · corrects deformity · maintains joint space · separates surfaces (TO)

Stage-wise treatment

TB hip – by stage (TO)

- Synovitis → ATT + TRACTION + REST → **mobilise when pain settles**
- Early arthritis → **ATT + TRACTION + SYNOVECTOMY / DEBRIDEMENT**
  (↑ vascularity → better ATT delivery; risks AVN, SCFE, iatrogenic fracture)
- Advanced arthritis → **ARTHROLYSIS + SUBTOTAL SYNOVECTOMY (keep posterior capsule) + TRACTION**
- Subluxation / dislocation → **EXCISION ARTHROPLASTY · ARTHRODESIS · THA**
  (children: ATT + traction can still heal – Sandhu)

Salvage options

	Girdlestone excision arthroplasty	Arthrodesis
For (TO)	**Squatting, cross-leg sitting** · one-time surgery	**Stable, painless hip** · good sitting · one-time
Against (TO)	**Trendelenburg gait** · **2–3 cm shortening** · stick outdoors	Secondary arthritis of **spine, opposite hip, ipsilateral knee** · no floor activities

Fusion position (TO) **flexion 30° in adults**, **1° per year of age in children** (TO – verify: usual flexion 20–30°) · **ER 5–10°** · neutral abduction–adduction

Extra-articular fusion (TO) **Brittain ischiofemoral** · **Abbott–Lucas two-stage** fusion

THA (TO) **safe in healed TB** · increasingly accepted in **active TB under ATT cover** · e.g. ATT 3 mo pre-op + 15 mo post-op, or 2 wk pre-op + 12 mo post-op

X vs Y

	Synovitis stage	Arthritis stages
Attitude	**FABER**	**FADIR**
Length	Apparent lengthening	Apparent → true shortening
Joint space	Preserved	↓
Treatment	ATT + traction	+ debridement / arthrolysis / salvage

How it's asked

"Commonest initial focus of TB hip?" → **acetabular roof** (TO)

– – Active space – –

"Night cries in TB hip — cause?"	→ **relaxation of protective muscle spasm during sleep** (TO)
"Attitude in synovitis stage?"	→ **flexion, abduction, external rotation — apparent lengthening** (TO) ✗ Wrong options: FADIR (arthritis stages)
"Phemister triad?"	→ **juxta-articular osteoporosis, peripheral erosion, joint-space narrowing** (TO)
"Shanmugasundaram type with worst prognosis?"	→ **wandering acetabulum (29%)** (TO)
"Salvage allowing squatting / cross-leg sitting?"	→ **Girdlestone excision arthroplasty** (TO) ✗ Wrong options: arthrodesis (no floor activities)